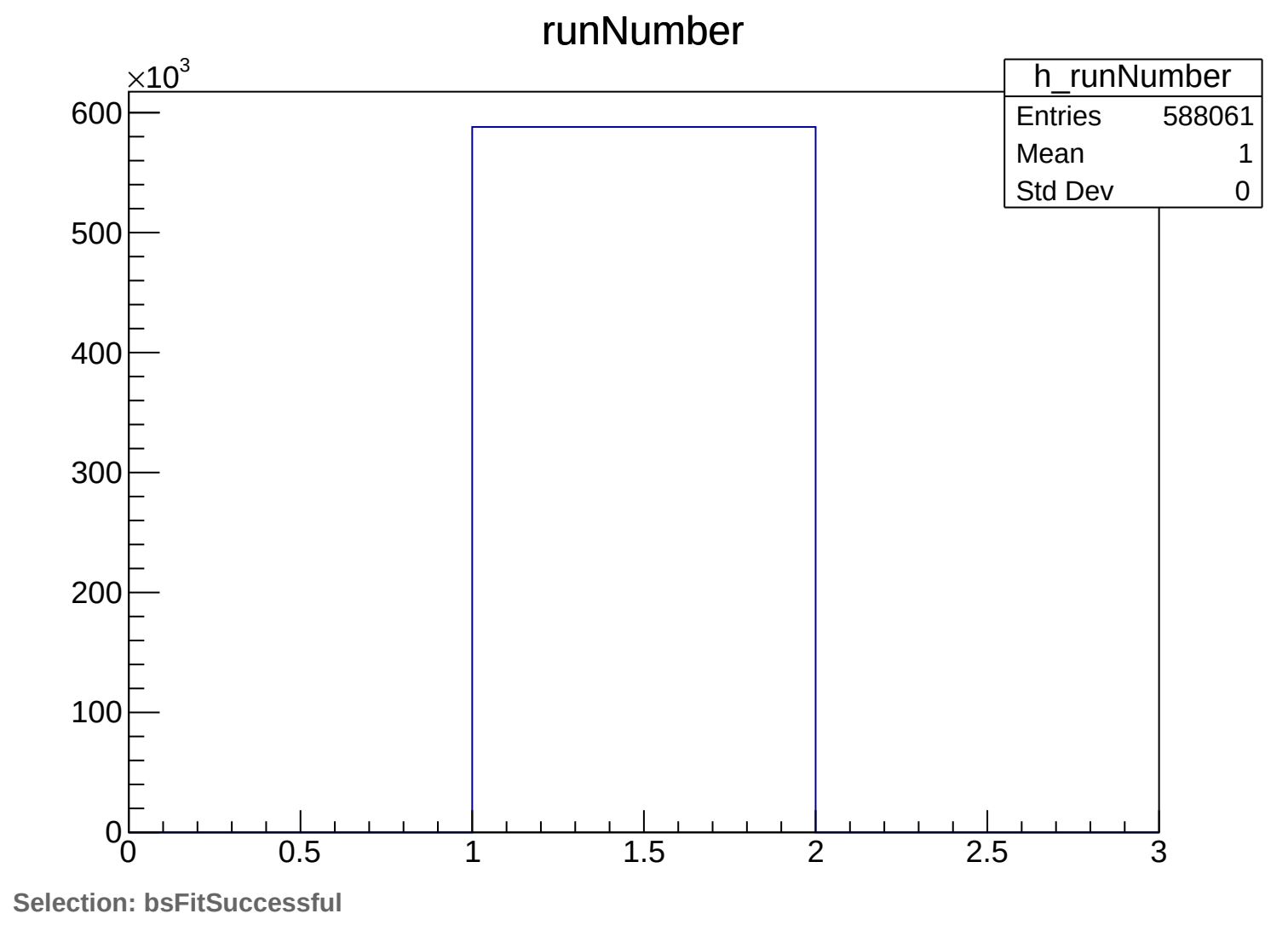
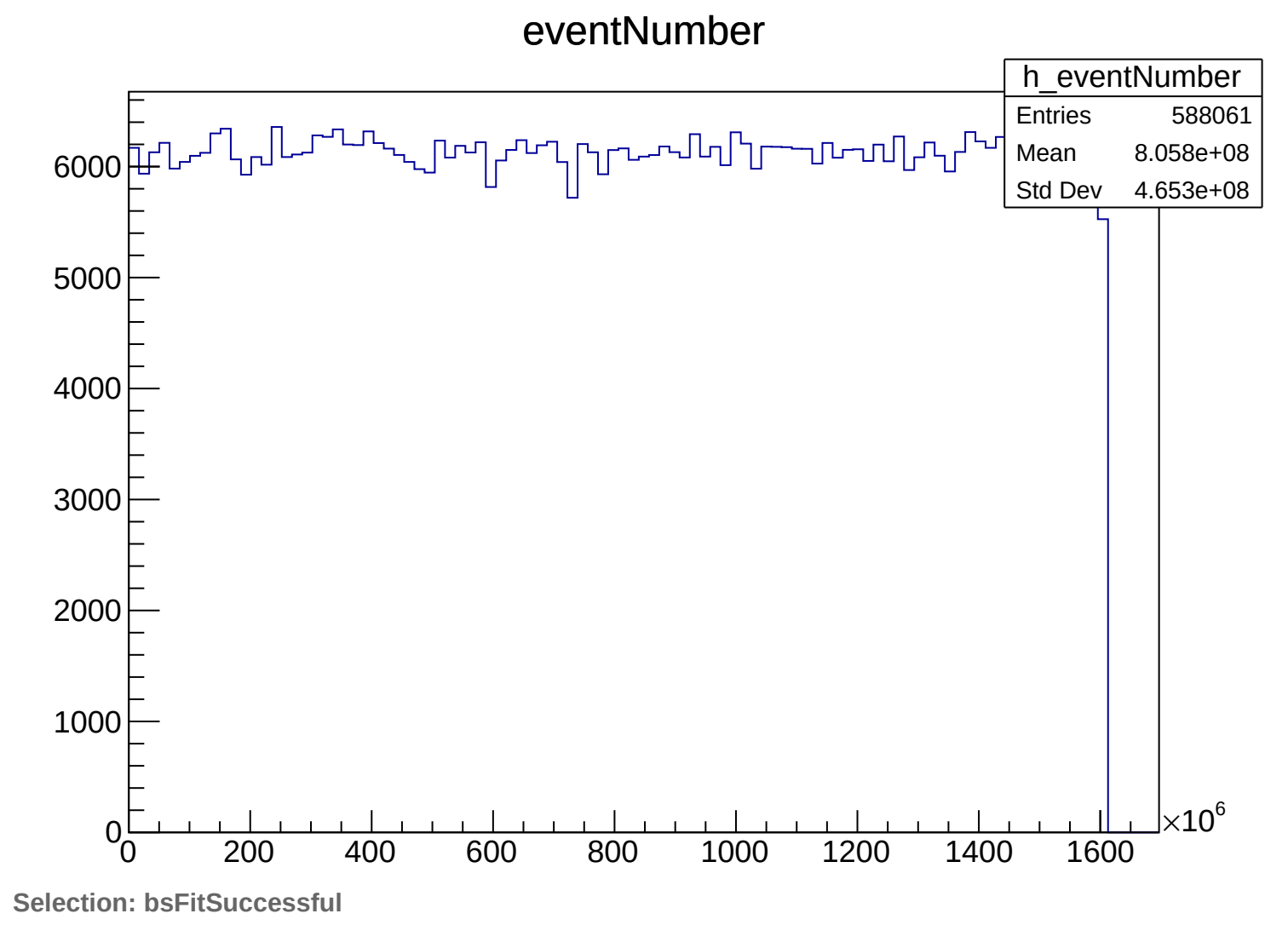
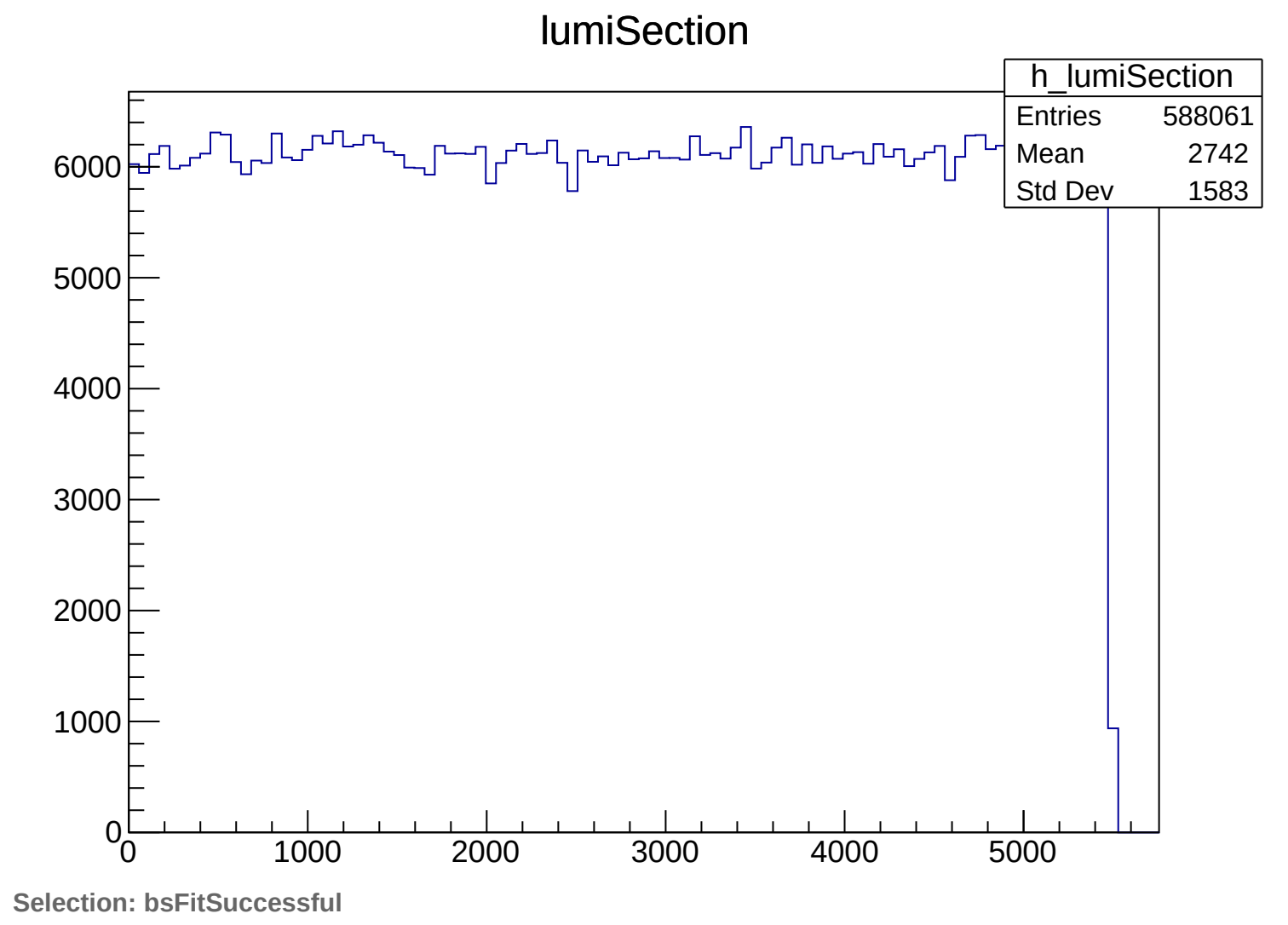
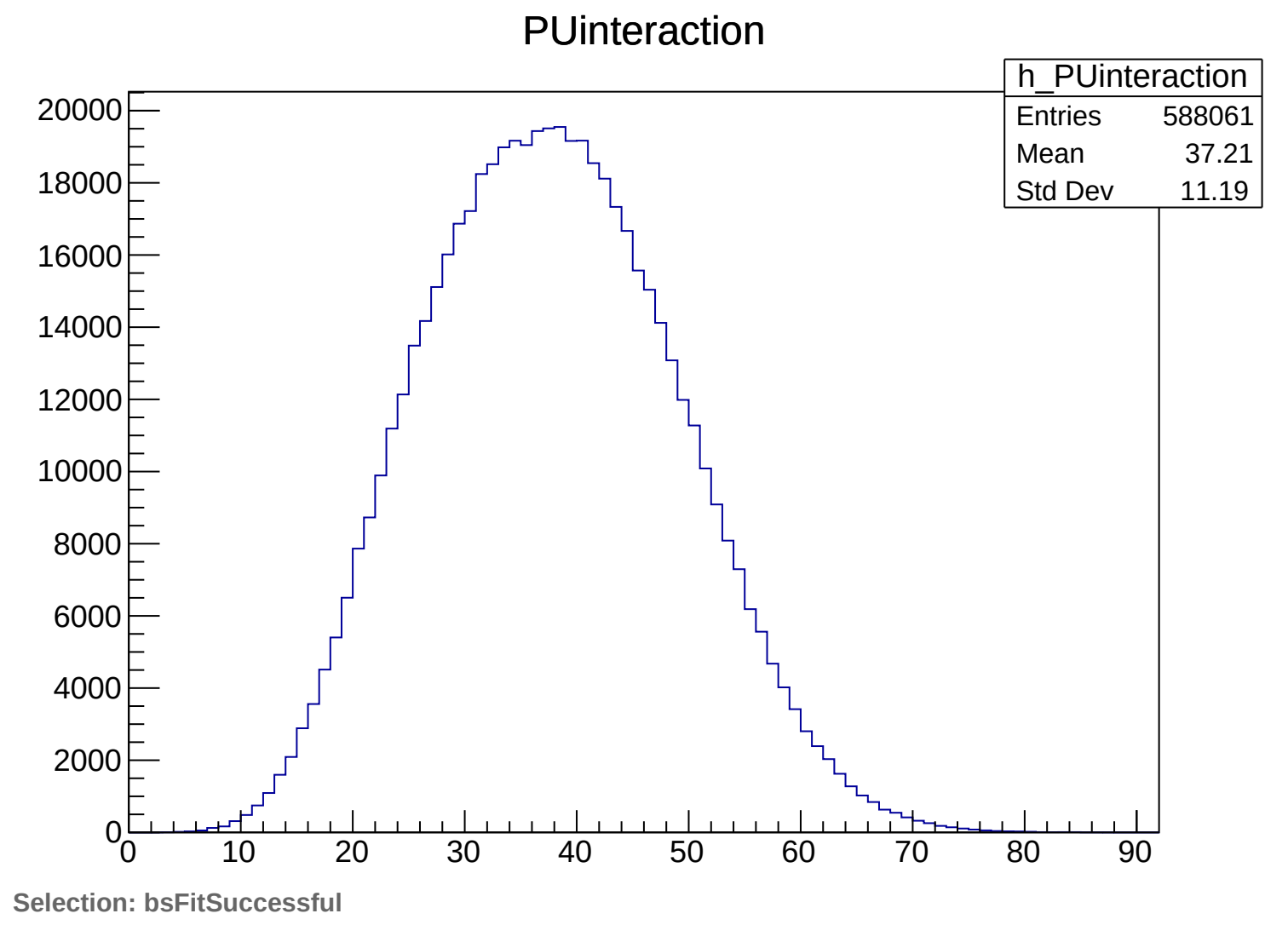
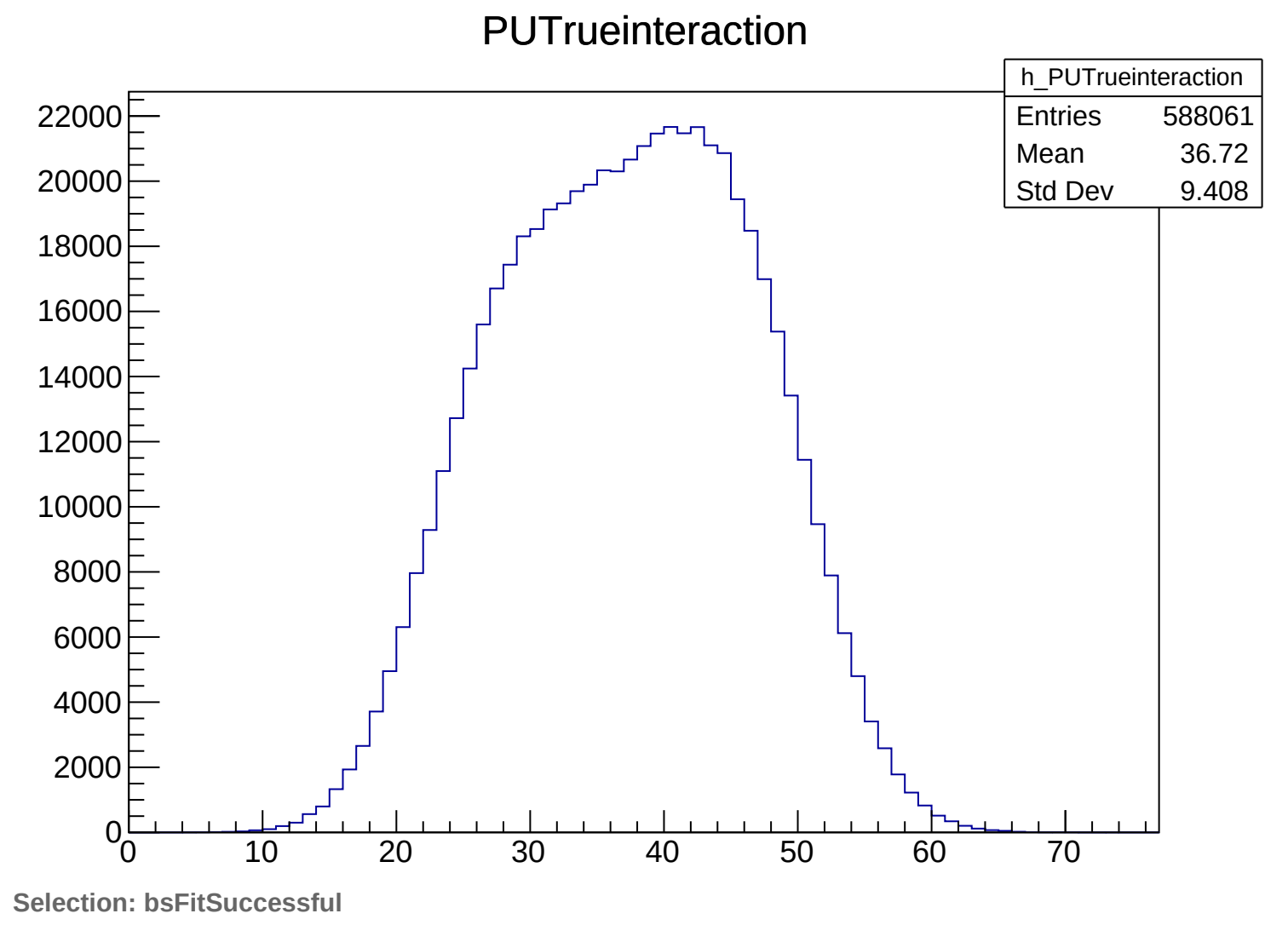
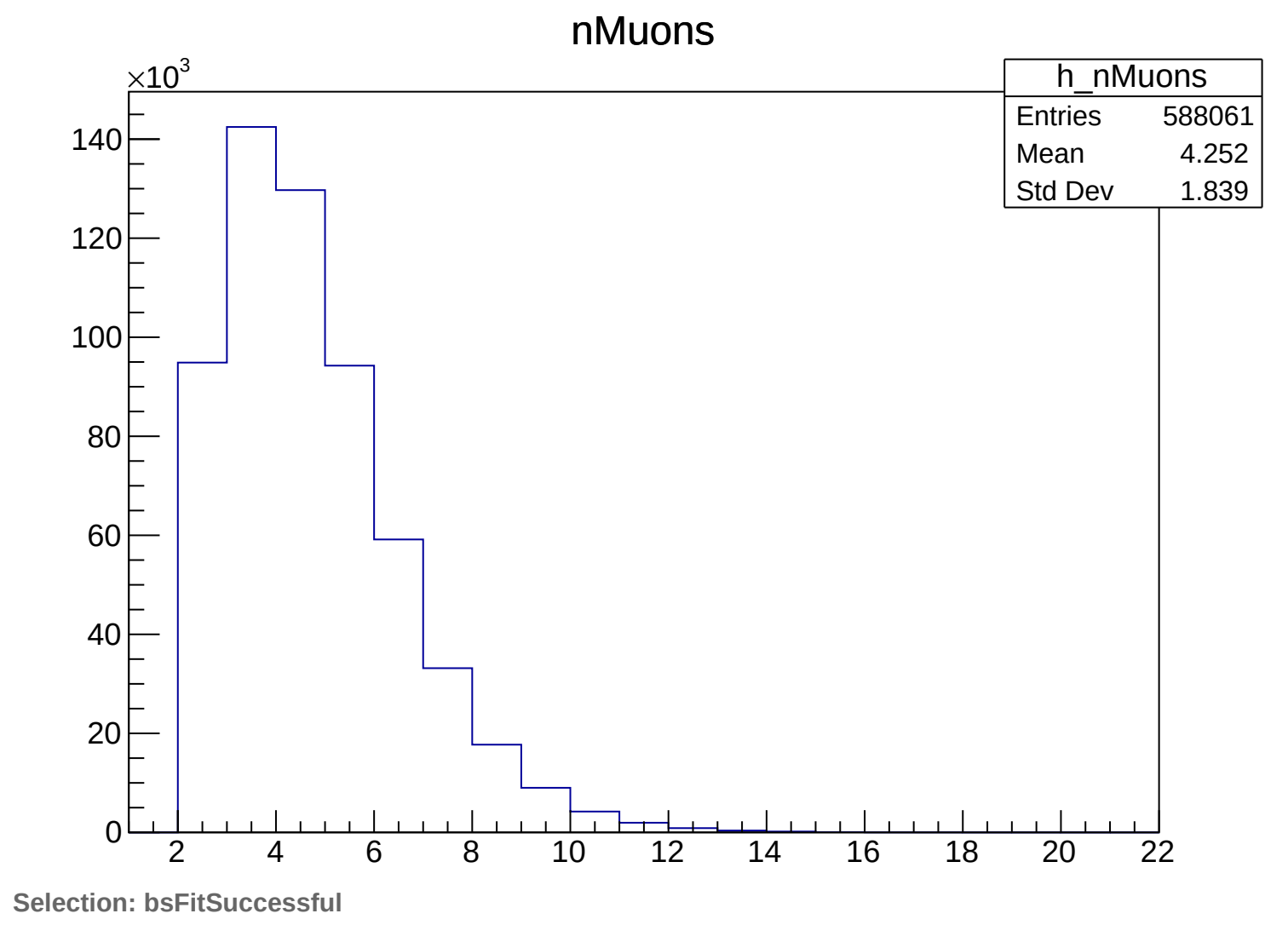

runNumber
h_runNumber
Entries 588061
Mean 1
Std Dev 0
×10^3
600
500
400
300
200
100
0
0
0.5
1
1.5
2
2.5
3
Selection: bsFitSuccessful

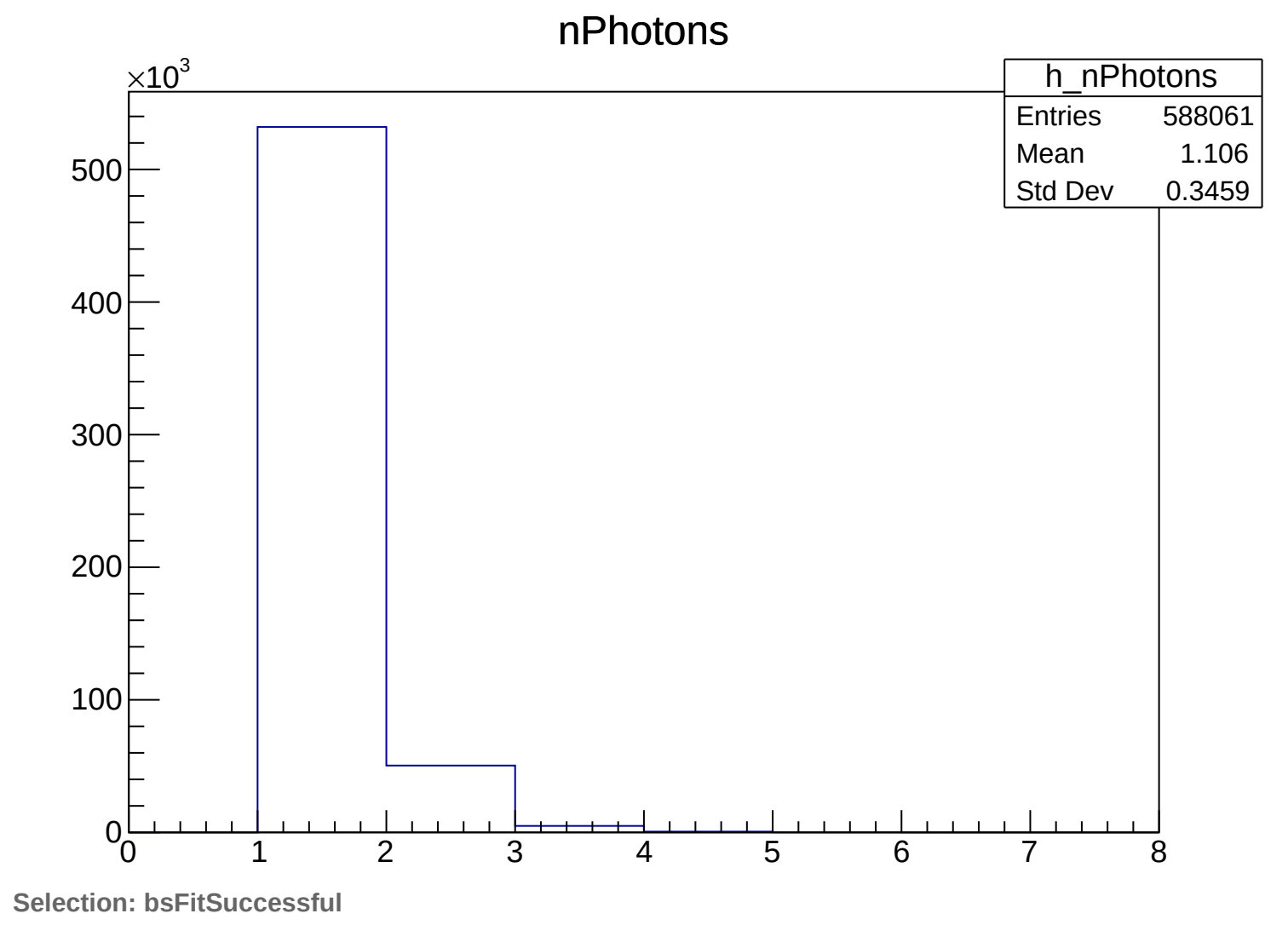

eventNumber
h_eventNumber
Entries 588061
Mean 8.058e+08
Std Dev 4.653e+08
6000
5000
4000
3000
2000
1000
0
0
200
400
600
800
1000
1200
1400
1600
×10^6
Selection: bsFitSuccessful

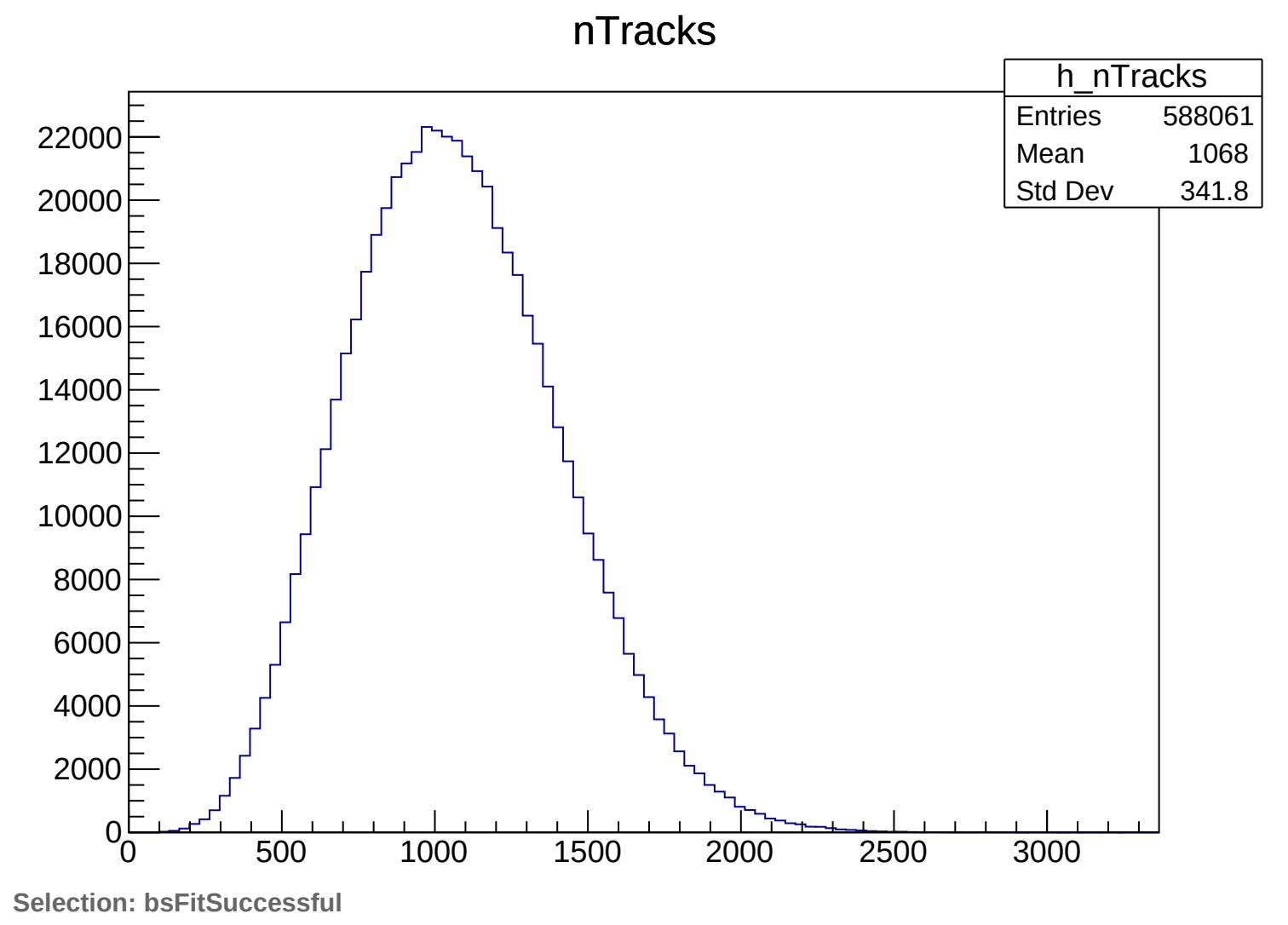

lumiSection
h_lumiSection
Entries 588061
Mean 2742
Std Dev 1583
6000
5000
4000
3000
2000
1000
0
0
1000
2000
3000
4000
5000
Selection: bsFitSuccessful

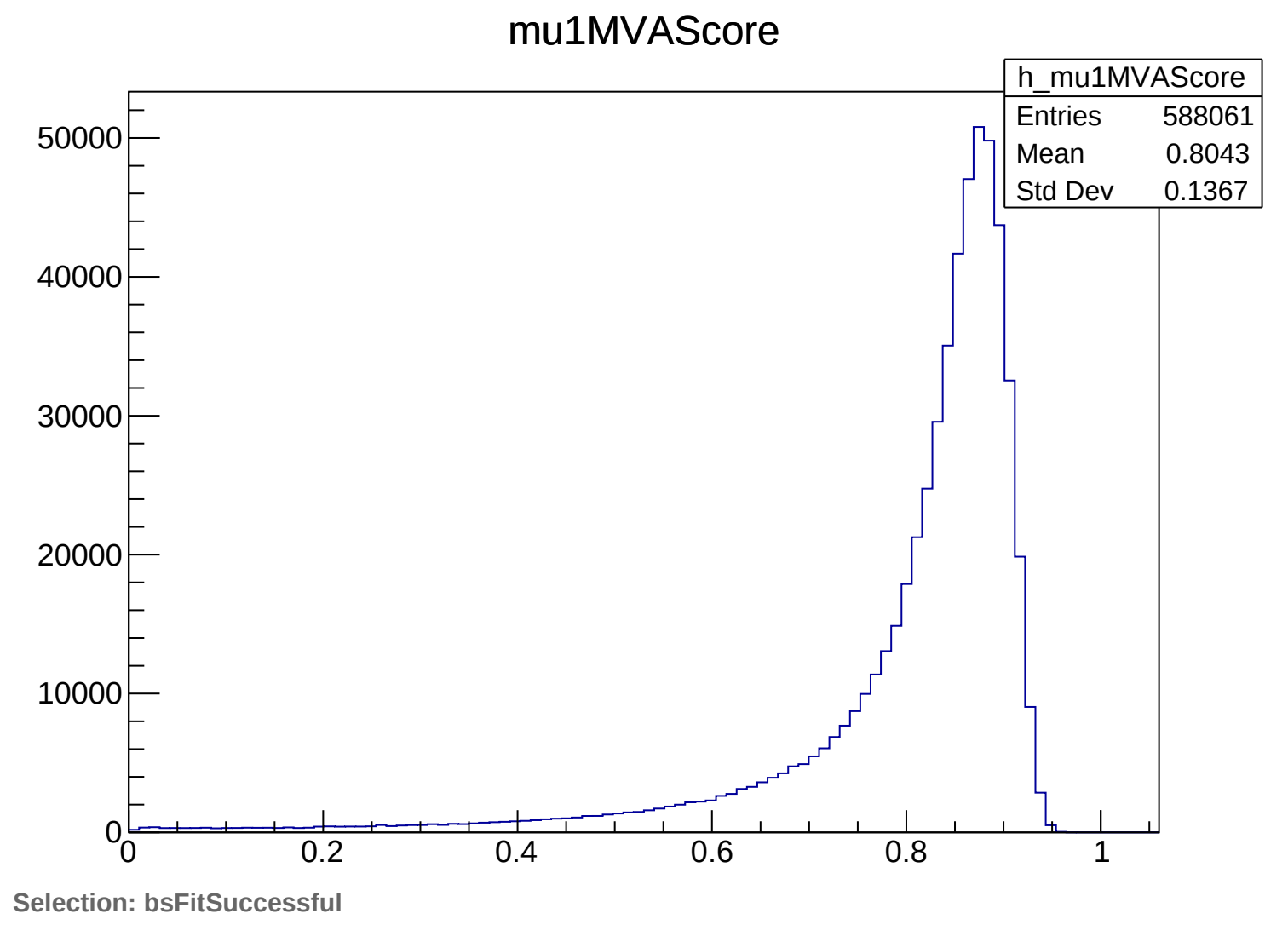

PUinteraction
h_PUinteraction
Entries 588061
Mean 37.21
Std Dev 11.19
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
10
20
30
40
50
60
70
80
90
Selection: bsFitSuccessful

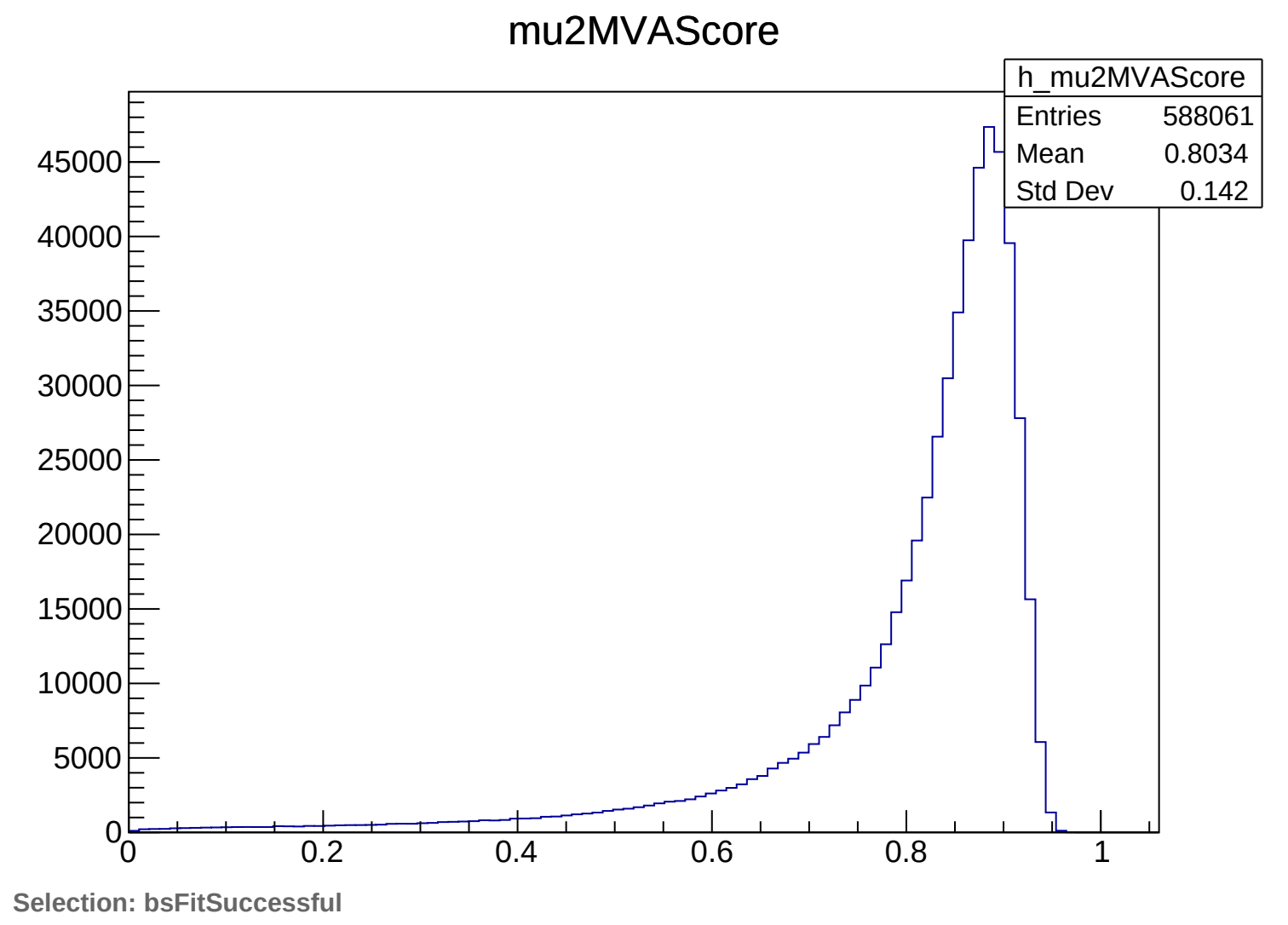

PUTrueinteraction
h_PUTrueinteraction
Entries 588061
Mean 36.72
Std Dev 9.408
22000
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
10
20
30
40
50
60
70
Selection: bsFitSuccessful

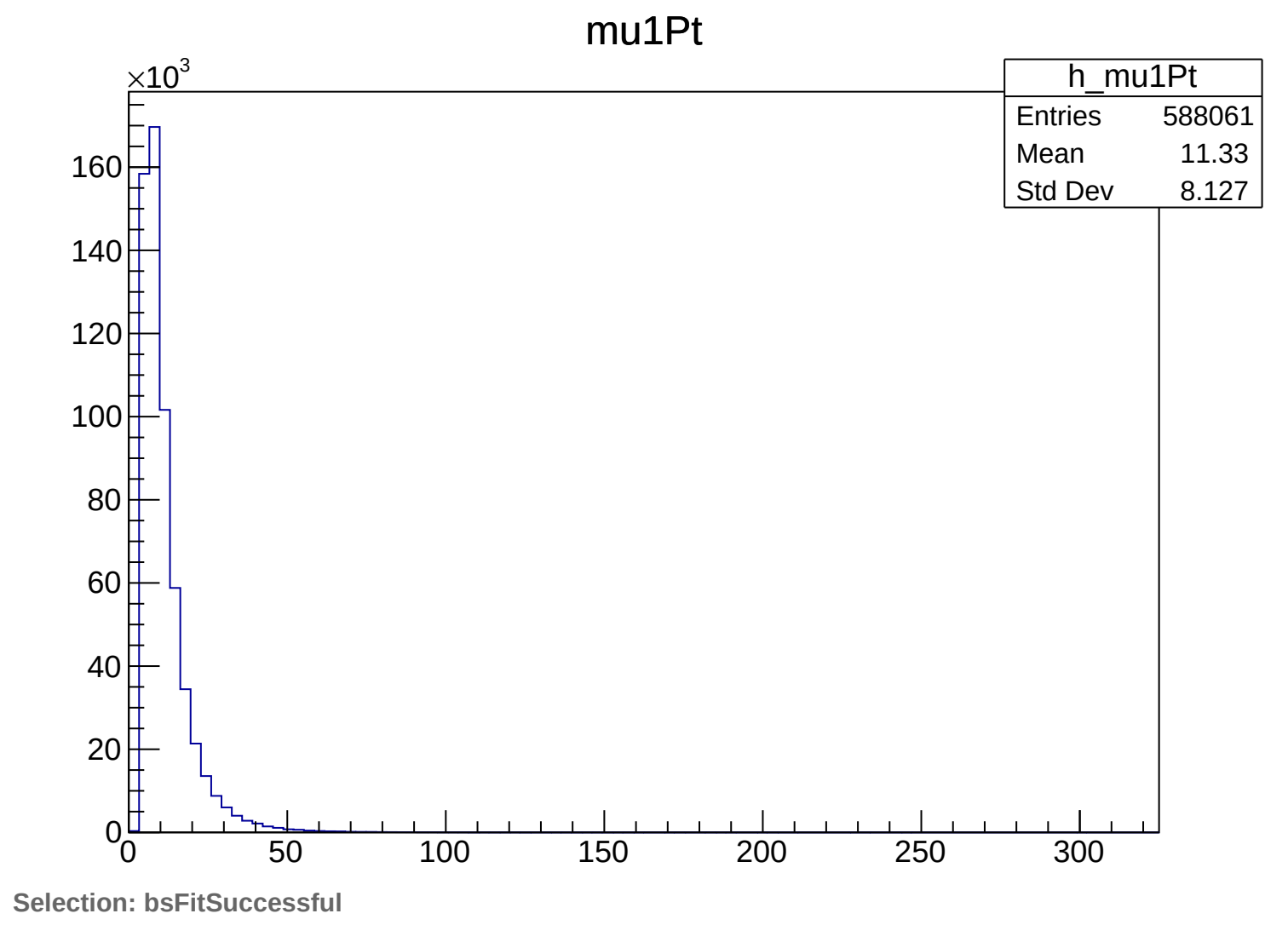

nMuons
×10^3
h_nMuons
Entries 588061
Mean 4.252
Std Dev 1.839
140
120
100
80
60
40
20
0
2
4
6
8
10
12
14
16
18
20
22
Selection: bsFitSuccessful

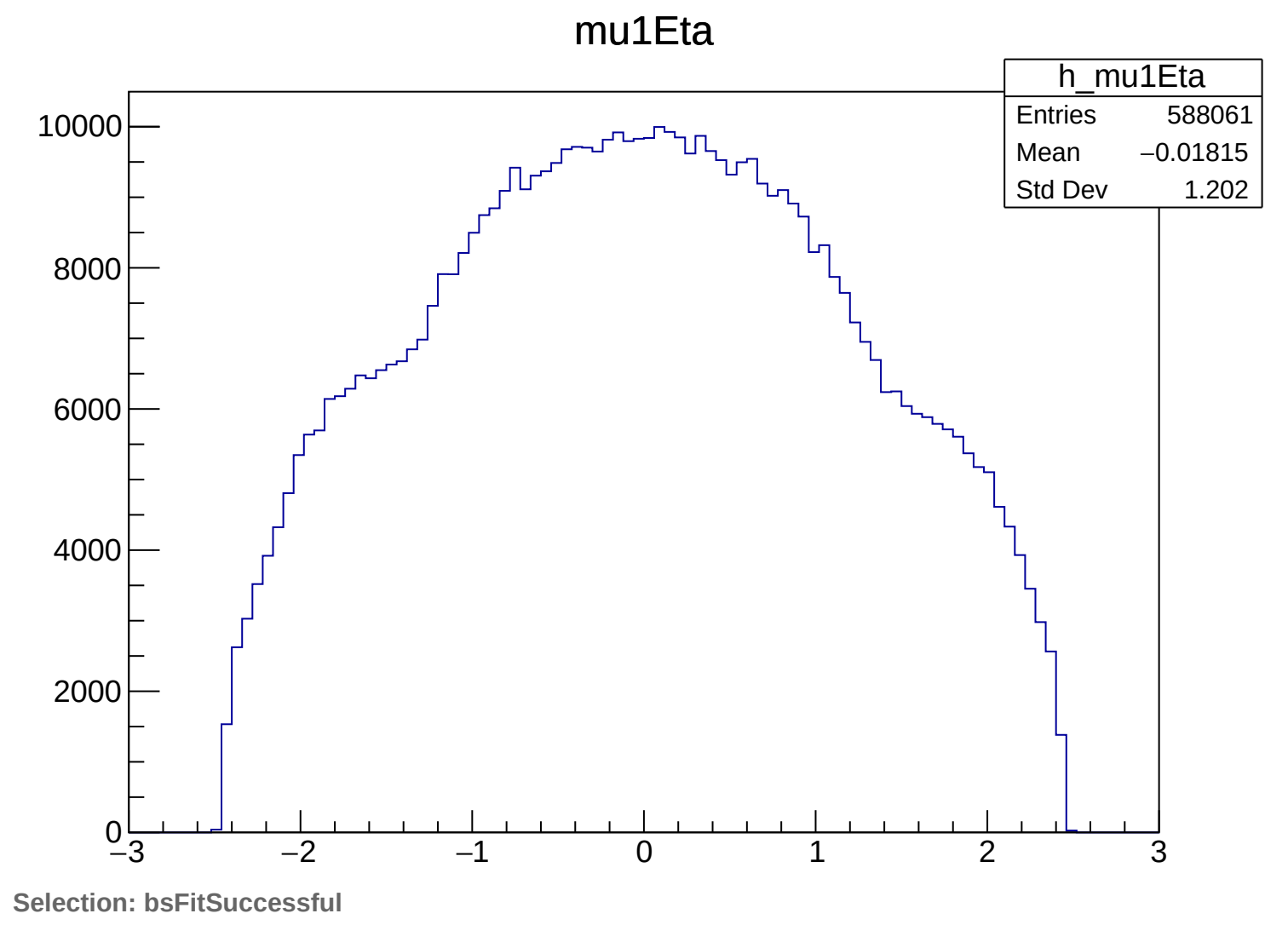

nPhotons
h_nPhotons
Entries 588061
Mean 1.106
Std Dev 0.3459
×10^3
500
400
300
200
100
0
0
1
2
3
4
5
6
7
8
Selection: bsFitSuccessful

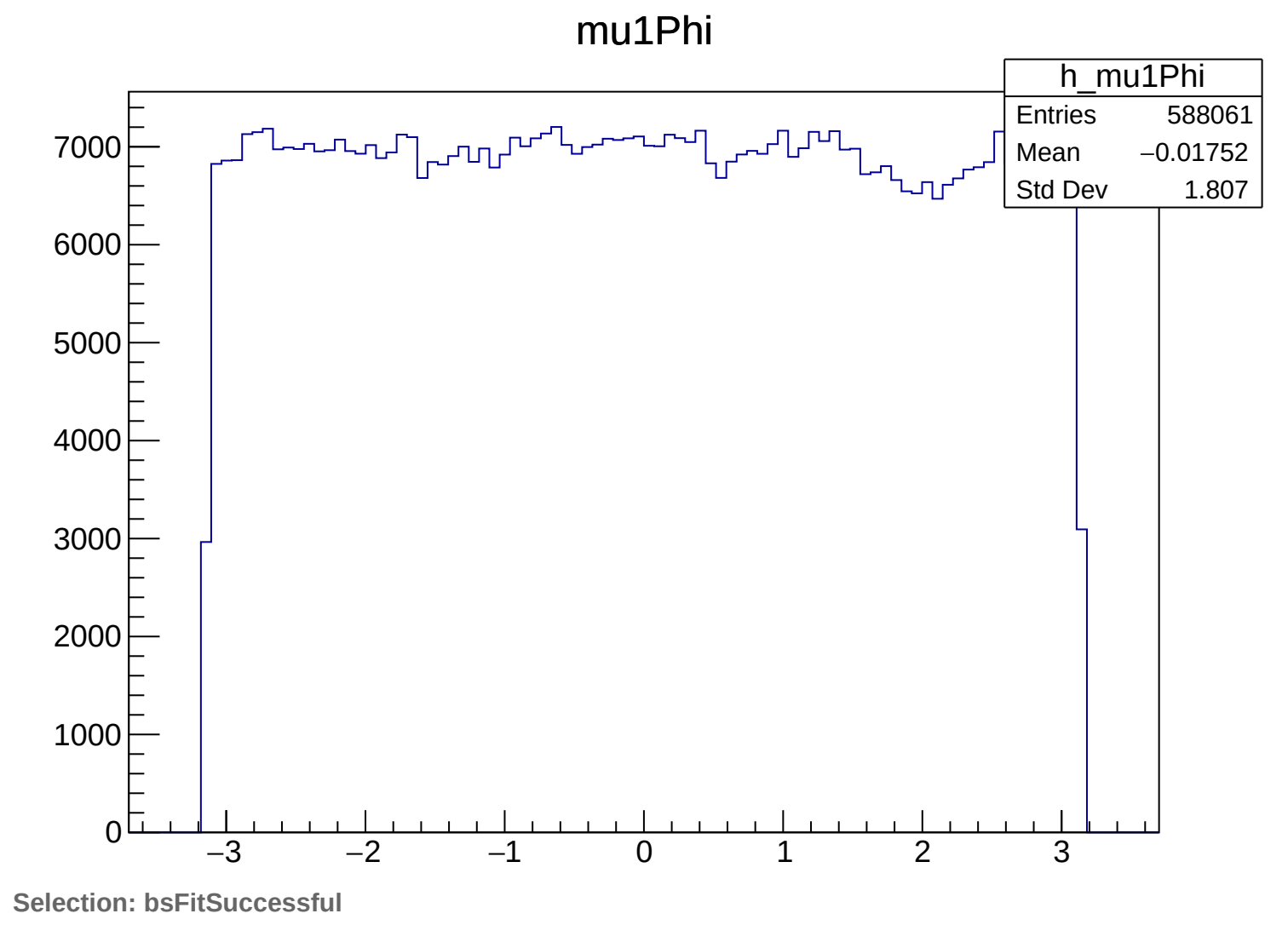

nTracks
h_nTracks
Entries 588061
Mean 1068
Std Dev 341.8
22000
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
500
1000
1500
2000
2500
3000
Selection: bsFitSuccessful

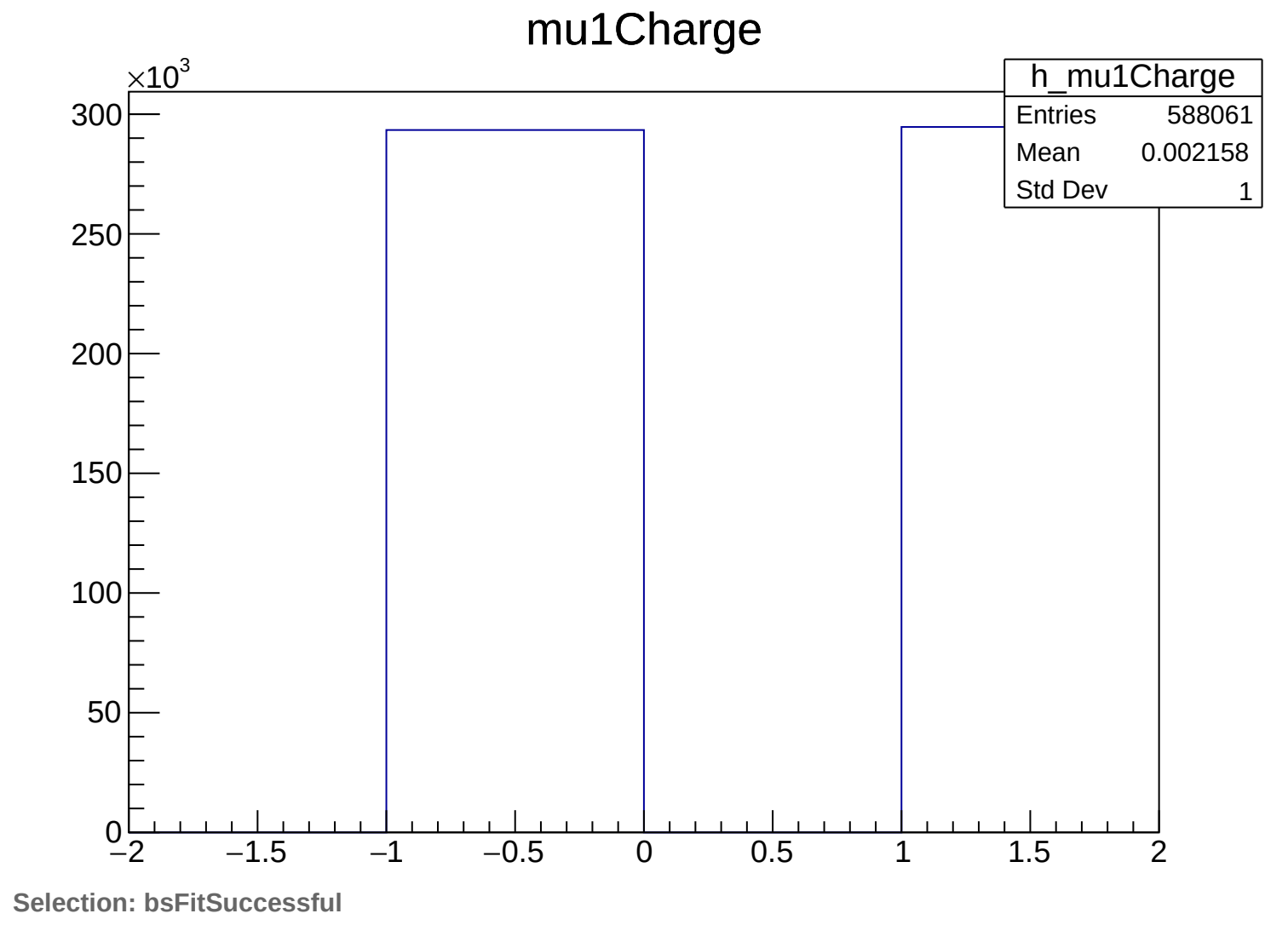

mu1MVAScore
h_mu1MVAScore
Entries 588061
Mean 0.8043
Std Dev 0.1367
50000
40000
30000
20000
10000
0
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful

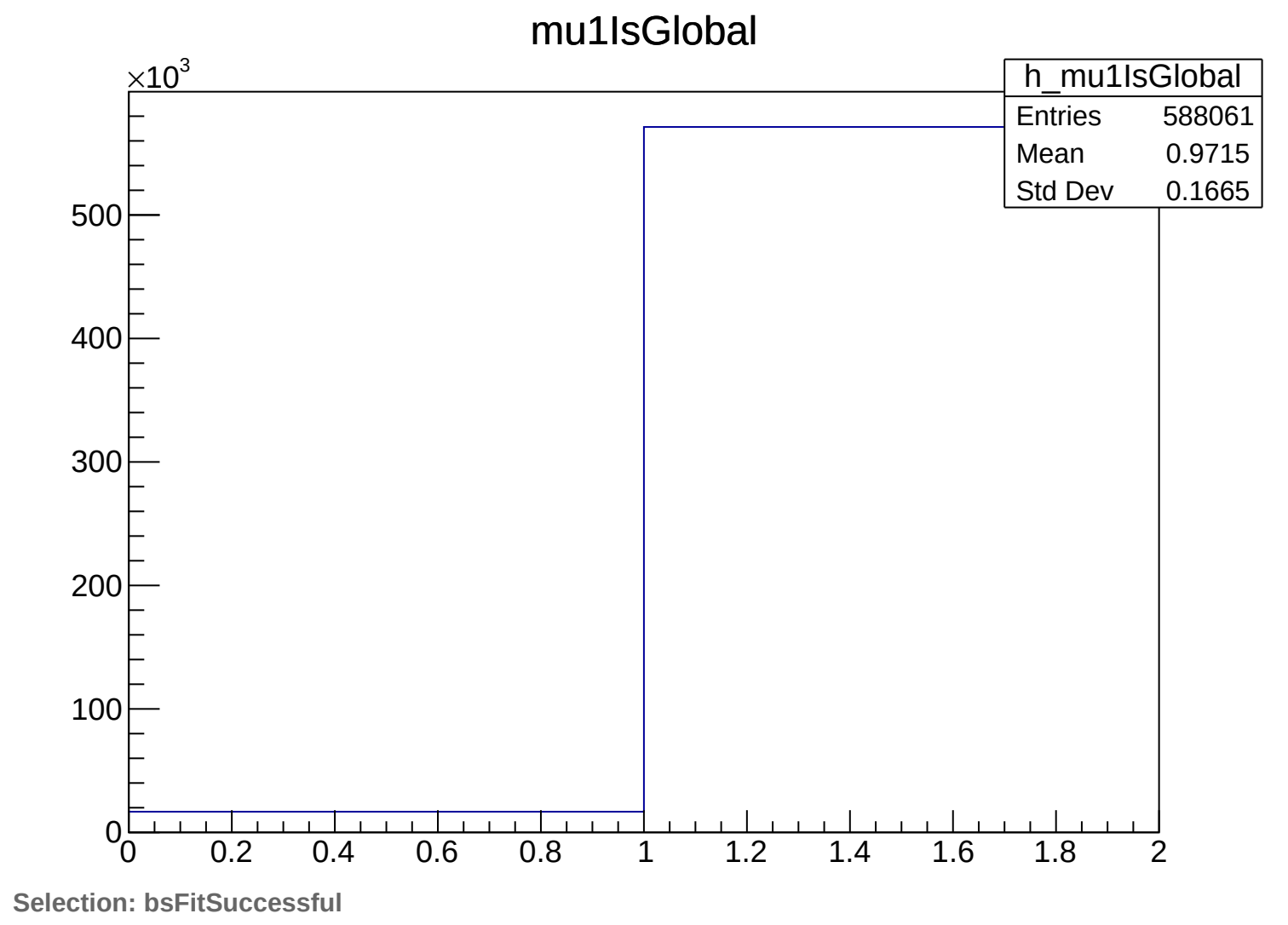

mu2MVAScore
h_mu2MVAScore
Entries 588061
Mean 0.8034
Std Dev 0.142
45000
40000
35000
30000
25000
20000
15000
10000
5000
0
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful

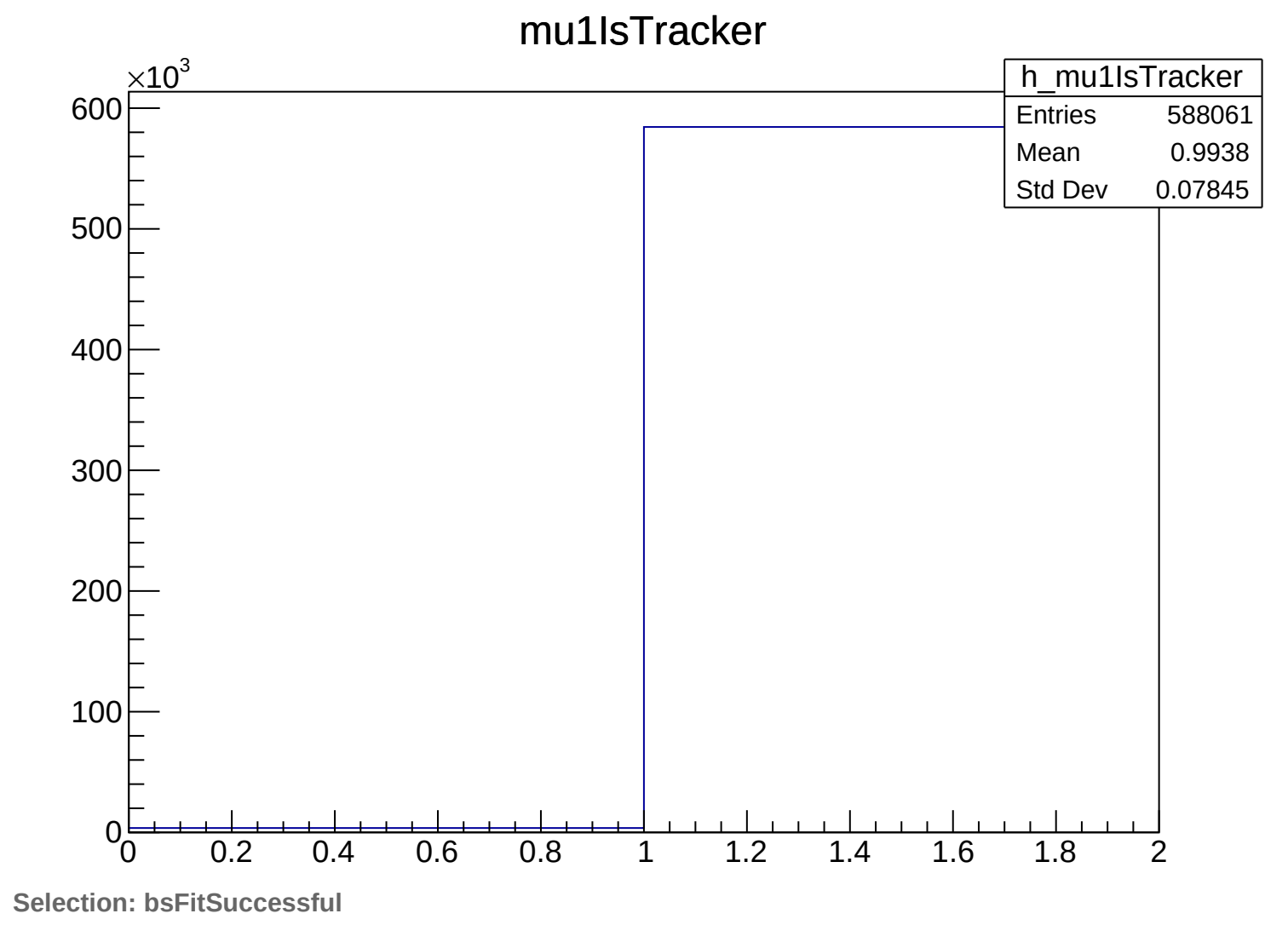

mu1Pt
×10^3
h_mu1Pt
Entries 588061
Mean 11.33
Std Dev 8.127
160
140
120
100
80
60
40
20
0
0
50
100
150
200
250
300
Selection: bsFitSuccessful

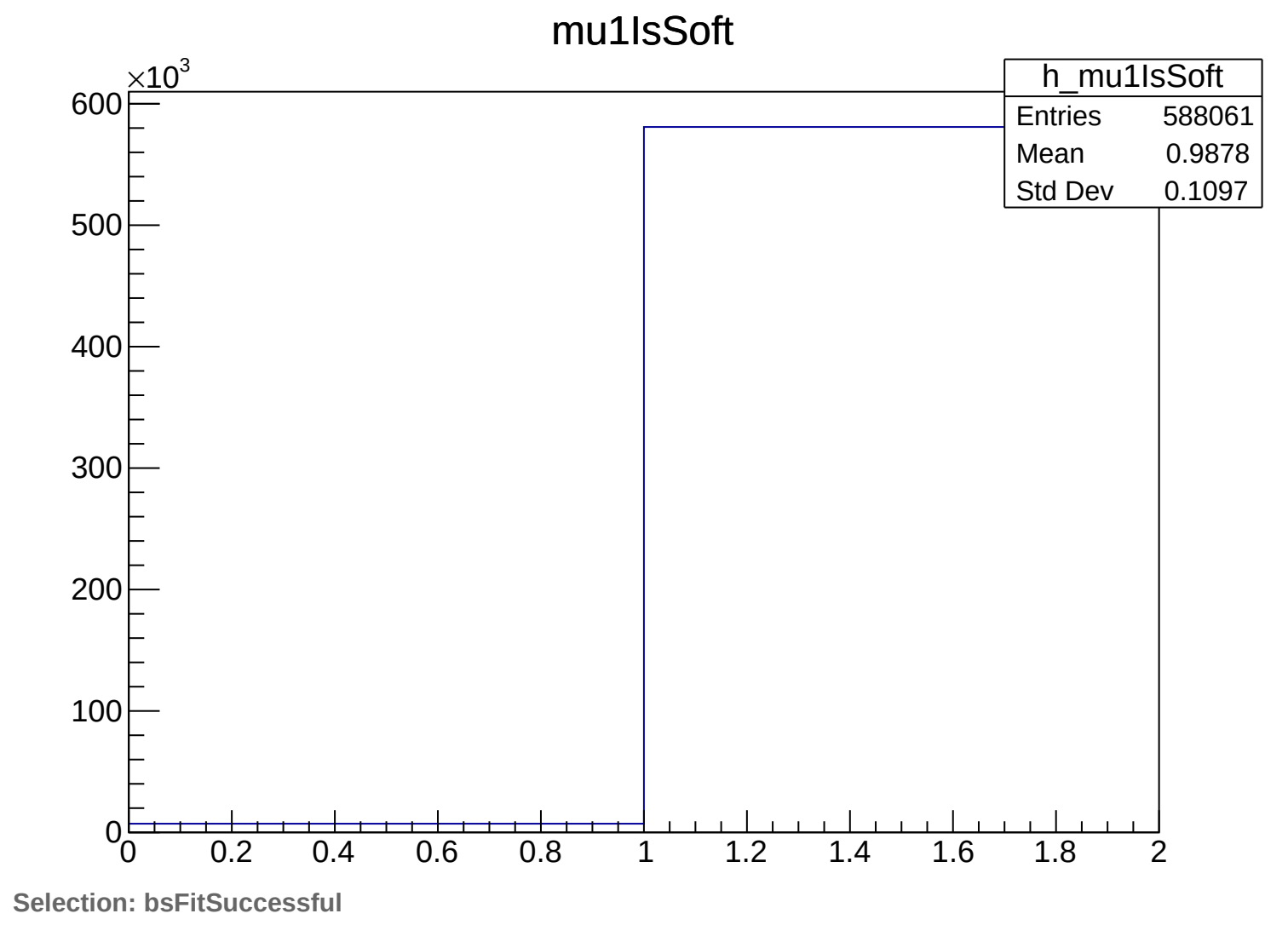

mu1Eta
h_mu1Eta
Entries 588061
Mean −0.01815
Std Dev 1.202
10000
8000
6000
4000
2000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

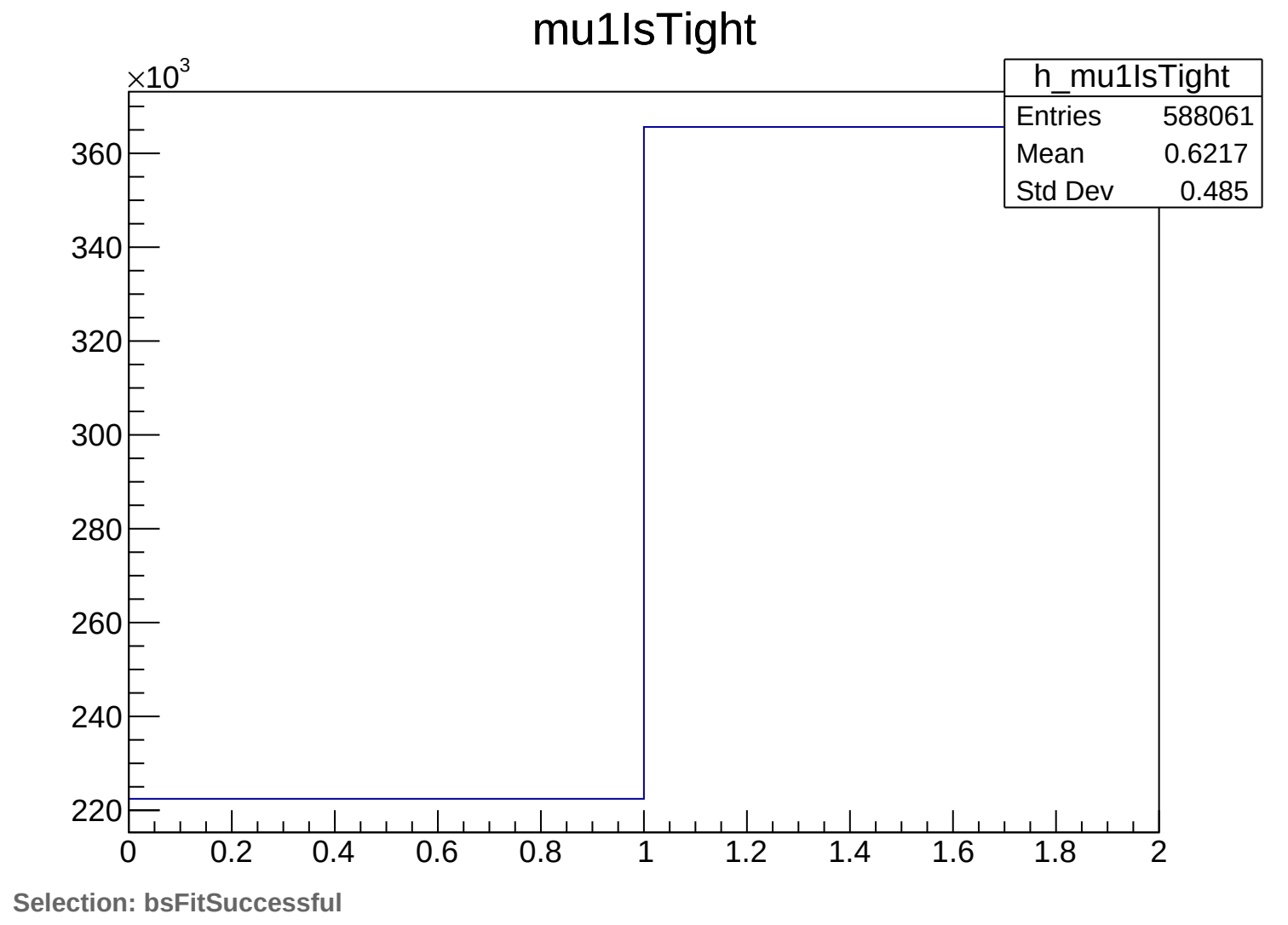

mu1Phi
h_mu1Phi
Entries 588061
Mean −0.01752
Std Dev 1.807
7000
6000
5000
4000
3000
2000
1000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

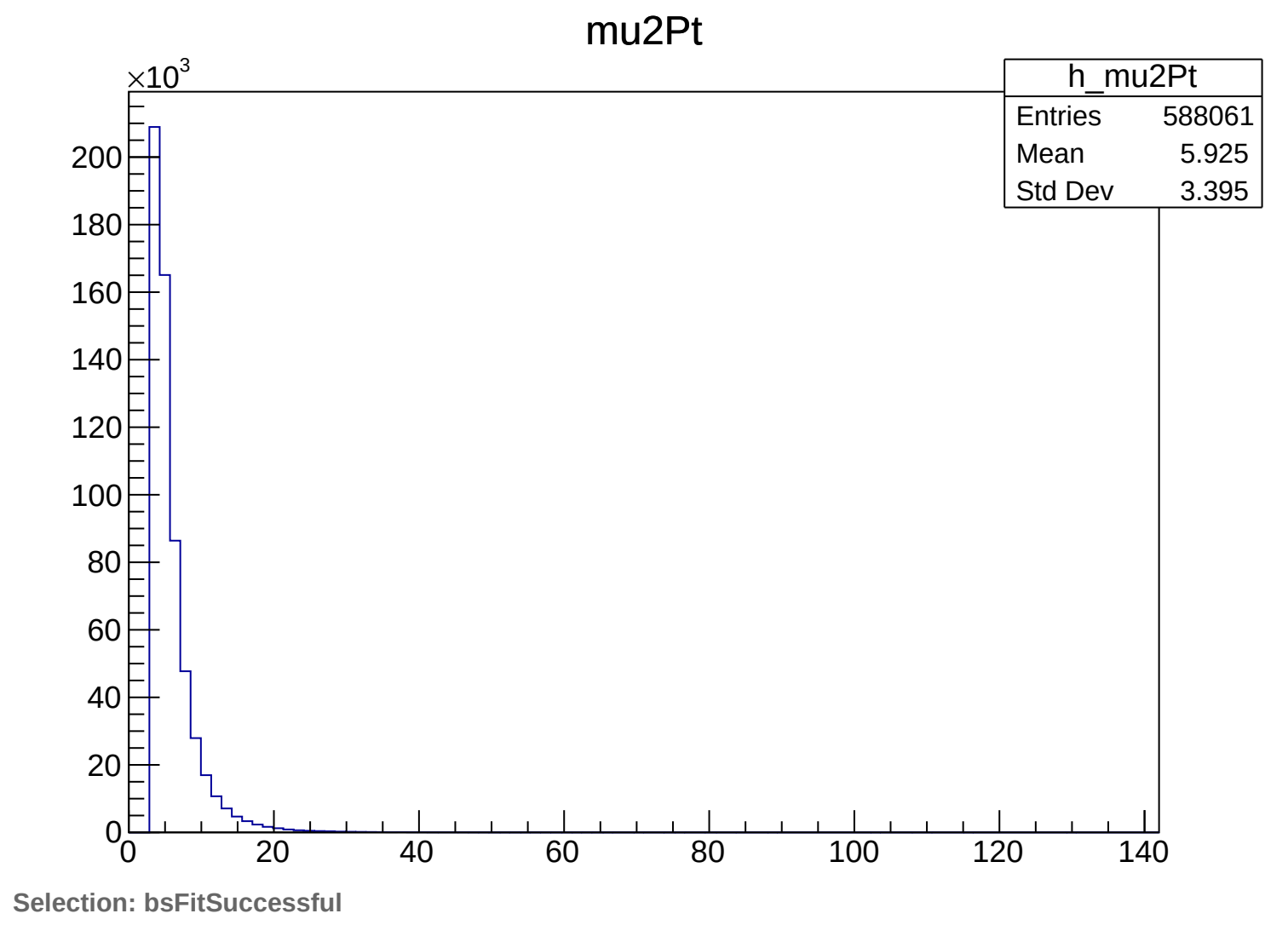

mu1Charge
h_mu1Charge
Entries 588061
Mean 0.002158
Std Dev 1
×10^3
300
250
200
150
100
50
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2
Selection: bsFitSuccessful

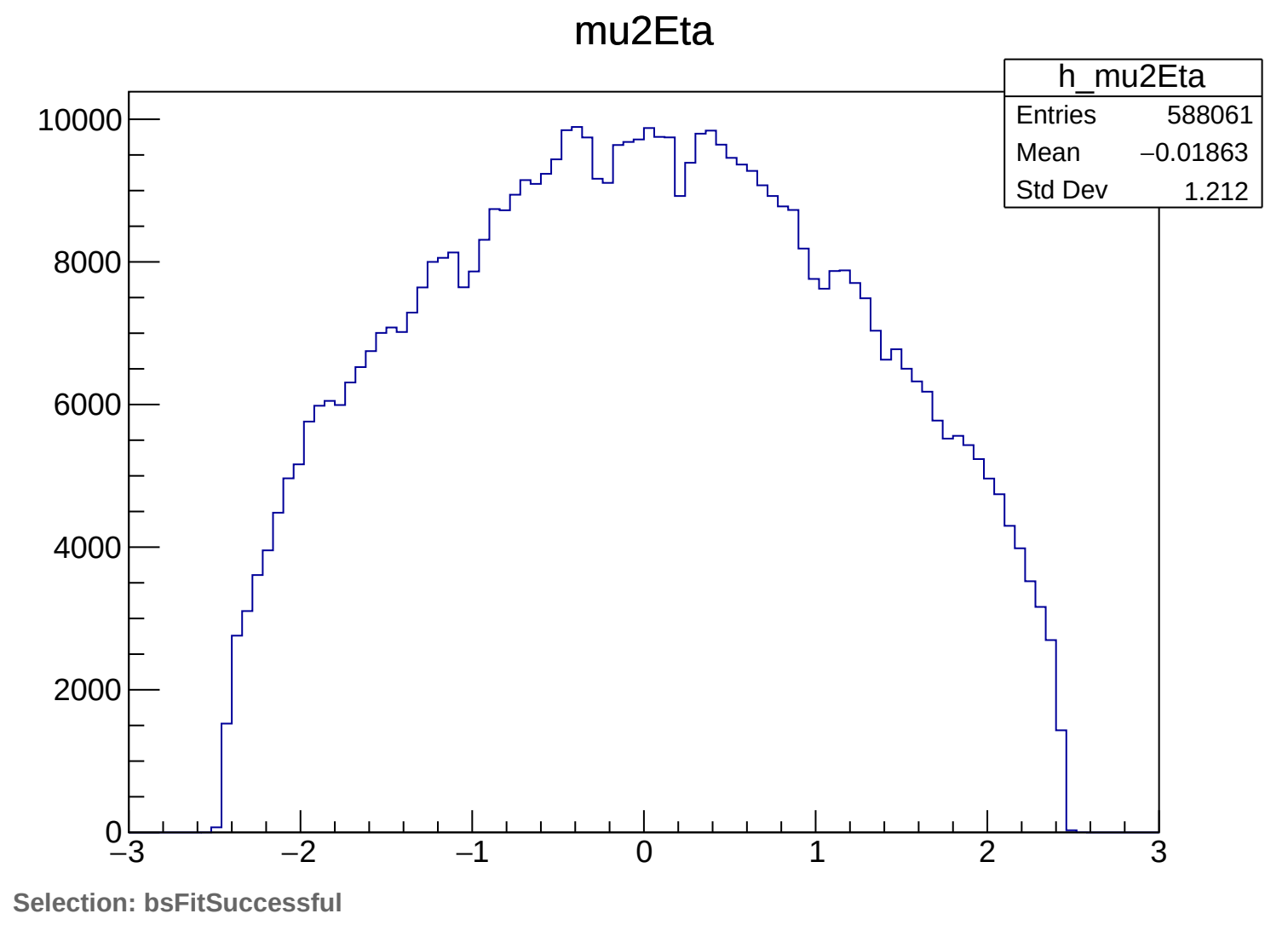

mu1IsGlobal
×10^3
h_mu1IsGlobal
Entries 588061
Mean 0.9715
Std Dev 0.1665
500
400
300
200
100
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

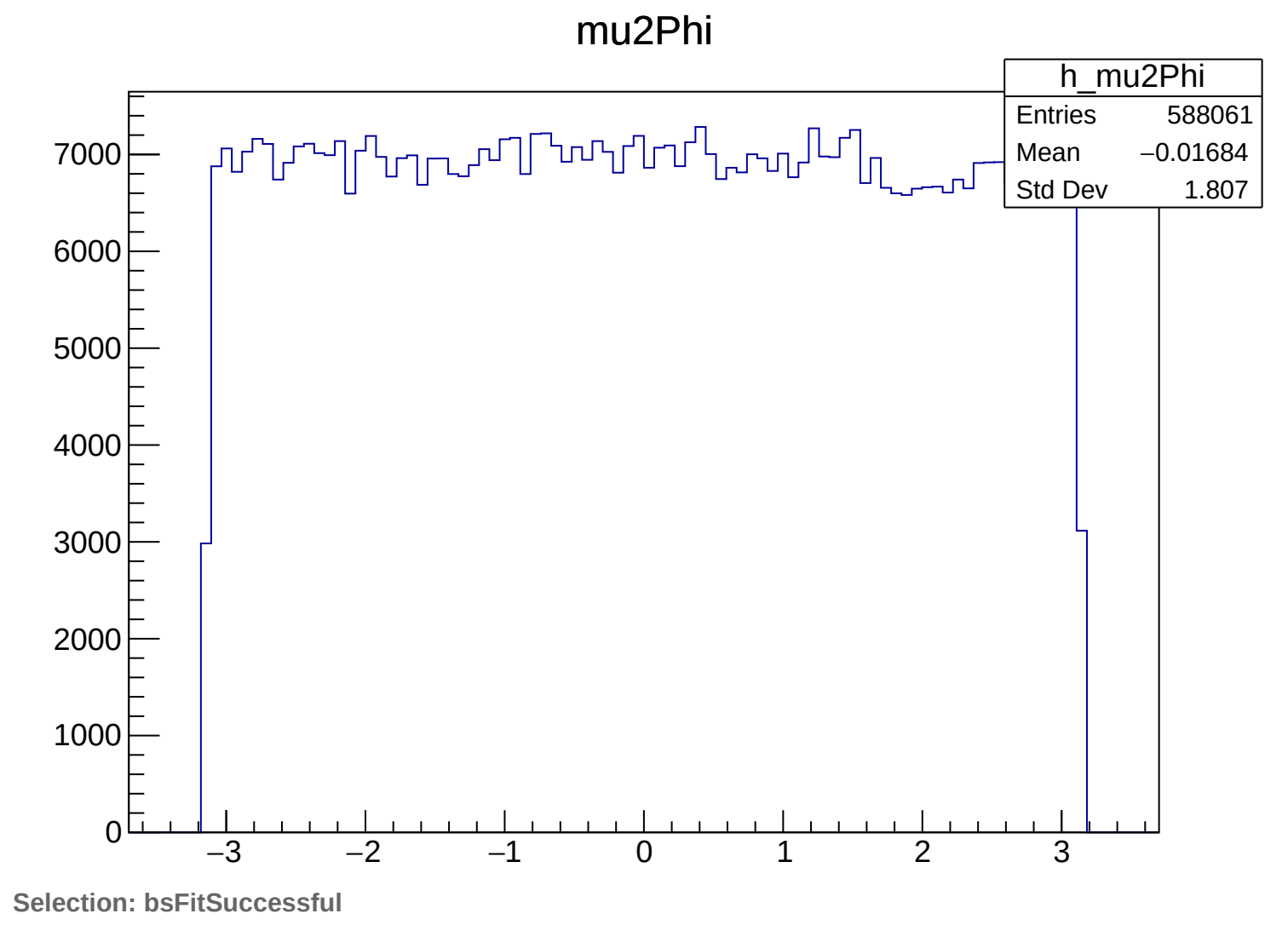

mu1IsTracker
h_mu1IsTracker
Entries 588061
Mean 0.9938
Std Dev 0.07845
×10^3
600
500
400
300
200
100
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

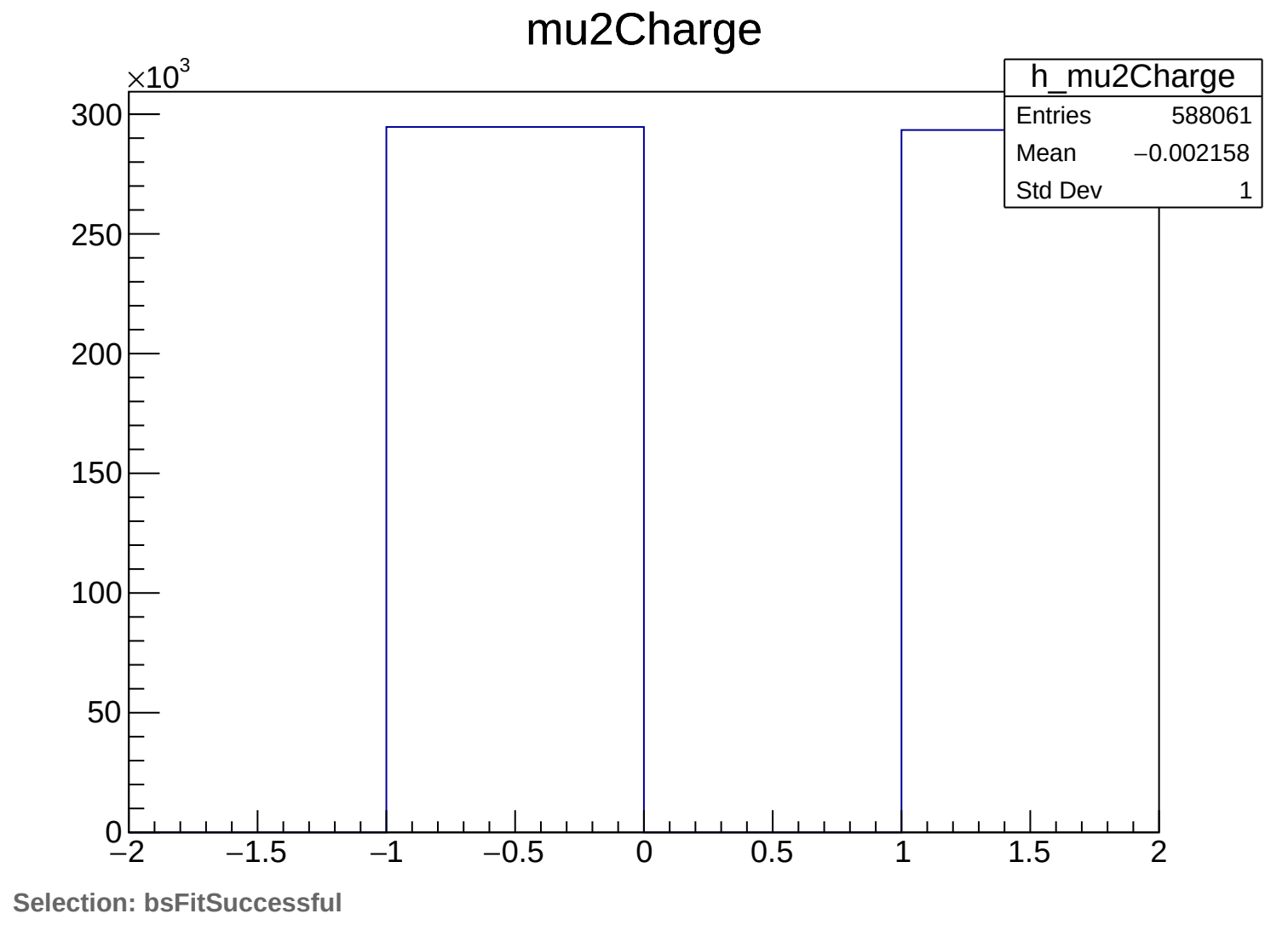

mu1IsSoft
h_mu1IsSoft
Entries 588061
Mean 0.9878
Std Dev 0.1097
×10^3
600
500
400
300
200
100
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

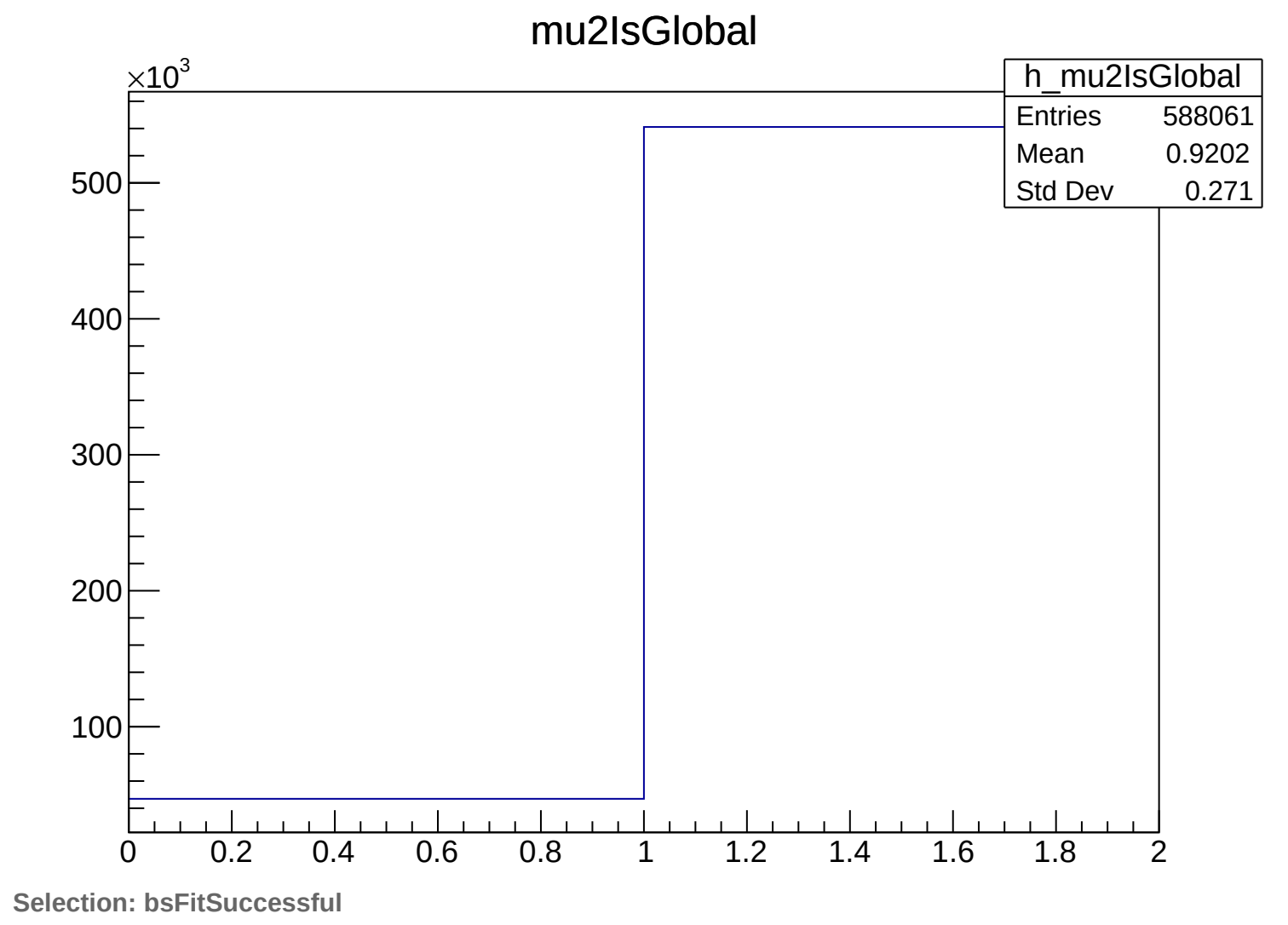

mu1IsTight
×10^3
360
340
320
300
280
260
240
220
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
h_mu1IsTight
Entries 588061
Mean 0.6217
Std Dev 0.485
Selection: bsFitSuccessful

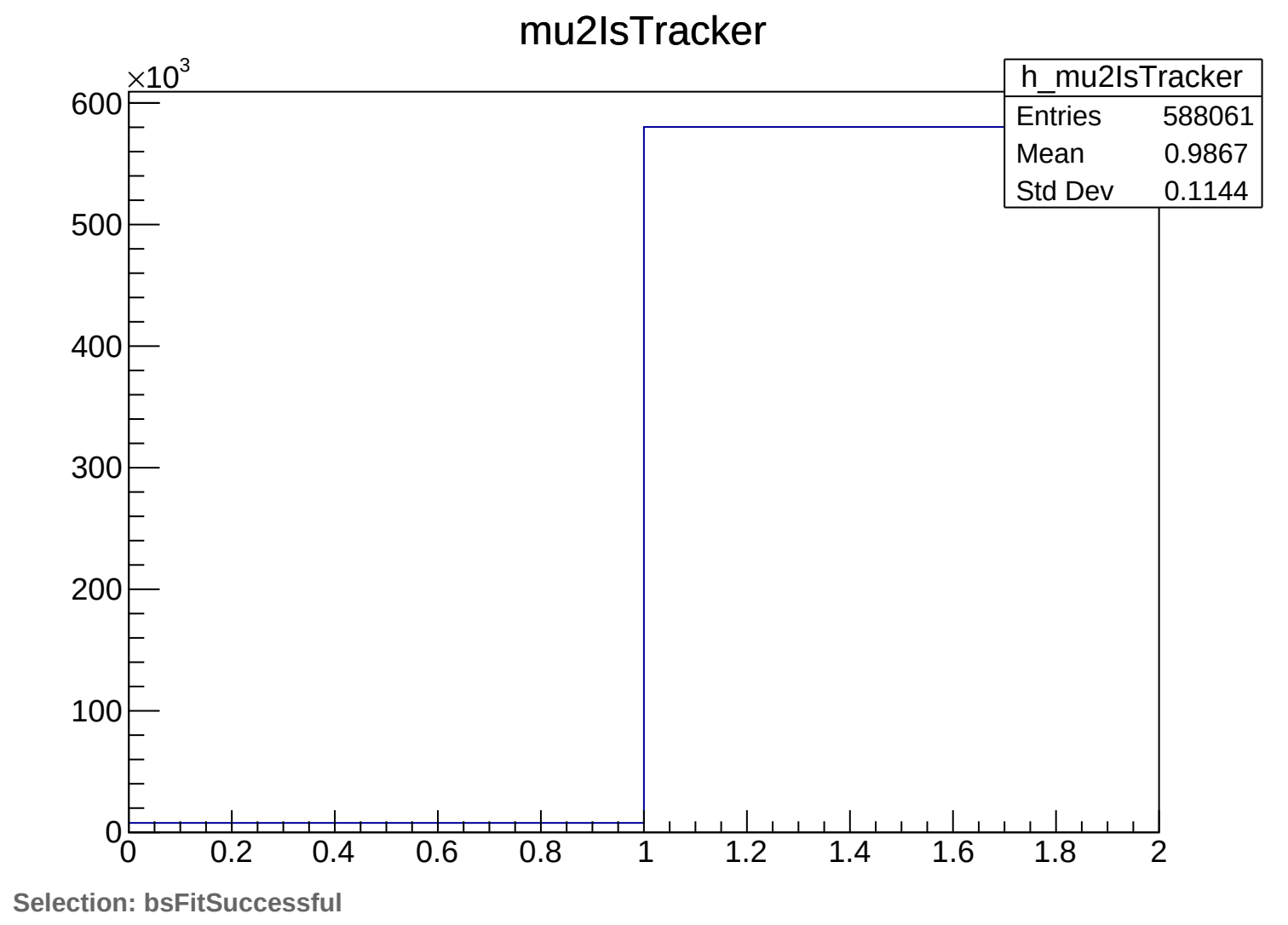

mu2Pt
×10^3
h_mu2Pt
Entries 588061
Mean 5.925
Std Dev 3.395
200
180
160
140
120
100
80
60
40
20
0
0
20
40
60
80
100
120
140
Selection: bsFitSuccessful

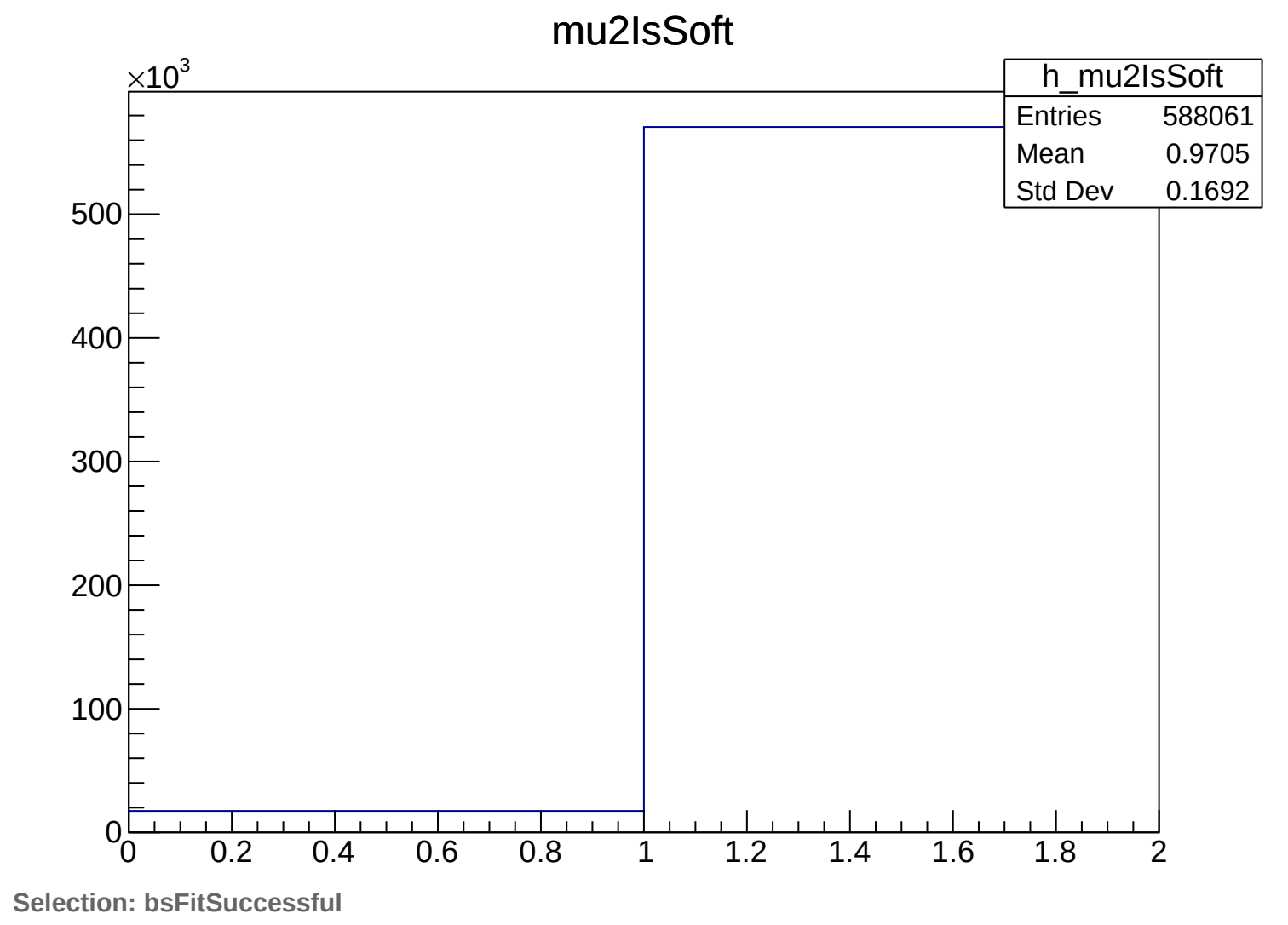

mu2Eta
h_mu2Eta
Entries 588061
Mean −0.01863
Std Dev 1.212
10000
8000
6000
4000
2000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

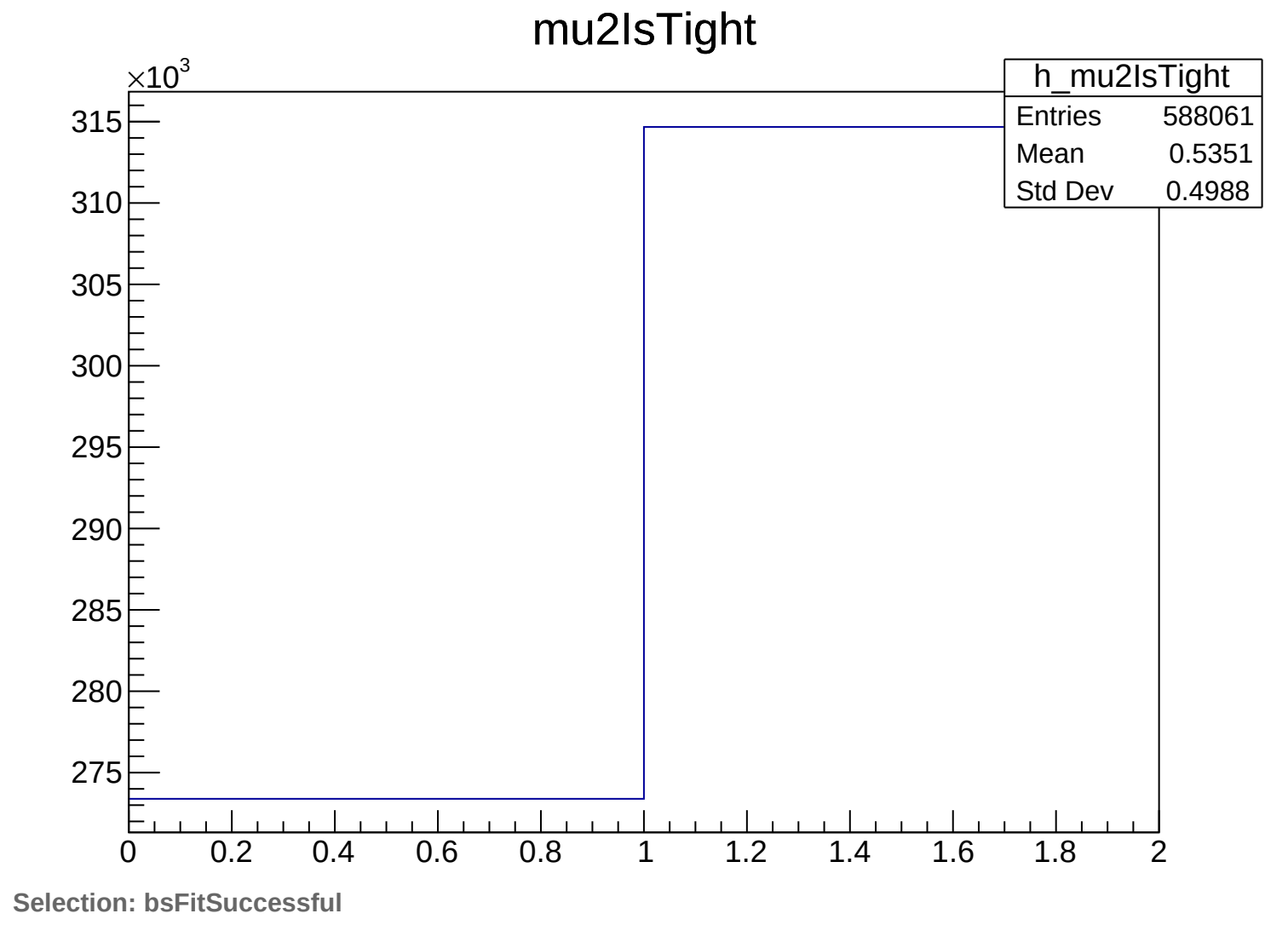

mu2Phi
h_mu2Phi
Entries 588061
Mean −0.01684
Std Dev 1.807
7000
6000
5000
4000
3000
2000
1000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

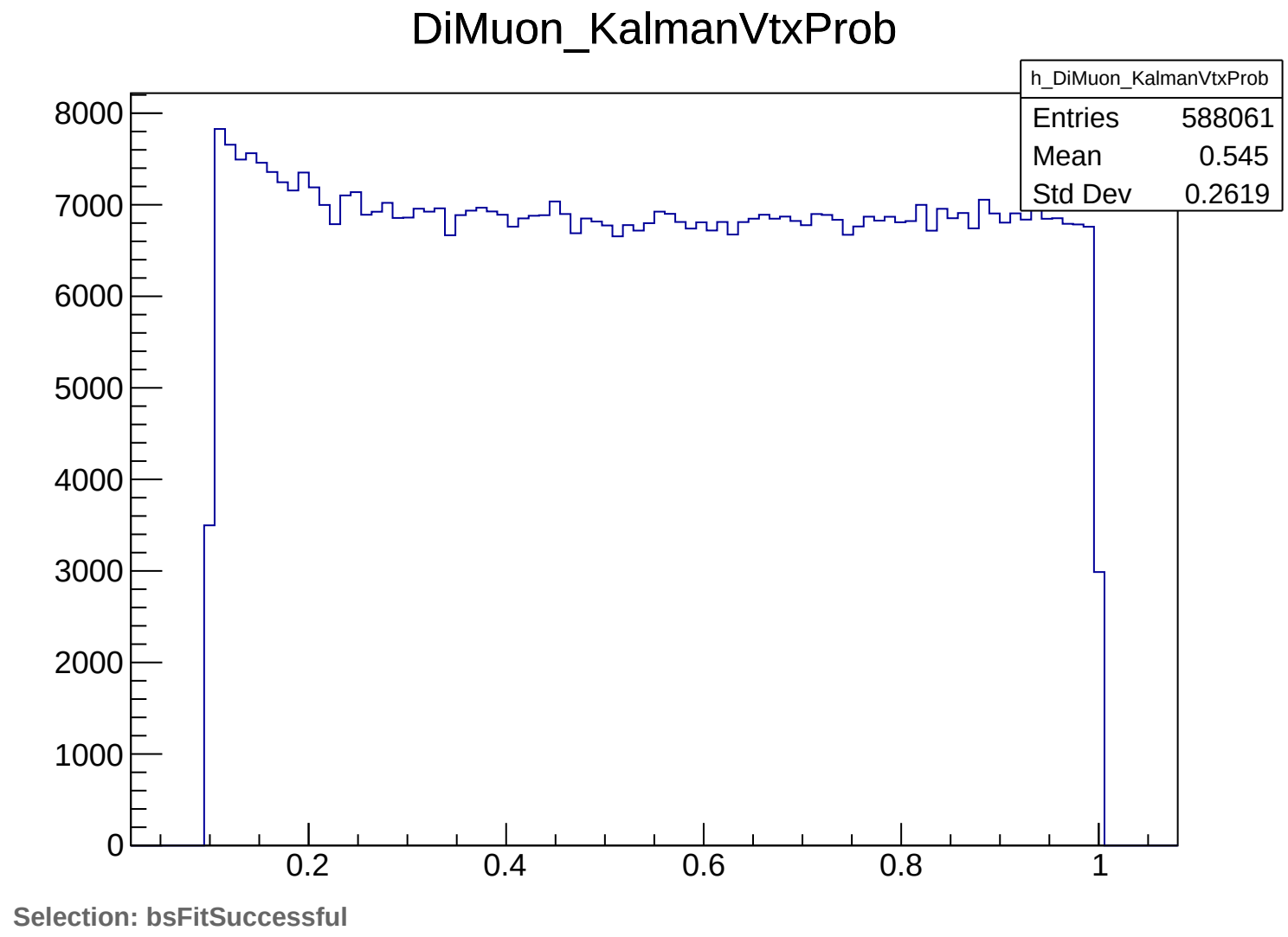

mu2Charge
h_mu2Charge
Entries 588061
Mean −0.002158
Std Dev 1
×10^3
300
250
200
150
100
50
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2
Selection: bsFitSuccessful

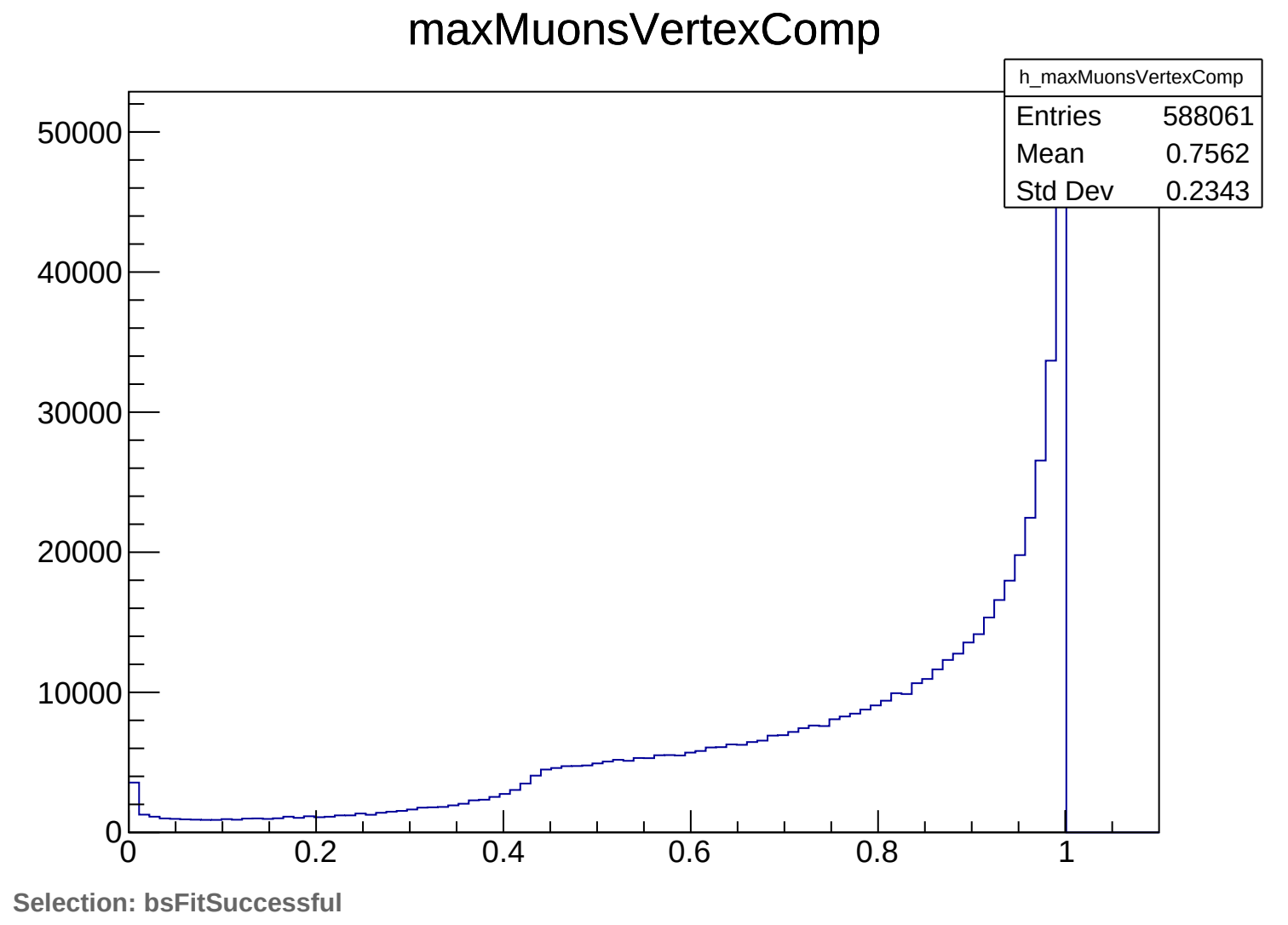

mu2IsGlobal
×10^3
h_mu2IsGlobal
Entries 588061
Mean 0.9202
Std Dev 0.271
500
400
300
200
100
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

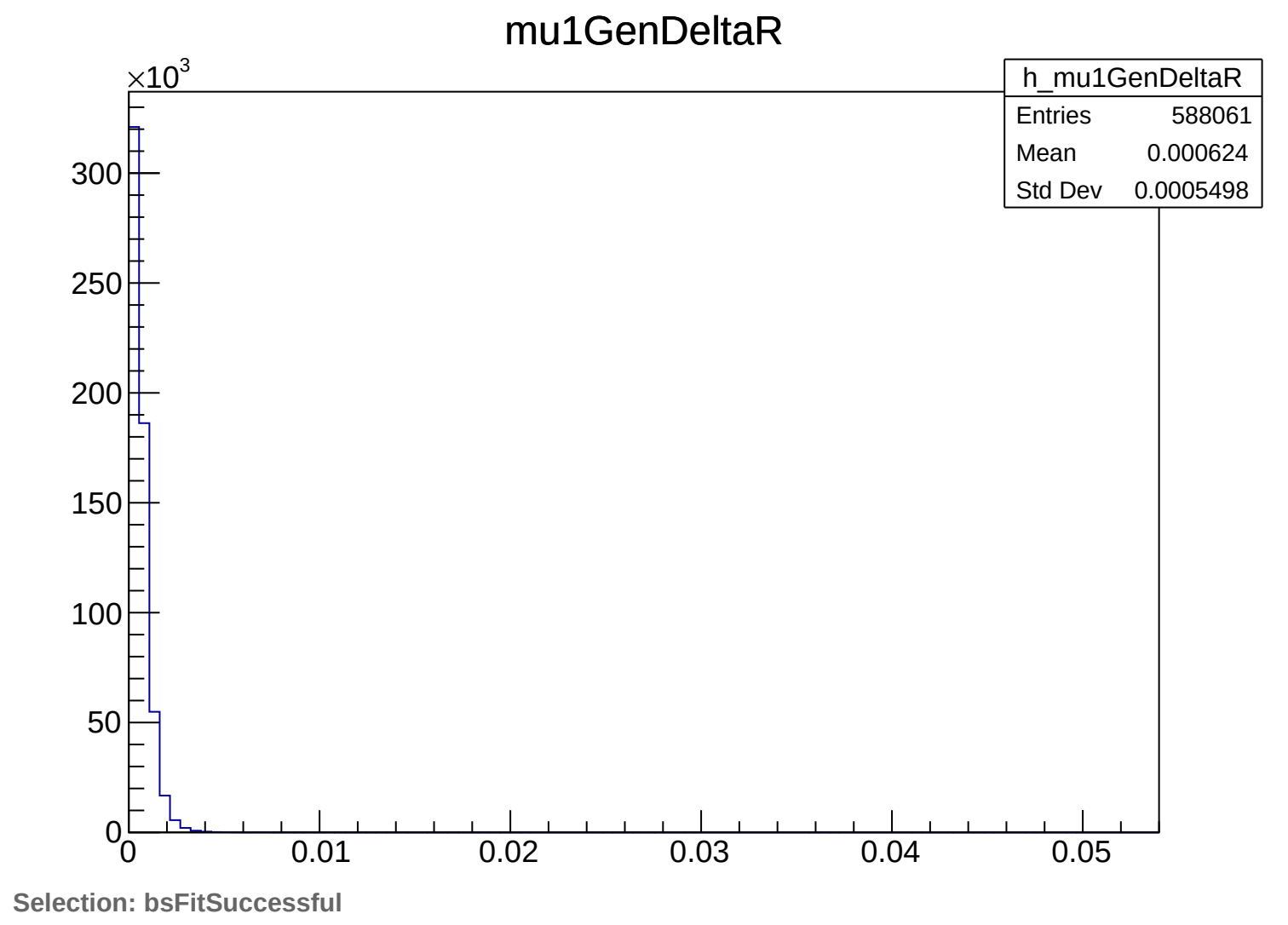

mu2IsTracker
h_mu2IsTracker
Entries 588061
Mean 0.9867
Std Dev 0.1144
×10^3
600
500
400
300
200
100
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

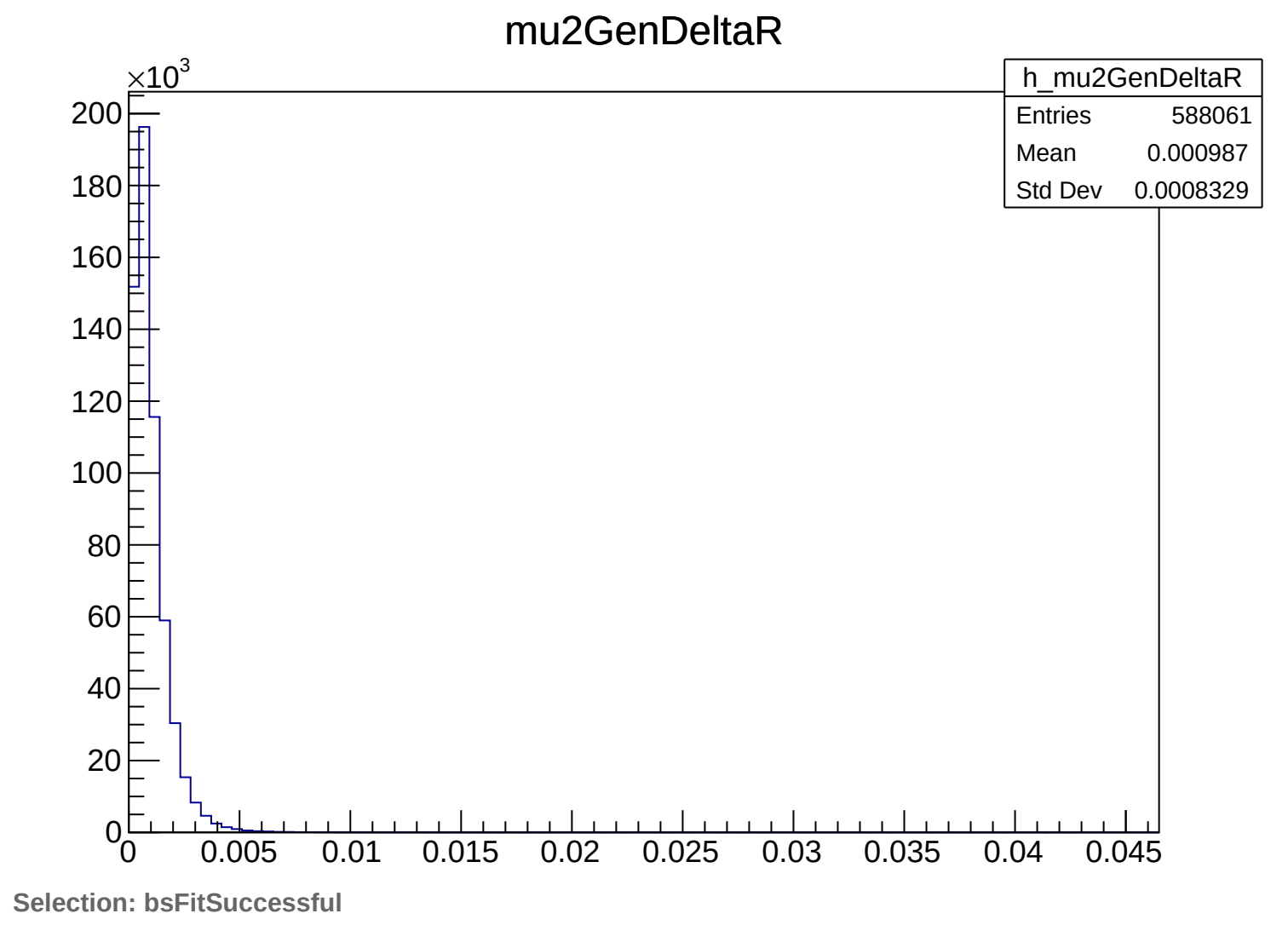

mu2IsSoft
h_mu2IsSoft
Entries 588061
Mean 0.9705
Std Dev 0.1692
×10^3
500
400
300
200
100
0
0 0.2 0.4 0.6 0.8 1 1.2 1.4 1.6 1.8 2
Selection: bsFitSuccessful

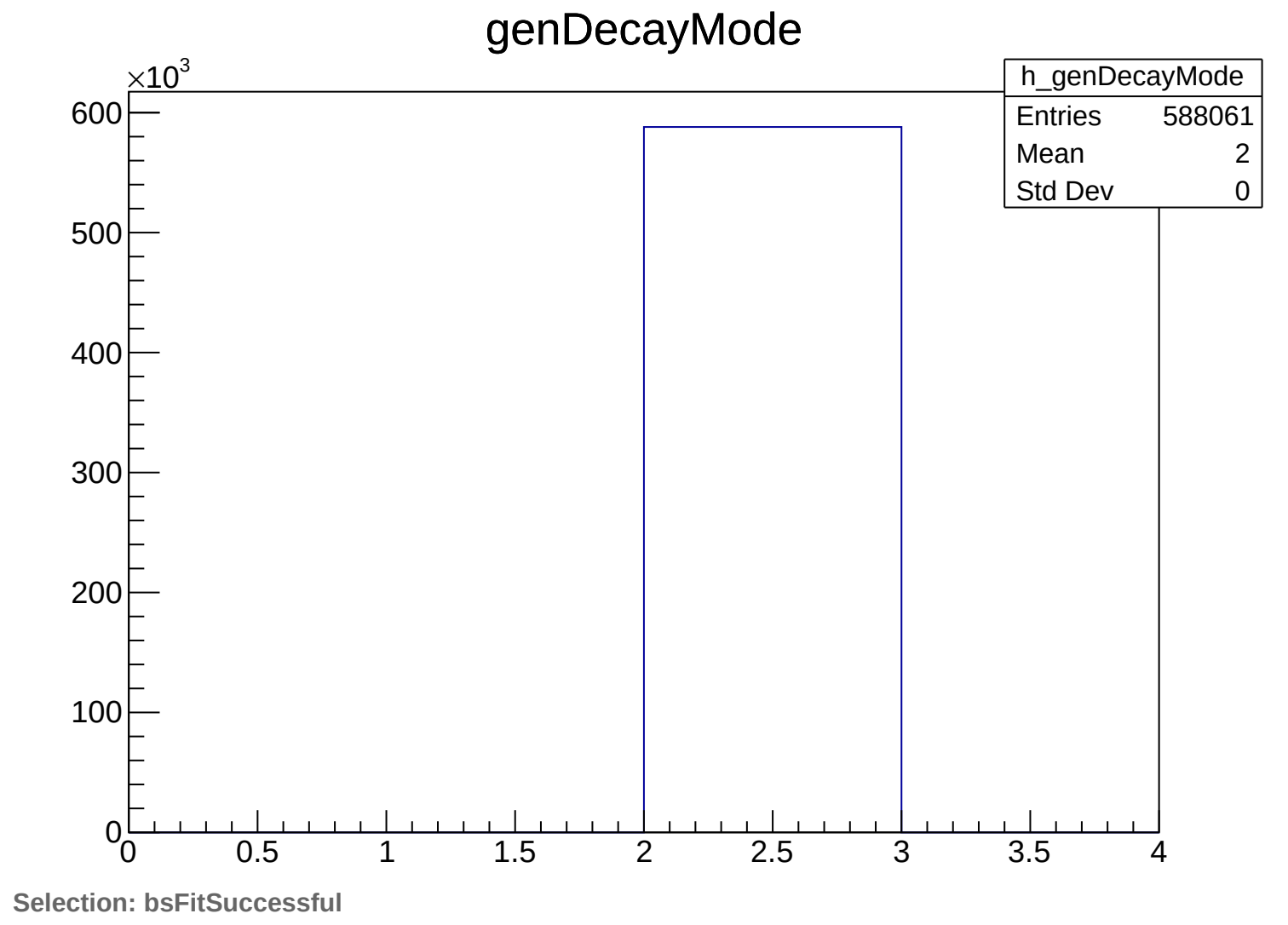

mu2IsTight
h_mu2IsTight
Entries 588061
Mean 0.5351
Std Dev 0.4988
×10^3
315
310
305
300
295
290
285
280
275
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

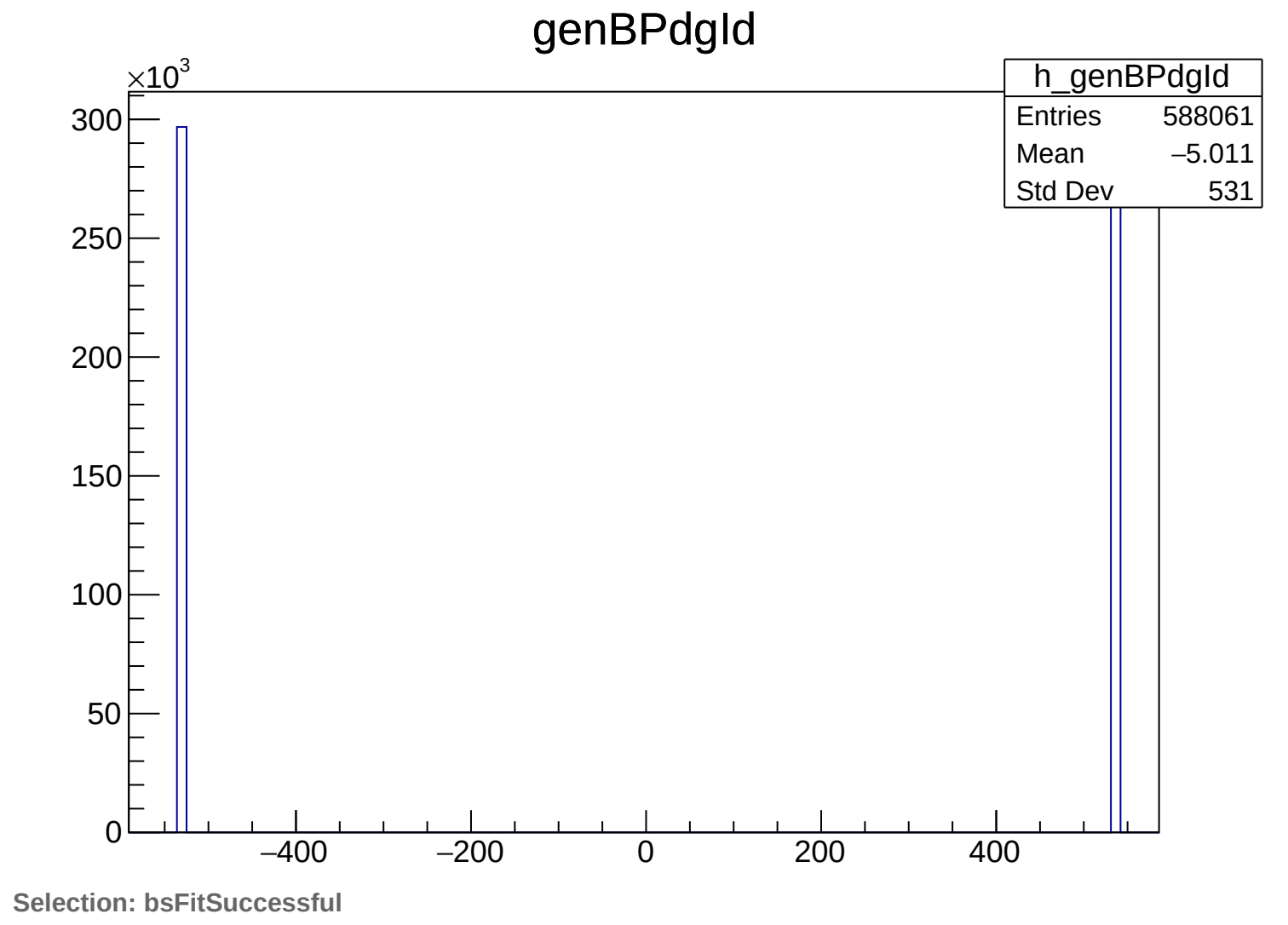

DiMuon_KalmanVtxProb
h_DiMuon_KalmanVtxProb
Entries 588061
Mean 0.545
Std Dev 0.2619
8000
7000
6000
5000
4000
3000
2000
1000
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful

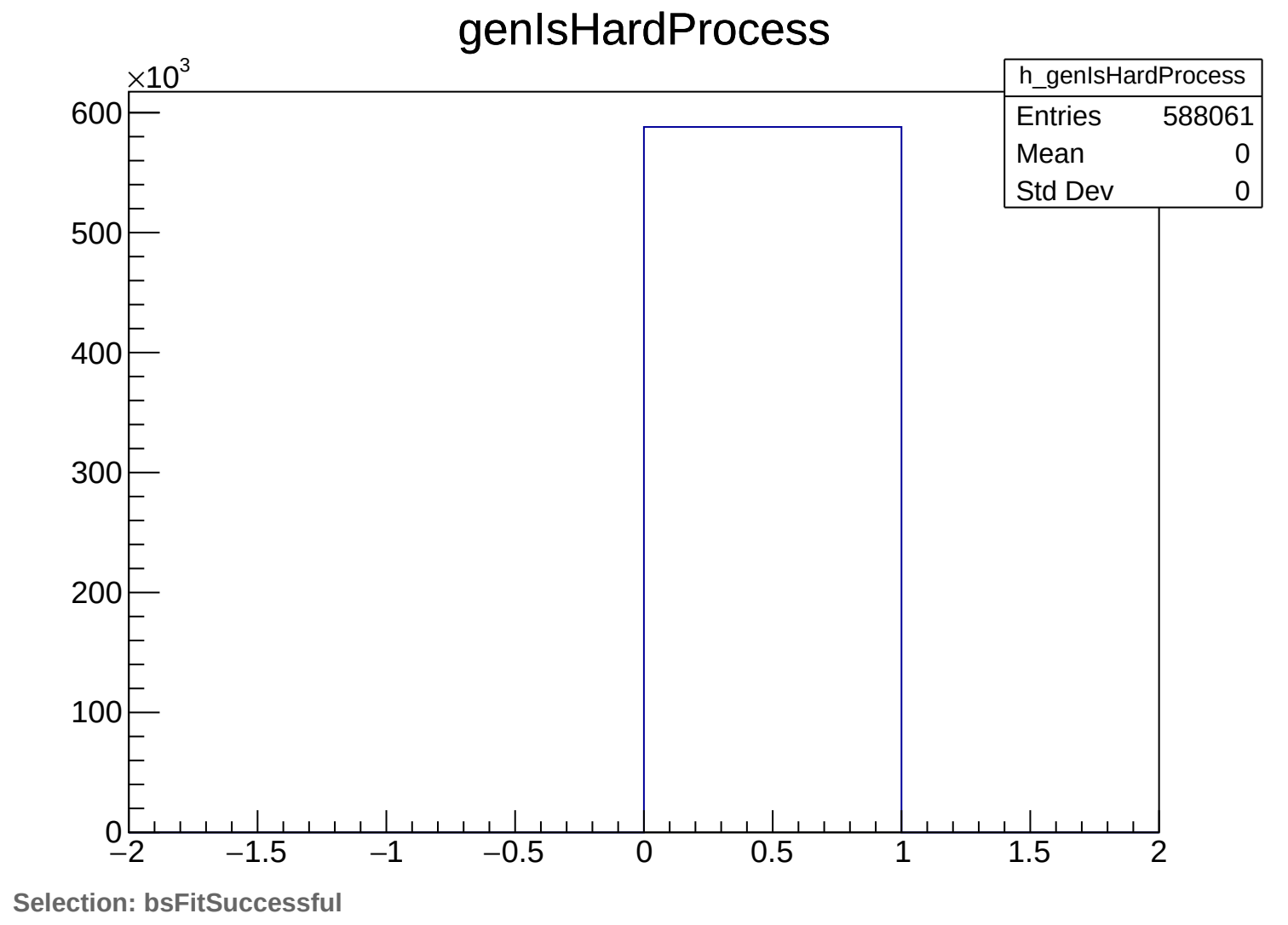

maxMuonsVertexComp
h_maxMuonsVertexComp
Entries 588061
Mean 0.7562
Std Dev 0.2343
50000
40000
30000
20000
10000
0
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful

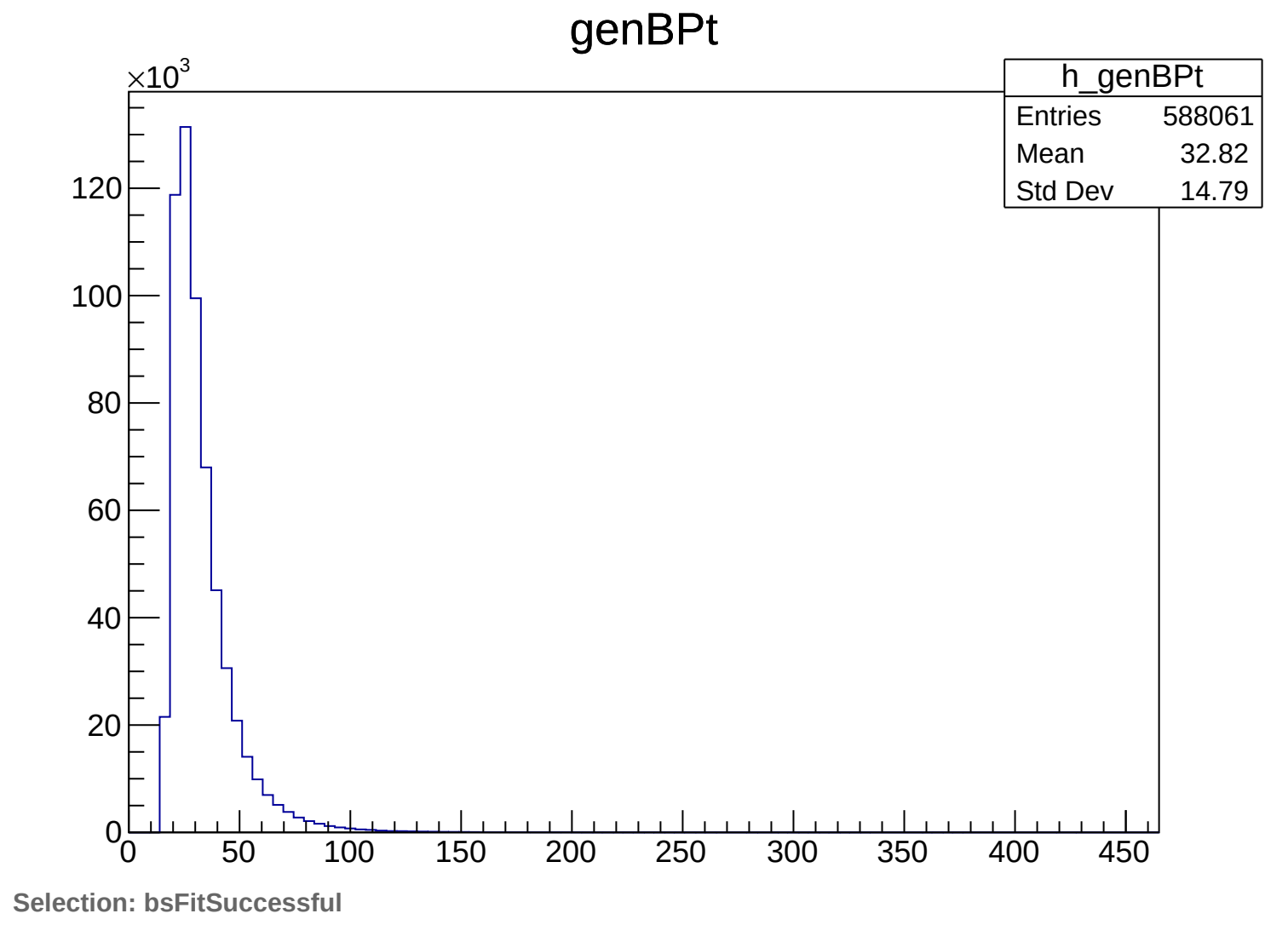

mu1GenDeltaR
×10^3
h_mu1GenDeltaR
Entries 588061
Mean 0.000624
Std Dev 0.0005498
300
250
200
150
100
50
0
0
0.01
0.02
0.03
0.04
0.05
Selection: bsFitSuccessful

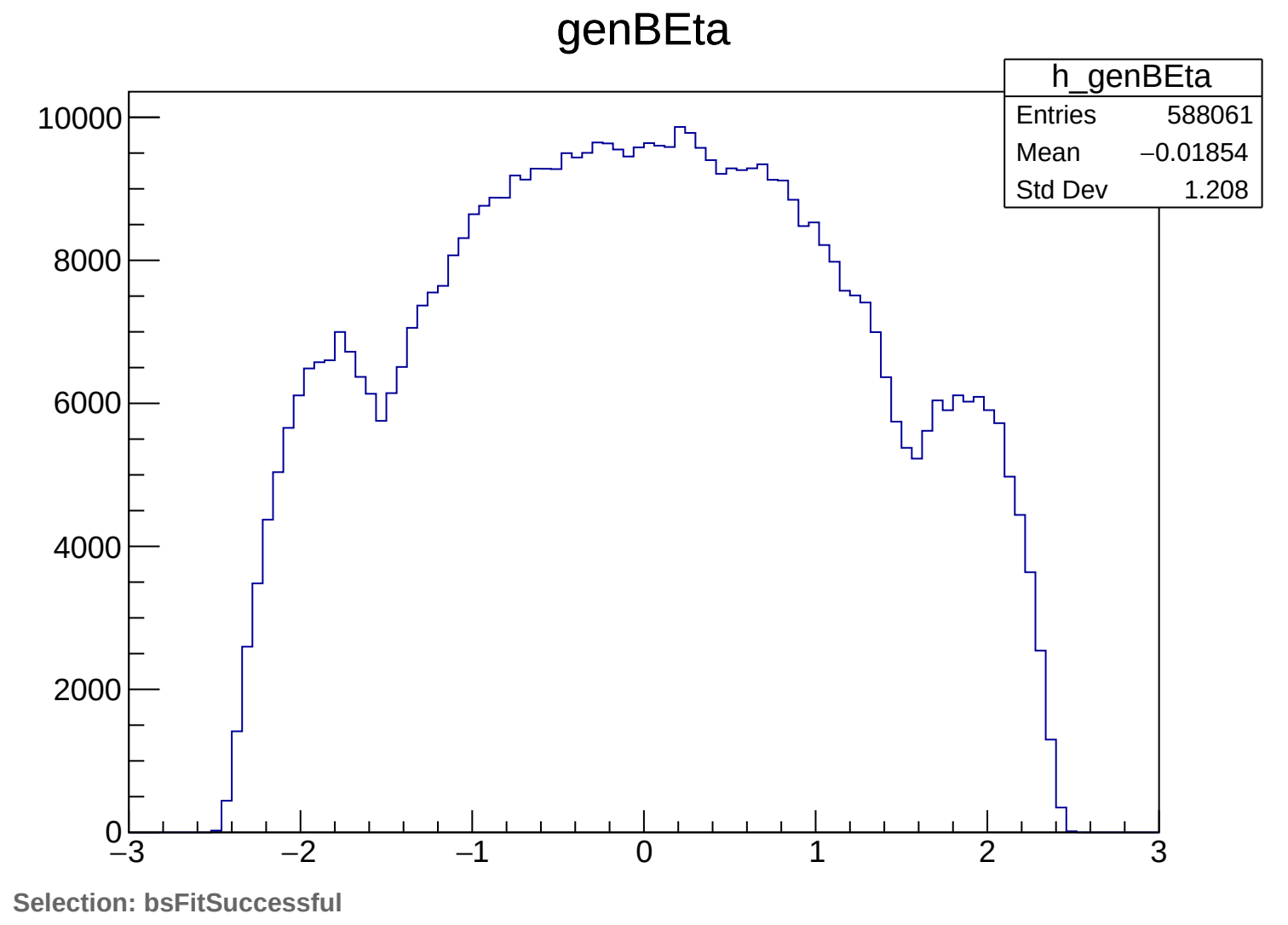

mu2GenDeltaR
×10^3
200
180
160
140
120
100
80
60
40
20
0
0
0.005
0.01
0.015
0.02
0.025
0.03
0.035
0.04
0.045
h_mu2GenDeltaR
Entries 588061
Mean 0.000987
Std Dev 0.0008329
Selection: bsFitSuccessful

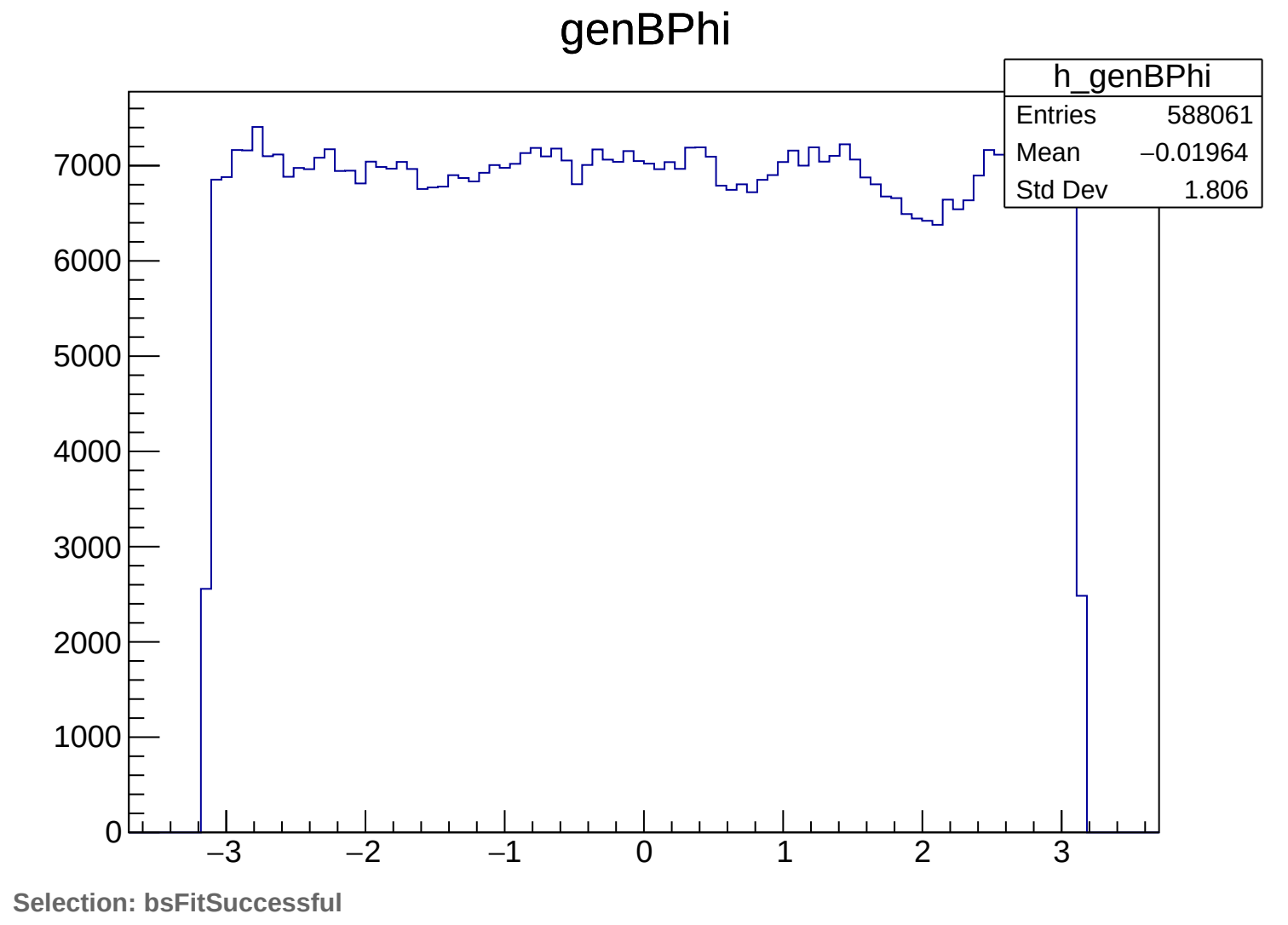

genDecayMode
×10^3
600
500
400
300
200
100
0
0
0.5
1
1.5
2
2.5
3
3.5
4
h_genDecayMode
Entries 588061
Mean 2
Std Dev 0
Selection: bsFitSuccessful

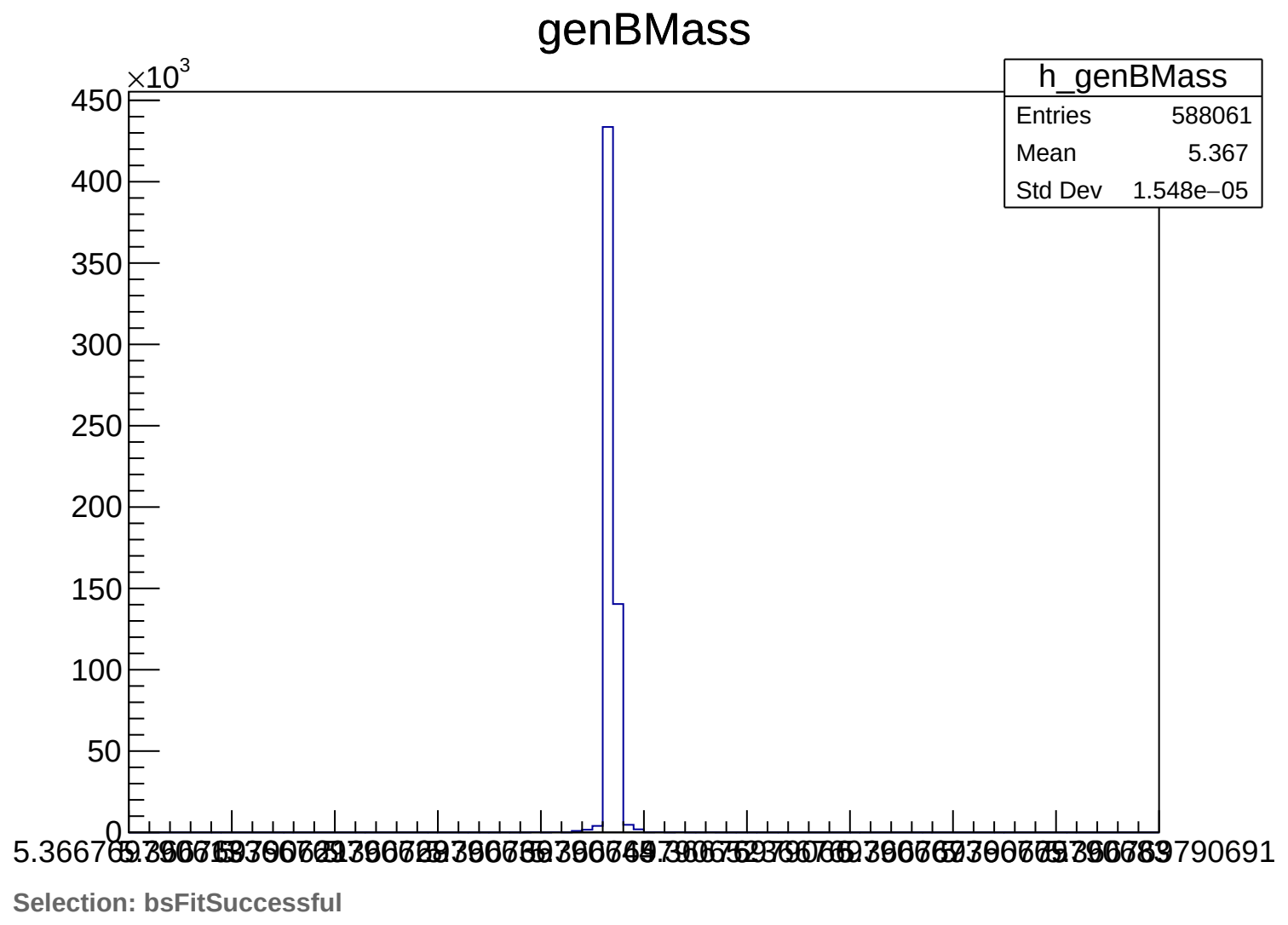

genBPdgId
×10^3
300
250
200
150
100
50
0
–400
–200
0
200
400
h_genBPdgId
Entries 588061
Mean –5.011
Std Dev 531
Selection: bsFitSuccessful

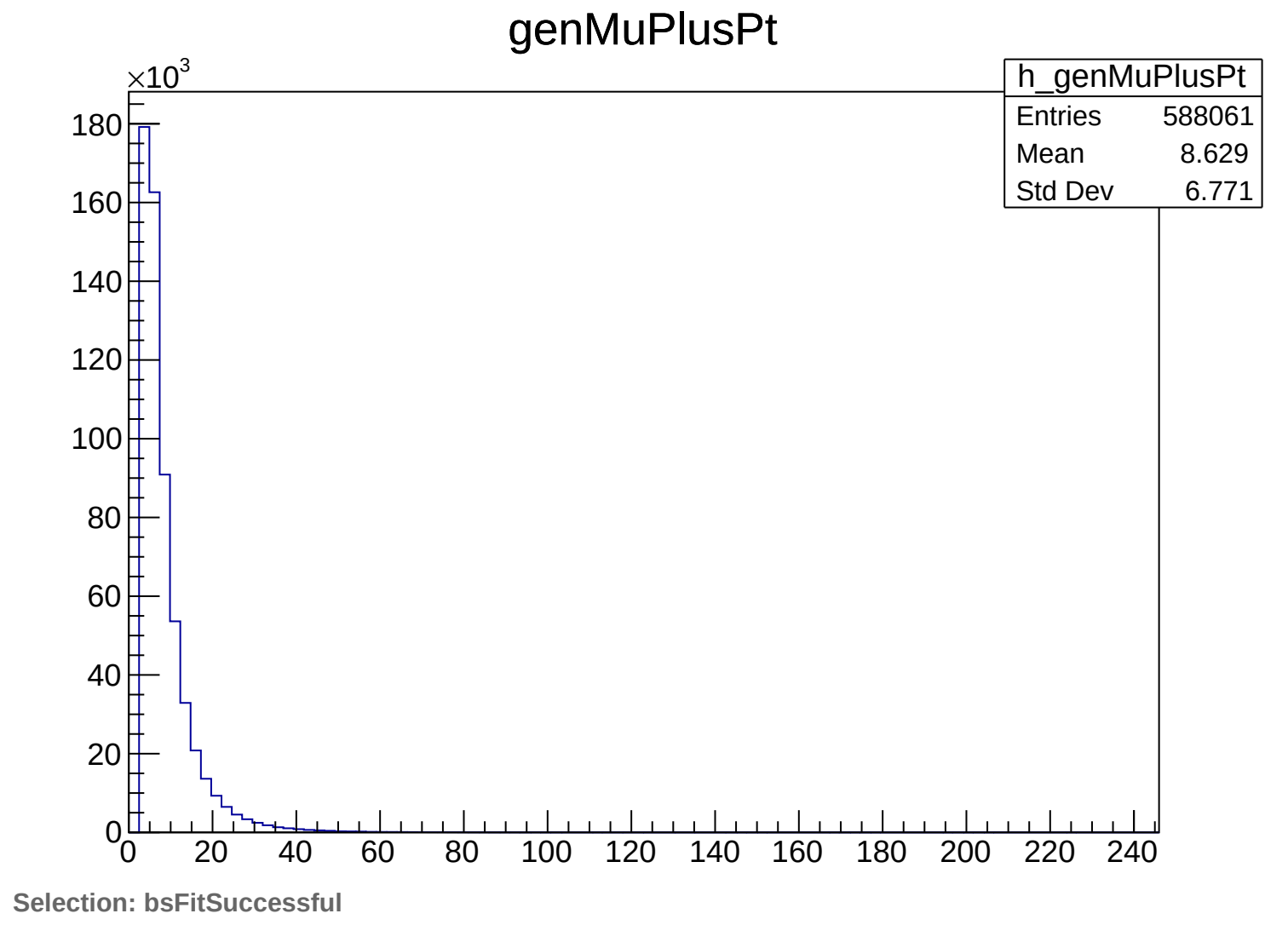

genIsHardProcess
×10^3
600
500
400
300
200
100
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2
h_genIsHardProcess
Entries 588061
Mean 0
Std Dev 0
Selection: bsFitSuccessful

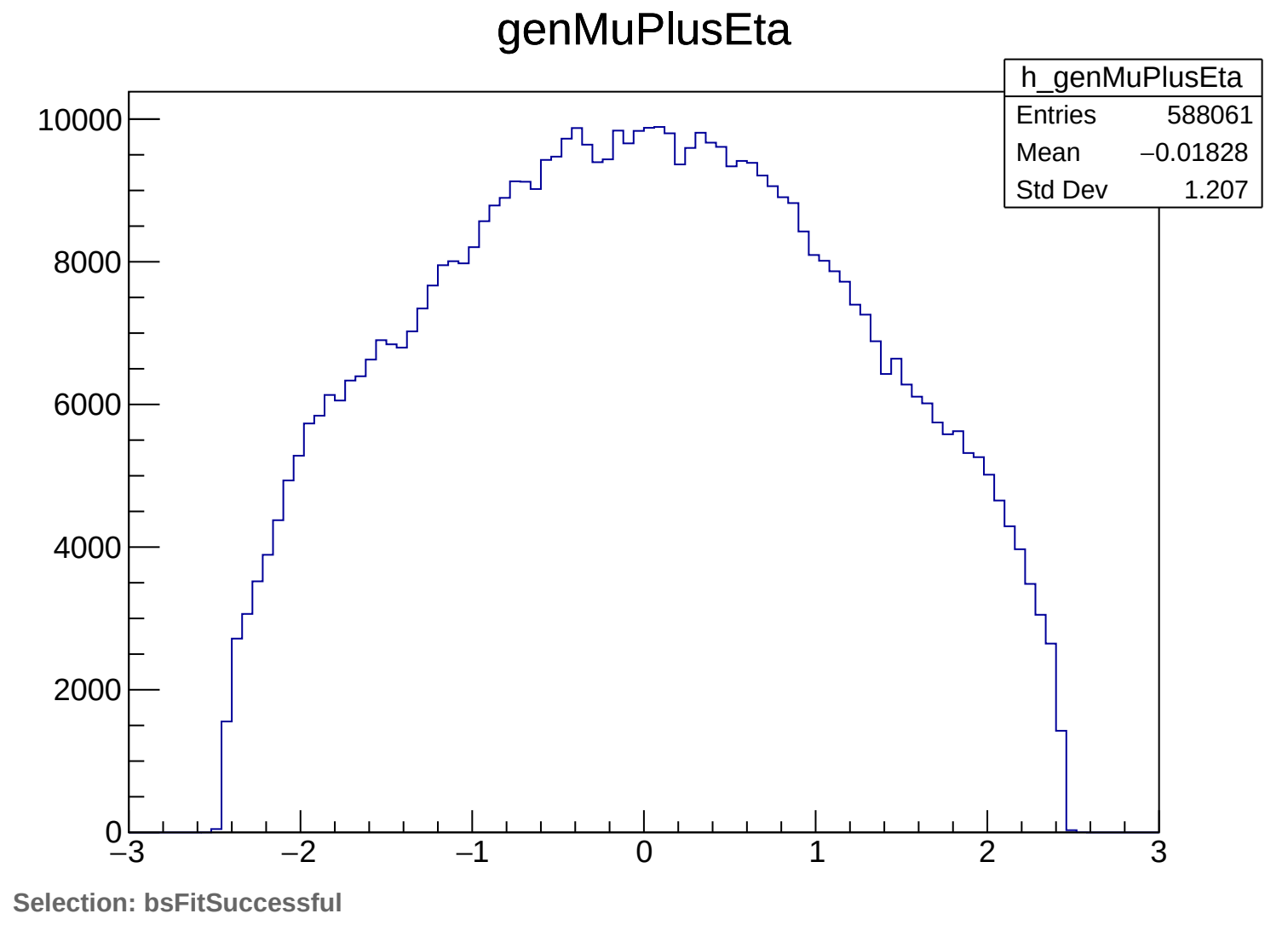

genBPt
h_genBPt
Entries 588061
Mean 32.82
Std Dev 14.79
×10^3
120
100
80
60
40
20
0
0
50
100
150
200
250
300
350
400
450
Selection: bsFitSuccessful

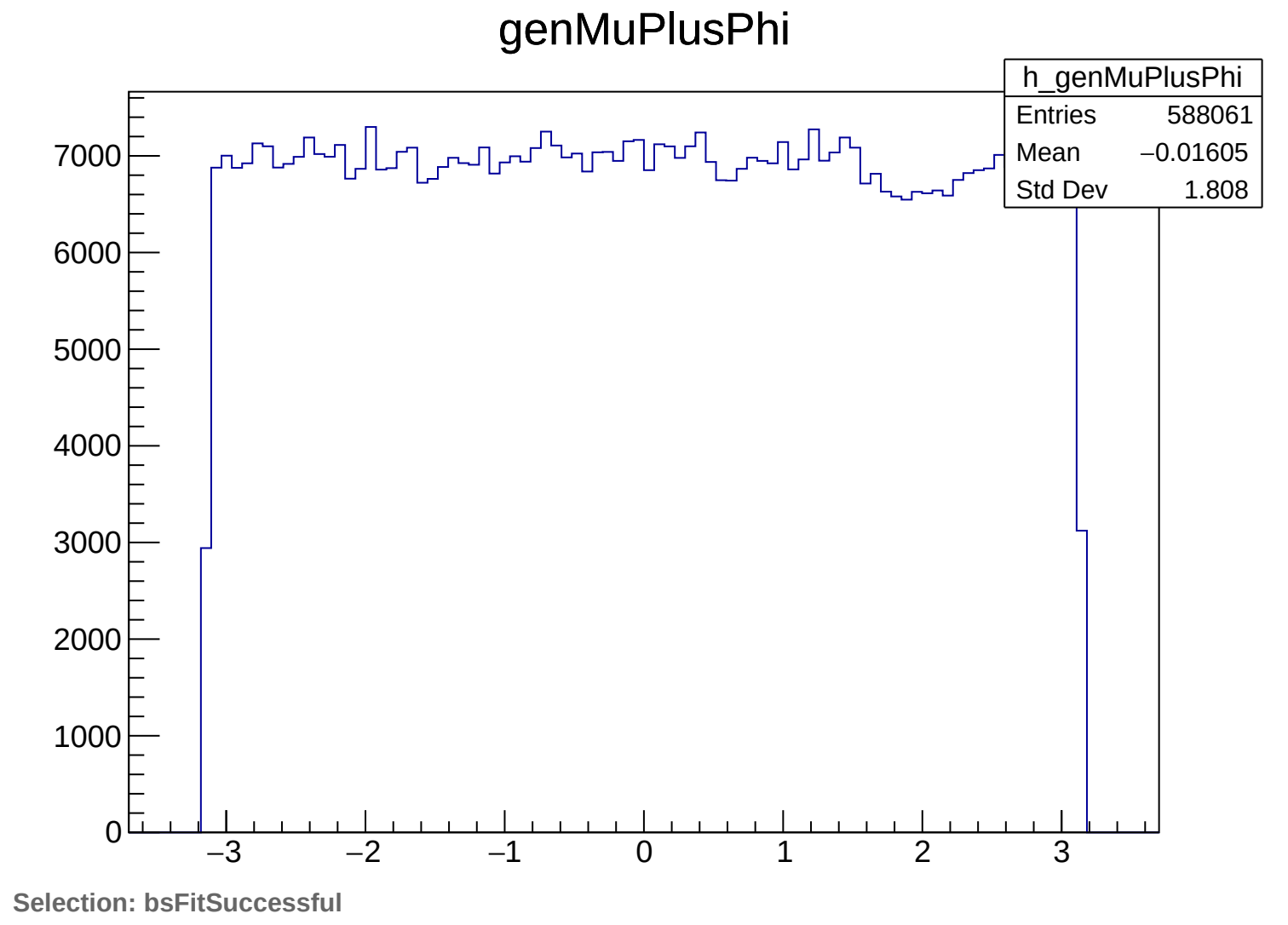

genBEta
h_genBEta
Entries 588061
Mean −0.01854
Std Dev 1.208
10000
8000
6000
4000
2000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

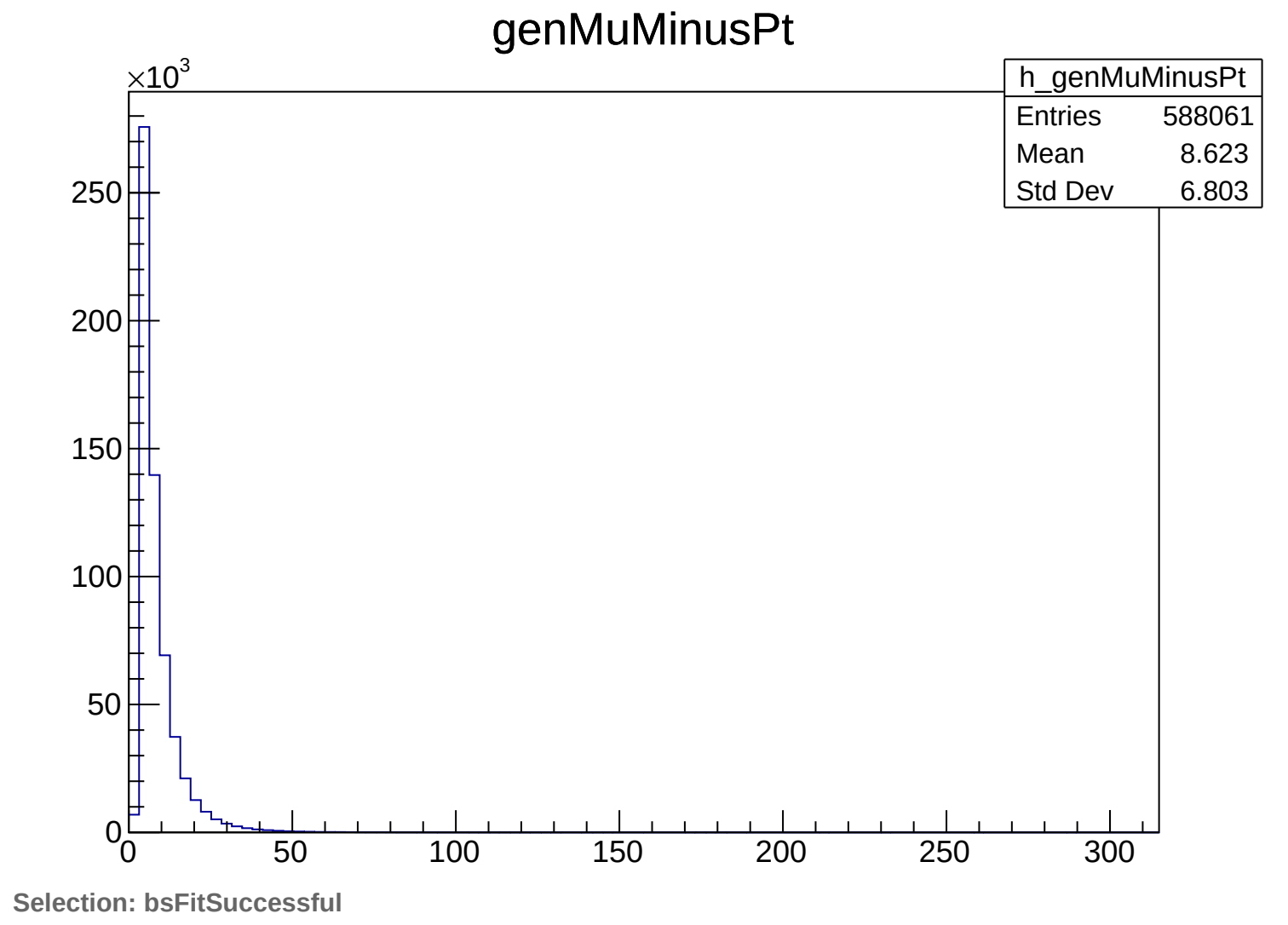

genBPhi
h_genBPhi
Entries 588061
Mean −0.01964
Std Dev 1.806
7000
6000
5000
4000
3000
2000
1000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

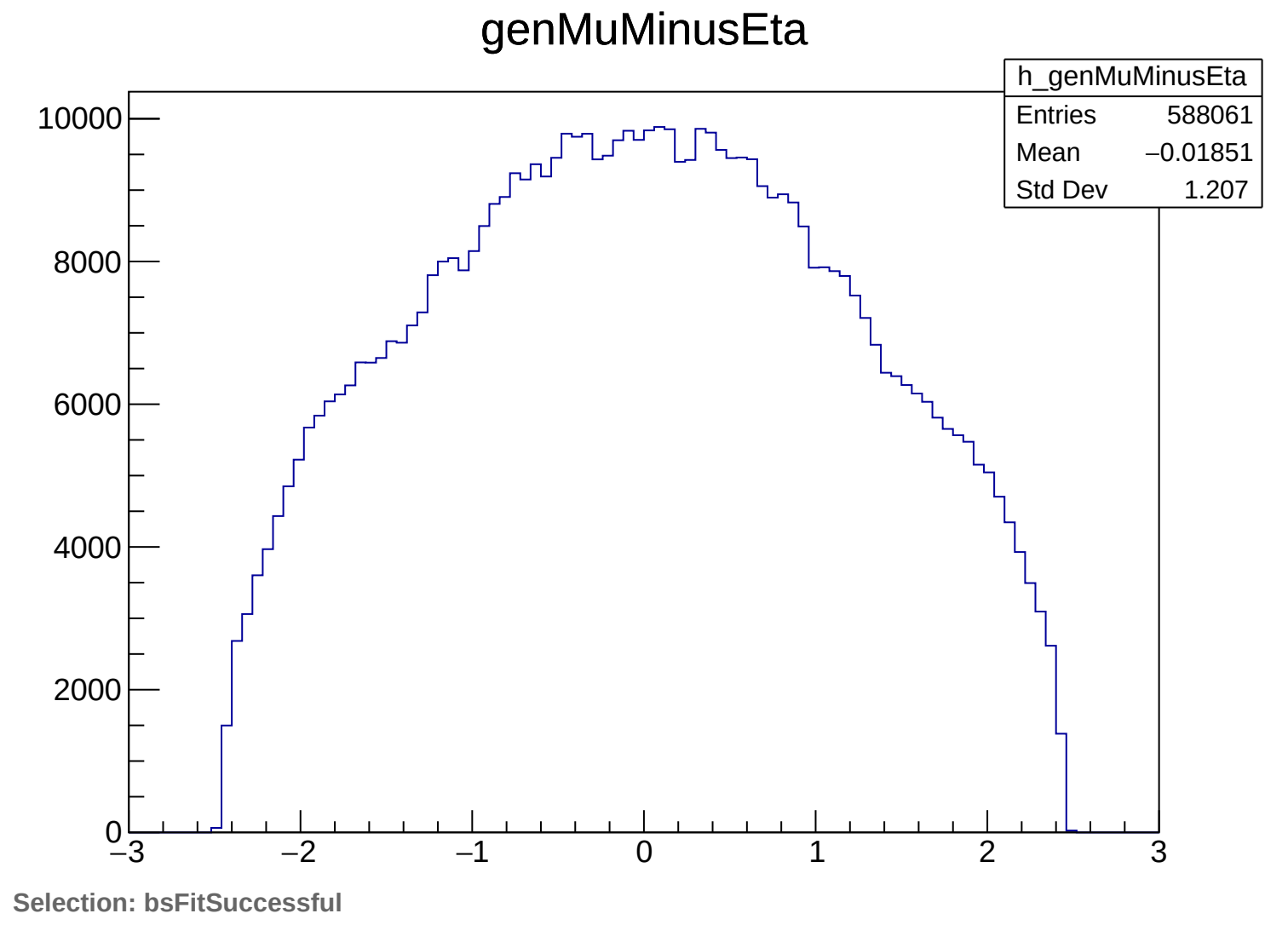

genBMass
h_genBMass
Entries 588061
Mean 5.367
Std Dev 1.548e−05
×10^3
450
400
350
300
250
200
150
100
50
0
5.366765
5.366790691
Selection: bsFitSuccessful

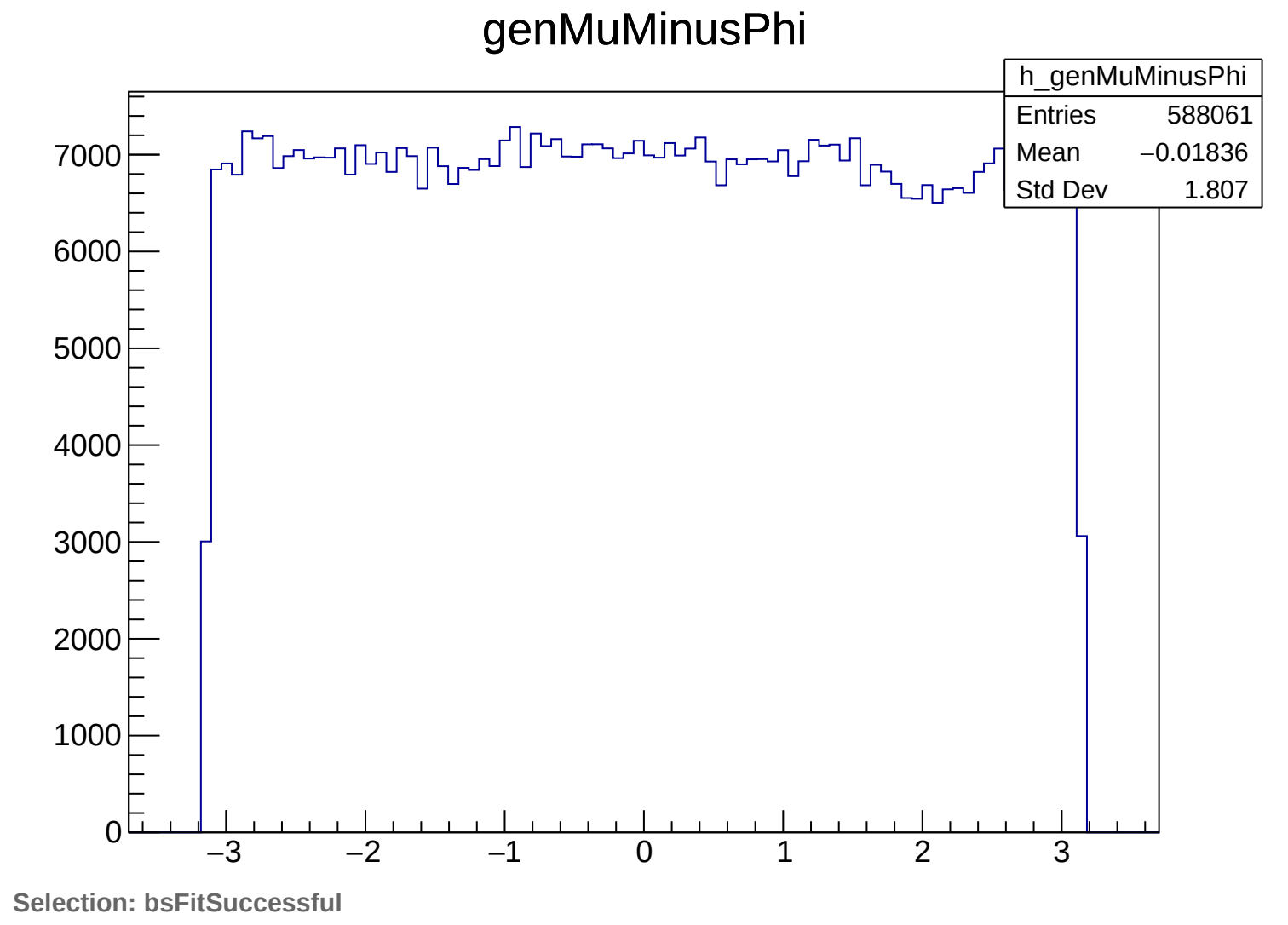

genMuPlusPt
×10^3
h_genMuPlusPt
Entries 588061
Mean 8.629
Std Dev 6.771
180
160
140
120
100
80
60
40
20
0
0
20
40
60
80
100
120
140
160
180
200
220
240
Selection: bsFitSuccessful

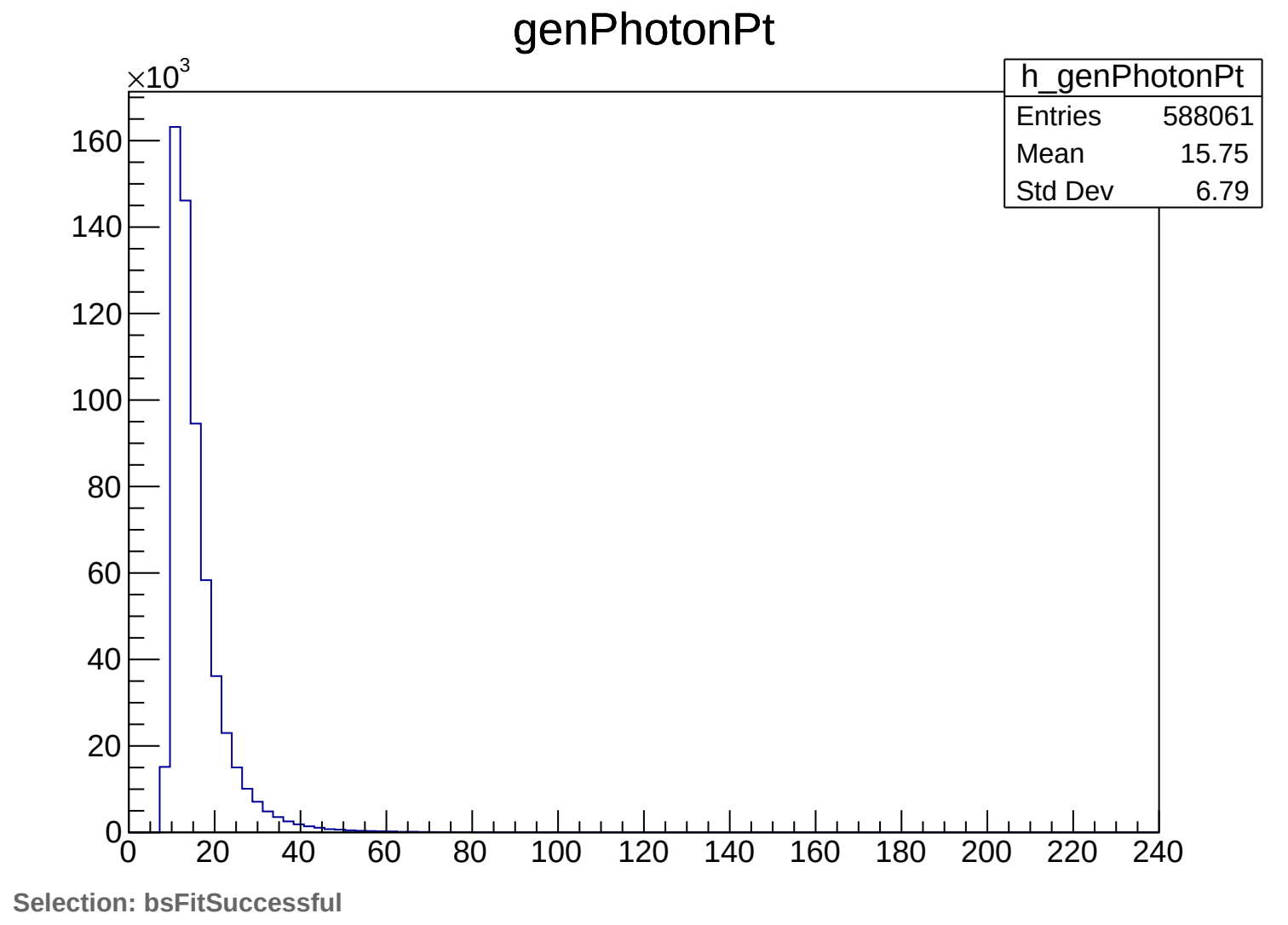

genMuPlusEta
h_genMuPlusEta
Entries 588061
Mean −0.01828
Std Dev 1.207
10000
8000
6000
4000
2000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

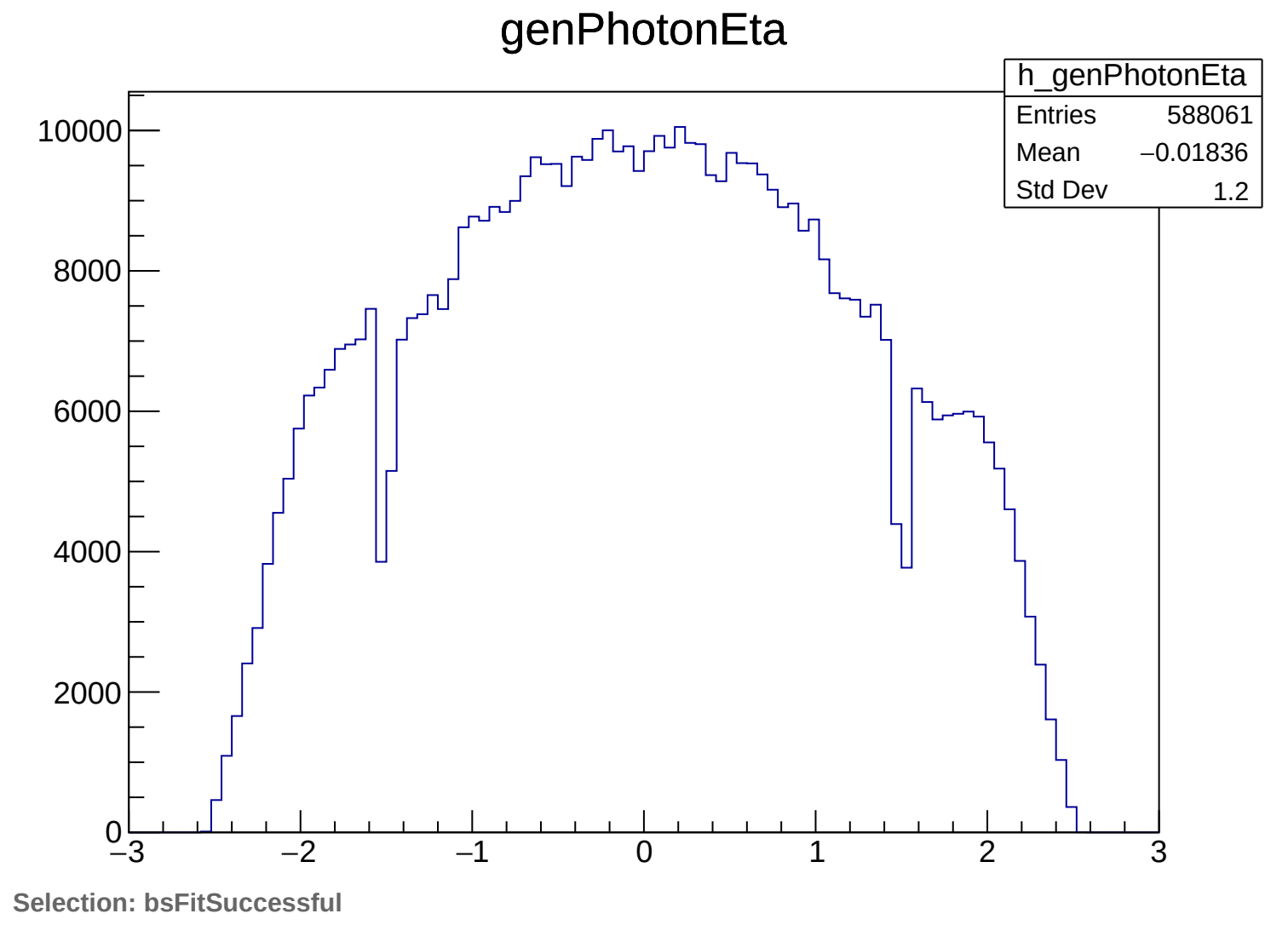

genMuPlusPhi
h_genMuPlusPhi
Entries 588061
Mean −0.01605
Std Dev 1.808
7000
6000
5000
4000
3000
2000
1000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

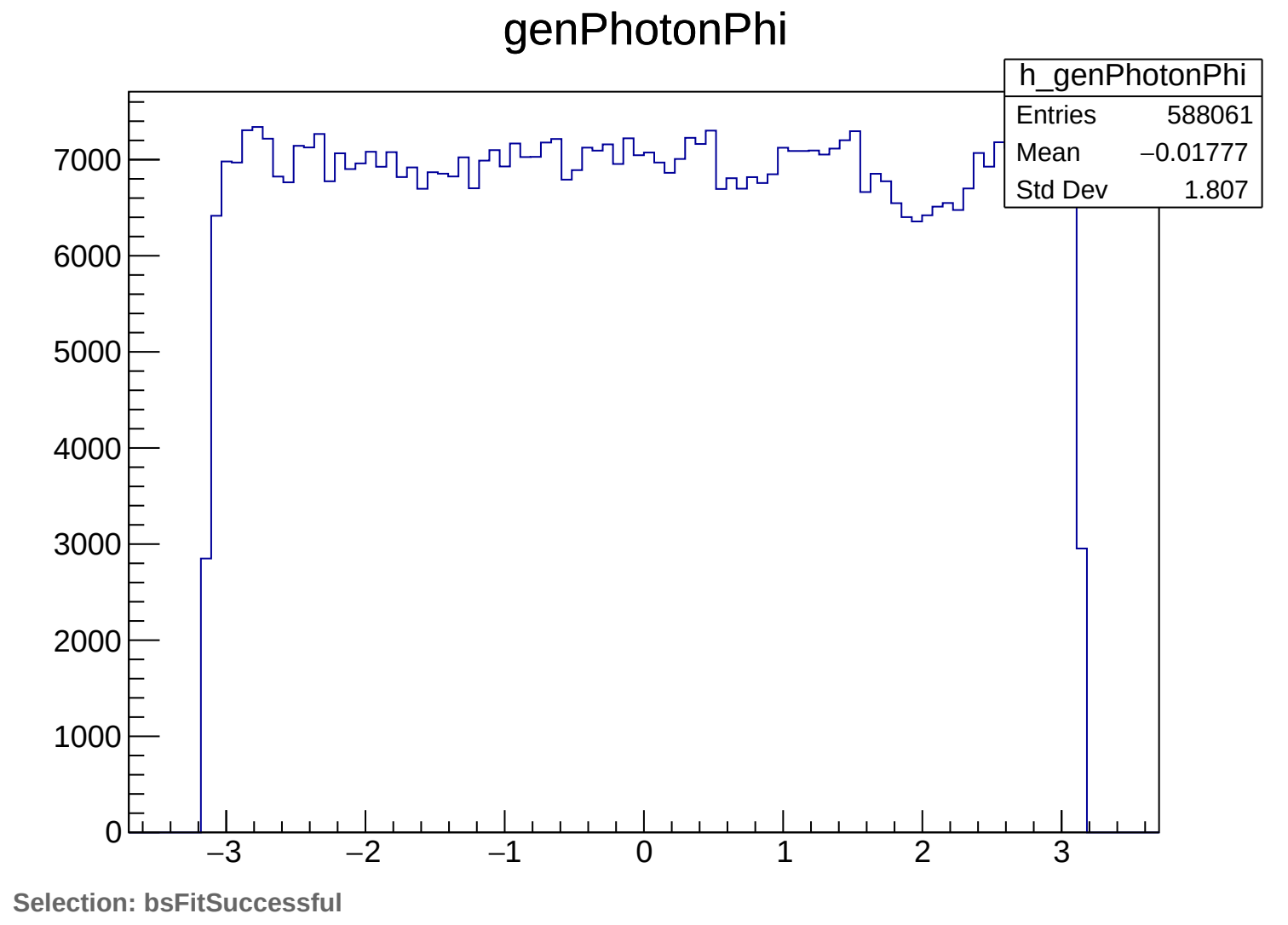

genMuMinusPt
×10^3
h_genMuMinusPt
Entries 588061
Mean 8.623
Std Dev 6.803
250
200
150
100
50
0
0
50
100
150
200
250
300
Selection: bsFitSuccessful

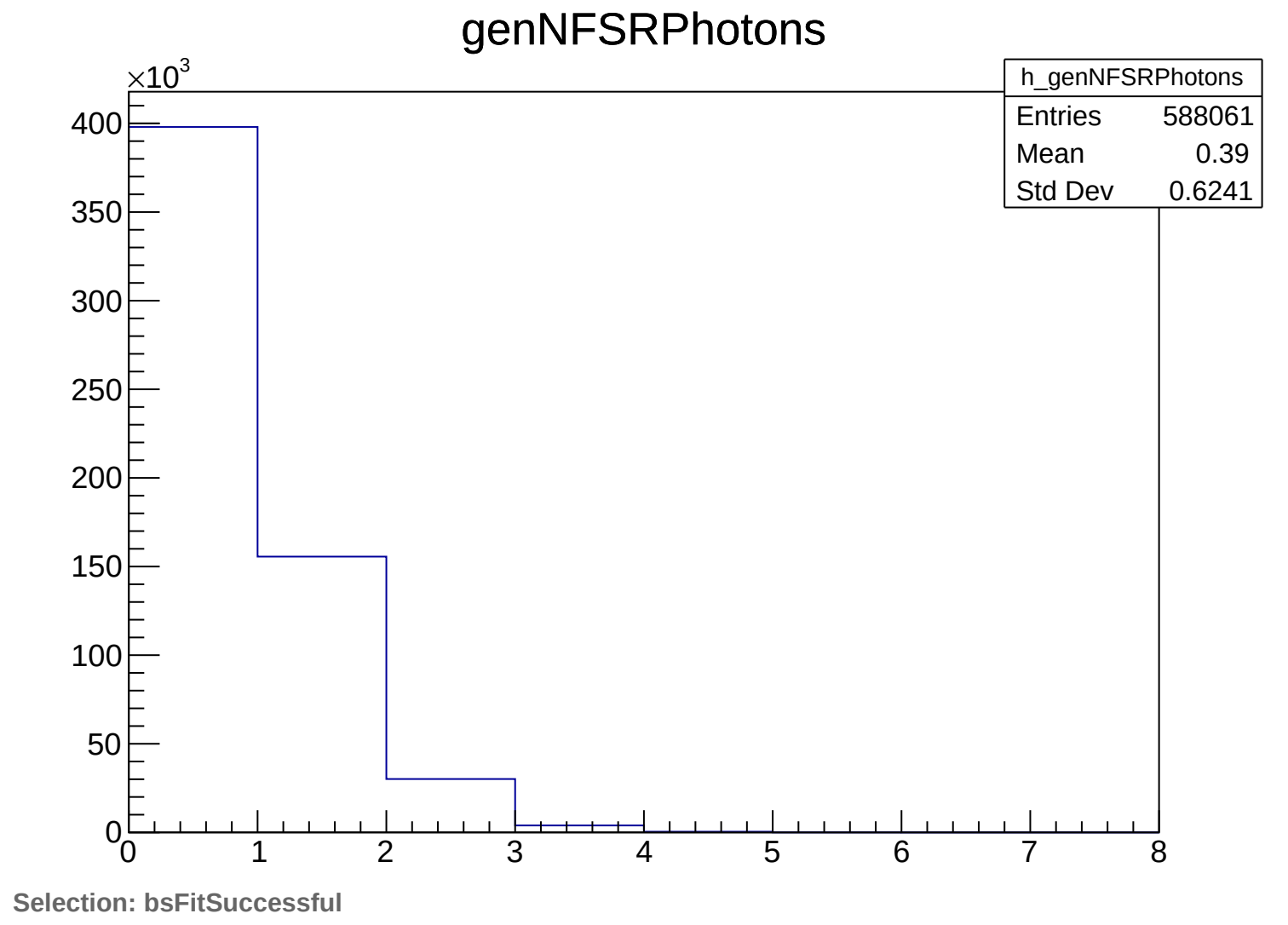

genMuMinusEta
h_genMuMinusEta
Entries 588061
Mean −0.01851
Std Dev 1.207
10000
8000
6000
4000
2000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

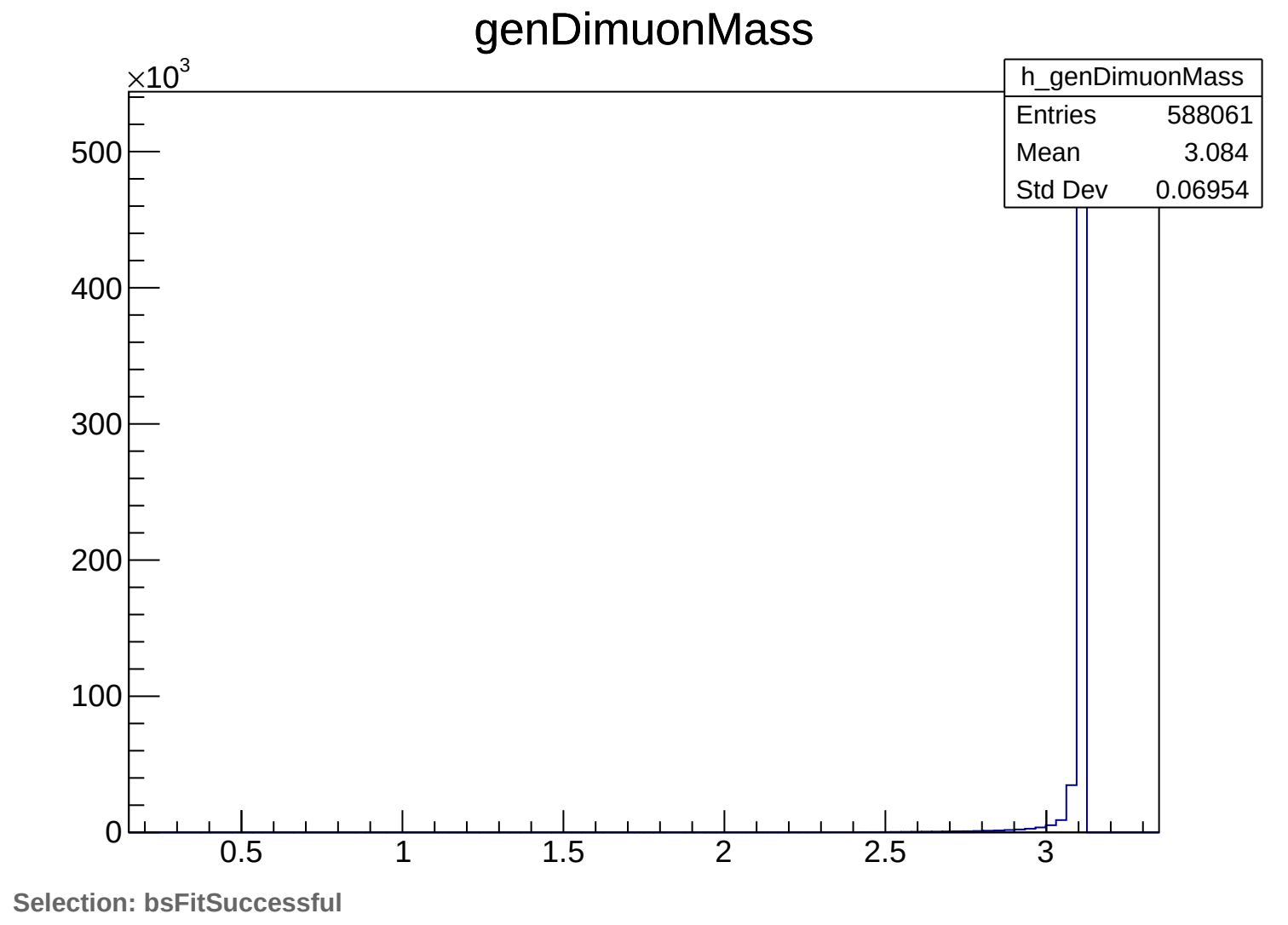

genMuMinusPhi
h_genMuMinusPhi
Entries 588061
Mean −0.01836
Std Dev 1.807
7000
6000
5000
4000
3000
2000
1000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful


genPhotonPt
×10^3
h_genPhotonPt
Entries 588061
Mean 15.75
Std Dev 6.79
160
140
120
100
80
60
40
20
0
0
20
40
60
80
100
120
140
160
180
200
220
240
Selection: bsFitSuccessful

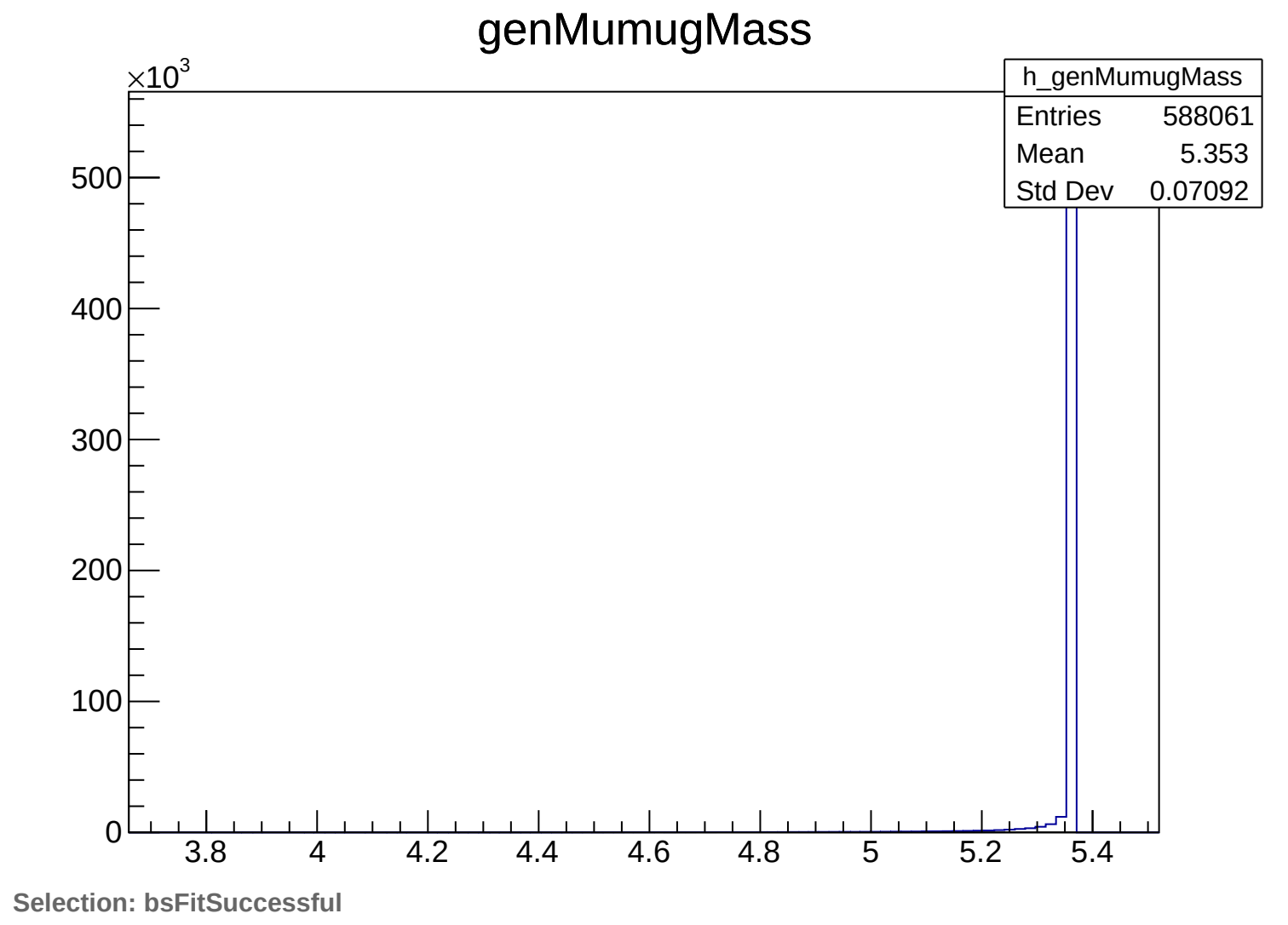

genPhotonEta
h_genPhotonEta
Entries 588061
Mean −0.01836
Std Dev 1.2
10000
8000
6000
4000
2000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

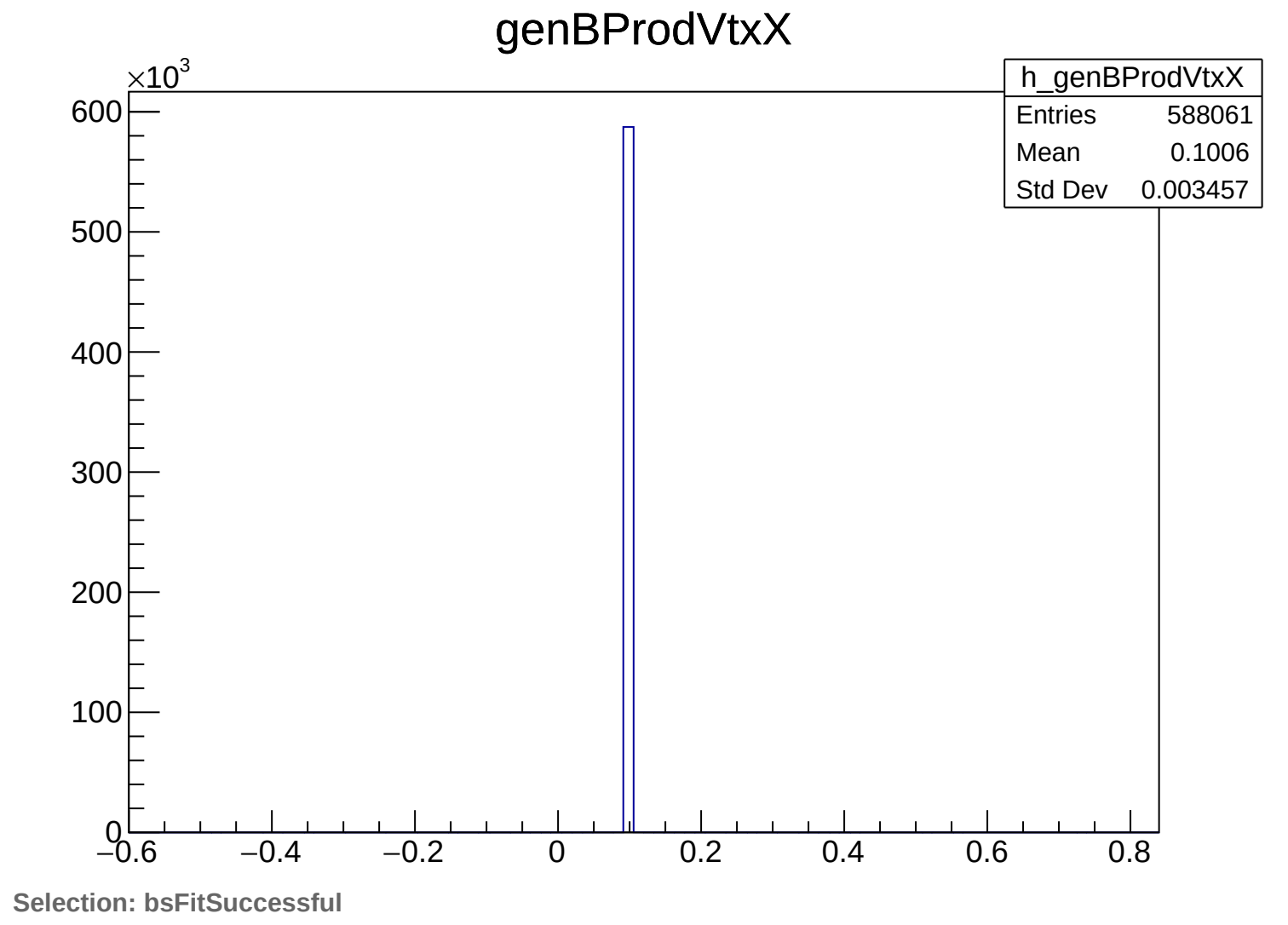

genPhotonPhi
h_genPhotonPhi
Entries 588061
Mean −0.01777
Std Dev 1.807
7000
6000
5000
4000
3000
2000
1000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

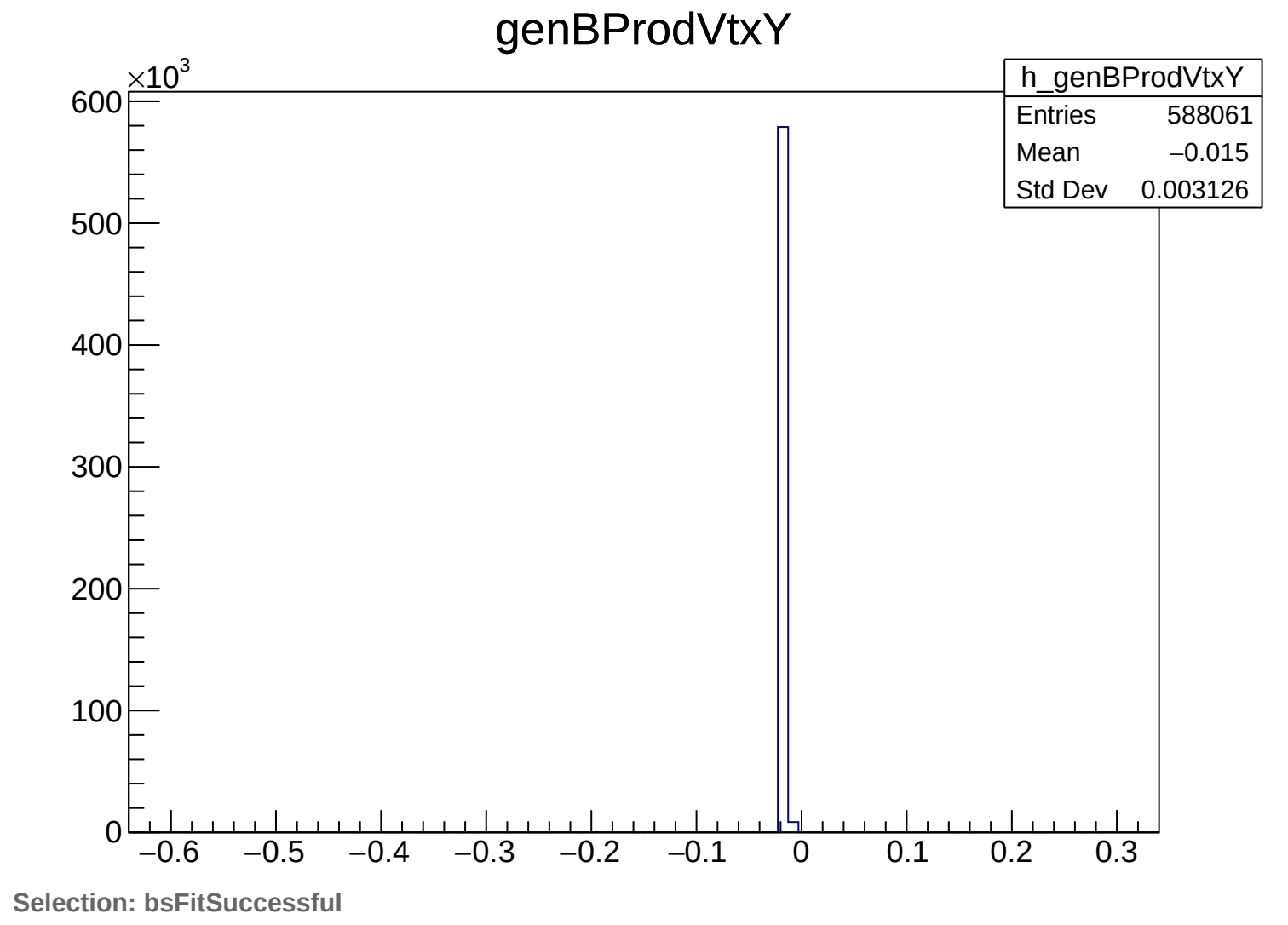

genNFSRPhotons
×10^3
h_genNFSRPhotons
Entries 588061
Mean 0.39
Std Dev 0.6241
400
350
300
250
200
150
100
50
0
0
1
2
3
4
5
6
7
8
Selection: bsFitSuccessful

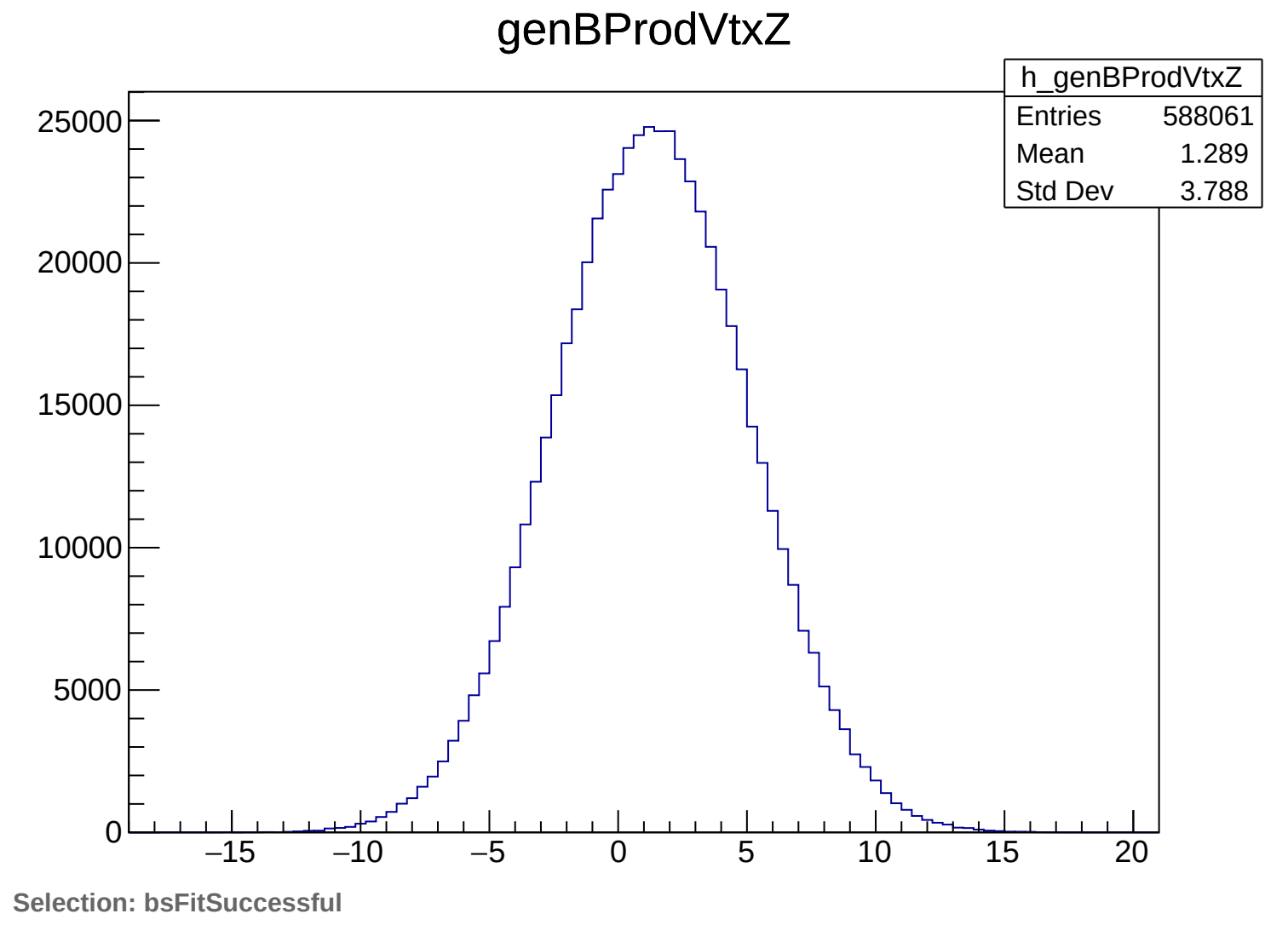


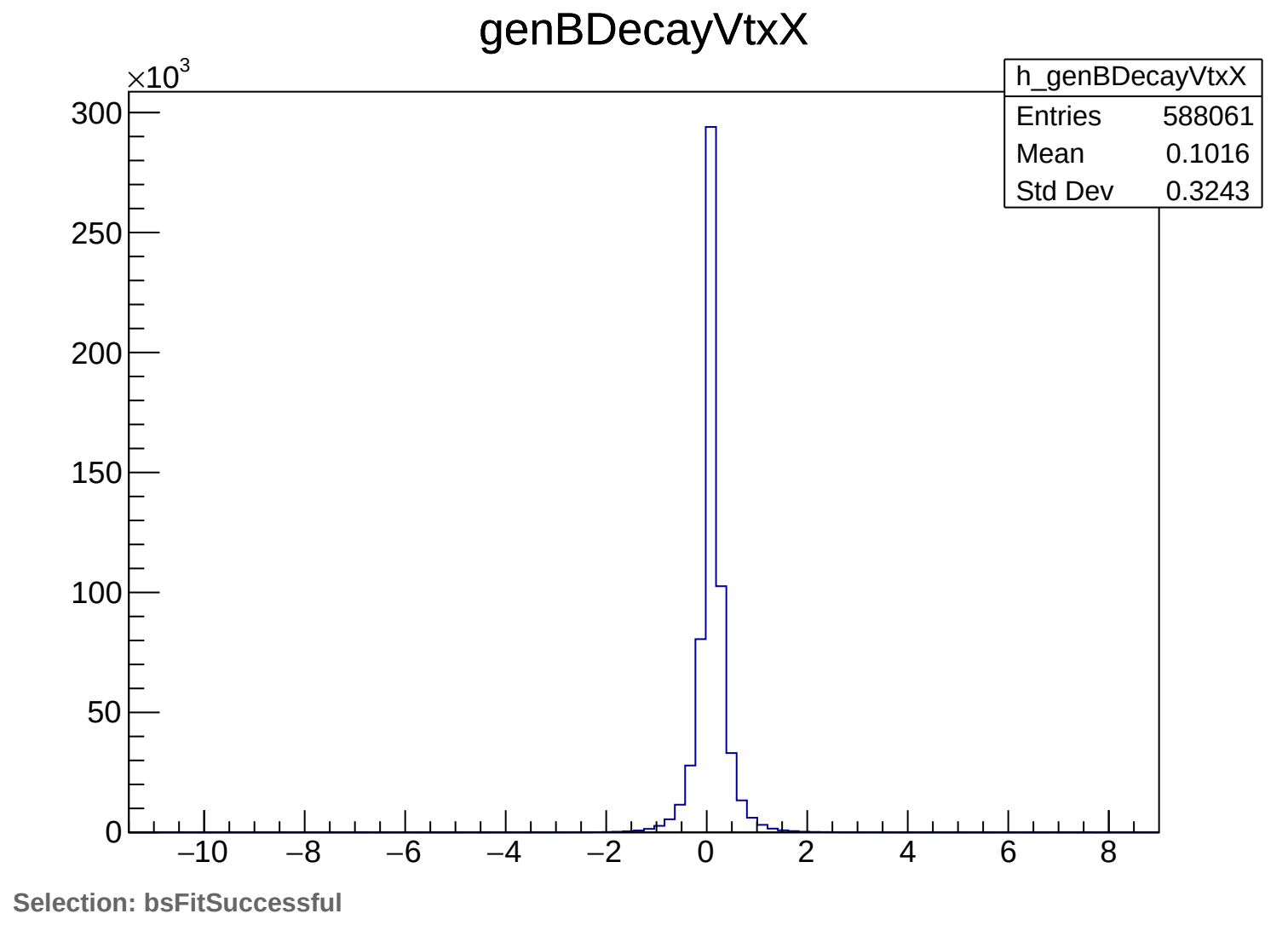
Selection: bsFitSuccessful

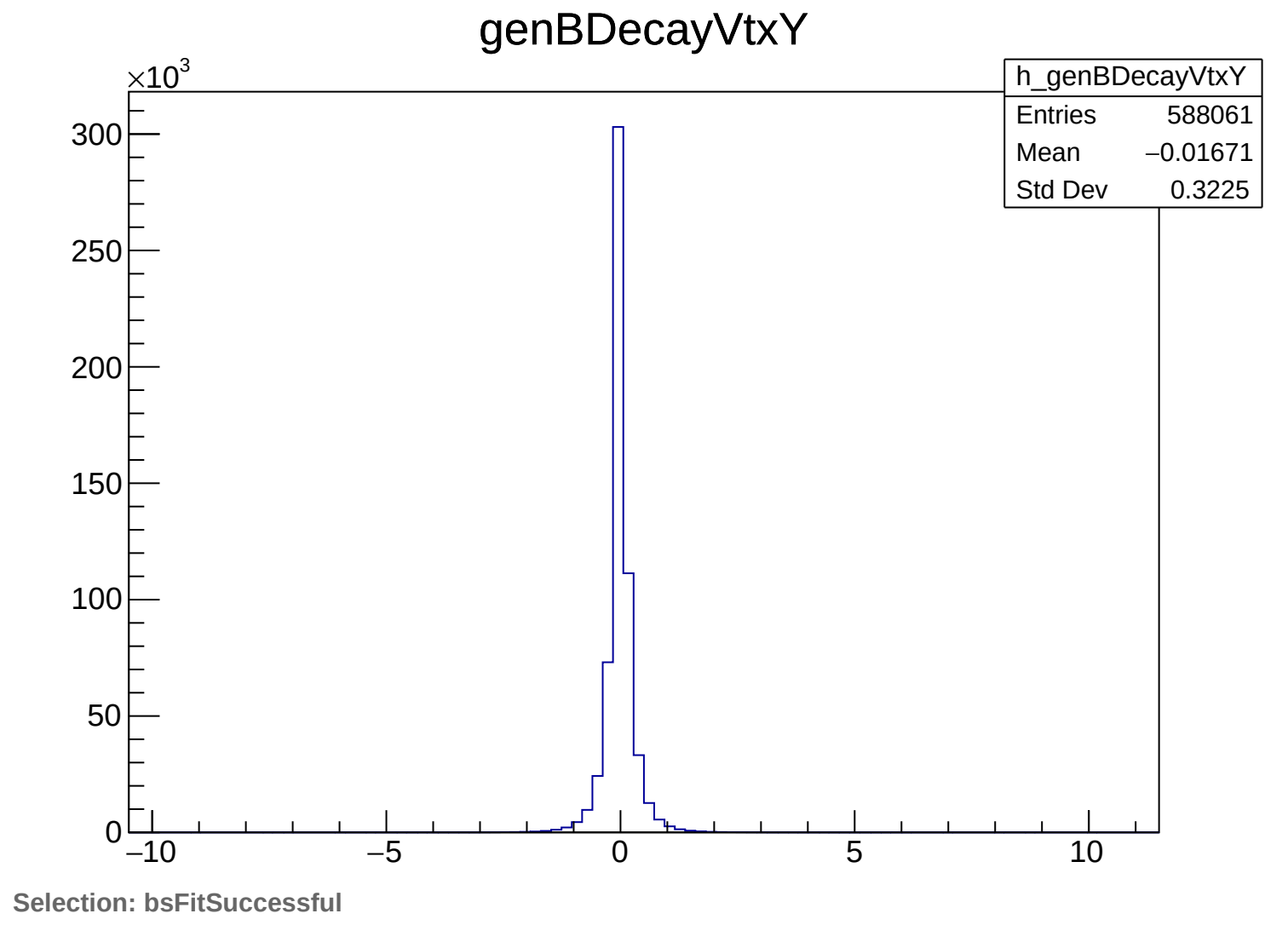

genMumugMass
×10^3
h_genMumugMass
Entries 588061
Mean 5.353
Std Dev 0.07092
500
400
300
200
100
0
3.8
4
4.2
4.4
4.6
4.8
5
5.2
5.4
Selection: bsFitSuccessful

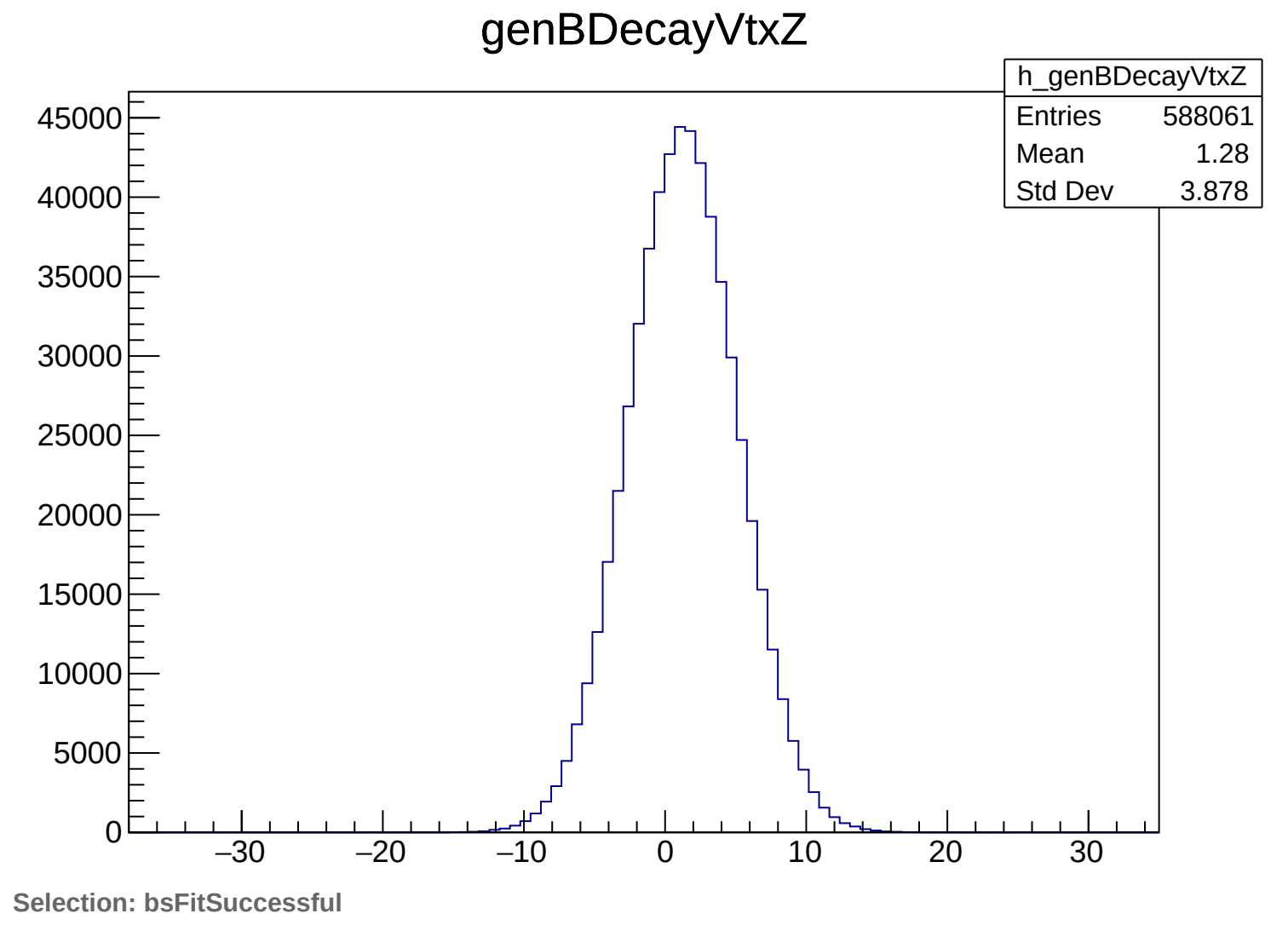

genBProdVtxX
×10^3
600
500
400
300
200
100
0
−0.6
−0.4
−0.2
0
0.2
0.4
0.6
0.8
h_genBProdVtxX
Entries 588061
Mean 0.1006
Std Dev 0.003457
Selection: bsFitSuccessful

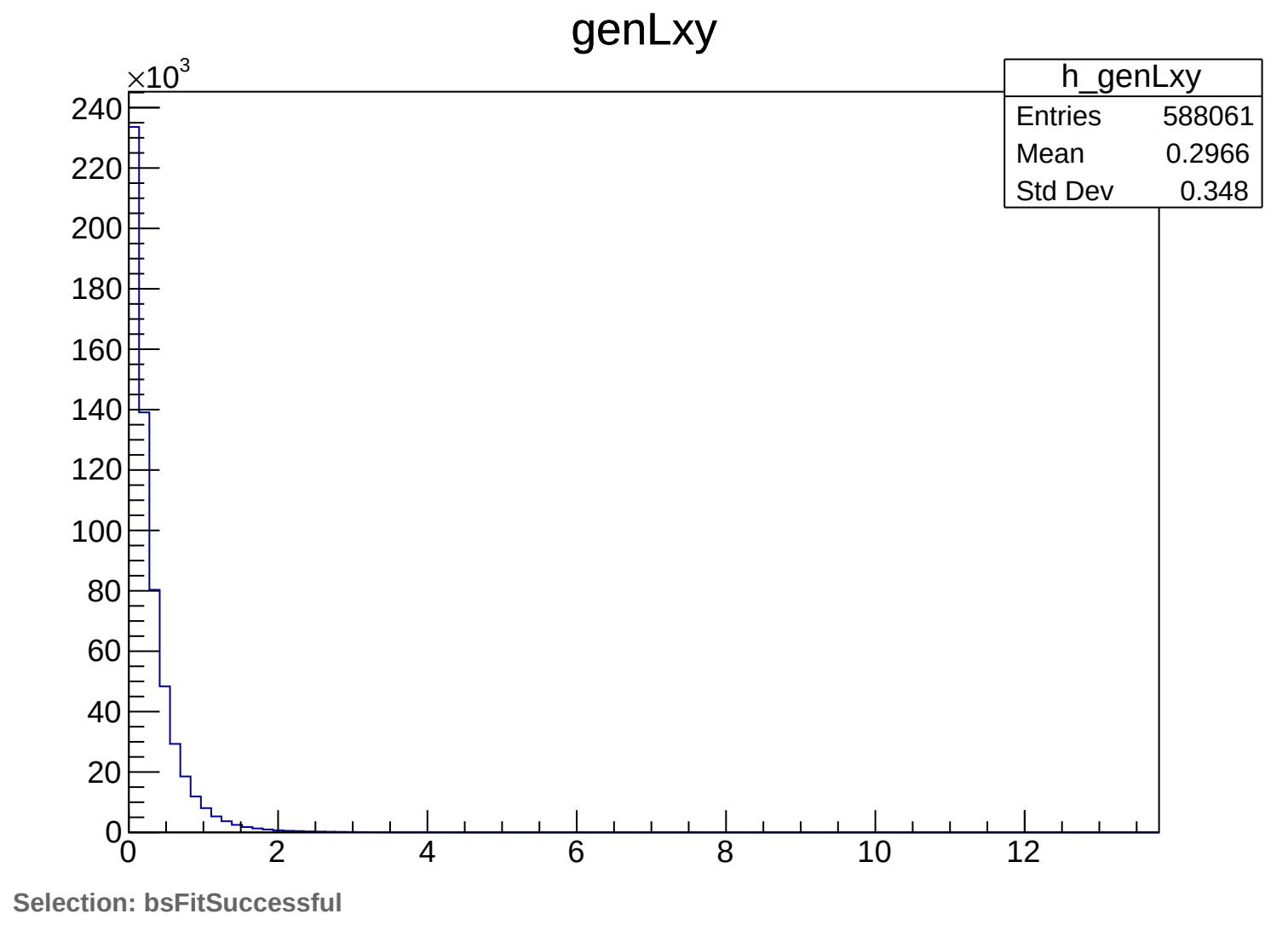

genBProdVtxY
×10^3
600
500
400
300
200
100
0
−0.6
−0.5
−0.4
−0.3
−0.2
−0.1
0
0.1
0.2
0.3
h_genBProdVtxY
Entries 588061
Mean −0.015
Std Dev 0.003126
Selection: bsFitSuccessful

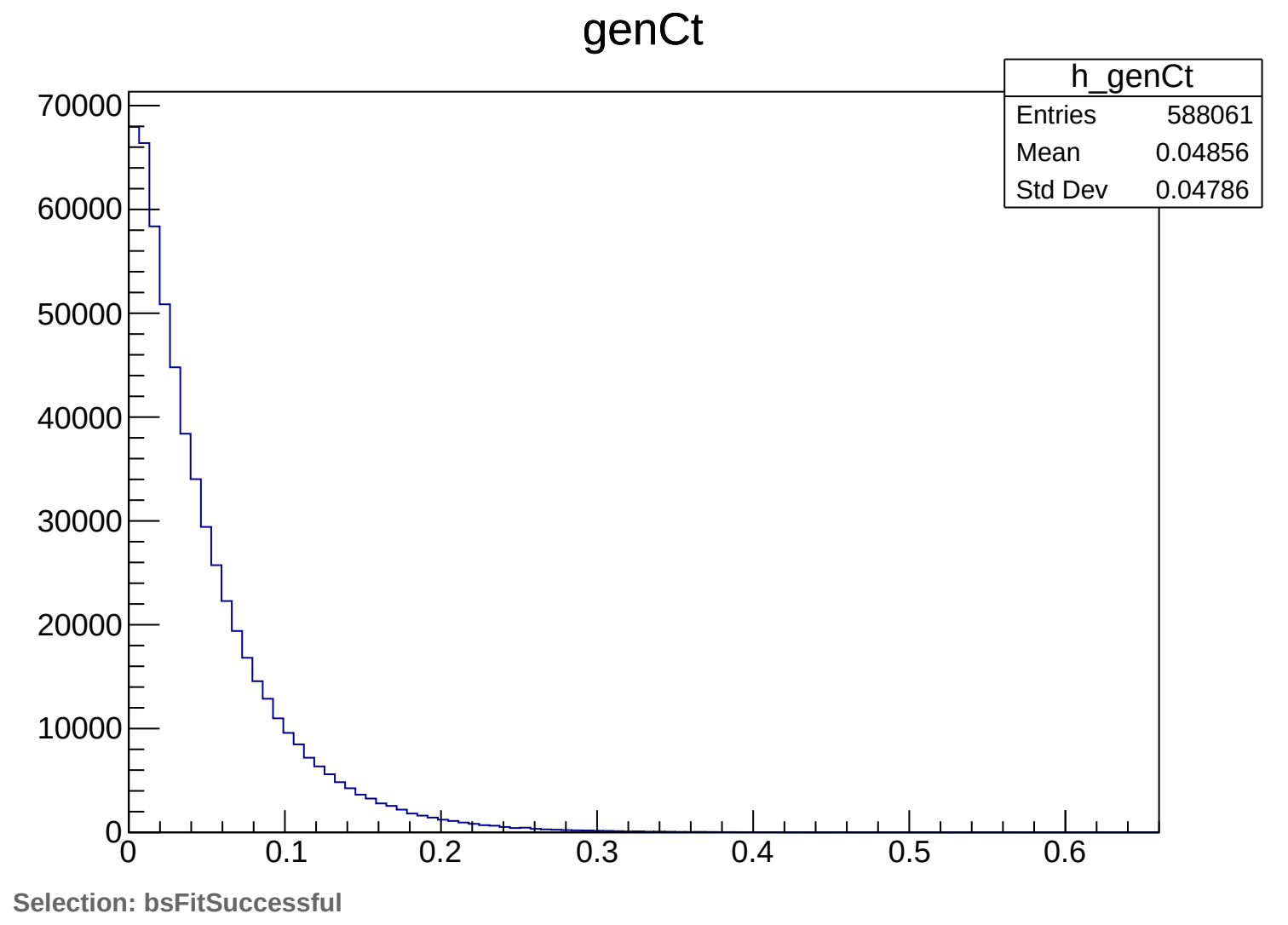

genBProdVtxZ
h_genBProdVtxZ
Entries 588061
Mean 1.289
Std Dev 3.788
25000
20000
15000
10000
5000
0
−15
−10
−5
0
5
10
15
20
Selection: bsFitSuccessful

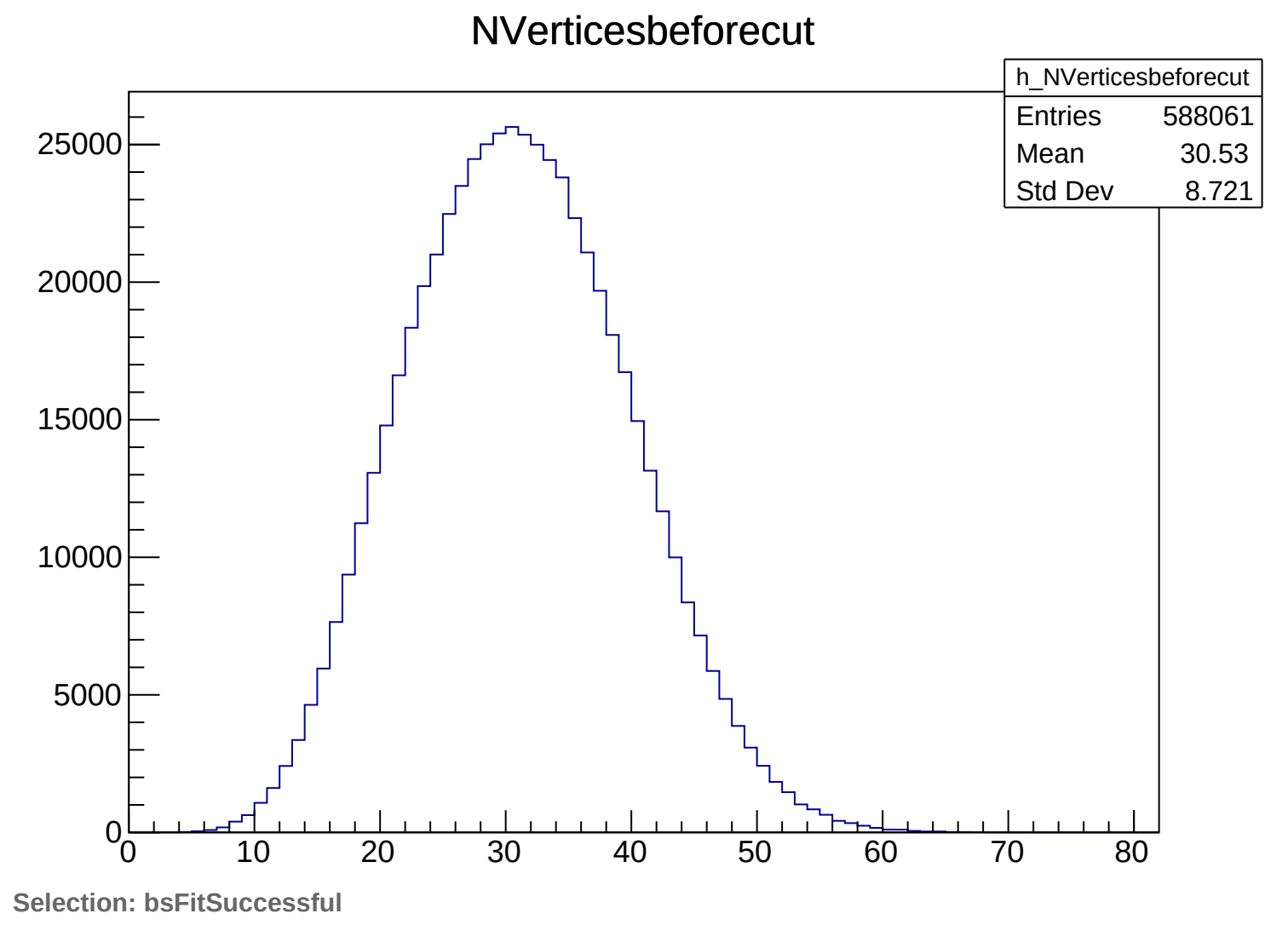

genBDecayVtxX
×10^3
300
250
200
150
100
50
0
−10
−8
−6
−4
−2
0
2
4
6
8
h_genBDecayVtxX
Entries 588061
Mean 0.1016
Std Dev 0.3243
Selection: bsFitSuccessful

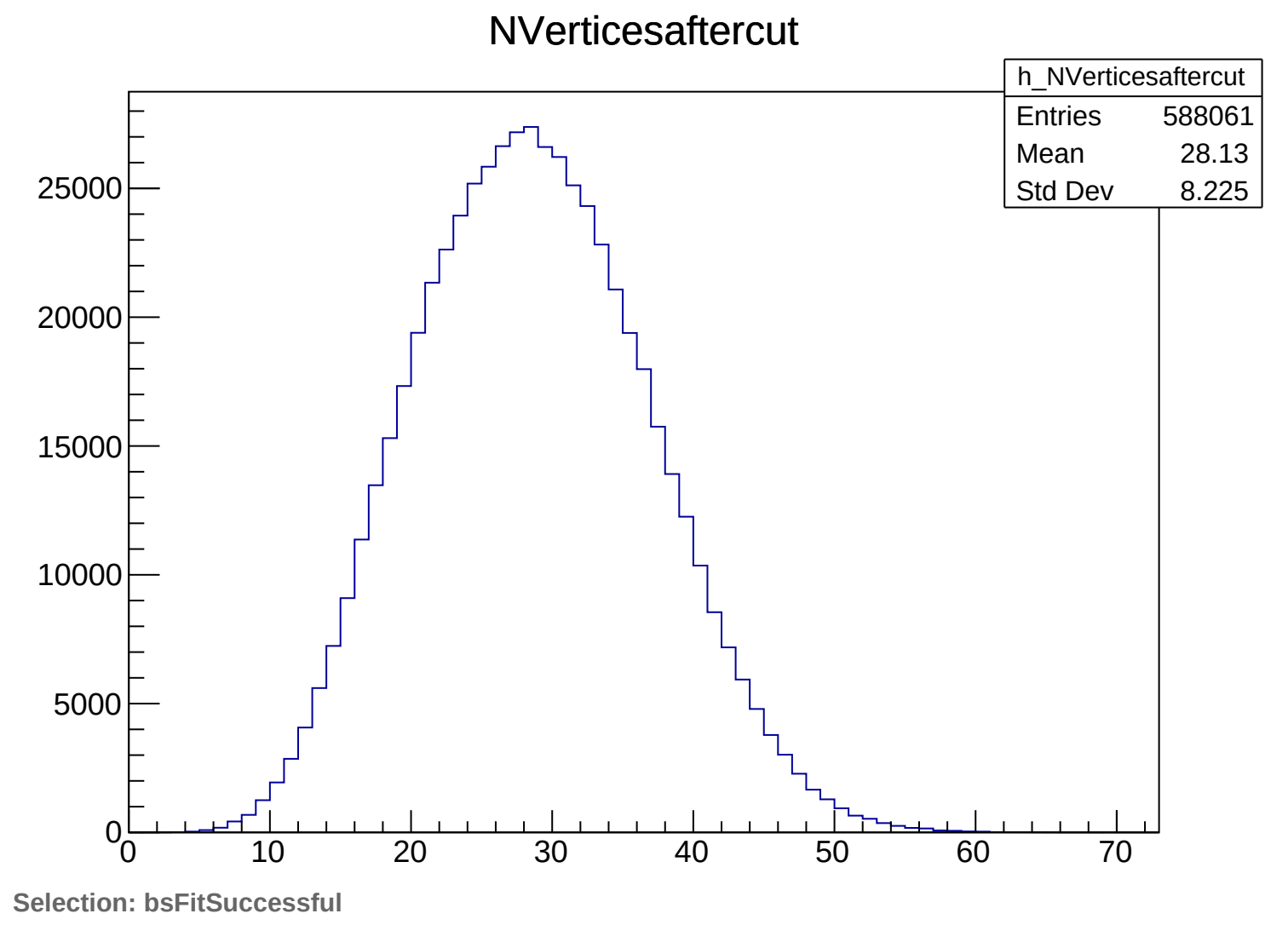

genBDecayVtxY
×10^3
300
250
200
150
100
50
0
−10
−5
0
5
10
h_genBDecayVtxY
Entries 588061
Mean −0.01671
Std Dev 0.3225
Selection: bsFitSuccessful

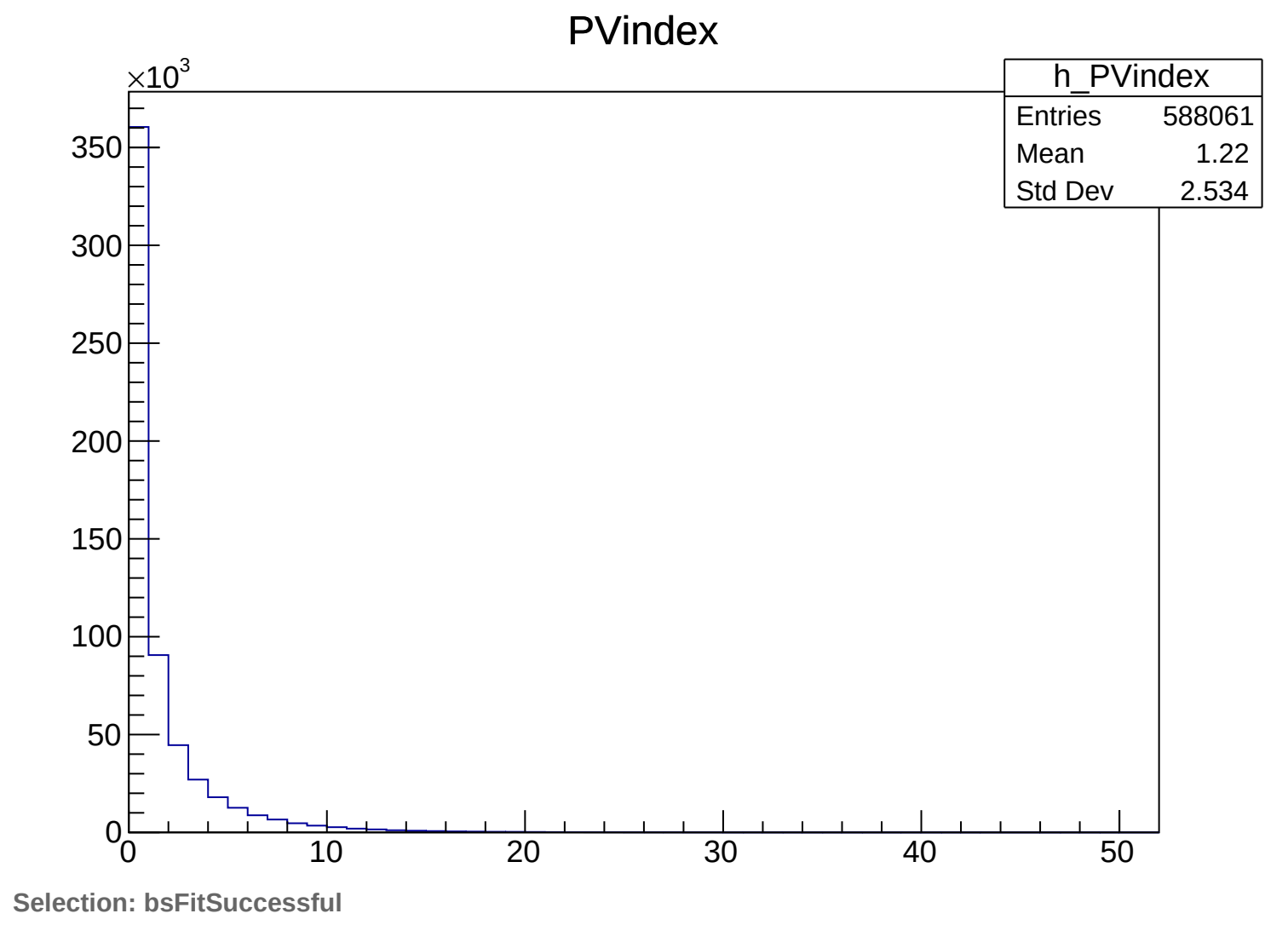

genBDecayVtxZ
h_genBDecayVtxZ
Entries 588061
Mean 1.28
Std Dev 3.878
45000
40000
35000
30000
25000
20000
15000
10000
5000
0
−30
−20
−10
0
10
20
30
Selection: bsFitSuccessful

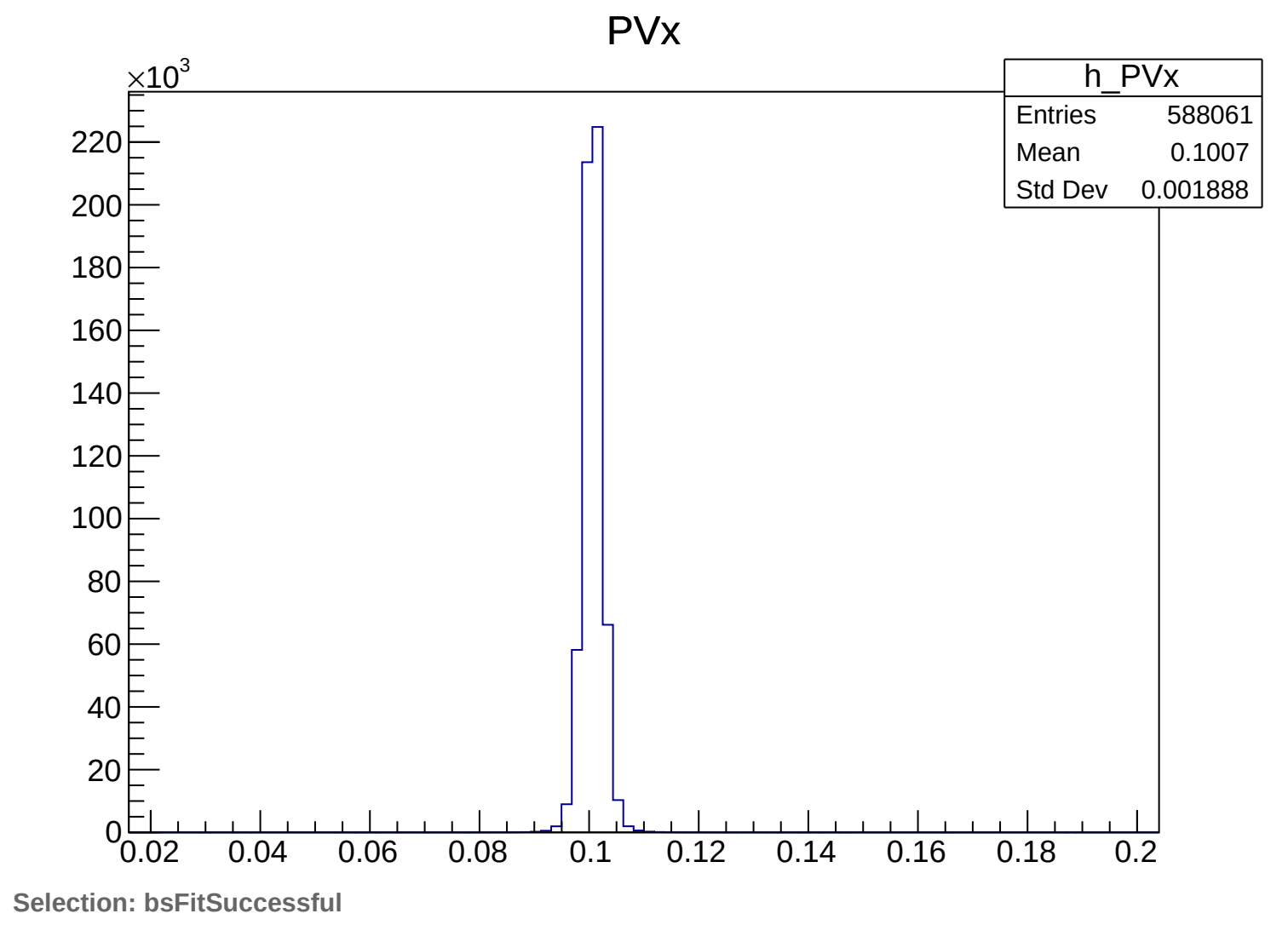

genLxy
×10^3
240
220
200
180
160
140
120
100
80
60
40
20
0
0
2
4
6
8
10
12
h_genLxy
Entries 588061
Mean 0.2966
Std Dev 0.348
Selection: bsFitSuccessful

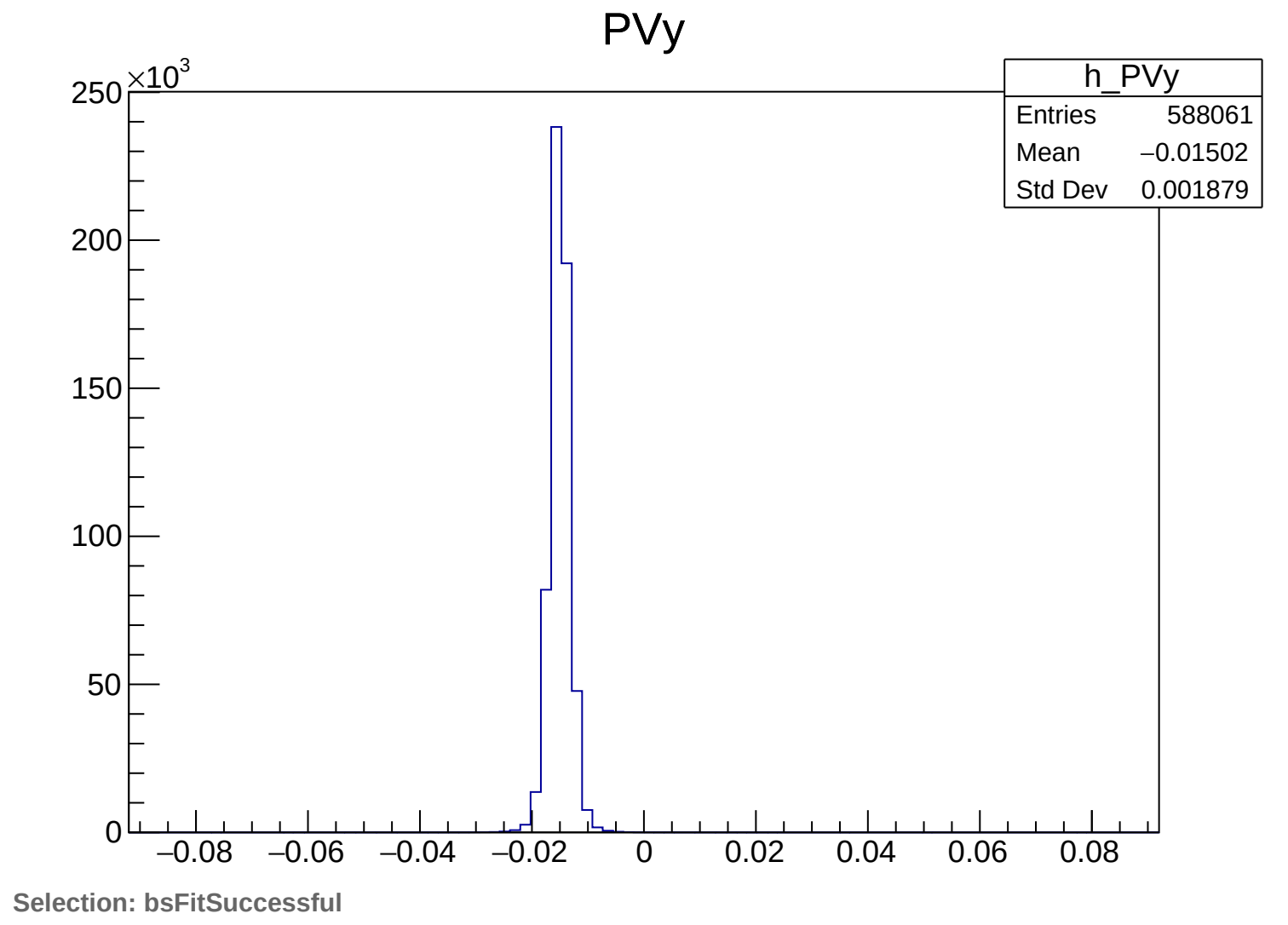

genCt
h_genCt
Entries 588061
Mean 0.04856
Std Dev 0.04786
70000
60000
50000
40000
30000
20000
10000
0
0
0.1
0.2
0.3
0.4
0.5
0.6
Selection: bsFitSuccessful

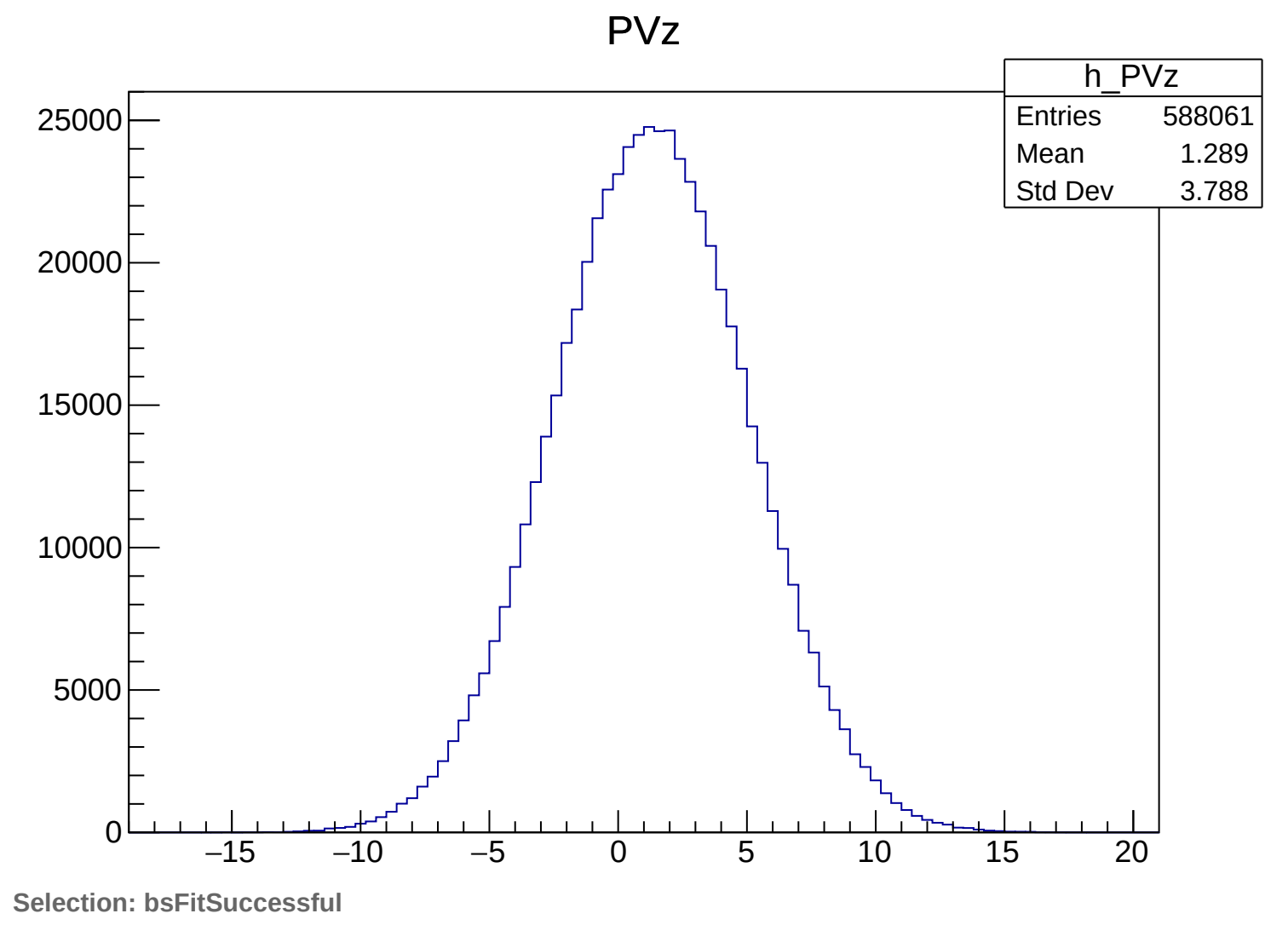

NVerticesbeforecut
h_NVerticesbeforecut
Entries 588061
Mean 30.53
Std Dev 8.721
25000
20000
15000
10000
5000
0
0
10
20
30
40
50
60
70
80
Selection: bsFitSuccessful

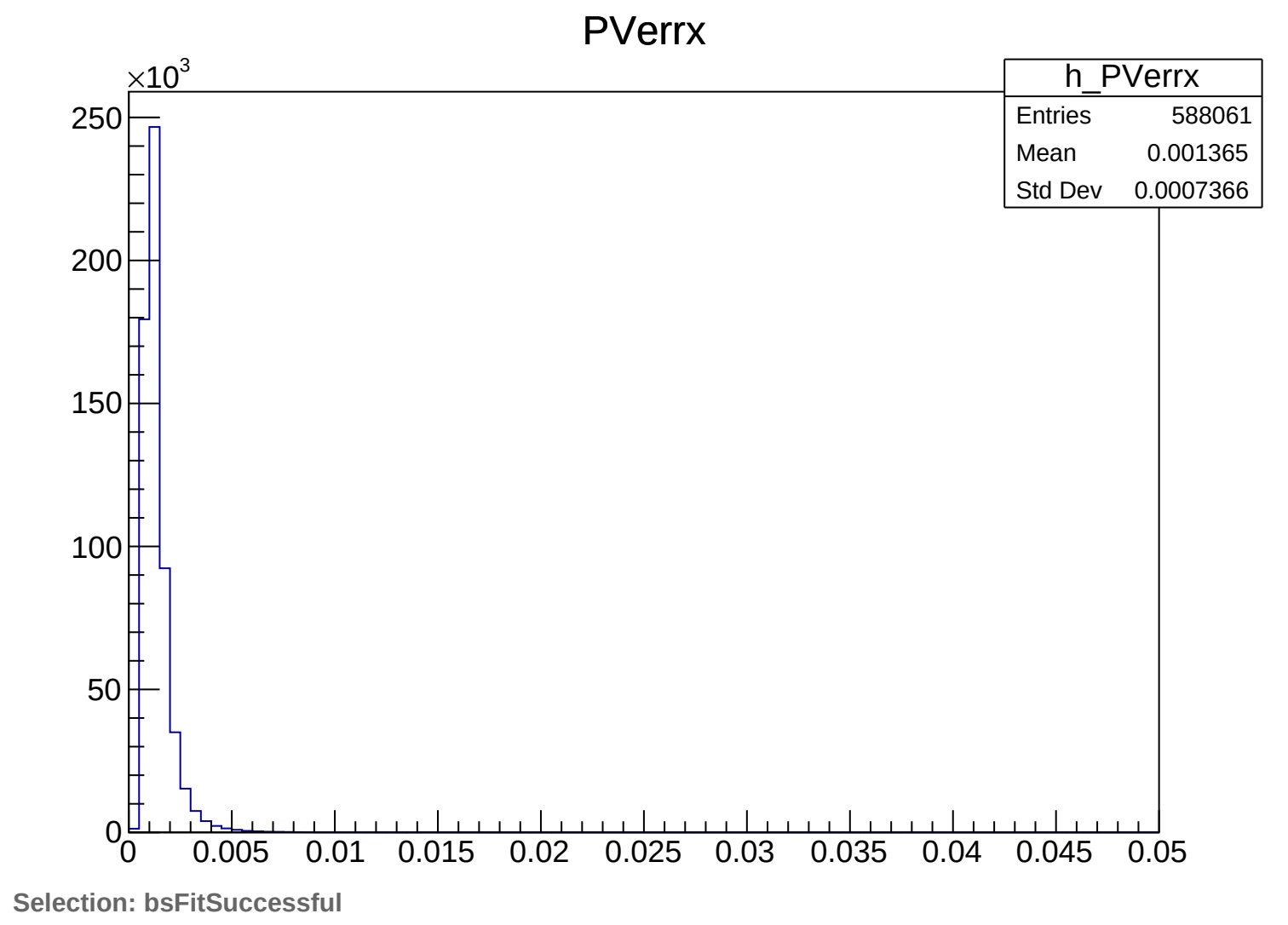

NVerticesaftercut
h_NVerticesaftercut
Entries 588061
Mean 28.13
Std Dev 8.225
25000
20000
15000
10000
5000
0
0
10
20
30
40
50
60
70
Selection: bsFitSuccessful

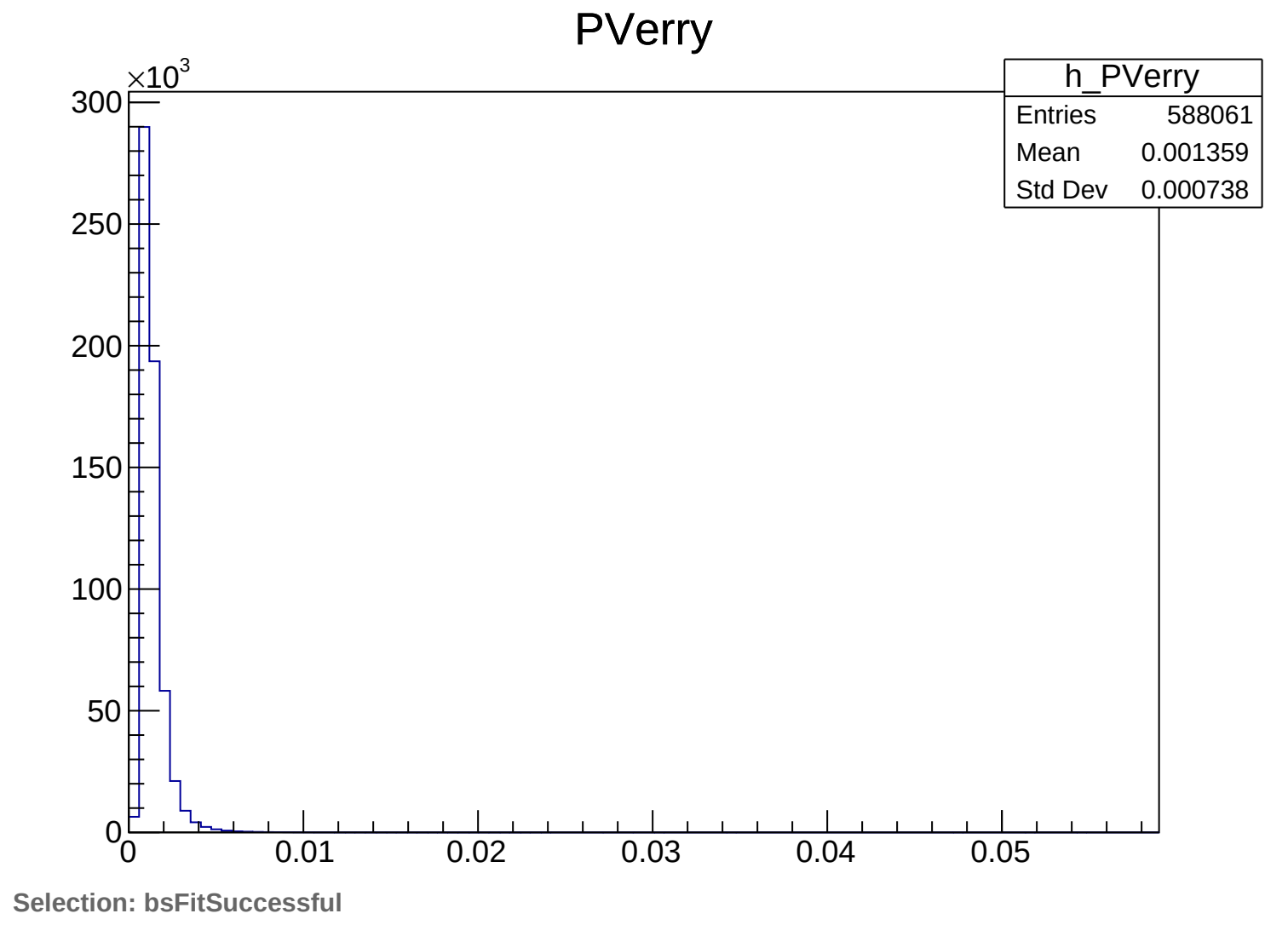

PVindex
h_PVindex
Entries 588061
Mean 1.22
Std Dev 2.534
×10^3
350
300
250
200
150
100
50
0
0
10
20
30
40
50
Selection: bsFitSuccessful

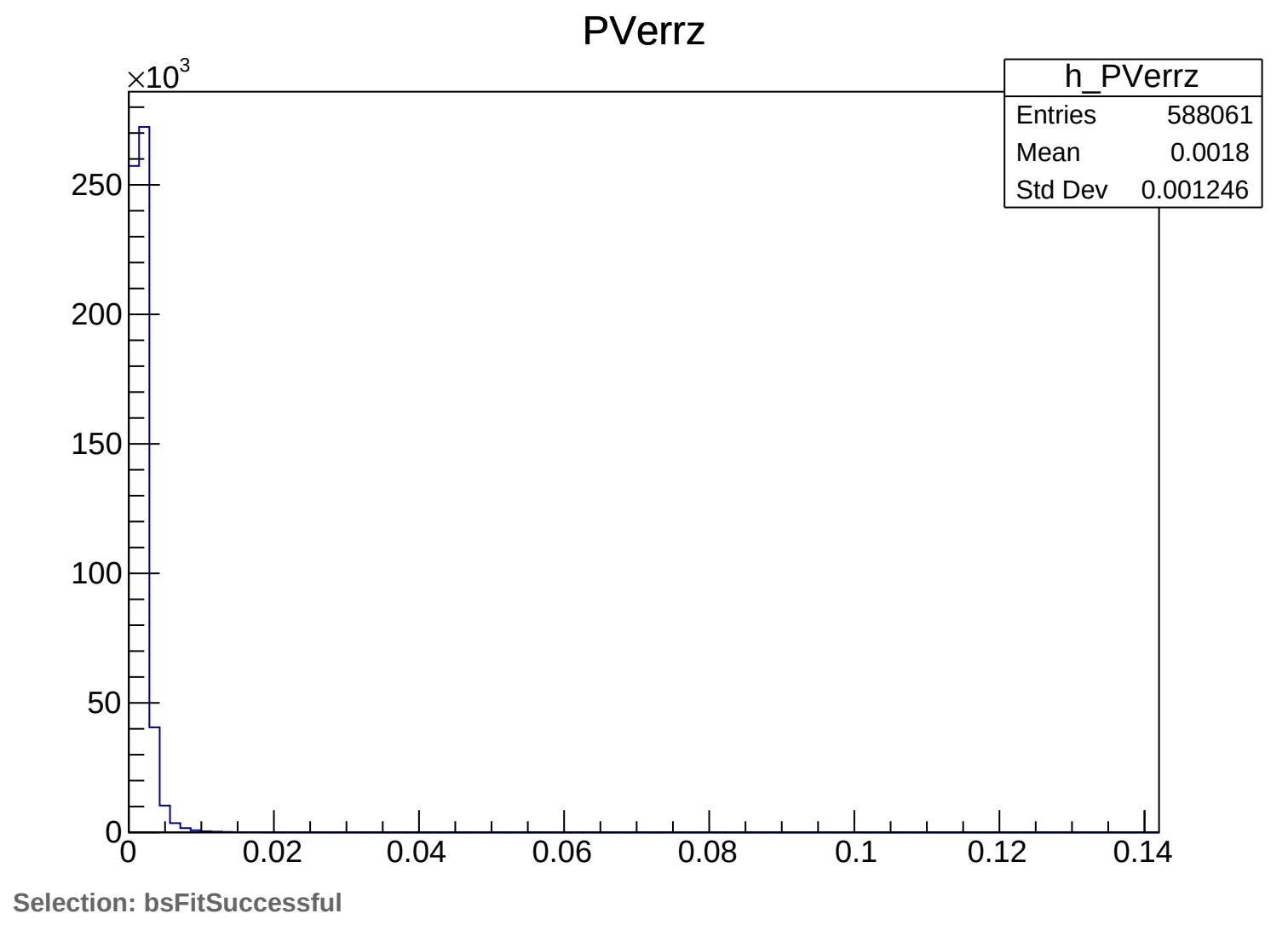

PVx
×10^3
h_PVx
Entries 588061
Mean 0.1007
Std Dev 0.001888
220
200
180
160
140
120
100
80
60
40
20
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
Selection: bsFitSuccessful

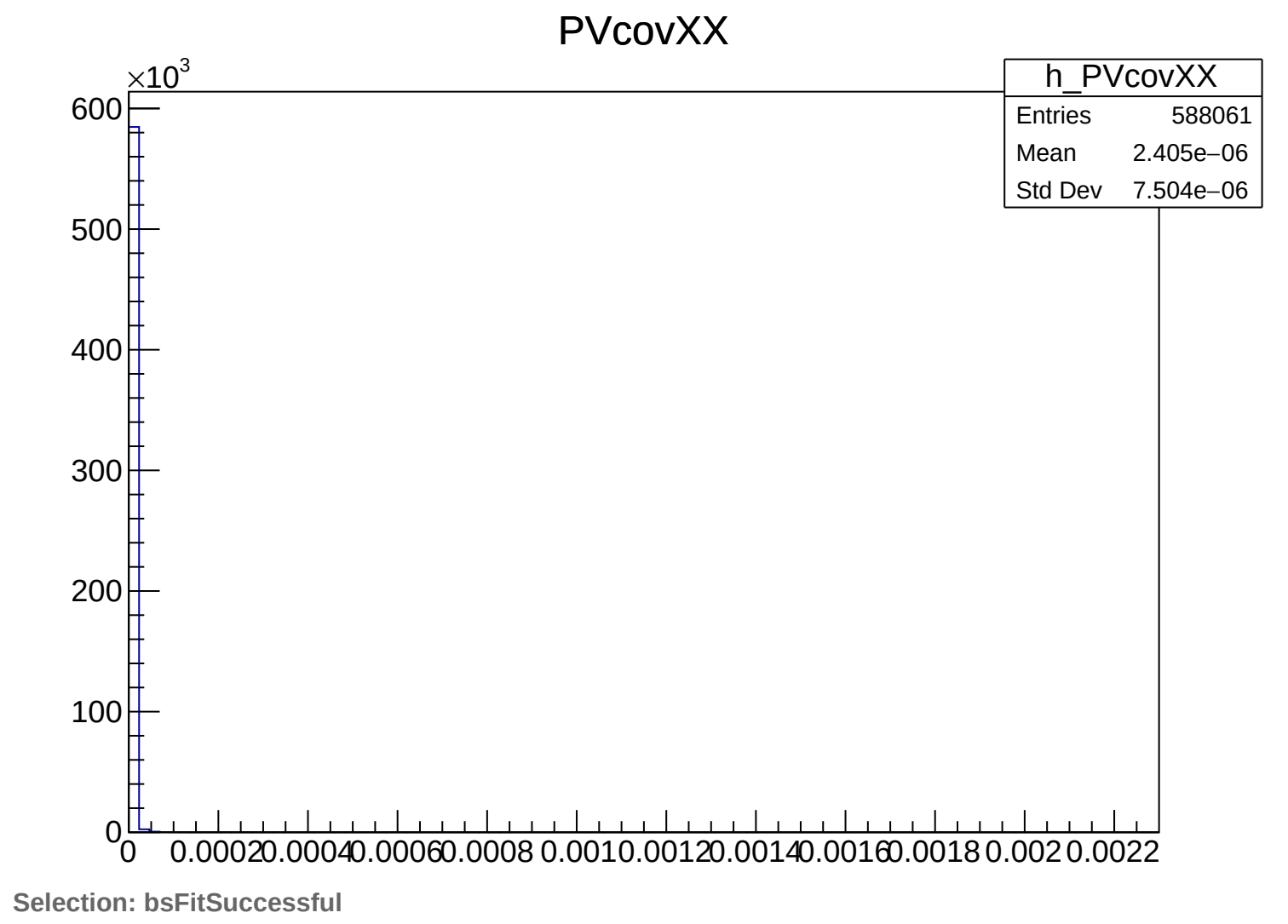

PVy
h_PVy
Entries 588061
Mean −0.01502
Std Dev 0.001879
×10^3
250
200
150
100
50
0
−0.08
−0.06
−0.04
−0.02
0
0.02
0.04
0.06
0.08
Selection: bsFitSuccessful

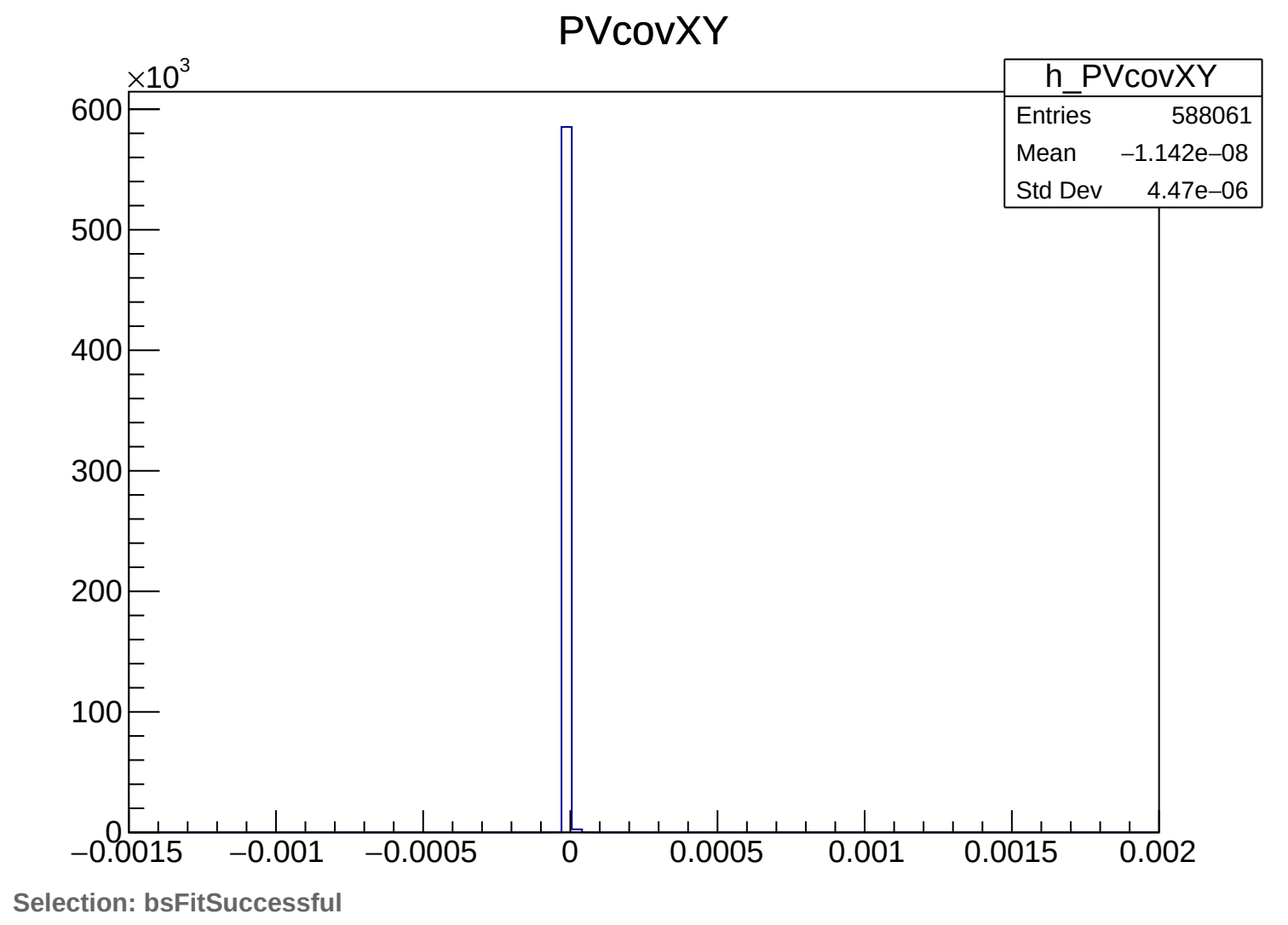

PVz
h_PVz
Entries 588061
Mean 1.289
Std Dev 3.788
25000
20000
15000
10000
5000
0
−15
−10
−5
0
5
10
15
20
Selection: bsFitSuccessful

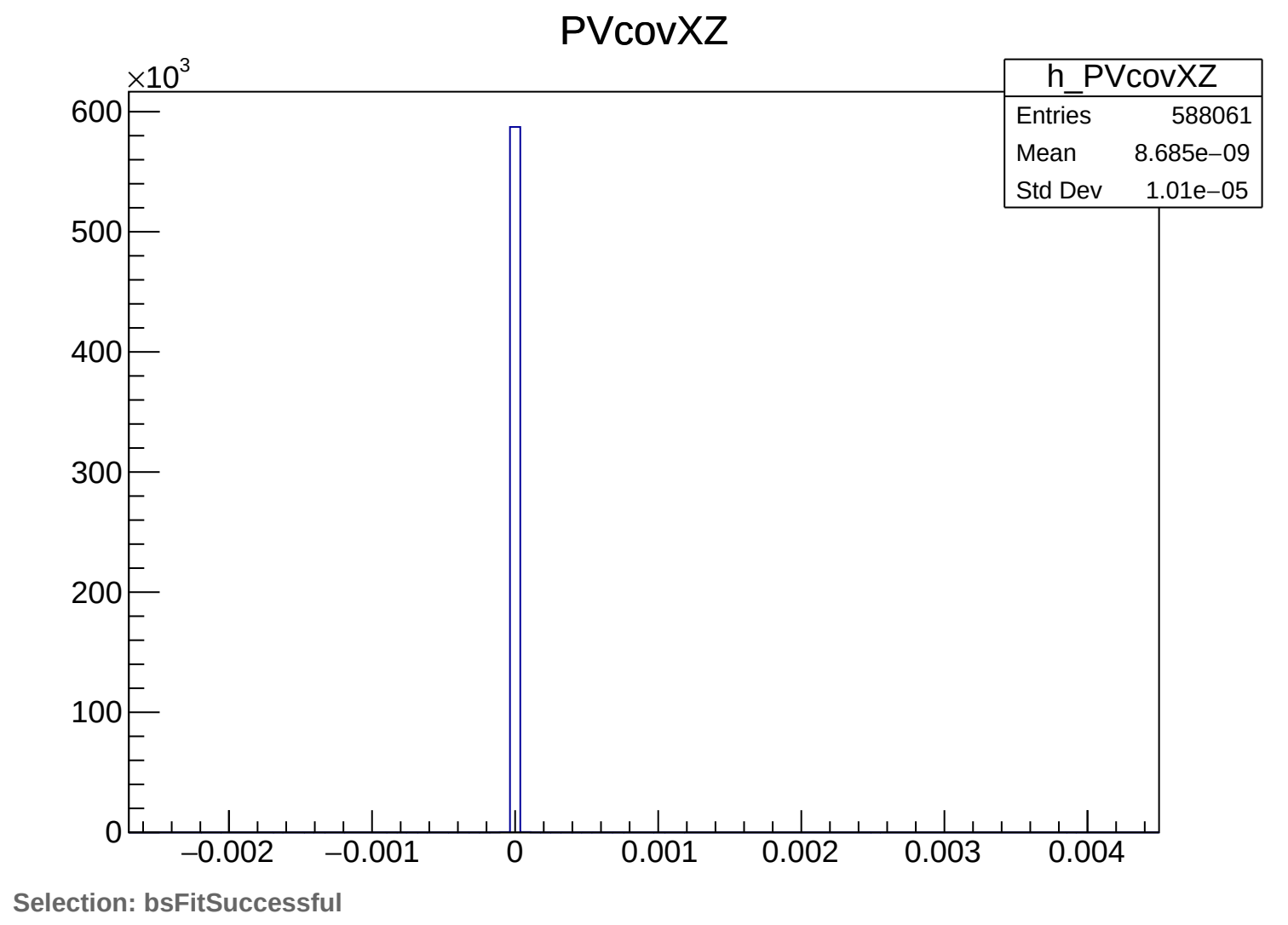

PVerrx
×10^3
h_PVerrx
Entries 588061
Mean 0.001365
Std Dev 0.0007366
250
200
150
100
50
0
0
0.005
0.01
0.015
0.02
0.025
0.03
0.035
0.04
0.045
0.05
Selection: bsFitSuccessful

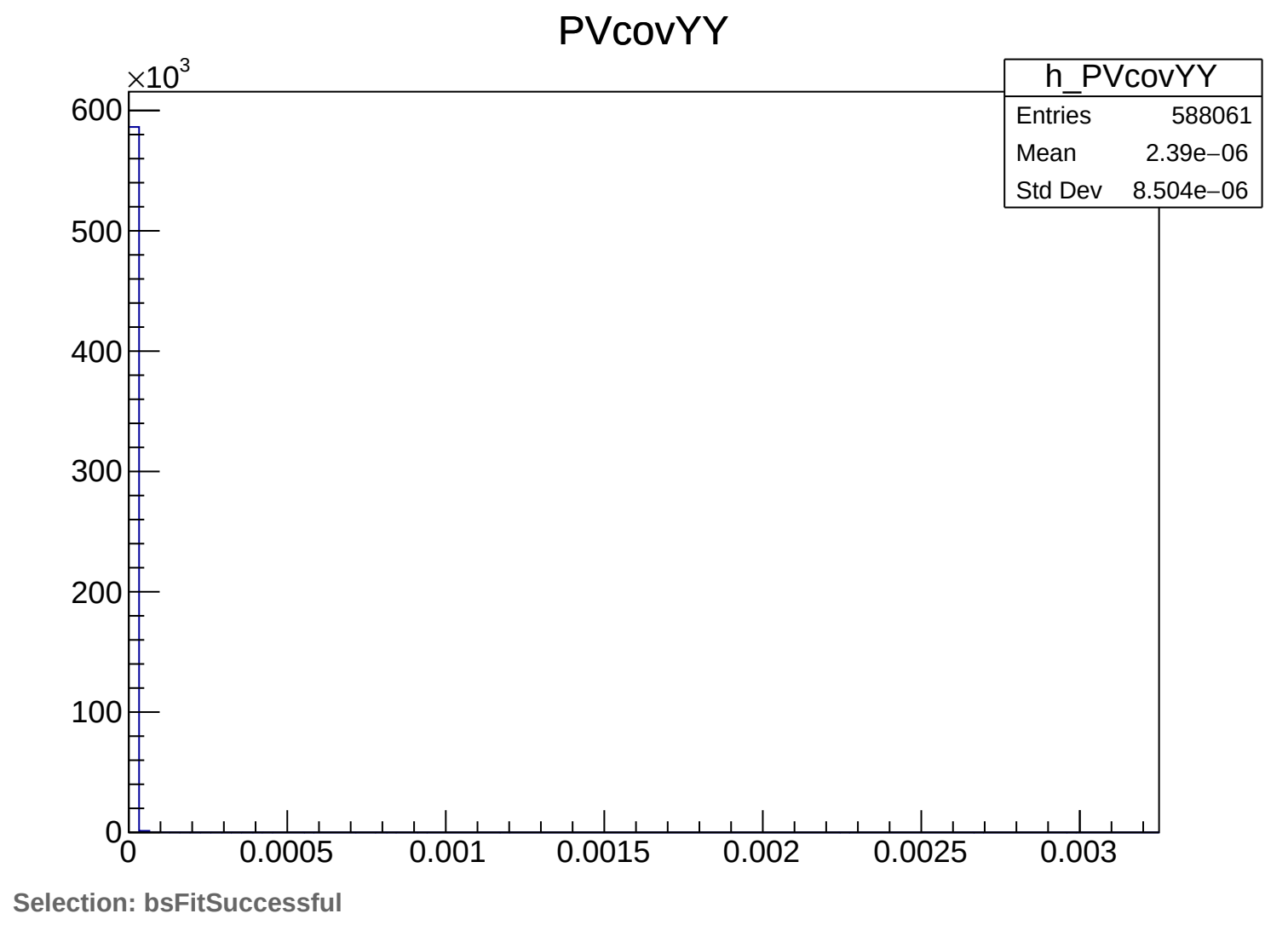

PVerry
×10^3
300
250
200
150
100
50
0
0
0.01
0.02
0.03
0.04
0.05
h_PVerry
Entries 588061
Mean 0.001359
Std Dev 0.000738
Selection: bsFitSuccessful

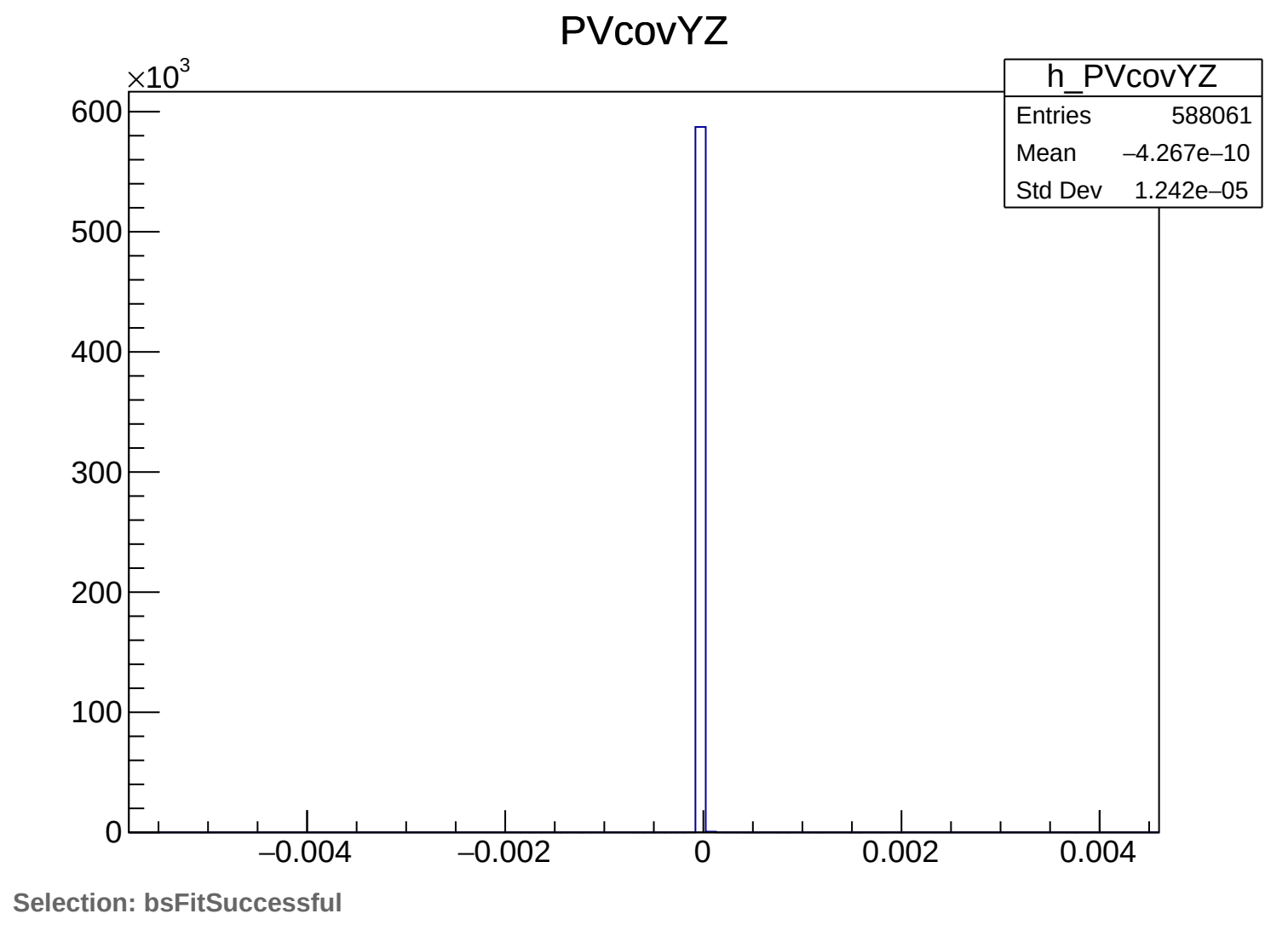

PVerrz
×10^3
h_PVerrz
Entries 588061
Mean 0.0018
Std Dev 0.001246
250
200
150
100
50
0
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
Selection: bsFitSuccessful

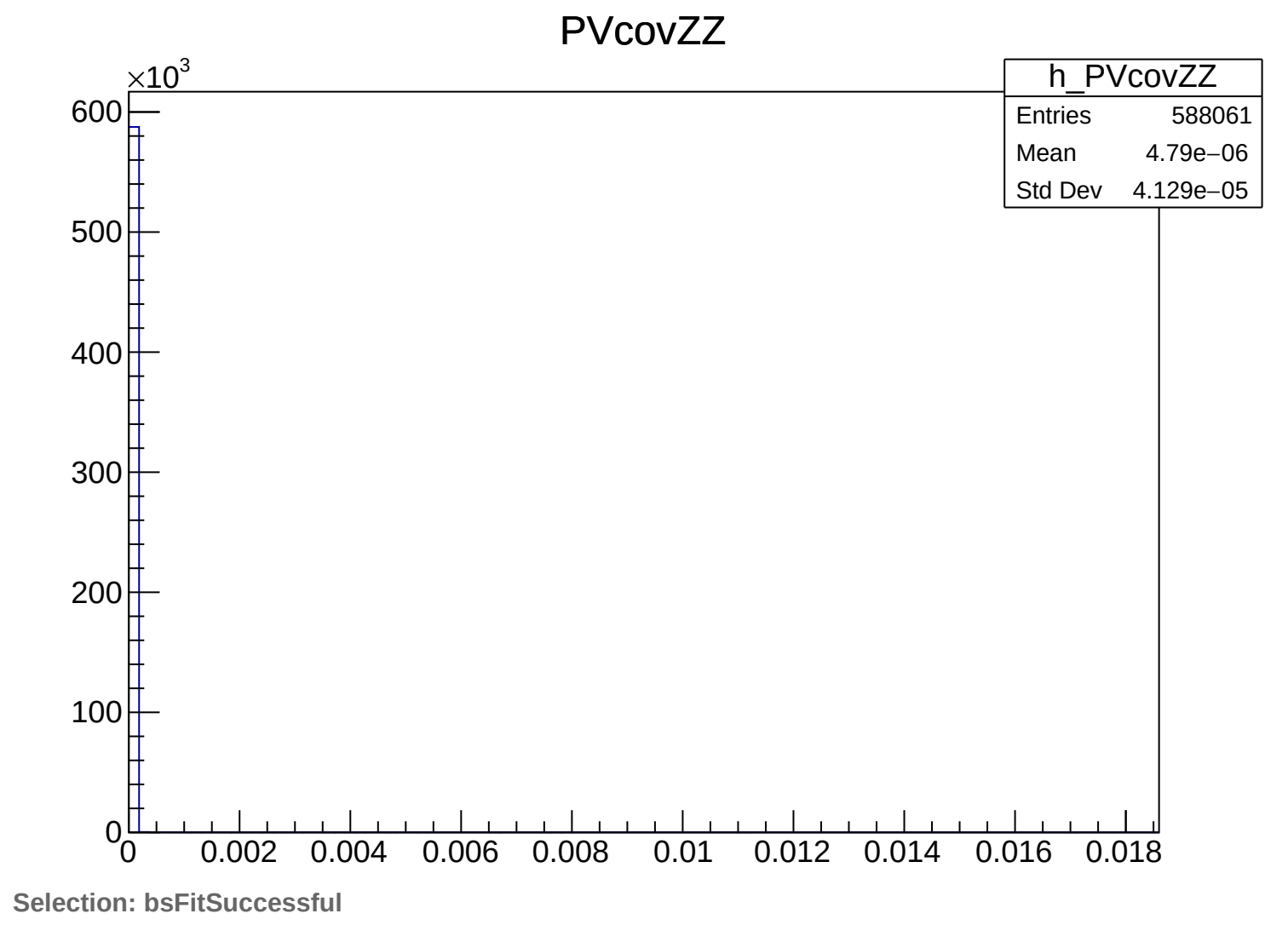

PVcovXX
h_PVcovXX
Entries 588061
Mean 2.405e−06
Std Dev 7.504e−06
×10^3
600
500
400
300
200
100
0
0 0.0002 0.0004 0.0006 0.0008 0.001 0.0012 0.0014 0.0016 0.0018 0.002 0.0022
Selection: bsFitSuccessful

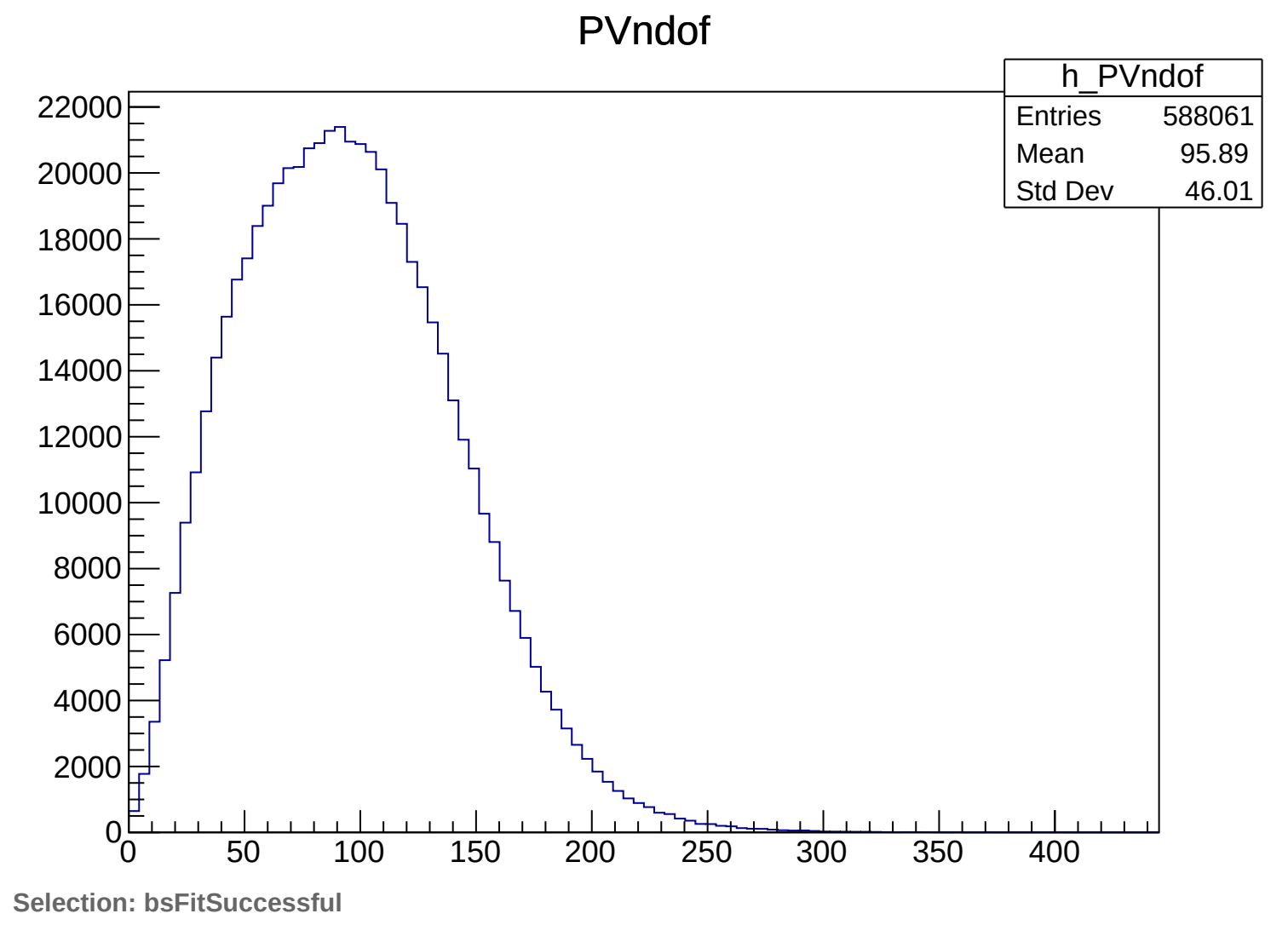

PVcovXY
h_PVcovXY
Entries 588061
Mean −1.142e−08
Std Dev 4.47e−06
×10^3
600
500
400
300
200
100
0
−0.0015
−0.001
−0.0005
0
0.0005
0.001
0.0015
0.002
Selection: bsFitSuccessful

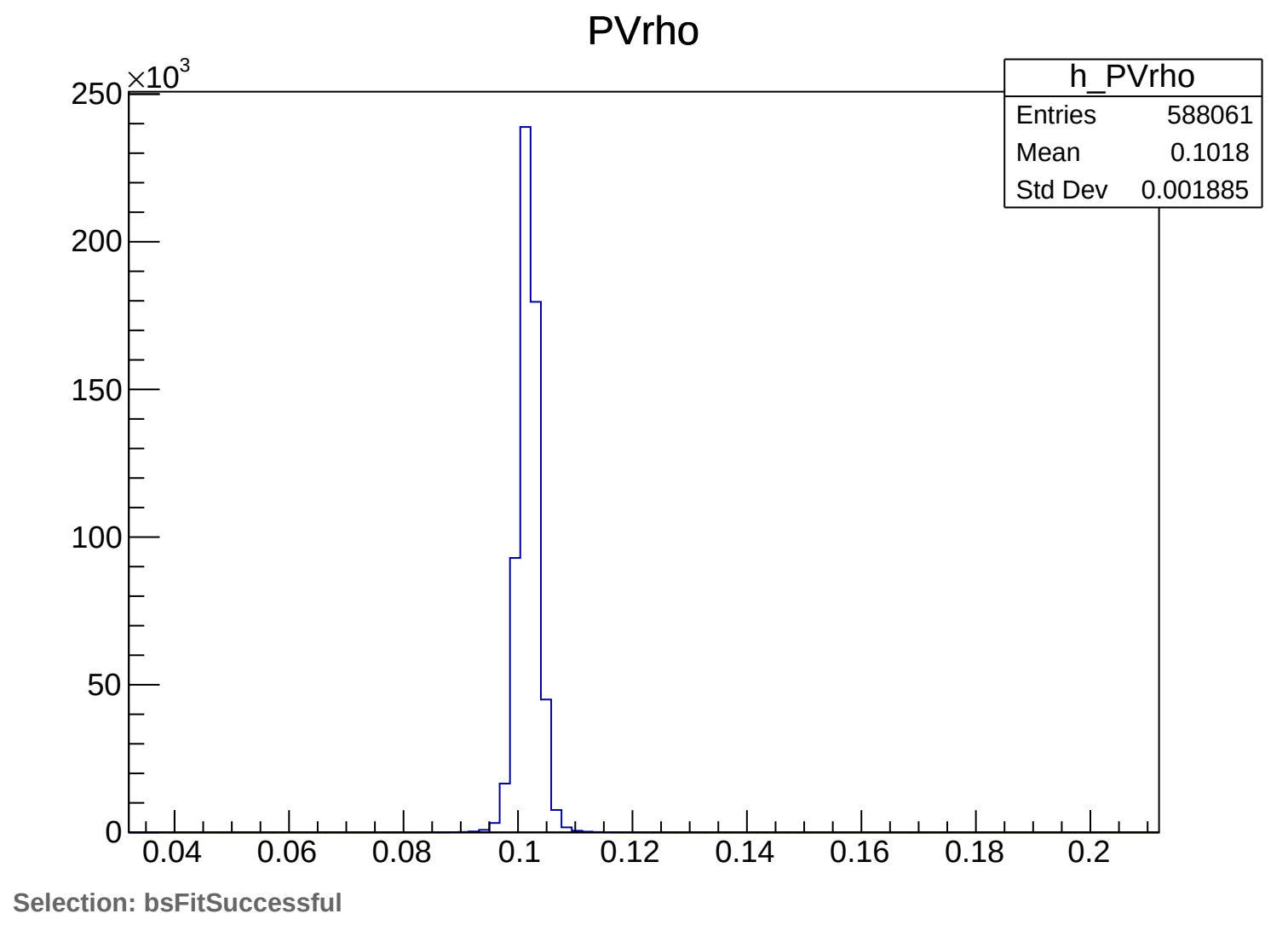

PVcovXZ
h_PVcovXZ
Entries 588061
Mean 8.685e−09
Std Dev 1.01e−05
×10^3
600
500
400
300
200
100
0
−0.002
−0.001
0
0.001
0.002
0.003
0.004
Selection: bsFitSuccessful

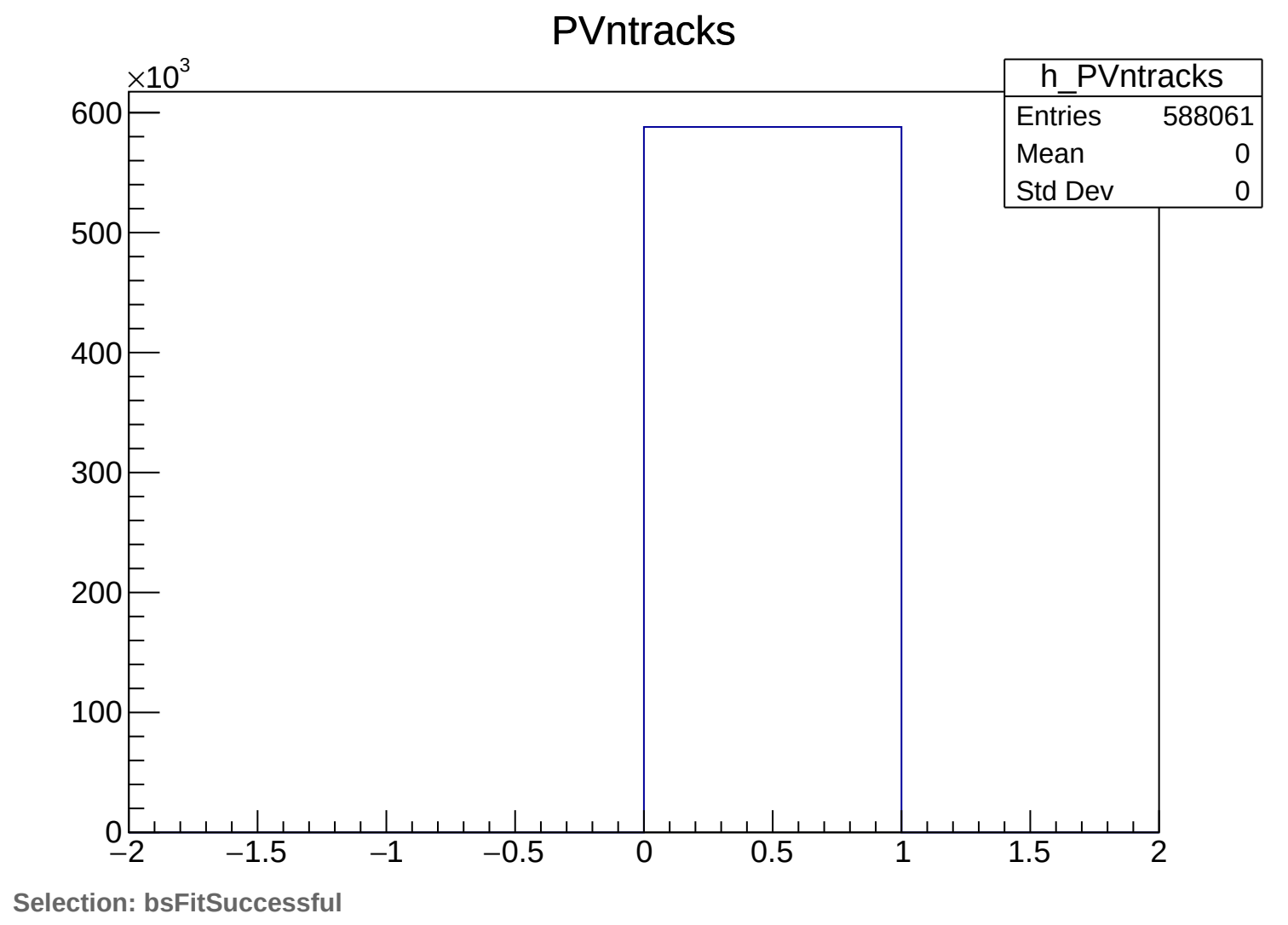

PVcovYY
h_PVcovYY
Entries 588061
Mean 2.39e−06
Std Dev 8.504e−06
×10^3
600
500
400
300
200
100
0
0
0.0005
0.001
0.0015
0.002
0.0025
0.003
Selection: bsFitSuccessful

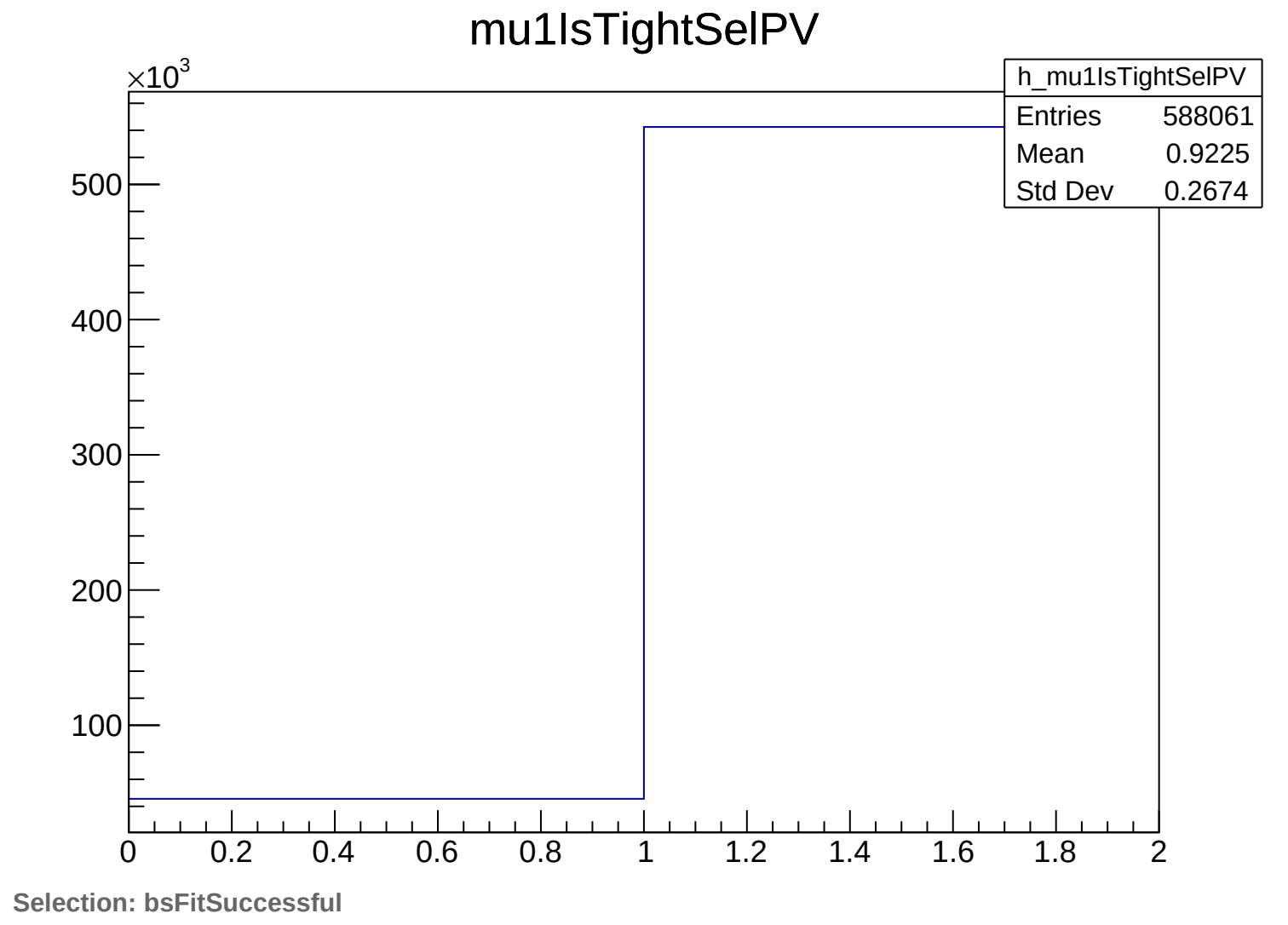

PVcovYZ
×10^3
600
500
400
300
200
100
0
–0.004
–0.002
0
0.002
0.004
h_PVcovYZ
Entries 588061
Mean –4.267e–10
Std Dev 1.242e–05
Selection: bsFitSuccessful

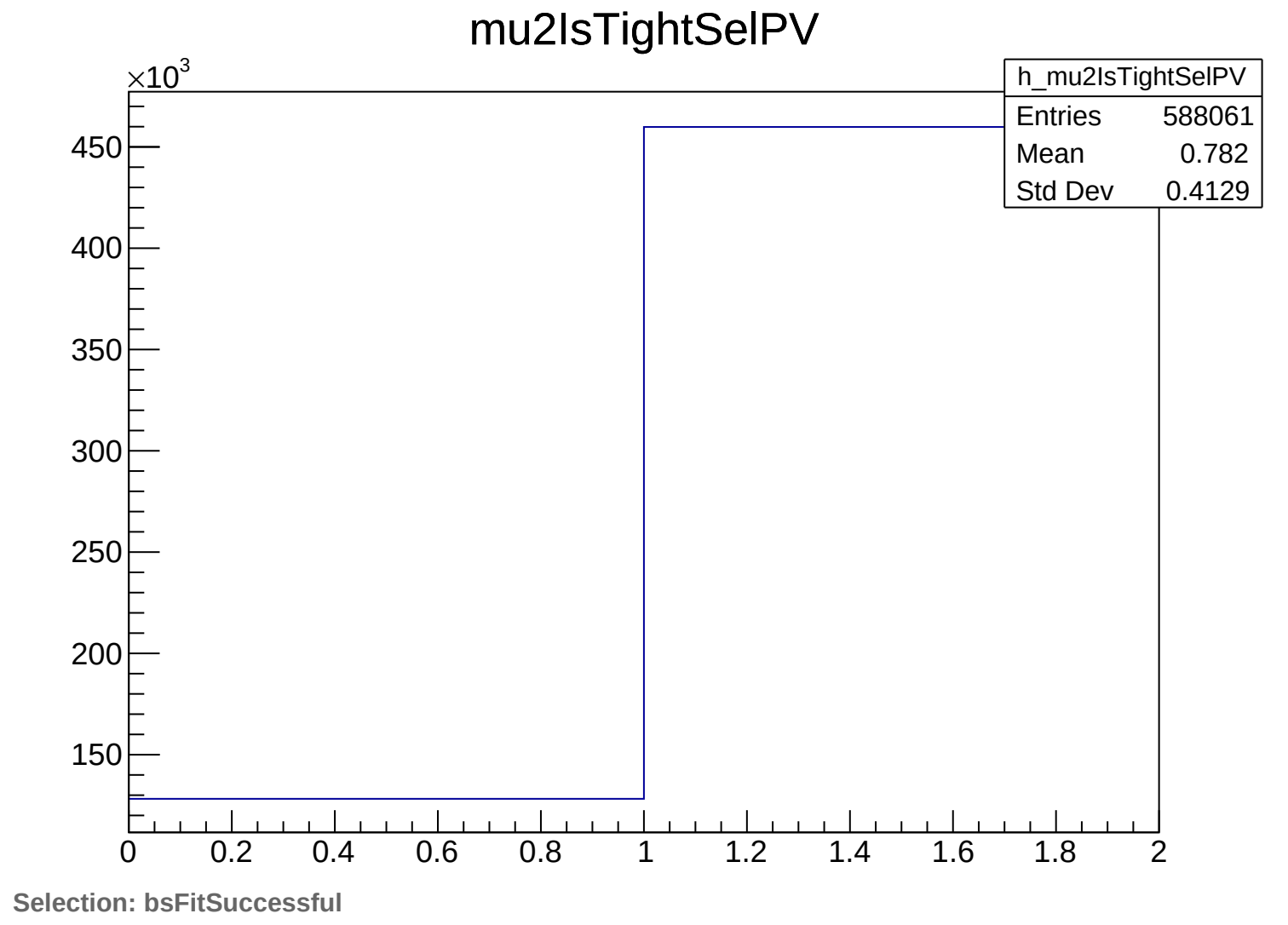

PVcovZZ
h_PVcovZZ
Entries 588061
Mean 4.79e−06
Std Dev 4.129e−05
×10^3
600
500
400
300
200
100
0
0
0.002
0.004
0.006
0.008
0.01
0.012
0.014
0.016
0.018
Selection: bsFitSuccessful

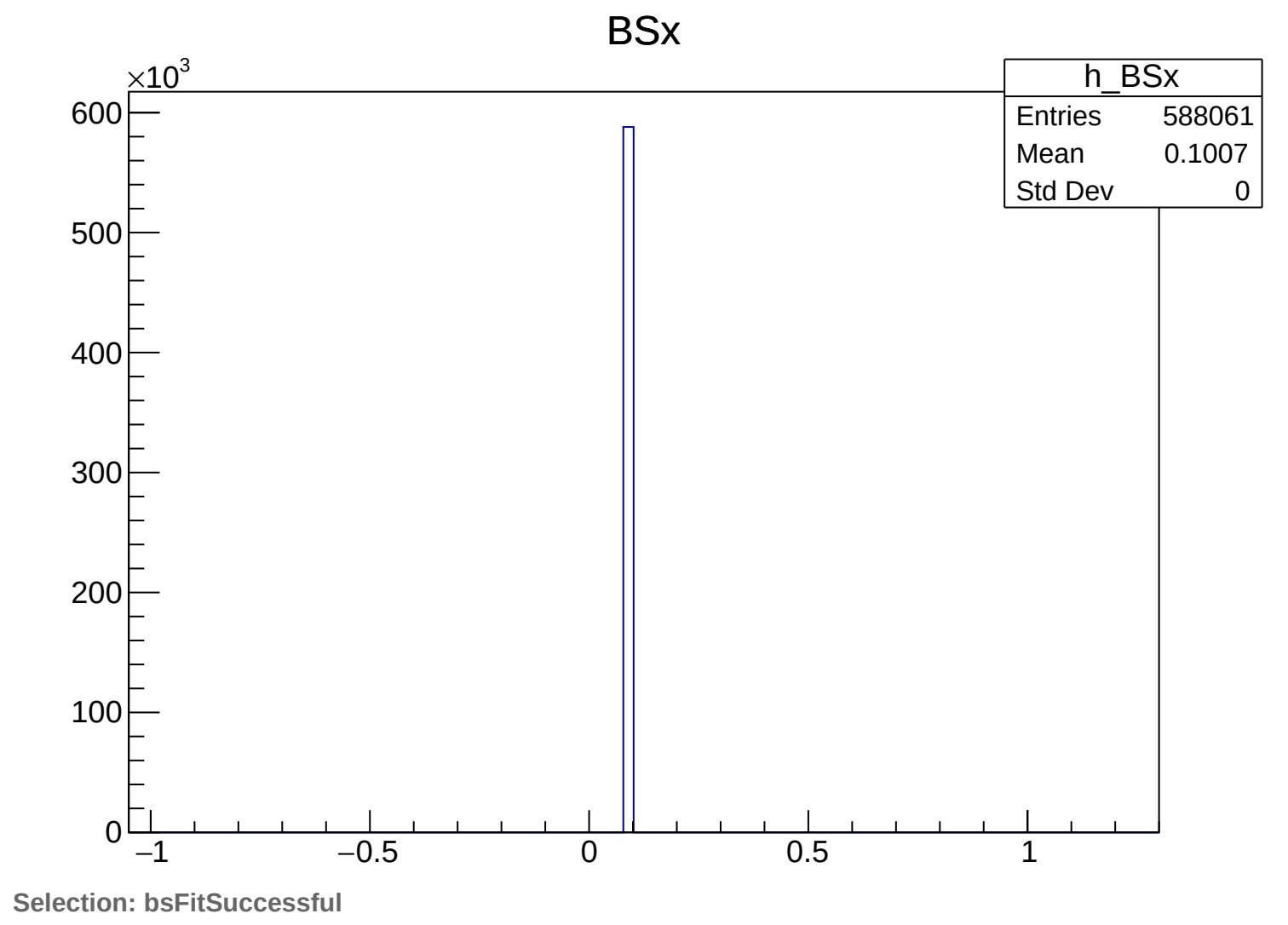

PVndof
h_PVndof
Entries 588061
Mean 95.89
Std Dev 46.01
22000
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
50
100
150
200
250
300
350
400
Selection: bsFitSuccessful

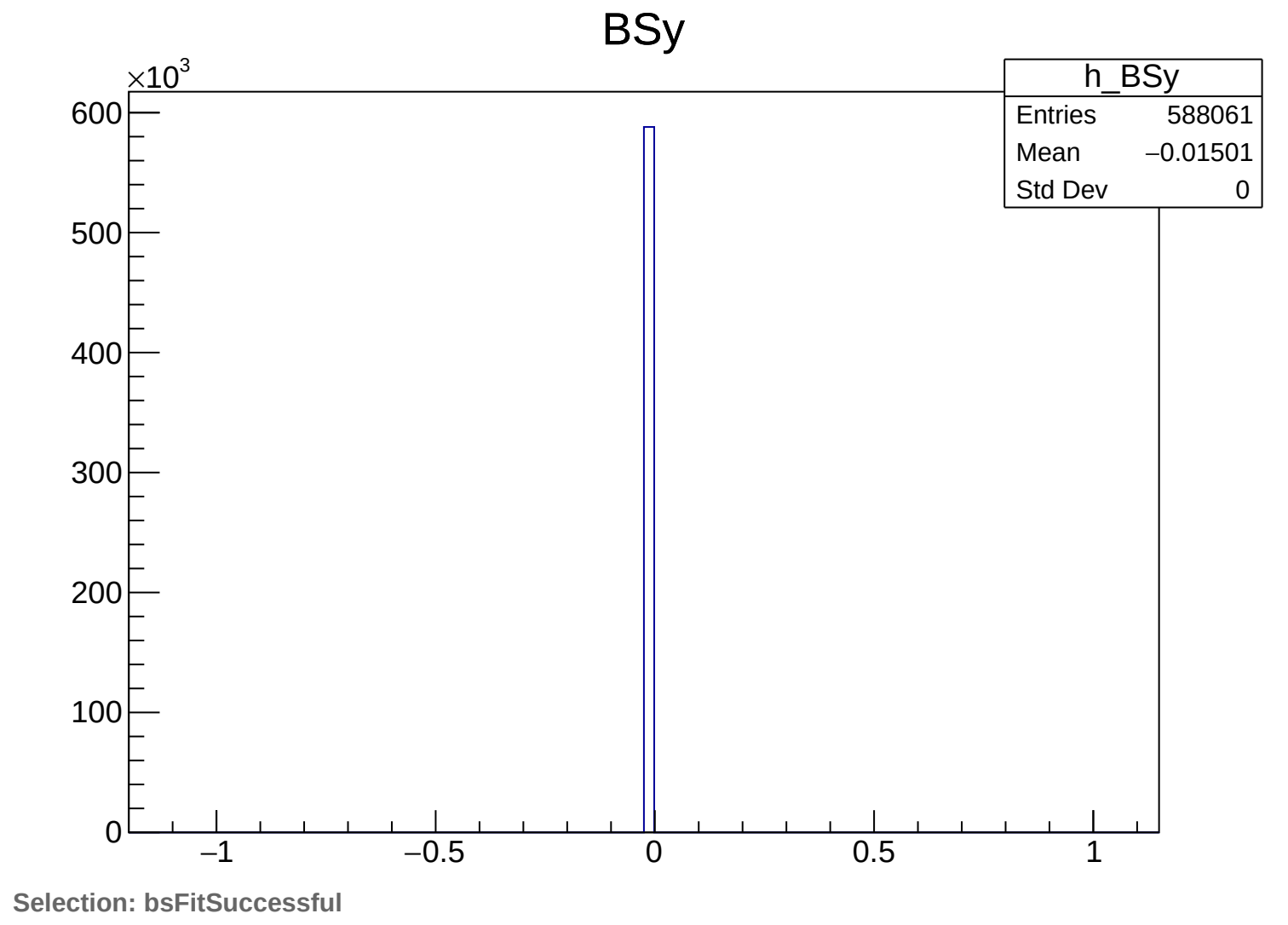

PVrho
h_PVrho
Entries 588061
Mean 0.1018
Std Dev 0.001885
×10^3
250
200
150
100
50
0
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
Selection: bsFitSuccessful

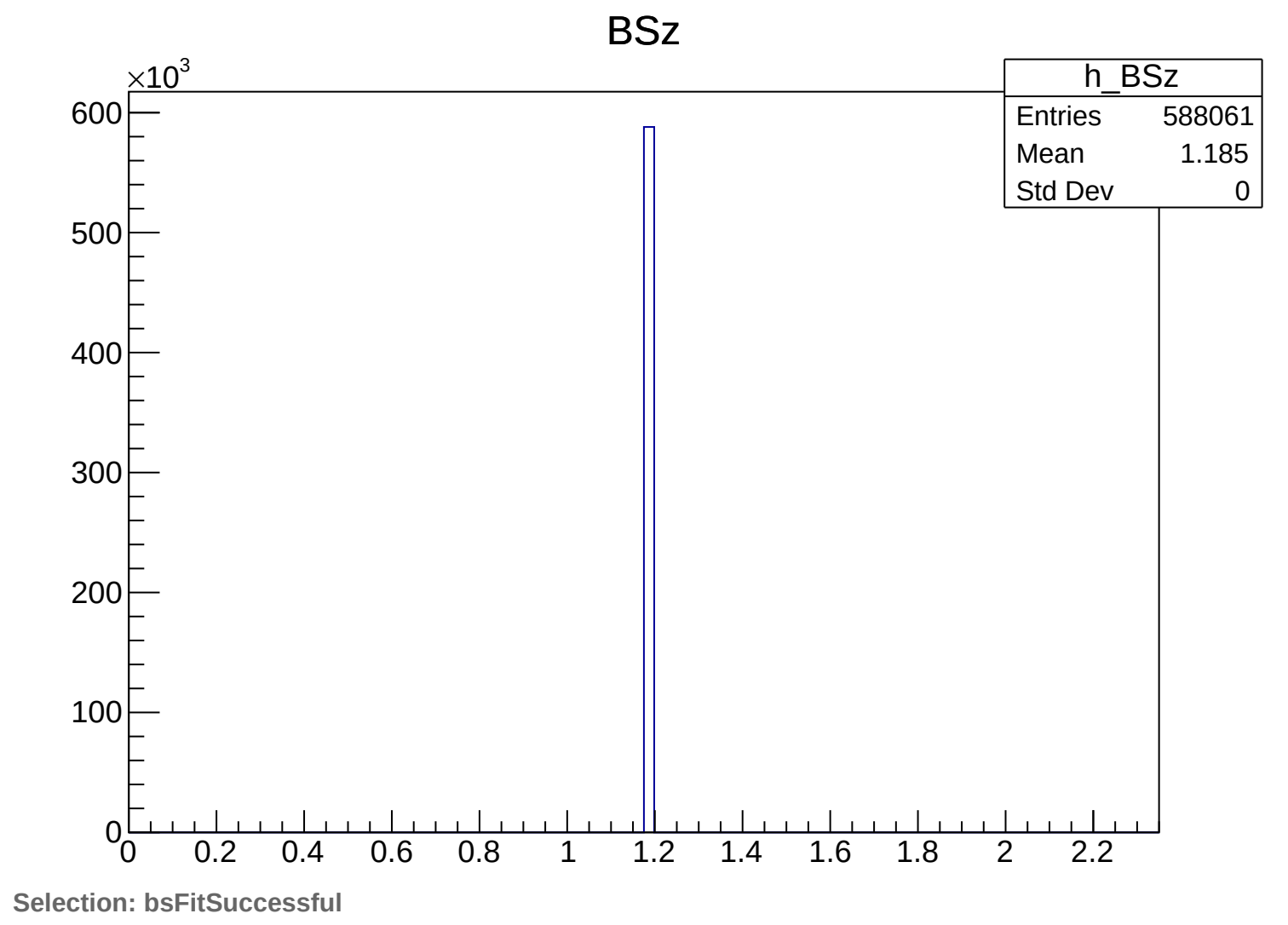

PVntracks
h_PVntracks
Entries 588061
Mean 0
Std Dev 0
×10^3
600
500
400
300
200
100
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2
Selection: bsFitSuccessful

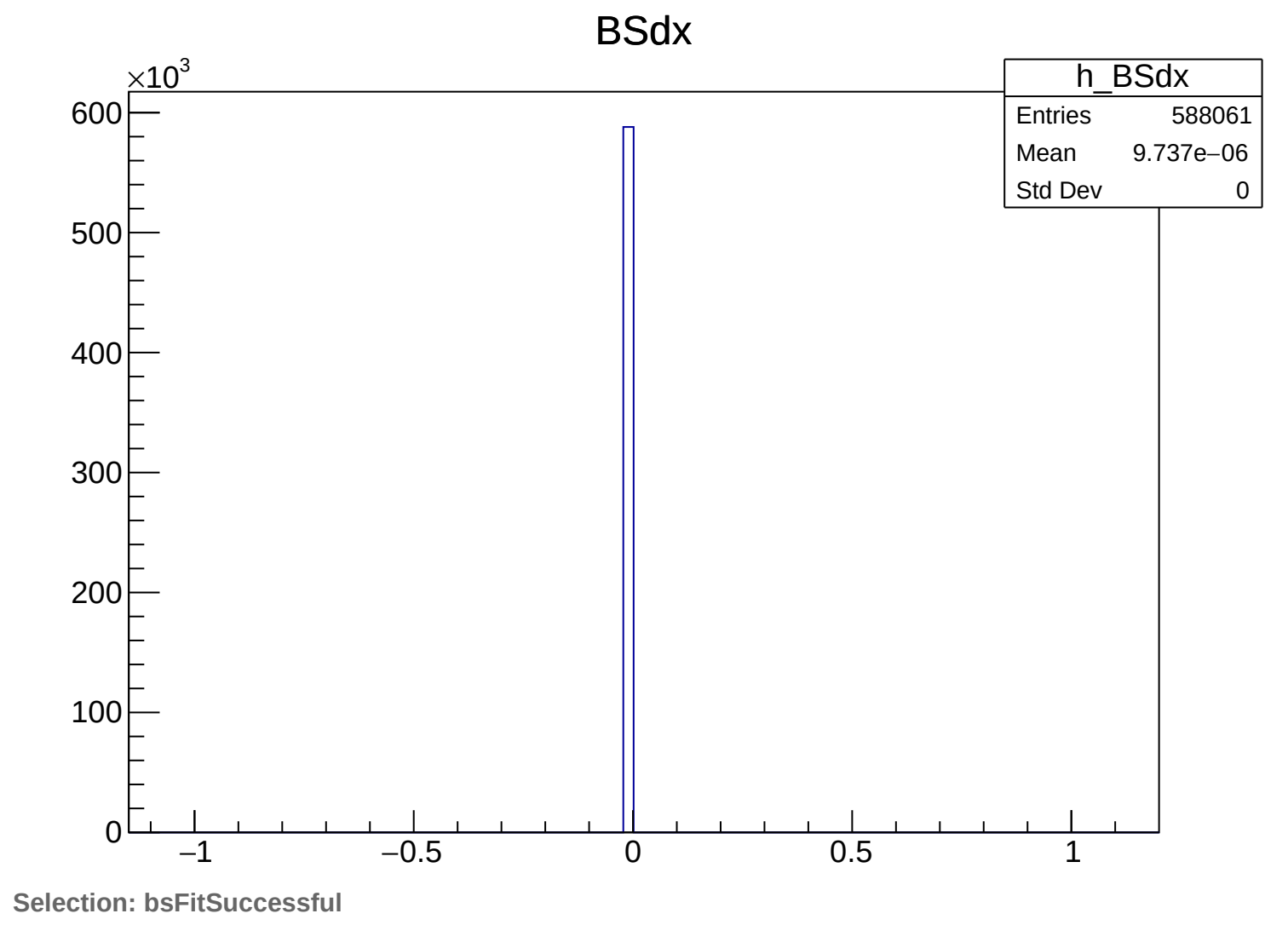

mu1IsTightSelPV
×10^3
h_mu1IsTightSelPV
Entries 588061
Mean 0.9225
Std Dev 0.2674
500
400
300
200
100
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

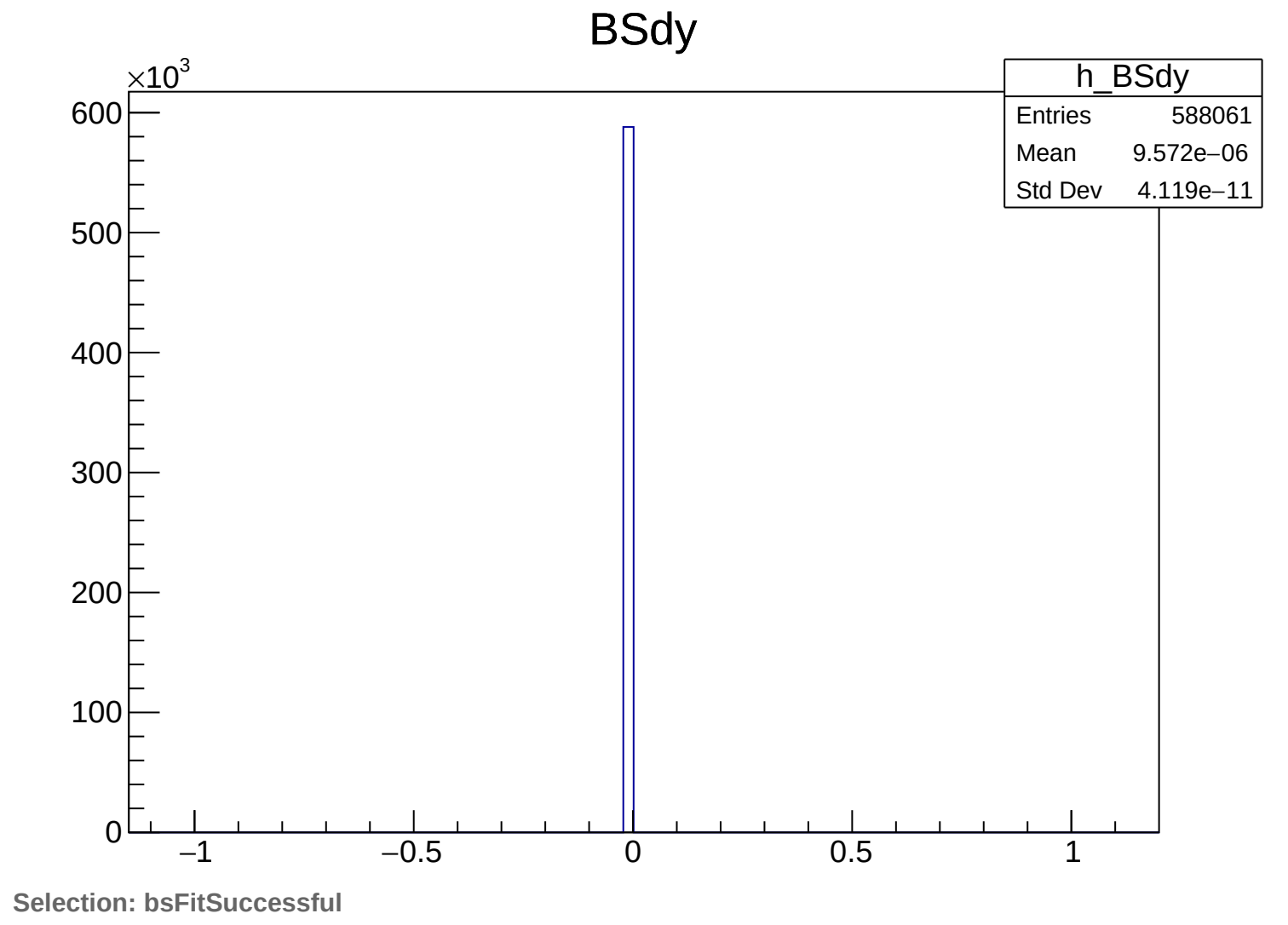

mu2IsTightSelPV
×10^3
450
400
350
300
250
200
150
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
h_mu2IsTightSelPV
Entries 588061
Mean 0.782
Std Dev 0.4129
Selection: bsFitSuccessful

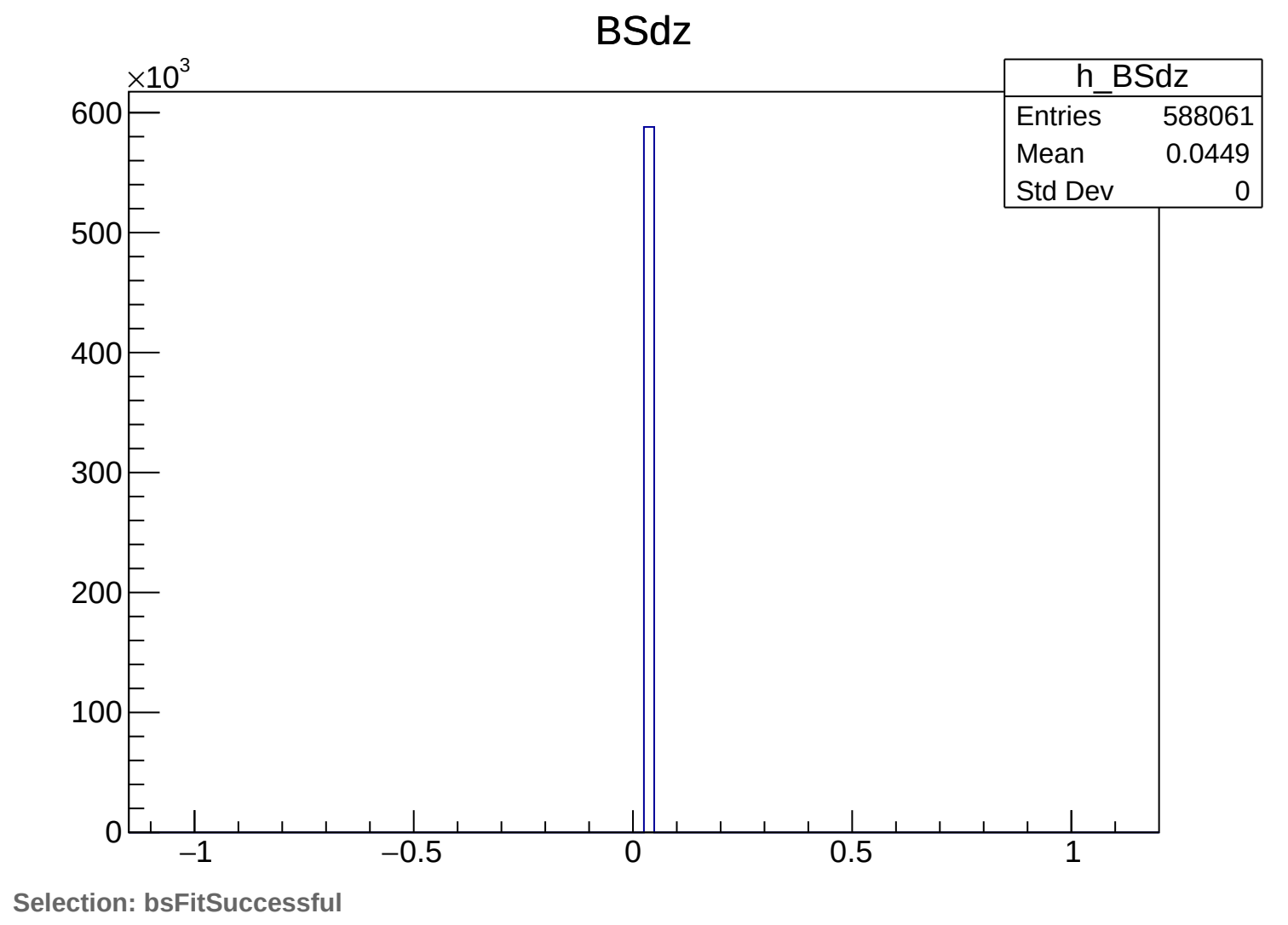

BSx
×10^3
600
500
400
300
200
100
0
−1
−0.5
0
0.5
1
h_BSx
Entries 588061
Mean 0.1007
Std Dev 0
Selection: bsFitSuccessful

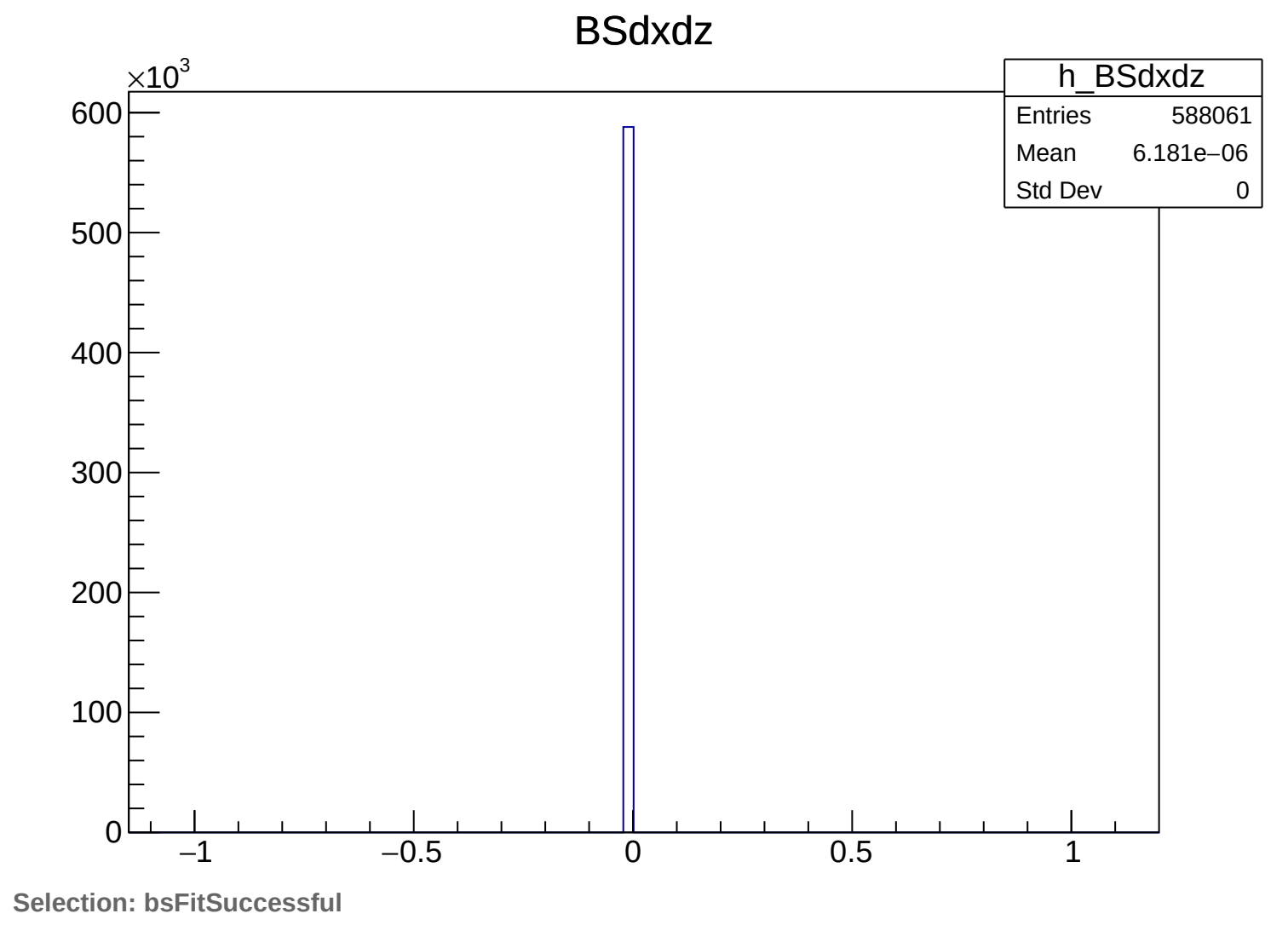

BSy
h_BSy
Entries 588061
Mean −0.01501
Std Dev 0
×10^3
600
500
400
300
200
100
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful

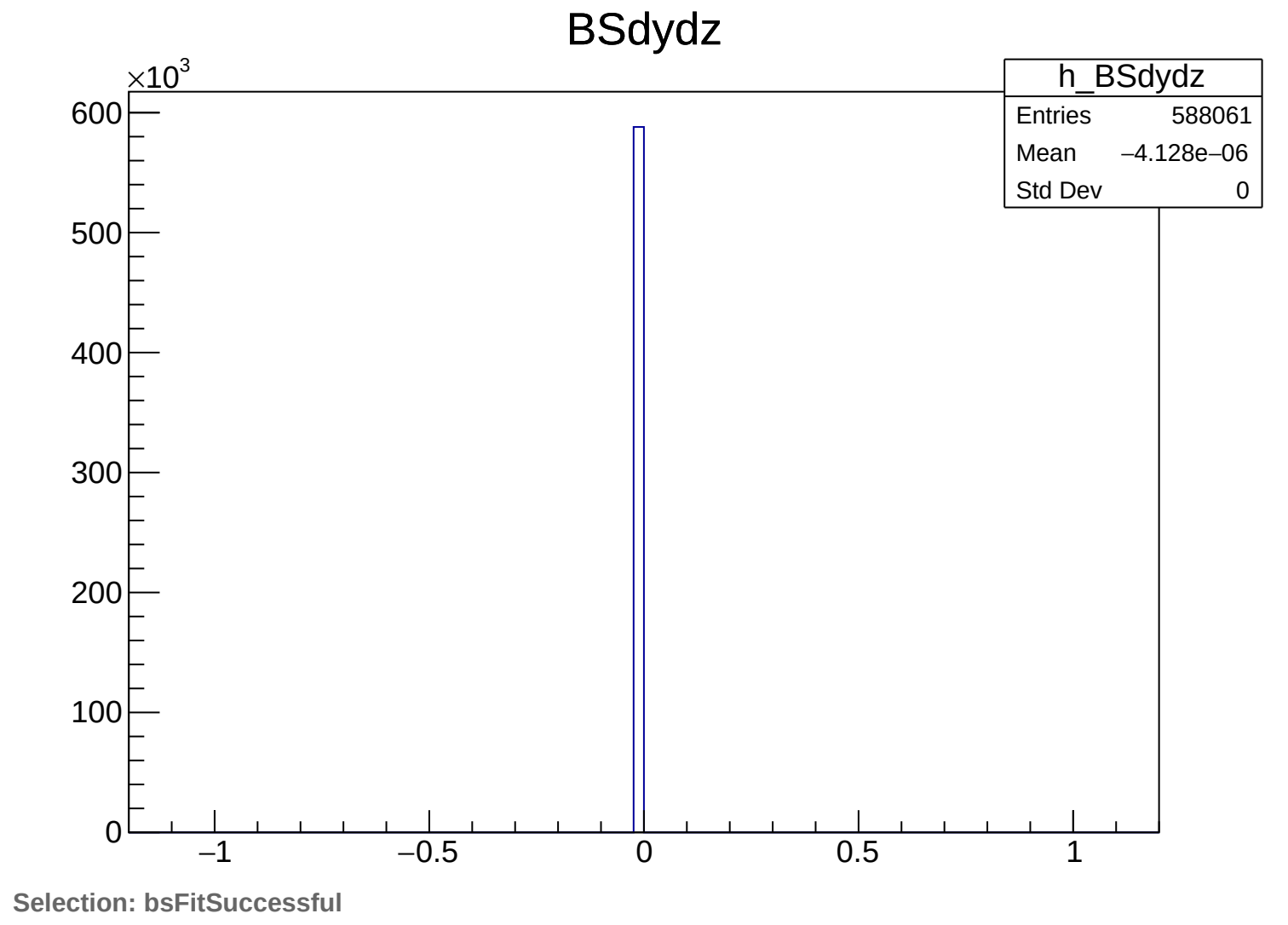

BSz
h_BSz
Entries 588061
Mean 1.185
Std Dev 0
×10^3
600
500
400
300
200
100
0
0 0.2 0.4 0.6 0.8 1 1.2 1.4 1.6 1.8 2 2.2
Selection: bsFitSuccessful

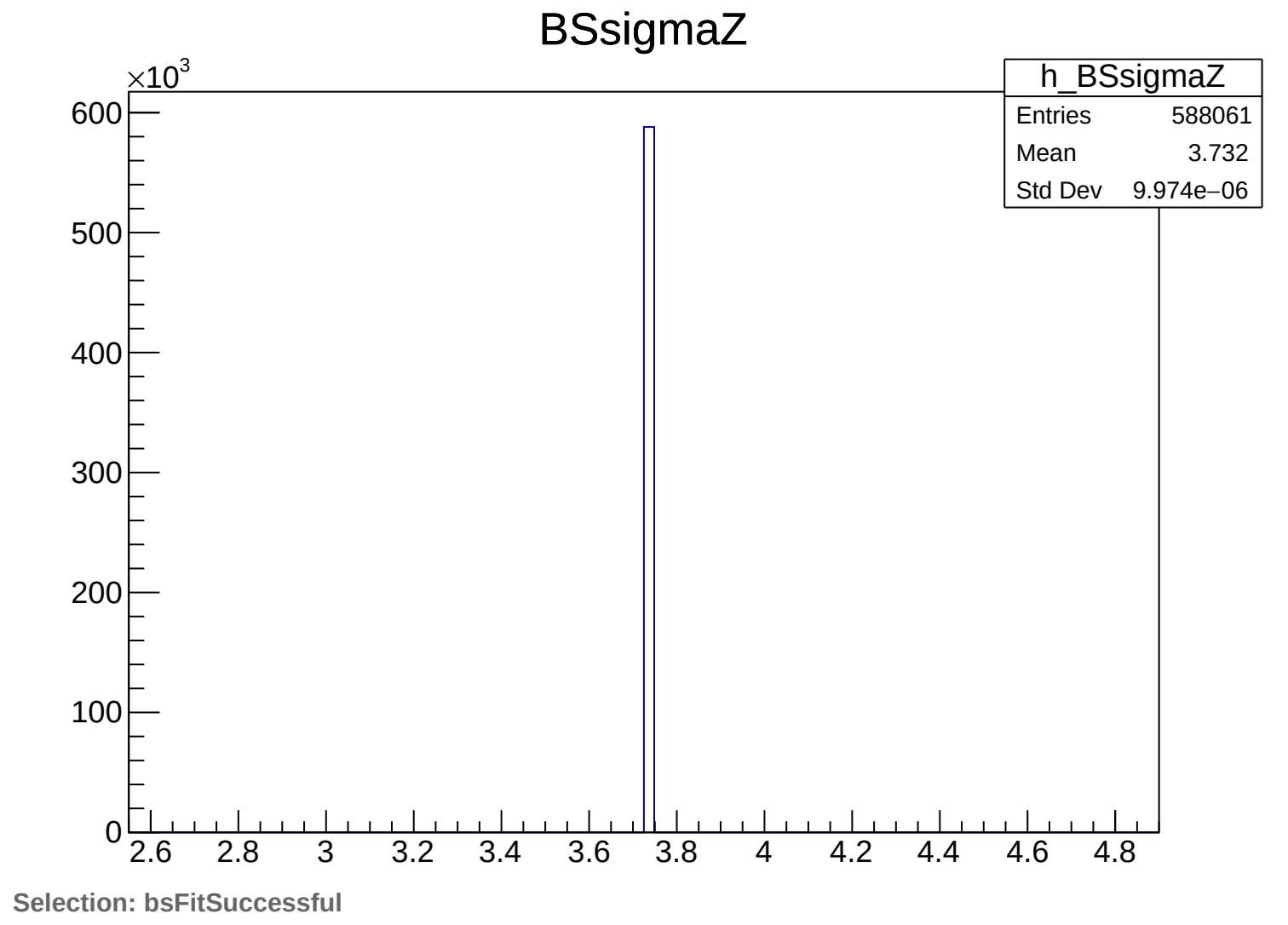

BSdx
h_BSdx
Entries 588061
Mean 9.737e−06
Std Dev 0
×10^3
600
500
400
300
200
100
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful

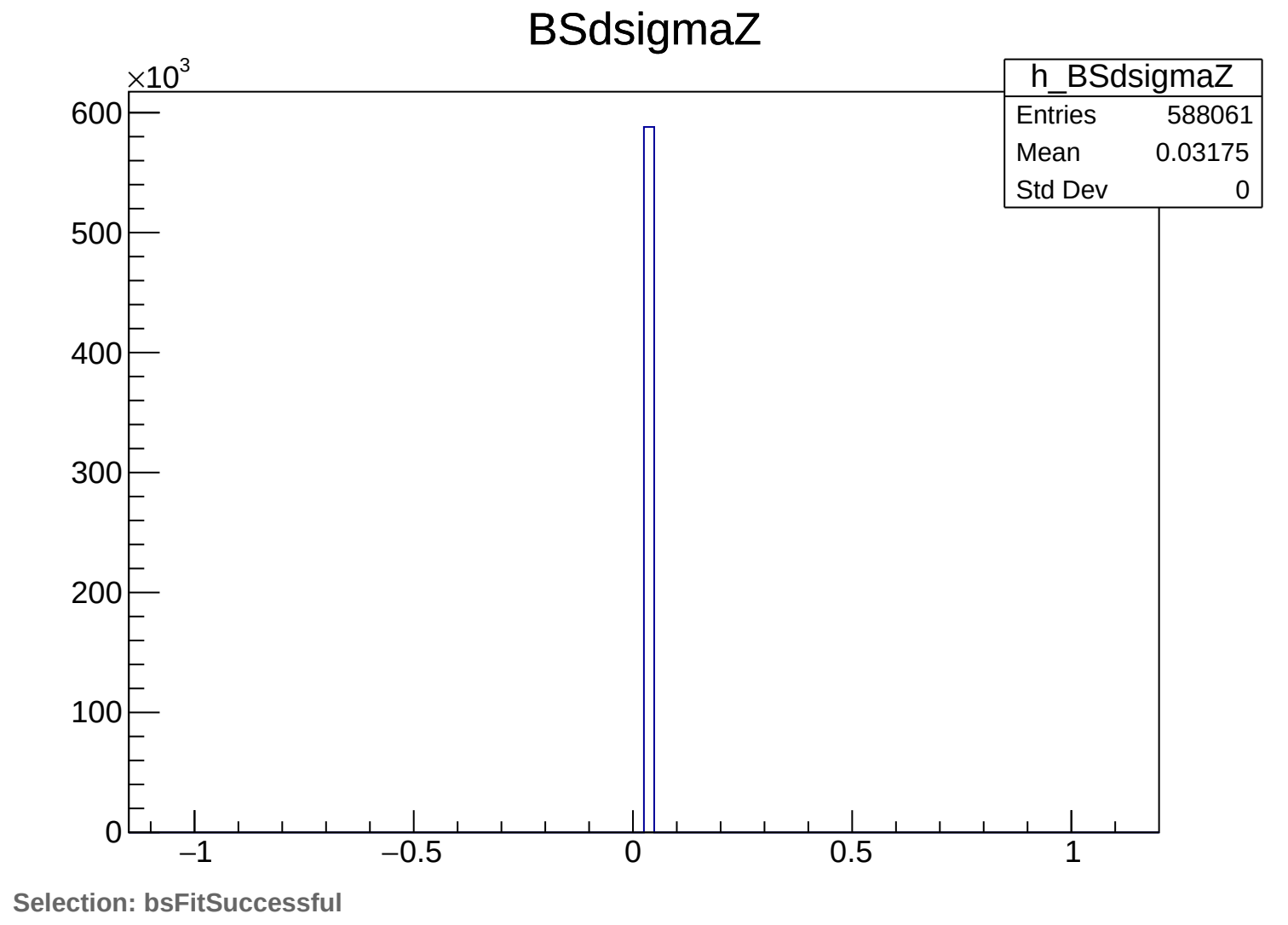

BSdy
h_BSdy
Entries 588061
Mean 9.572e−06
Std Dev 4.119e−11
×10^3
600
500
400
300
200
100
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful

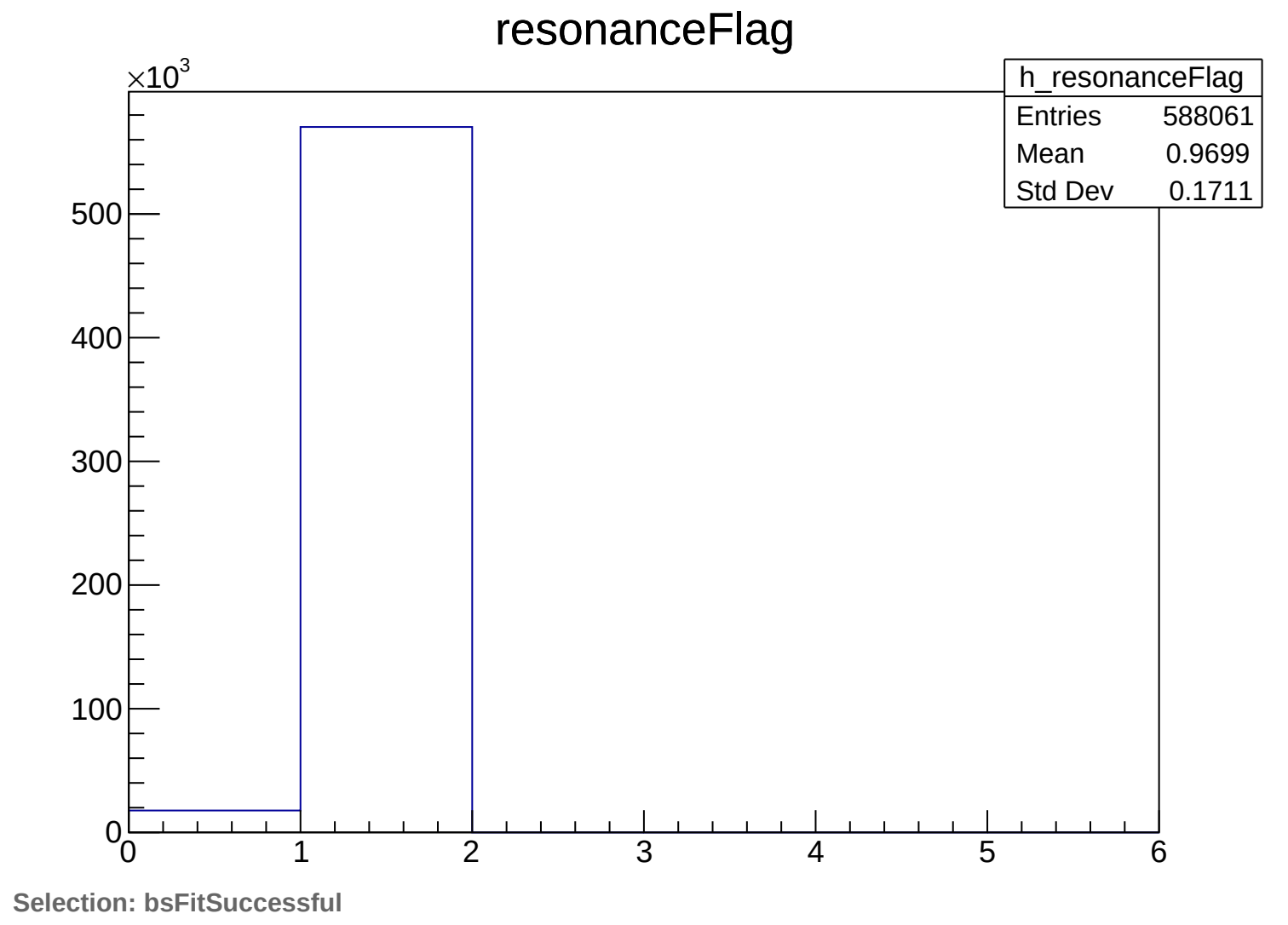

BSdz
×10^3
600
500
400
300
200
100
0
−1
−0.5
0
0.5
1
h_BSdz
Entries 588061
Mean 0.0449
Std Dev 0
Selection: bsFitSuccessful

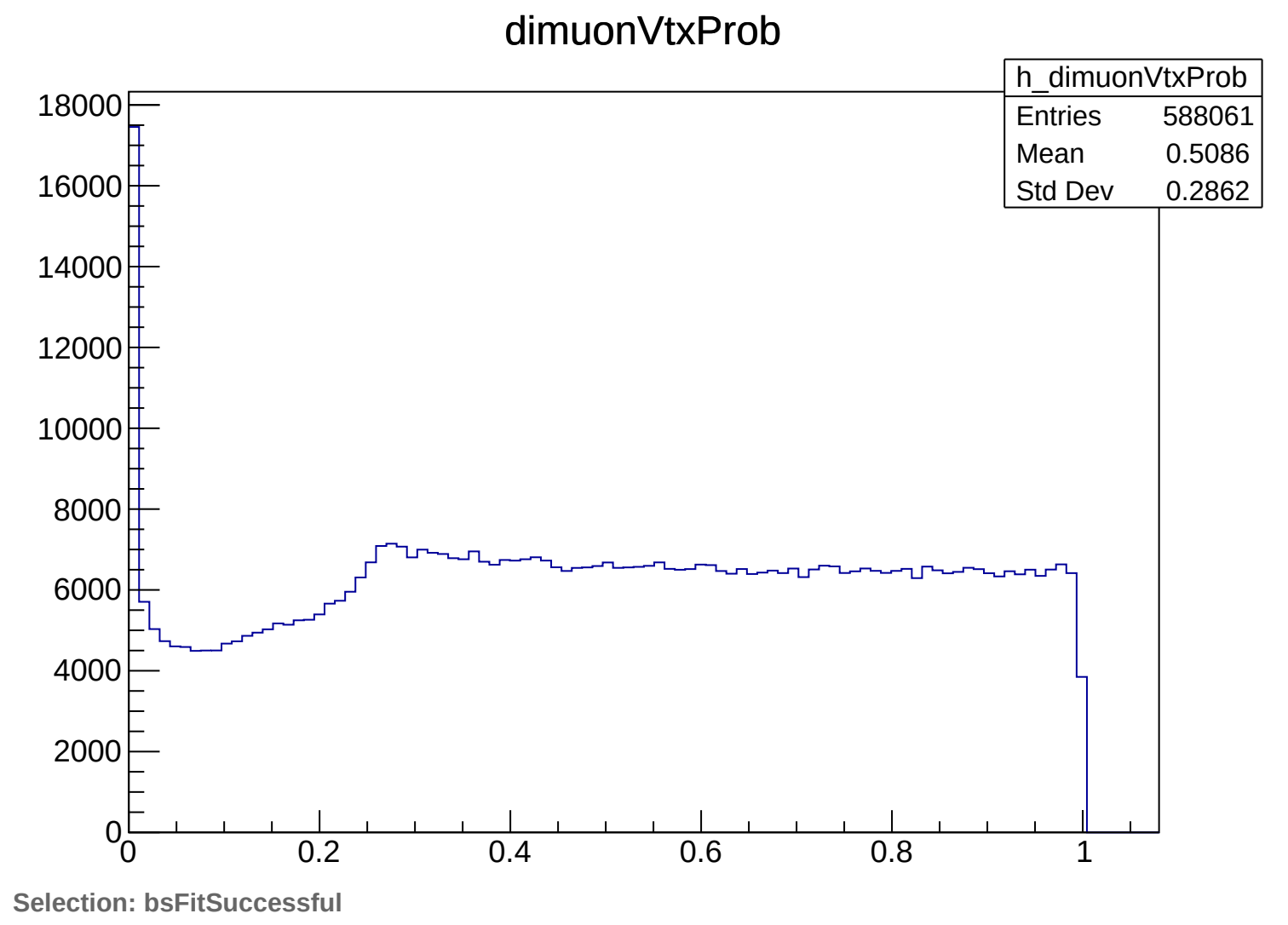

BSdxdz
h_BSdxdz
Entries 588061
Mean 6.181e−06
Std Dev 0
×10^3
600
500
400
300
200
100
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful

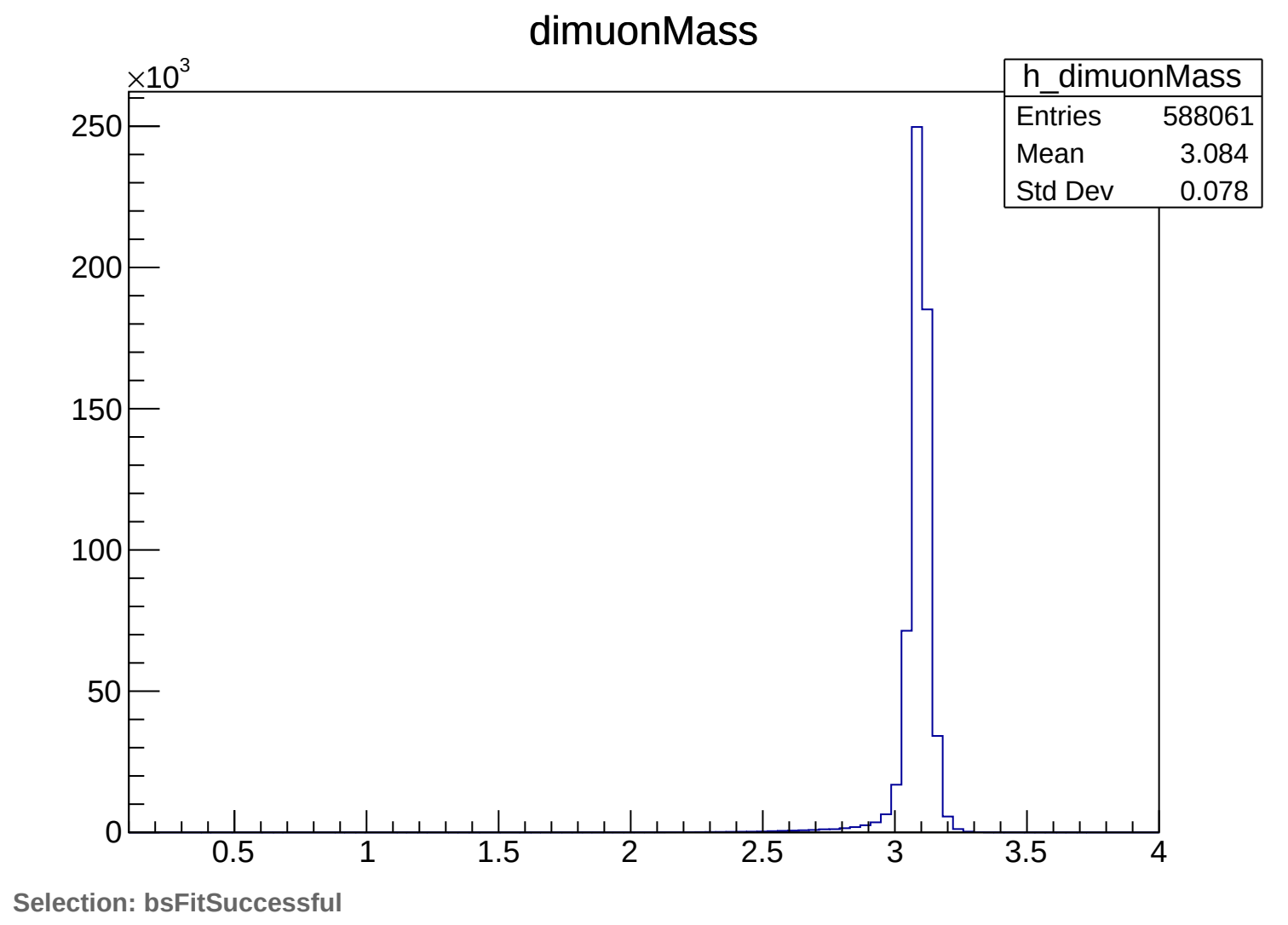

BSdydz
×10^3
600
500
400
300
200
100
0
−1
−0.5
0
0.5
1
h_BSdydz
Entries 588061
Mean −4.128e−06
Std Dev 0
Selection: bsFitSuccessful

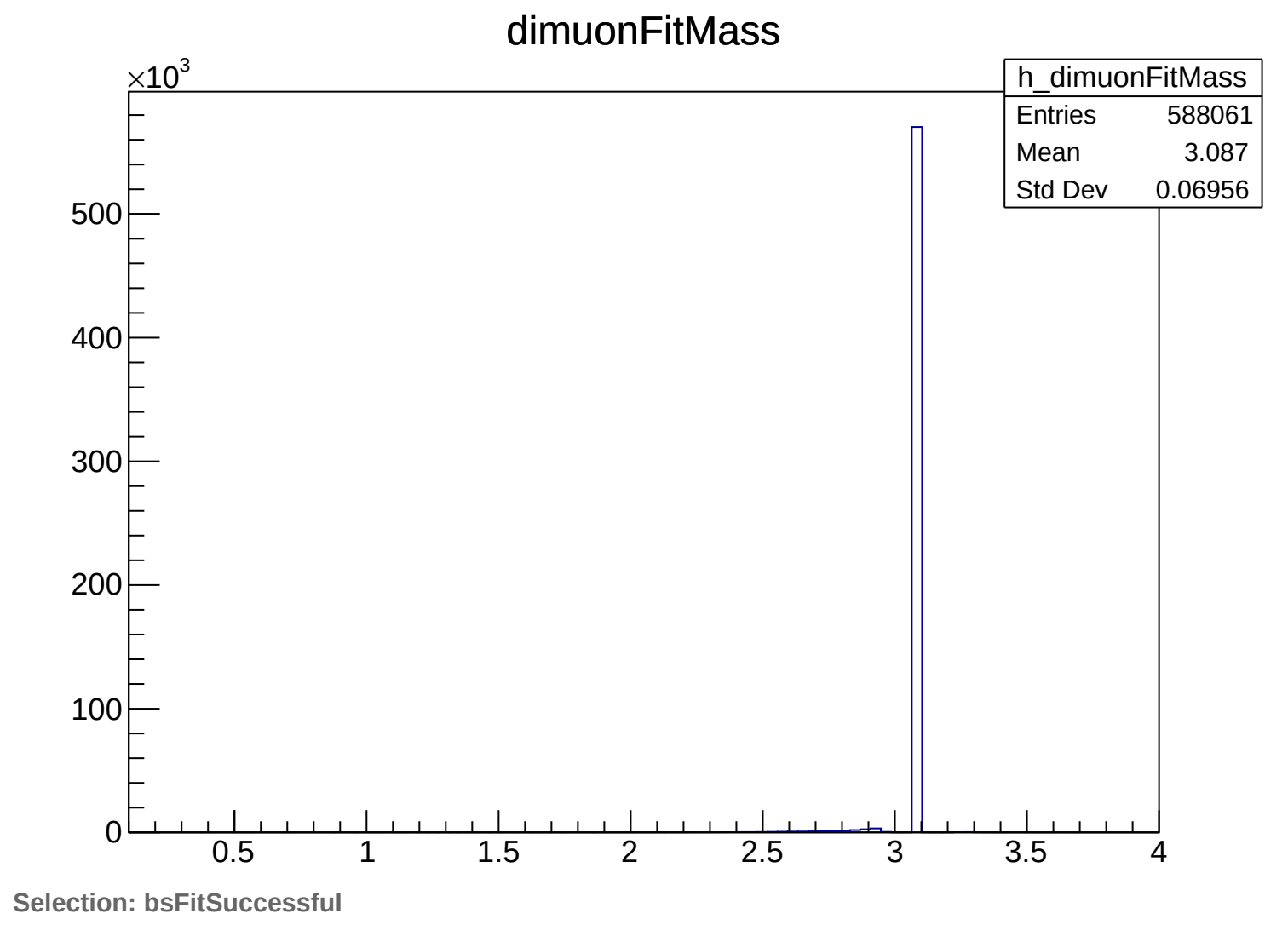

BSsigmaZ
h_BSsigmaZ
Entries 588061
Mean 3.732
Std Dev 9.974e−06
×10^3
600
500
400
300
200
100
0
2.6
2.8
3
3.2
3.4
3.6
3.8
4
4.2
4.4
4.6
4.8
Selection: bsFitSuccessful

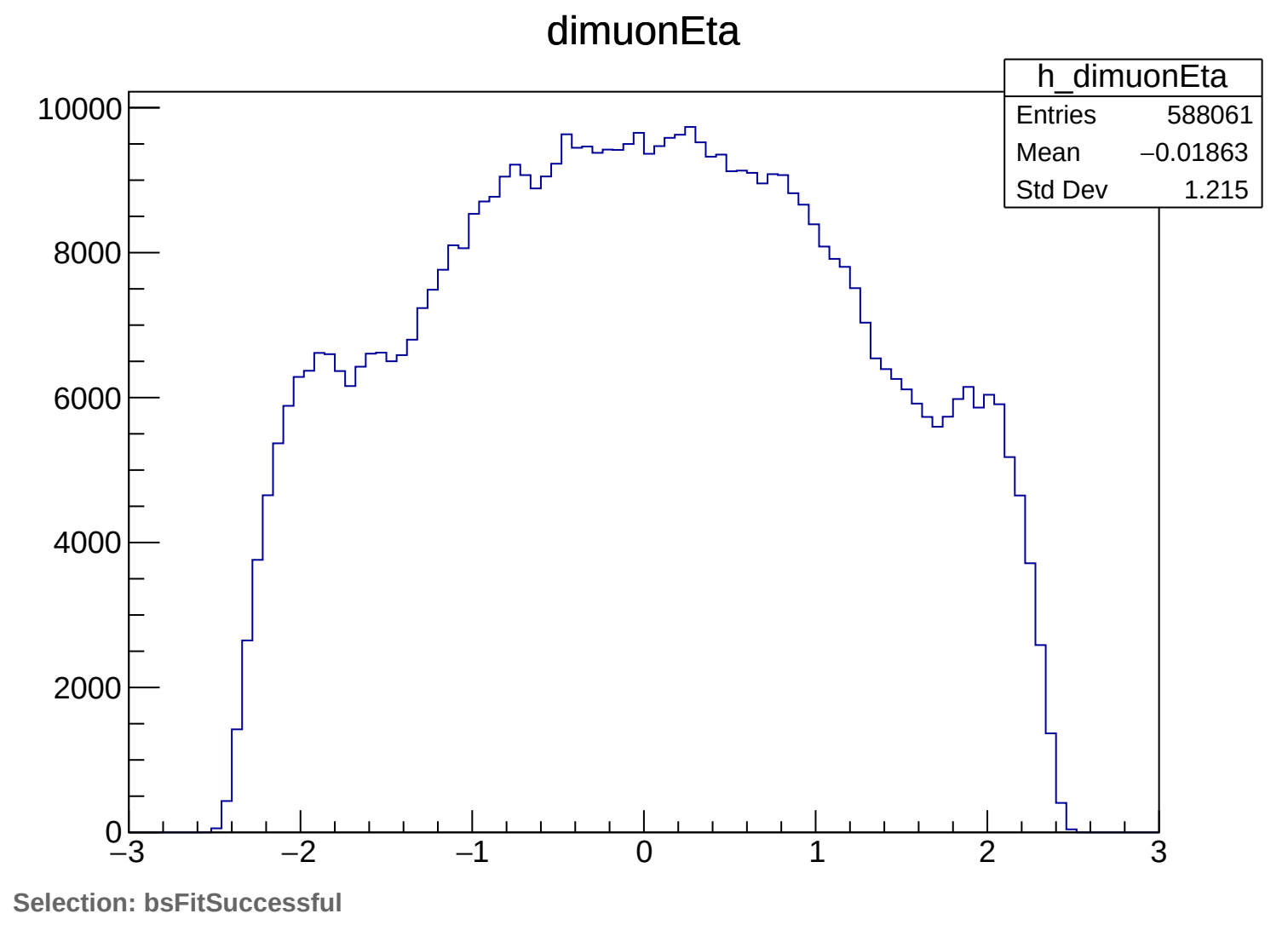

BSdsigmaZ
h_BSdsigmaZ
Entries 588061
Mean 0.03175
Std Dev 0
×10^3
600
500
400
300
200
100
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful

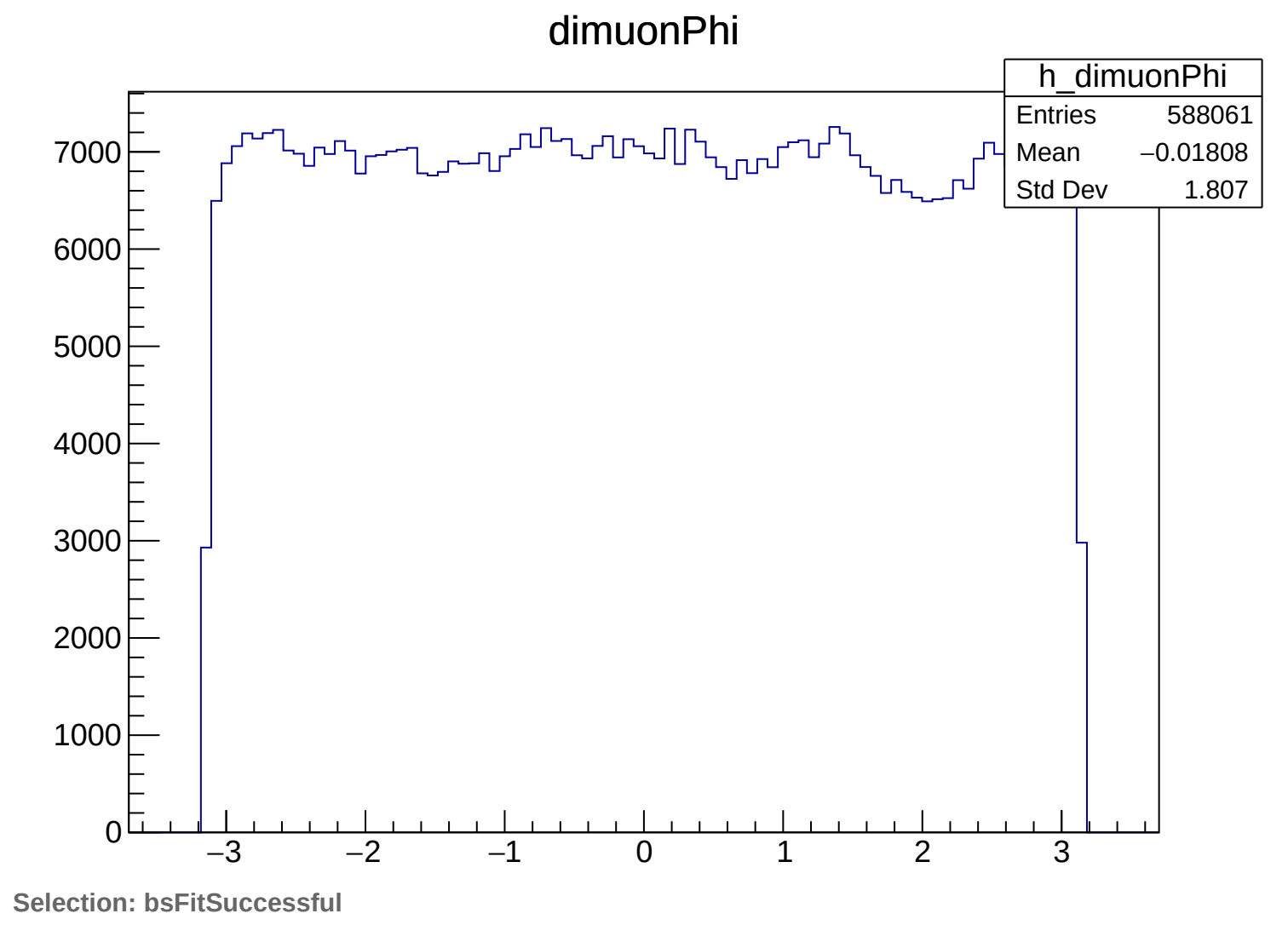

resonanceFlag
×10^3
h_resonanceFlag
Entries 588061
Mean 0.9699
Std Dev 0.1711
500
400
300
200
100
0
0
1
2
3
4
5
6
Selection: bsFitSuccessful

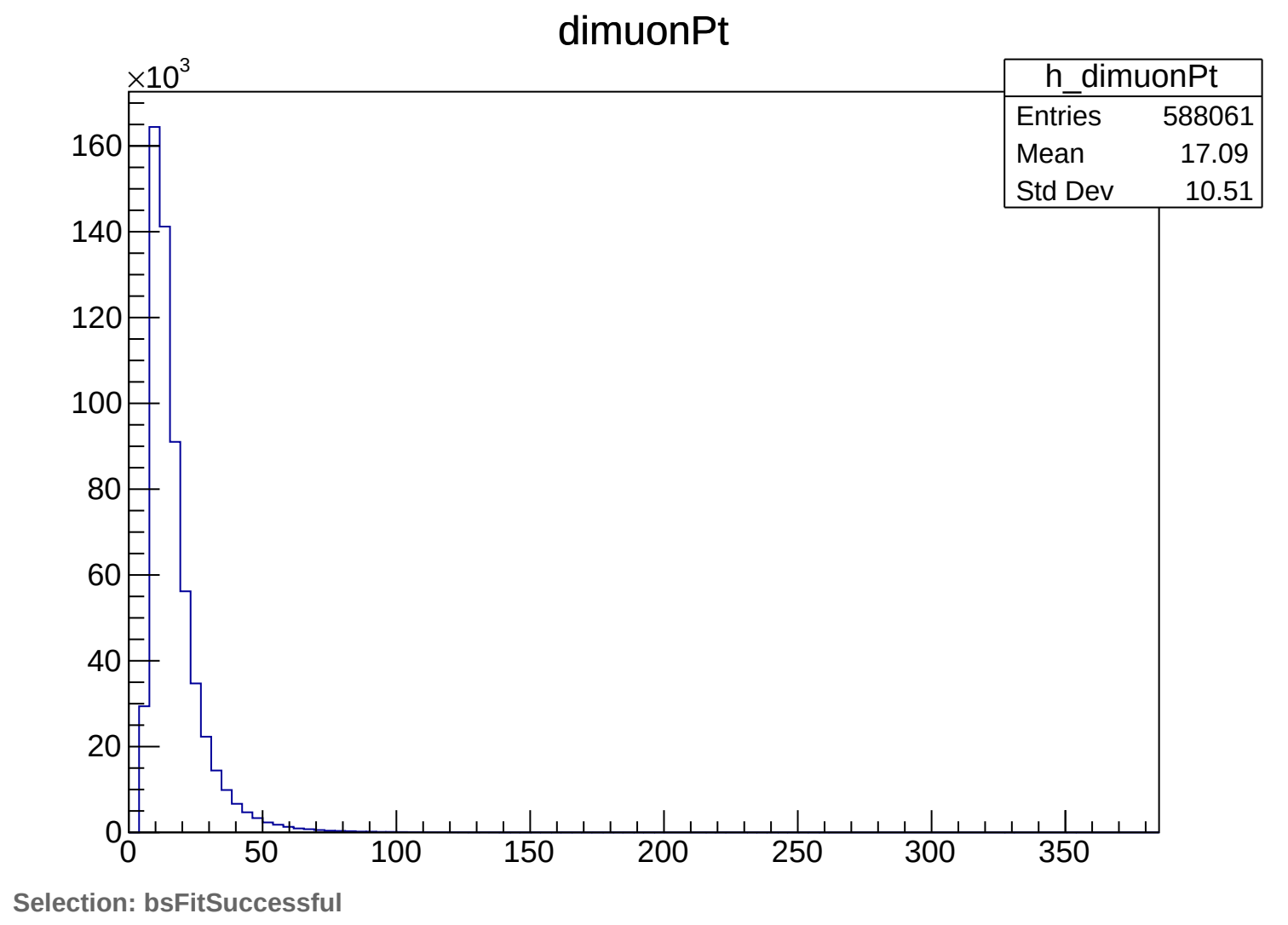

dimuonVtxProb
h_dimuonVtxProb
Entries 588061
Mean 0.5086
Std Dev 0.2862
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful

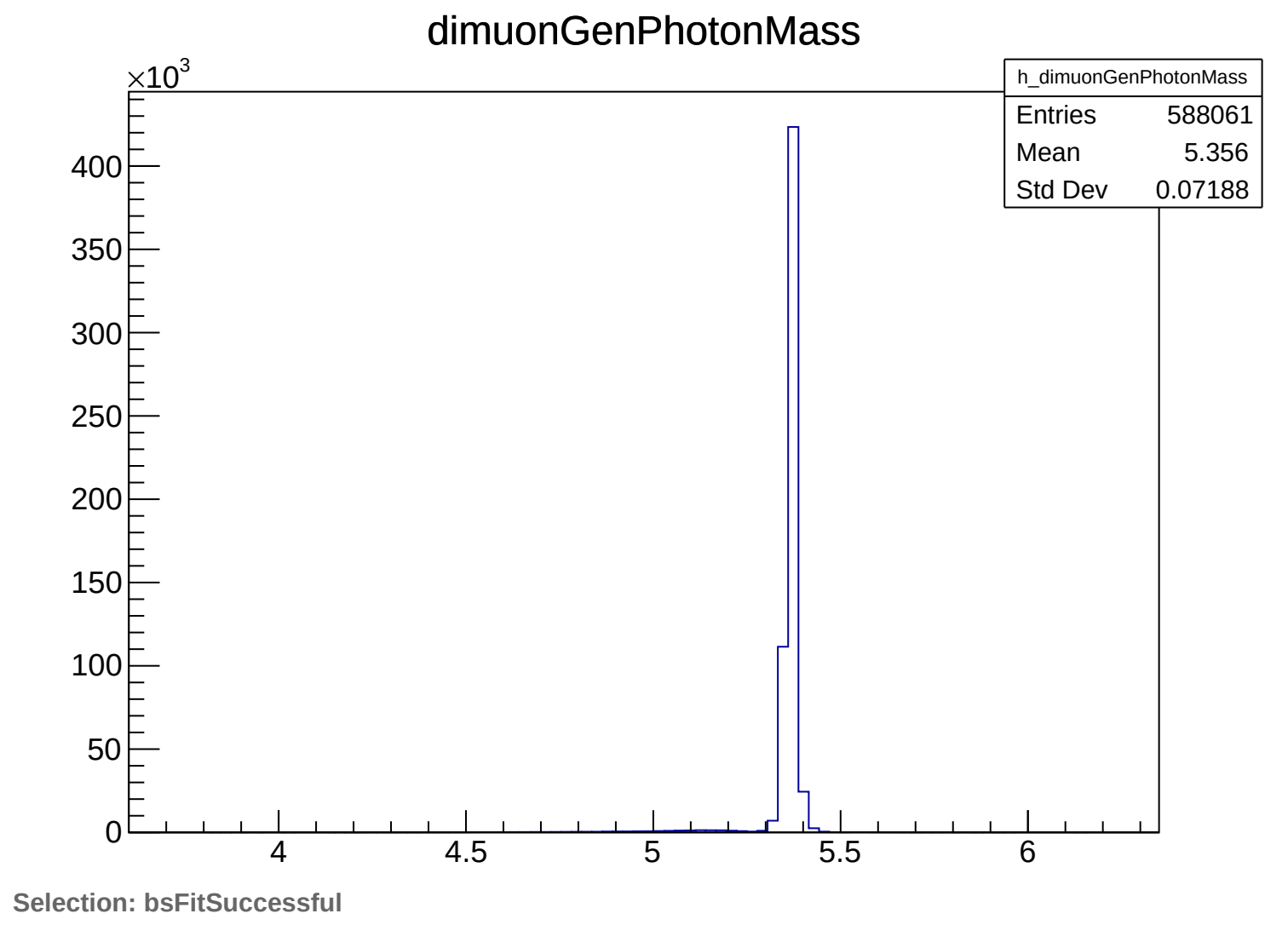

dimuonMass
×10^3
250
200
150
100
50
0
0.5
1
1.5
2
2.5
3
3.5
4
h_dimuonMass
Entries 588061
Mean 3.084
Std Dev 0.078
Selection: bsFitSuccessful

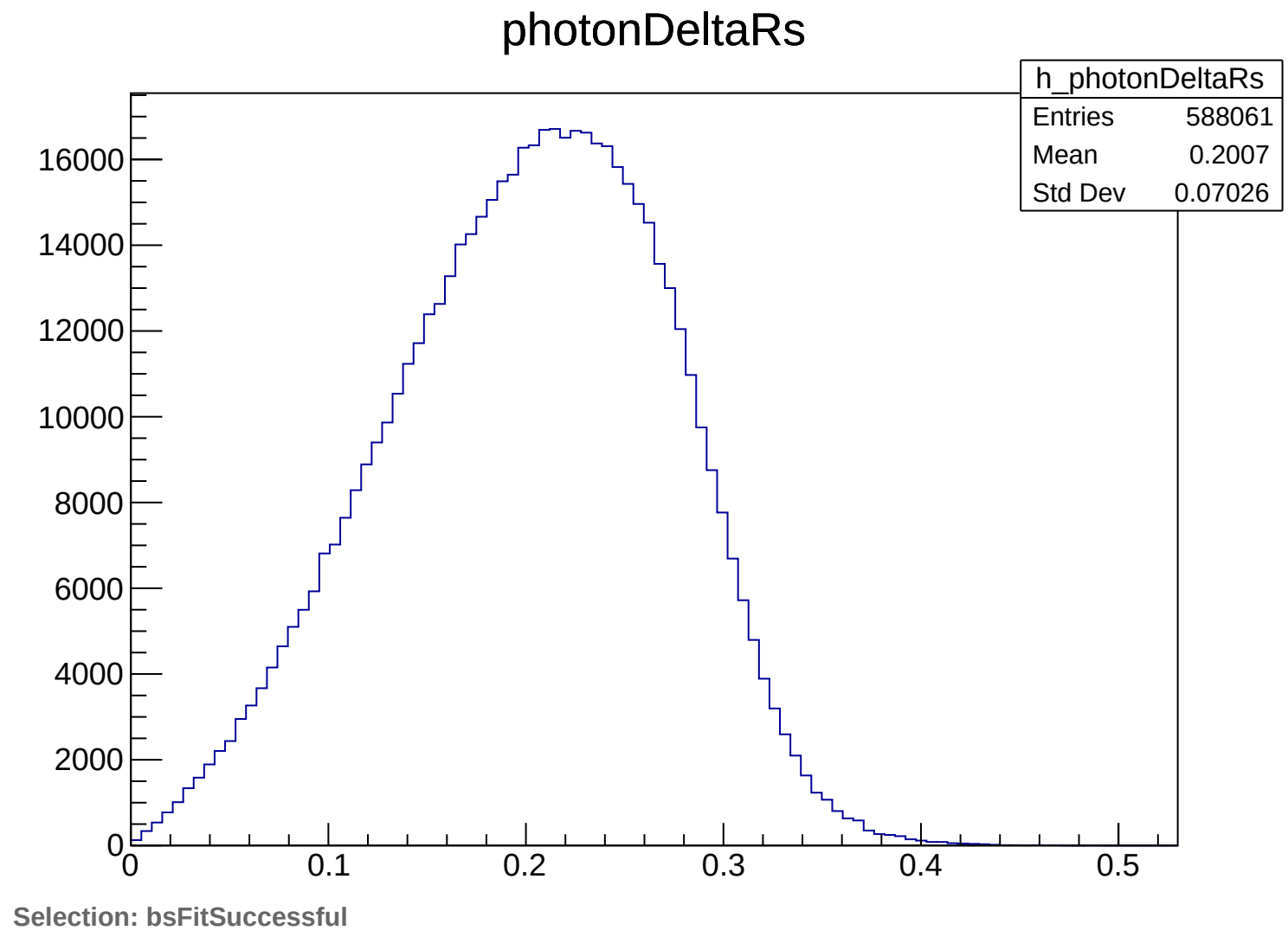

dimuonFitMass
h_dimuonFitMass
Entries 588061
Mean 3.087
Std Dev 0.06956
×10^3
500
400
300
200
100
0
0.5
1
1.5
2
2.5
3
3.5
4
Selection: bsFitSuccessful

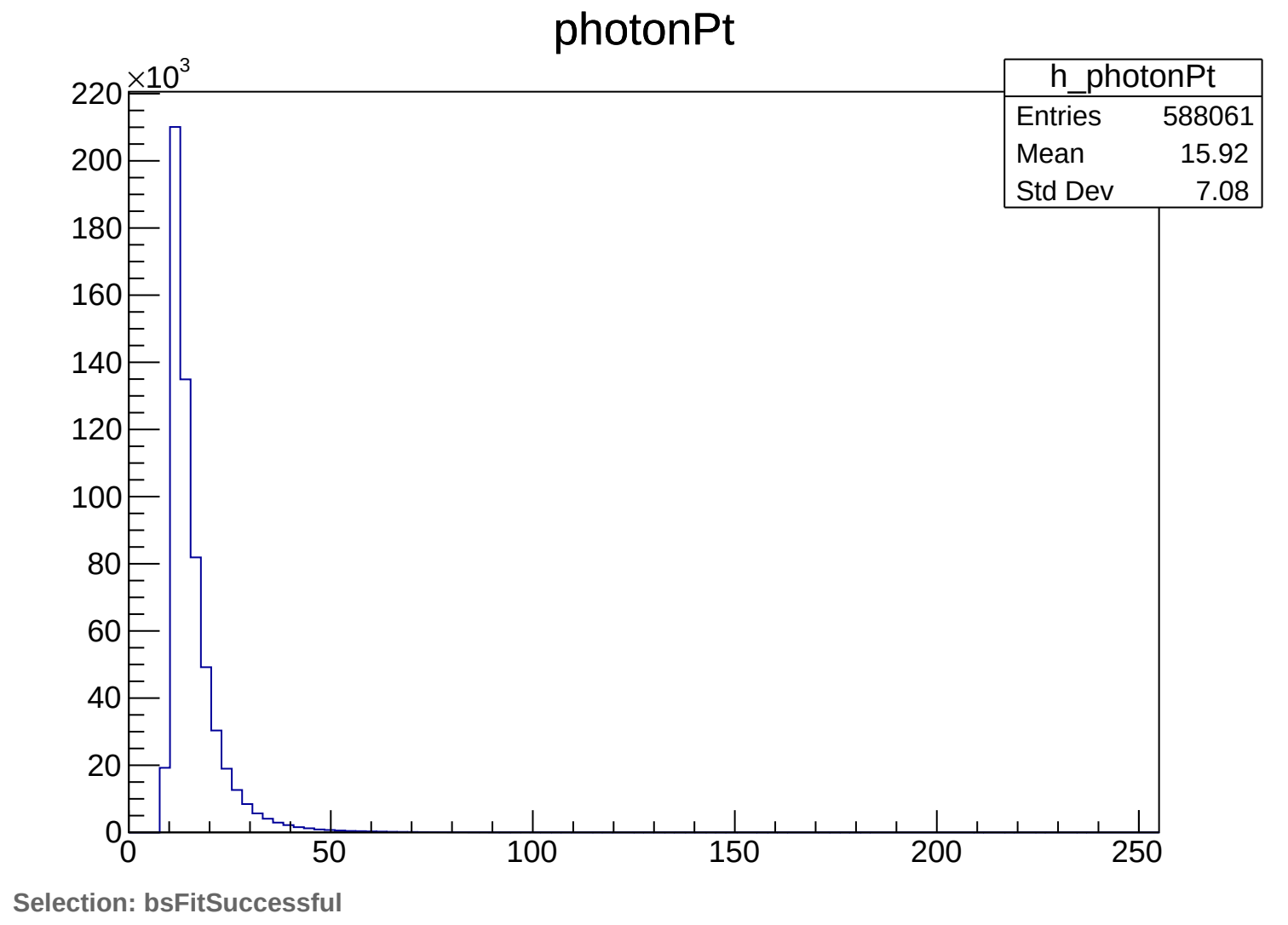

dimuonEta
h_dimuonEta
Entries 588061
Mean −0.01863
Std Dev 1.215
10000
8000
6000
4000
2000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

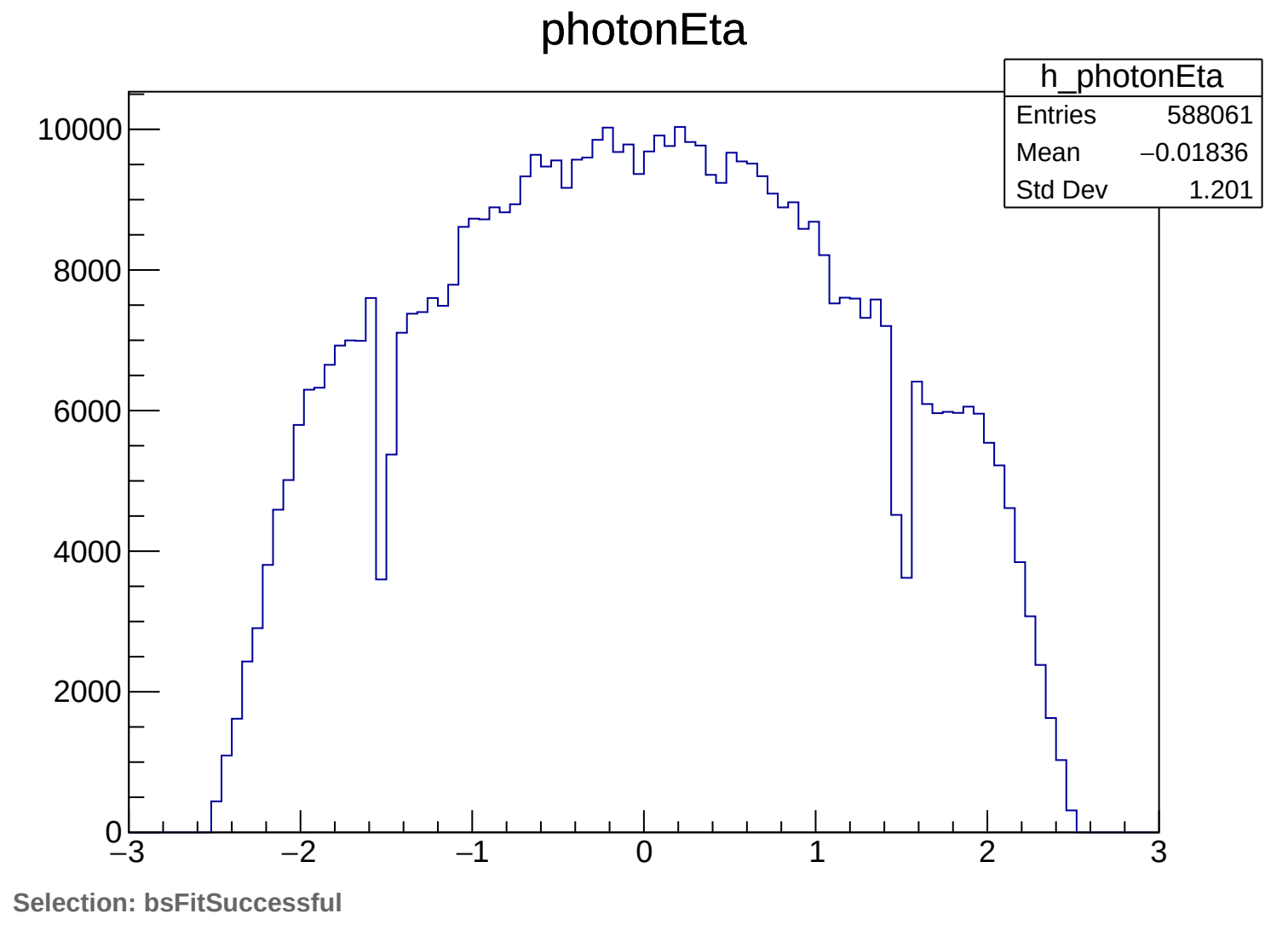

dimuonPhi
h_dimuonPhi
Entries 588061
Mean −0.01808
Std Dev 1.807
7000
6000
5000
4000
3000
2000
1000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

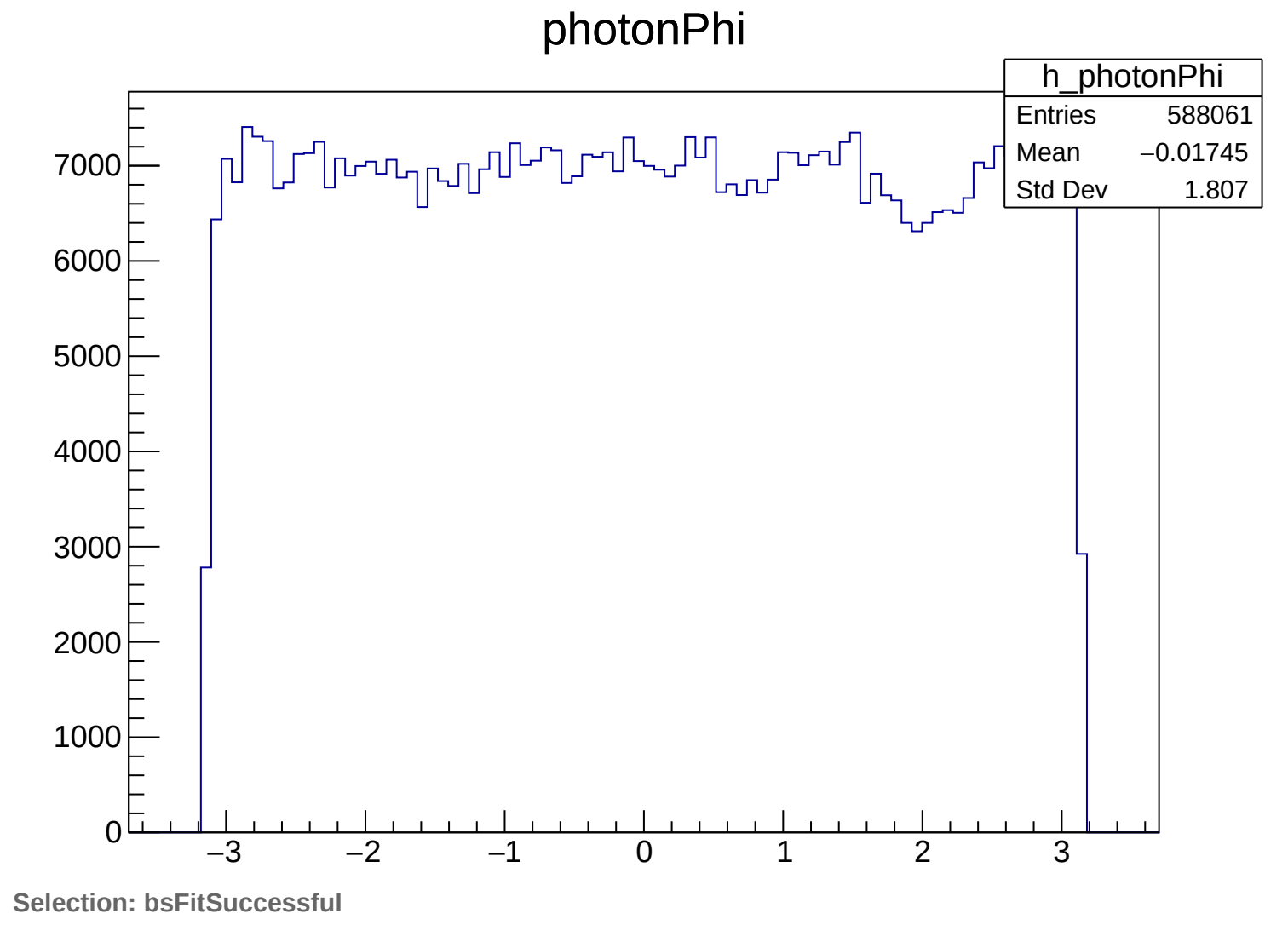

dimuonPt
×10^3
h_dimuonPt
Entries 588061
Mean 17.09
Std Dev 10.51
160
140
120
100
80
60
40
20
0
0
50
100
150
200
250
300
350
Selection: bsFitSuccessful

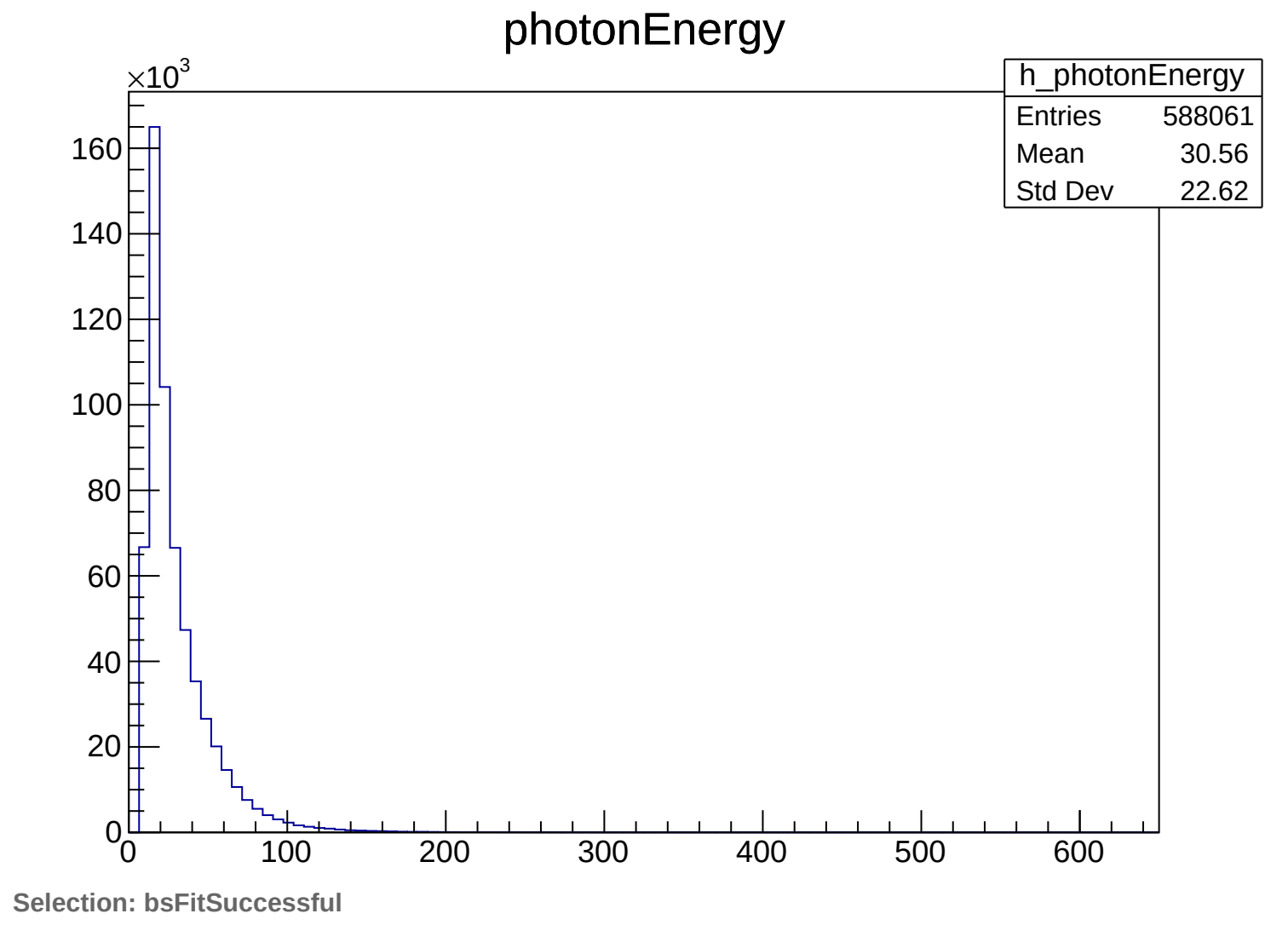

dimuonGenPhotonMass
×10^3
h_dimuonGenPhotonMass
Entries 588061
Mean 5.356
Std Dev 0.07188
400
350
300
250
200
150
100
50
0
4
4.5
5
5.5
6
Selection: bsFitSuccessful

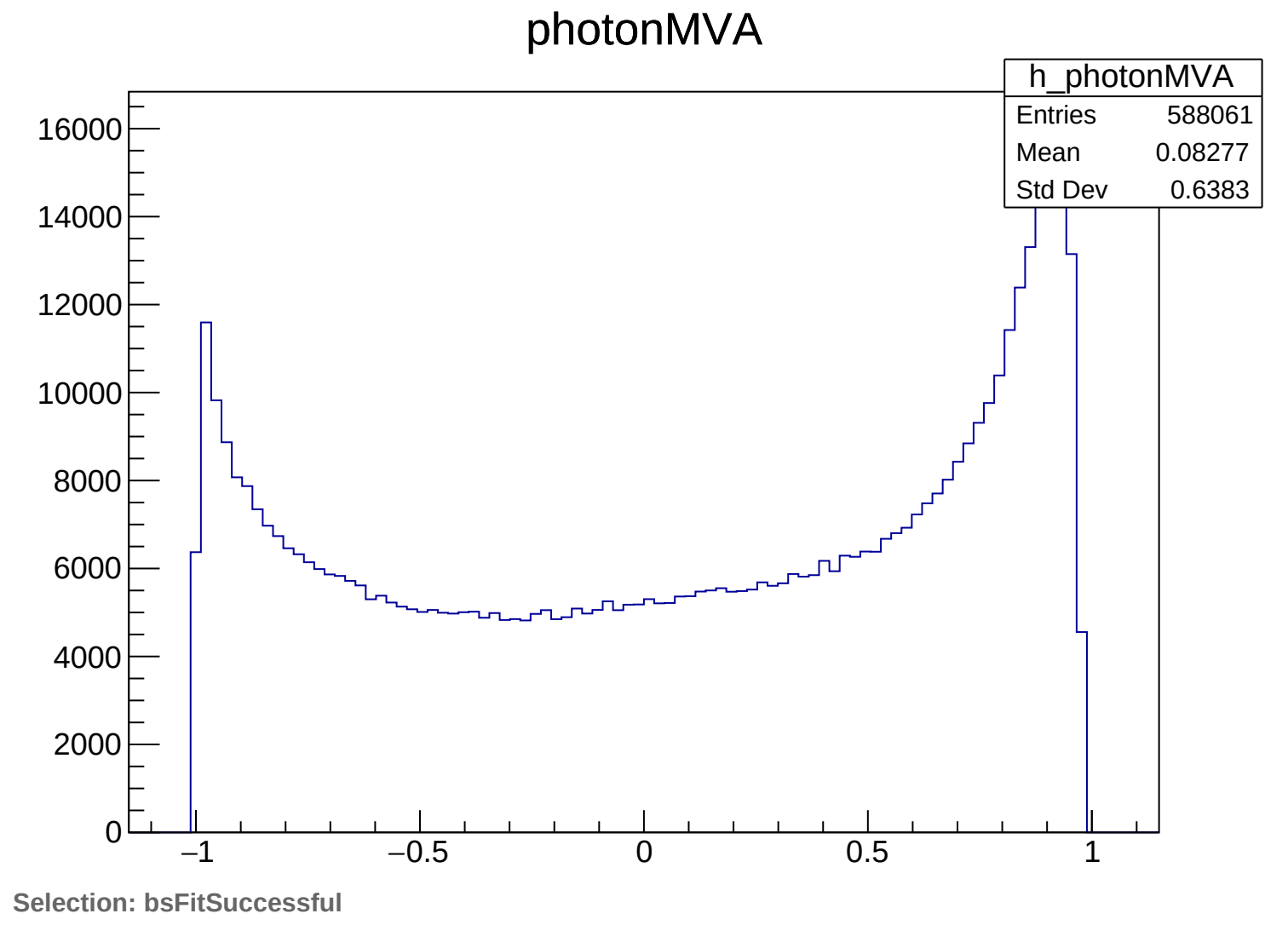

photonDeltaRs
h_photonDeltaRs
Entries 588061
Mean 0.2007
Std Dev 0.07026
16000
14000
12000
10000
8000
6000
4000
2000
0
0
0.1
0.2
0.3
0.4
0.5
Selection: bsFitSuccessful

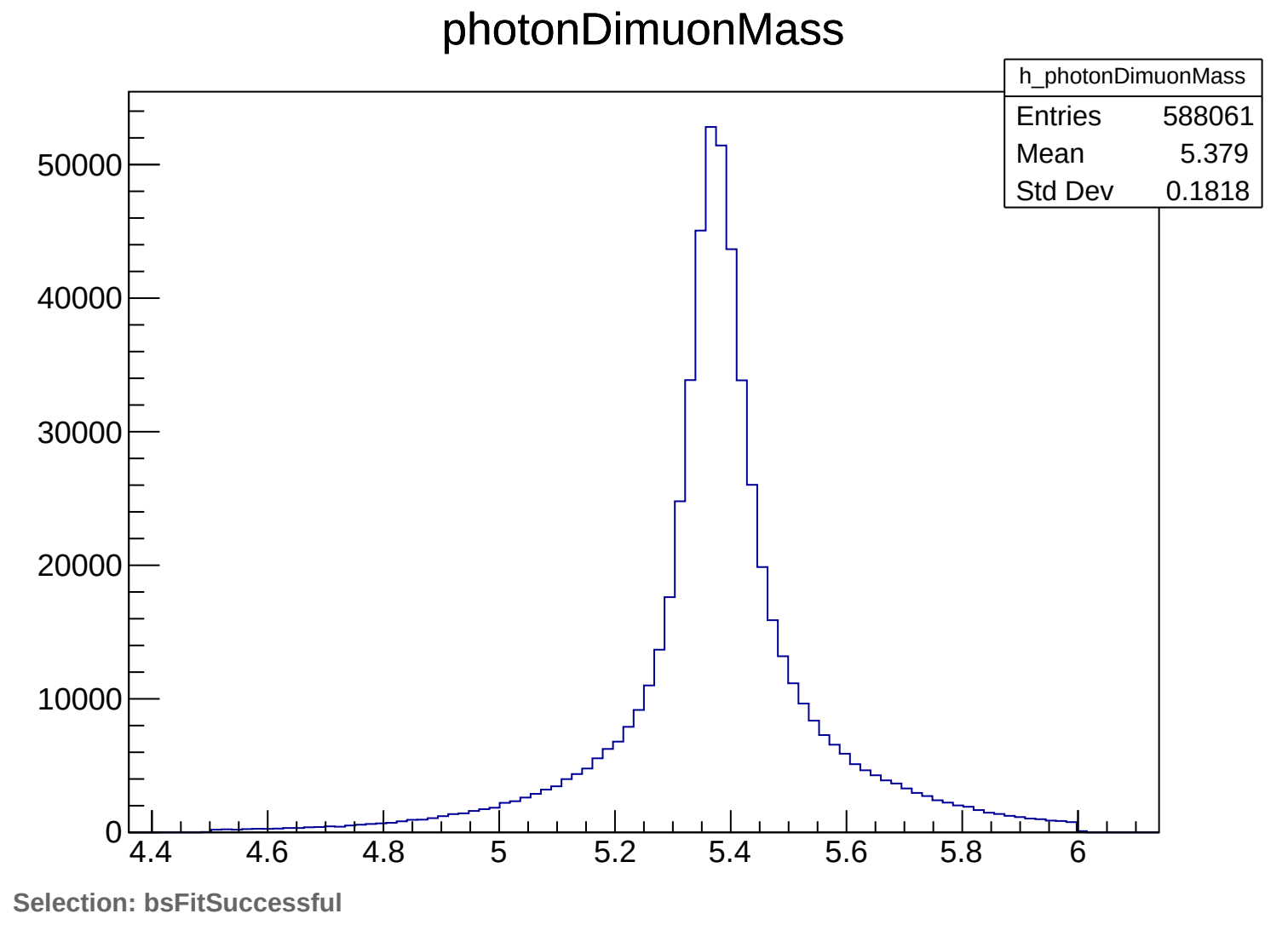

photonPt
h_photonPt
Entries 588061
Mean 15.92
Std Dev 7.08
×10^3
220
200
180
160
140
120
100
80
60
40
20
0
0
50
100
150
200
250
Selection: bsFitSuccessful

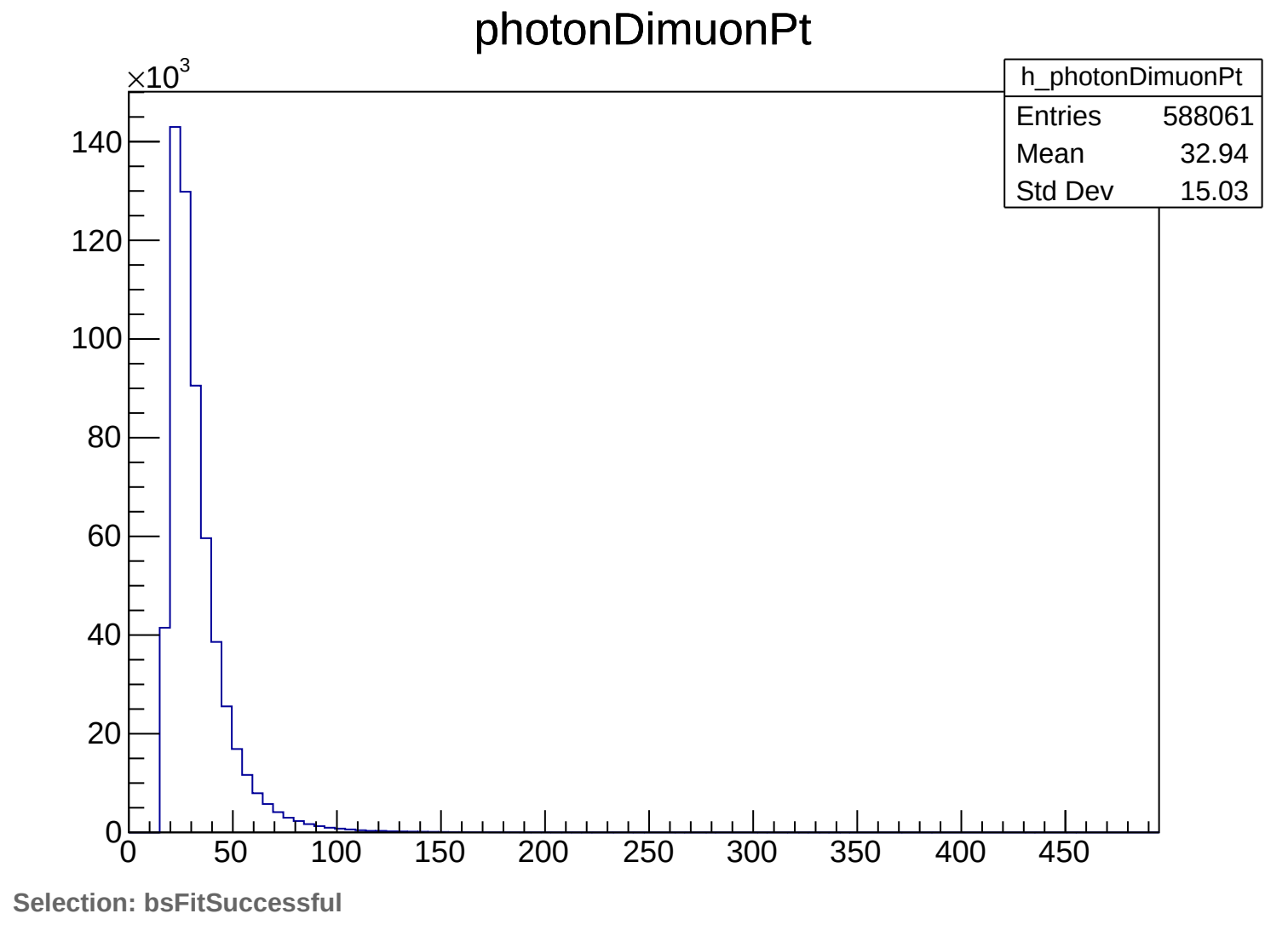

photonEta
h_photonEta
Entries 588061
Mean −0.01836
Std Dev 1.201
10000
8000
6000
4000
2000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

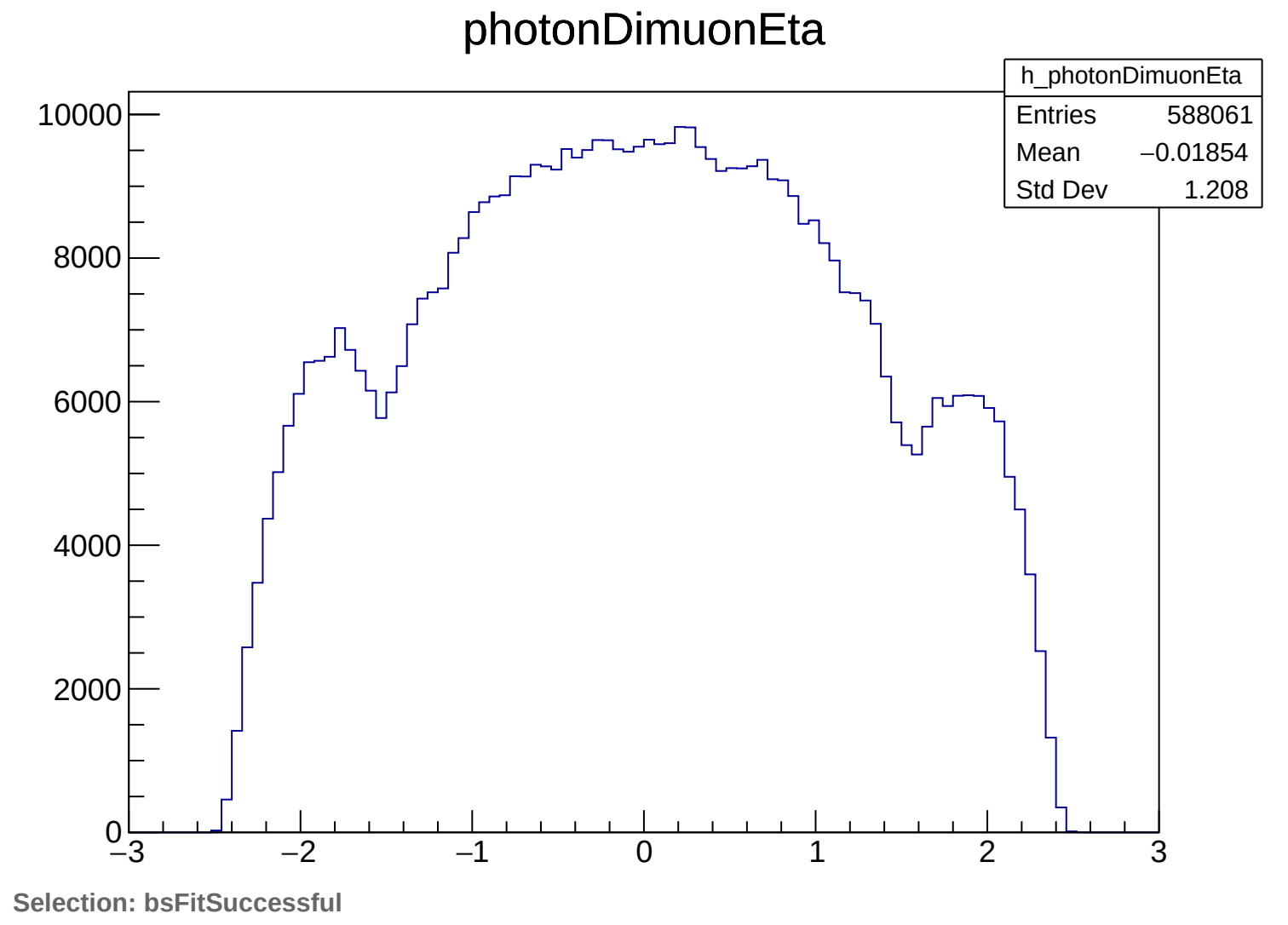

photonPhi
h_photonPhi
Entries 588061
Mean −0.01745
Std Dev 1.807
7000
6000
5000
4000
3000
2000
1000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

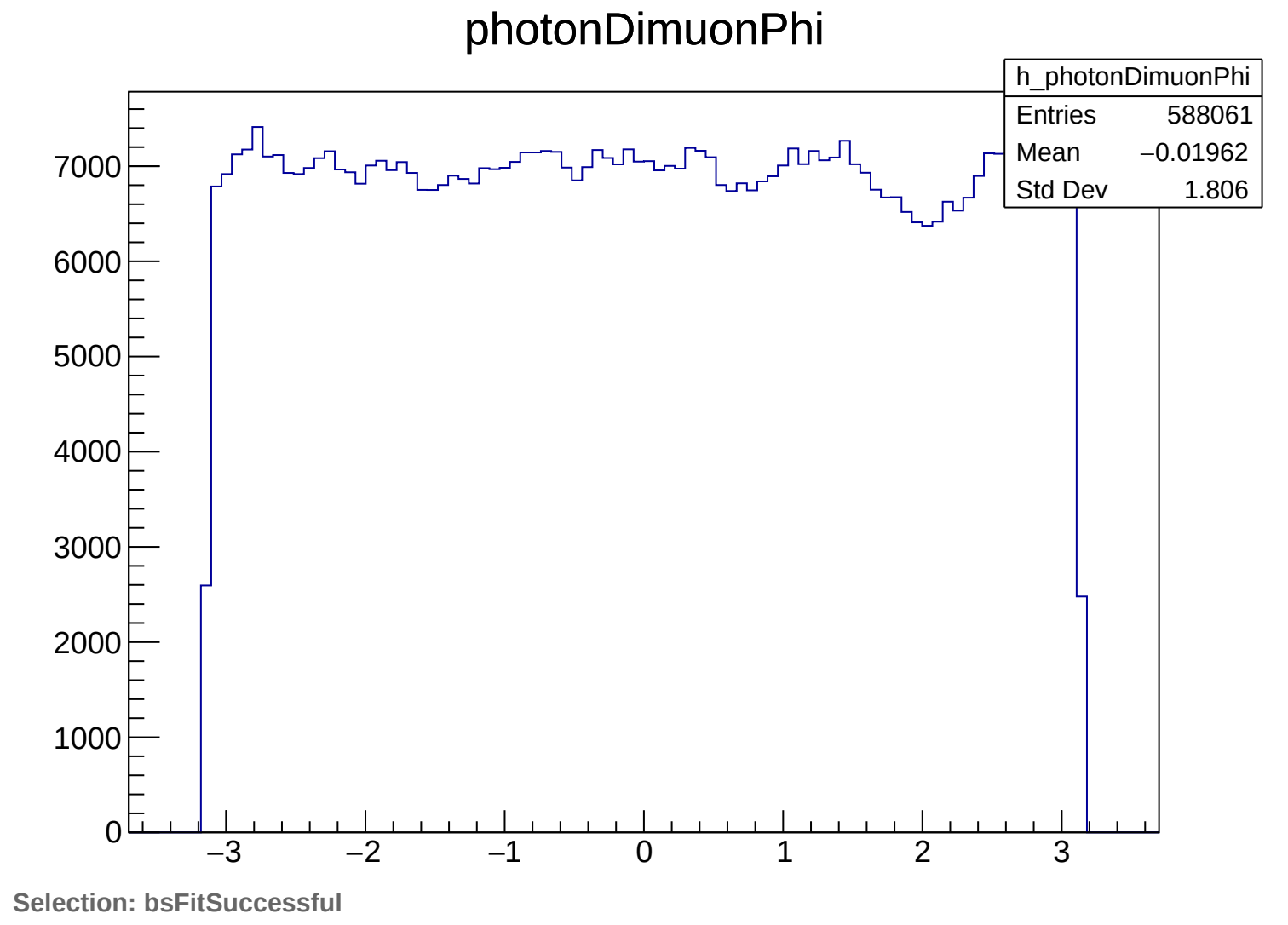

photonEnergy
×10^3
h_photonEnergy
Entries 588061
Mean 30.56
Std Dev 22.62
160
140
120
100
80
60
40
20
0
0
100
200
300
400
500
600
Selection: bsFitSuccessful

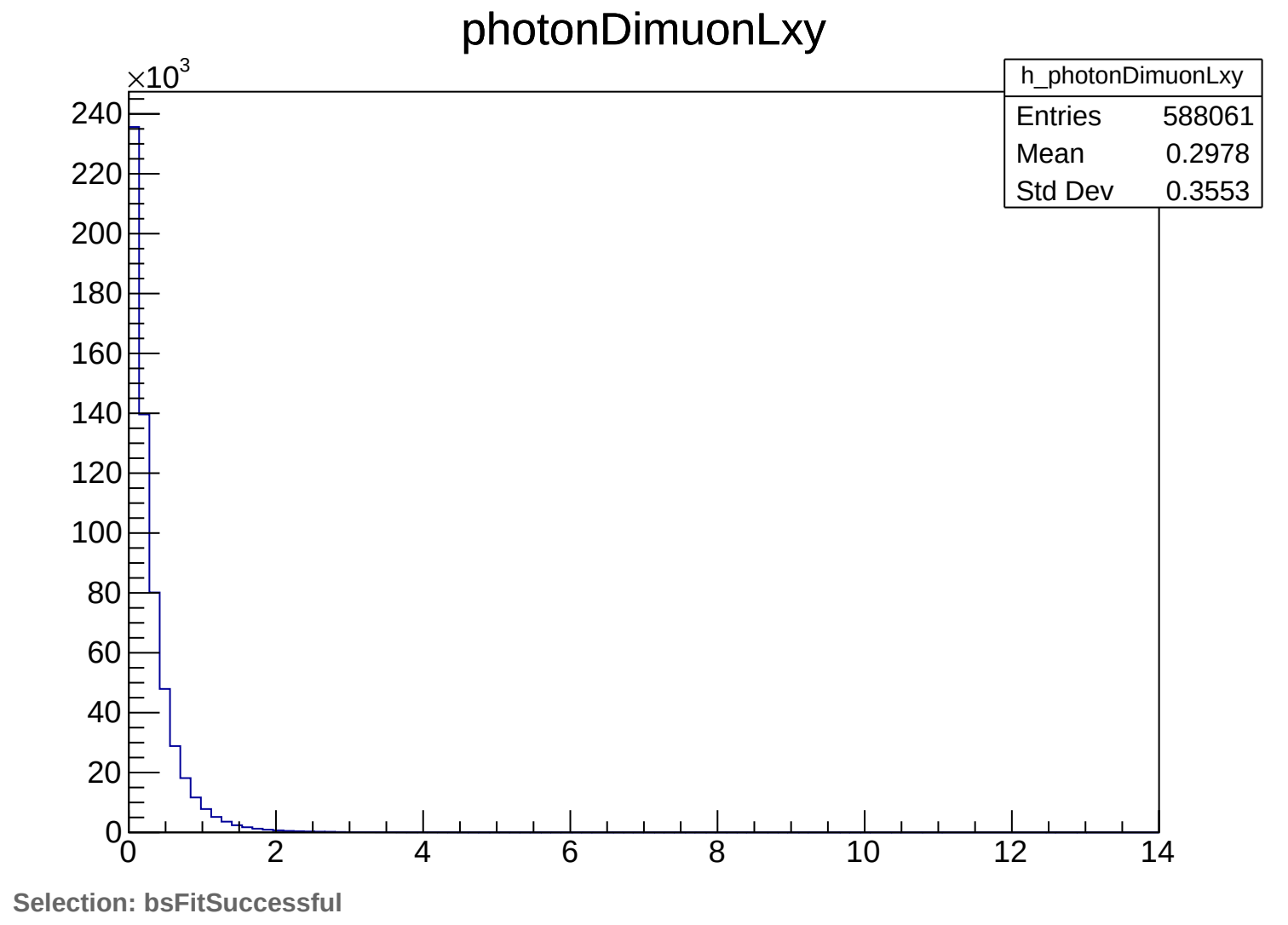

photonMVA
h_photonMVA
Entries 588061
Mean 0.08277
Std Dev 0.6383
16000
14000
12000
10000
8000
6000
4000
2000
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful

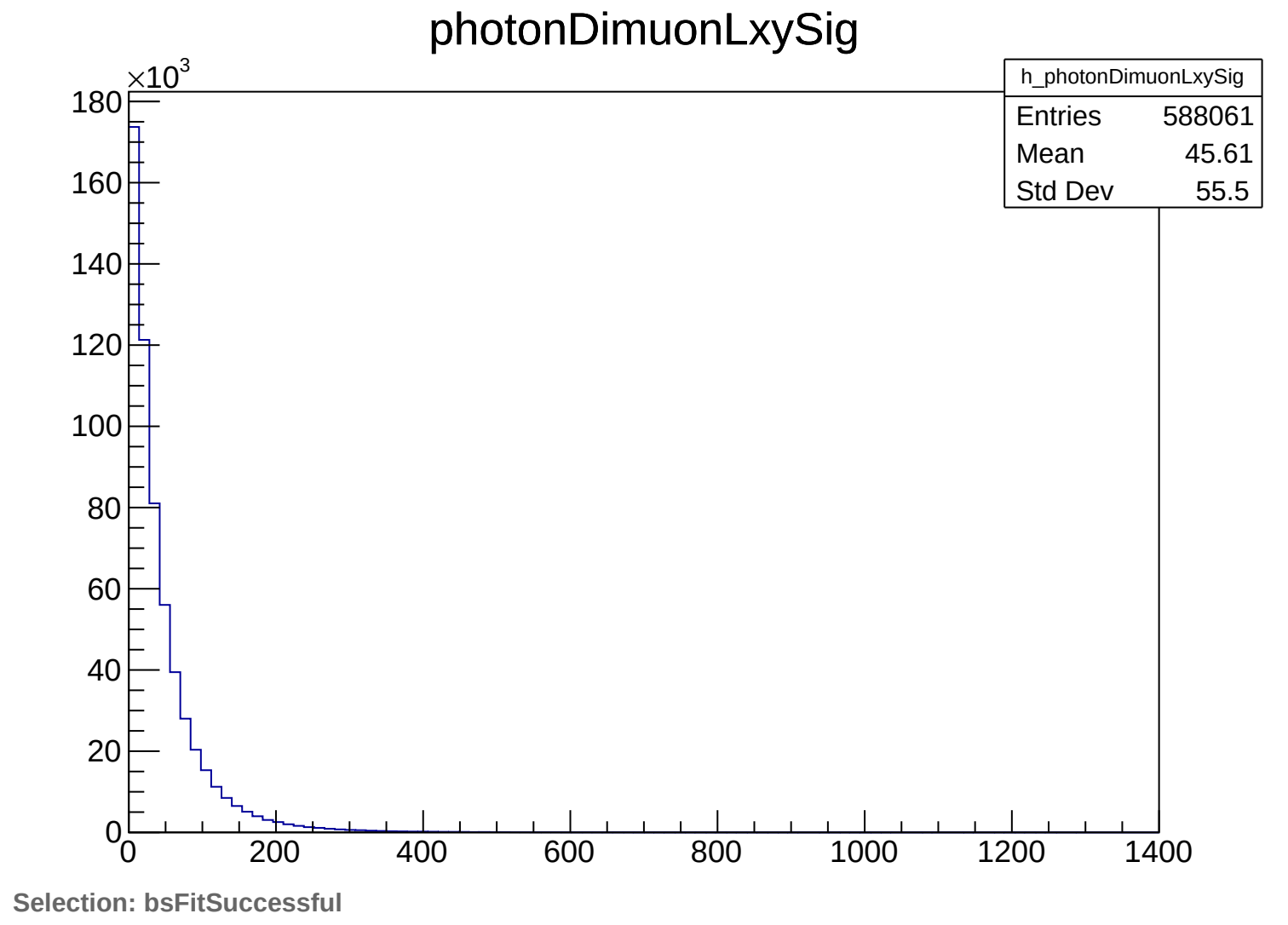

photonDimuonMass
h_photonDimuonMass
Entries 588061
Mean 5.379
Std Dev 0.1818
50000
40000
30000
20000
10000
0
4.4
4.6
4.8
5
5.2
5.4
5.6
5.8
6
Selection: bsFitSuccessful

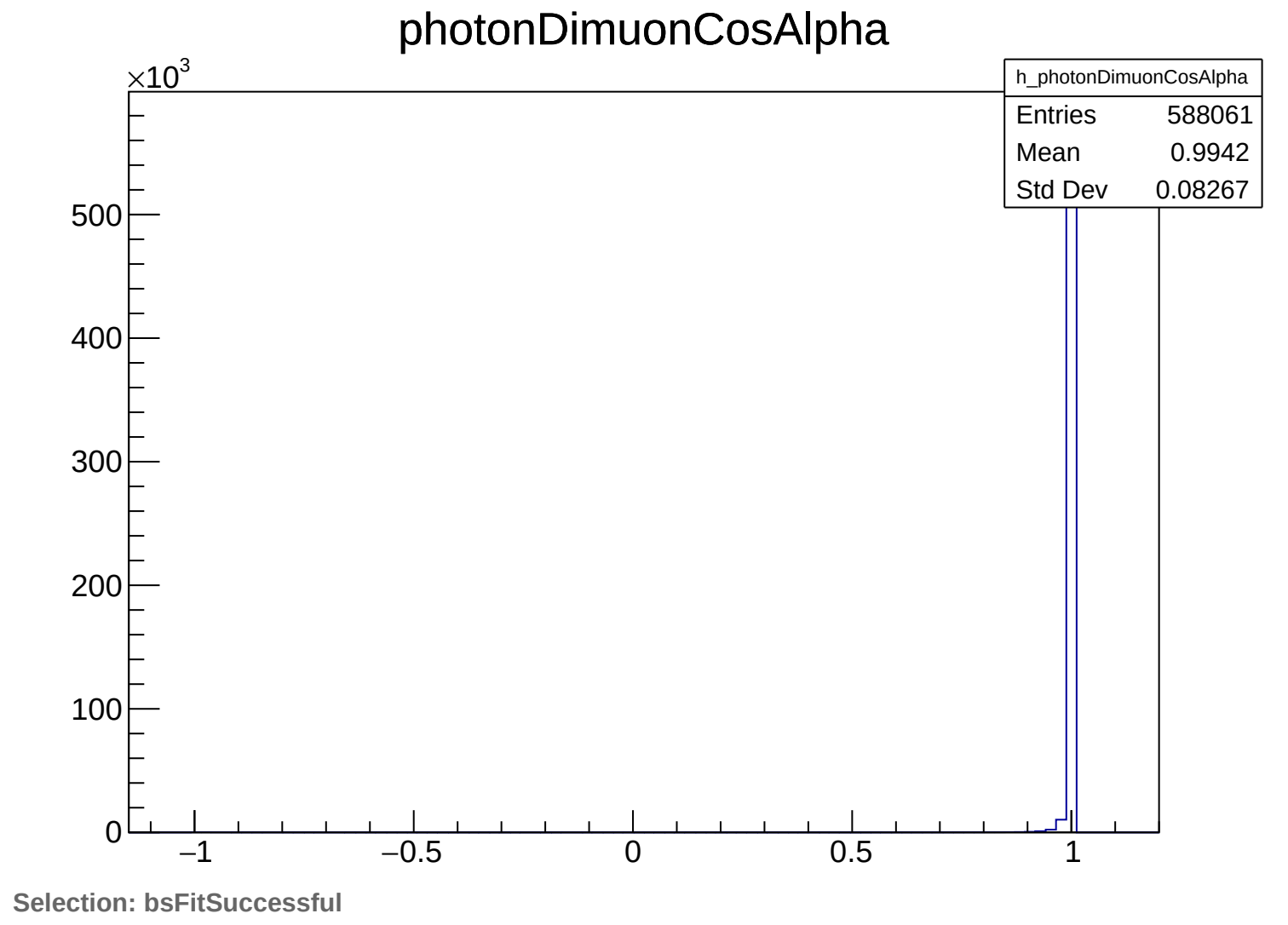

photonDimuonPt
×10^3
h_photonDimuonPt
Entries 588061
Mean 32.94
Std Dev 15.03
140
120
100
80
60
40
20
0
0
50
100
150
200
250
300
350
400
450
Selection: bsFitSuccessful

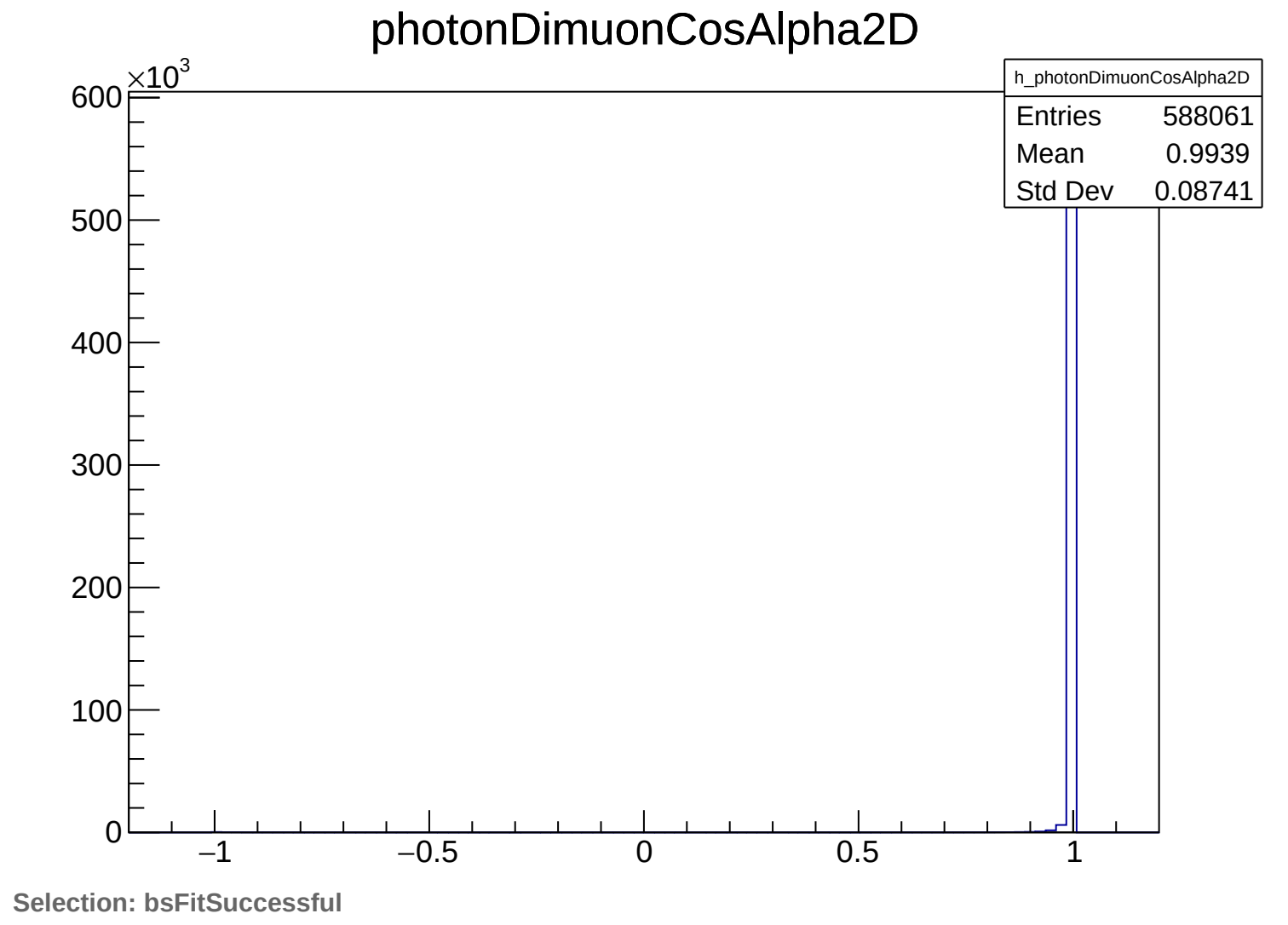

photonDimuonEta
h_photonDimuonEta
Entries 588061
Mean −0.01854
Std Dev 1.208
10000
8000
6000
4000
2000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

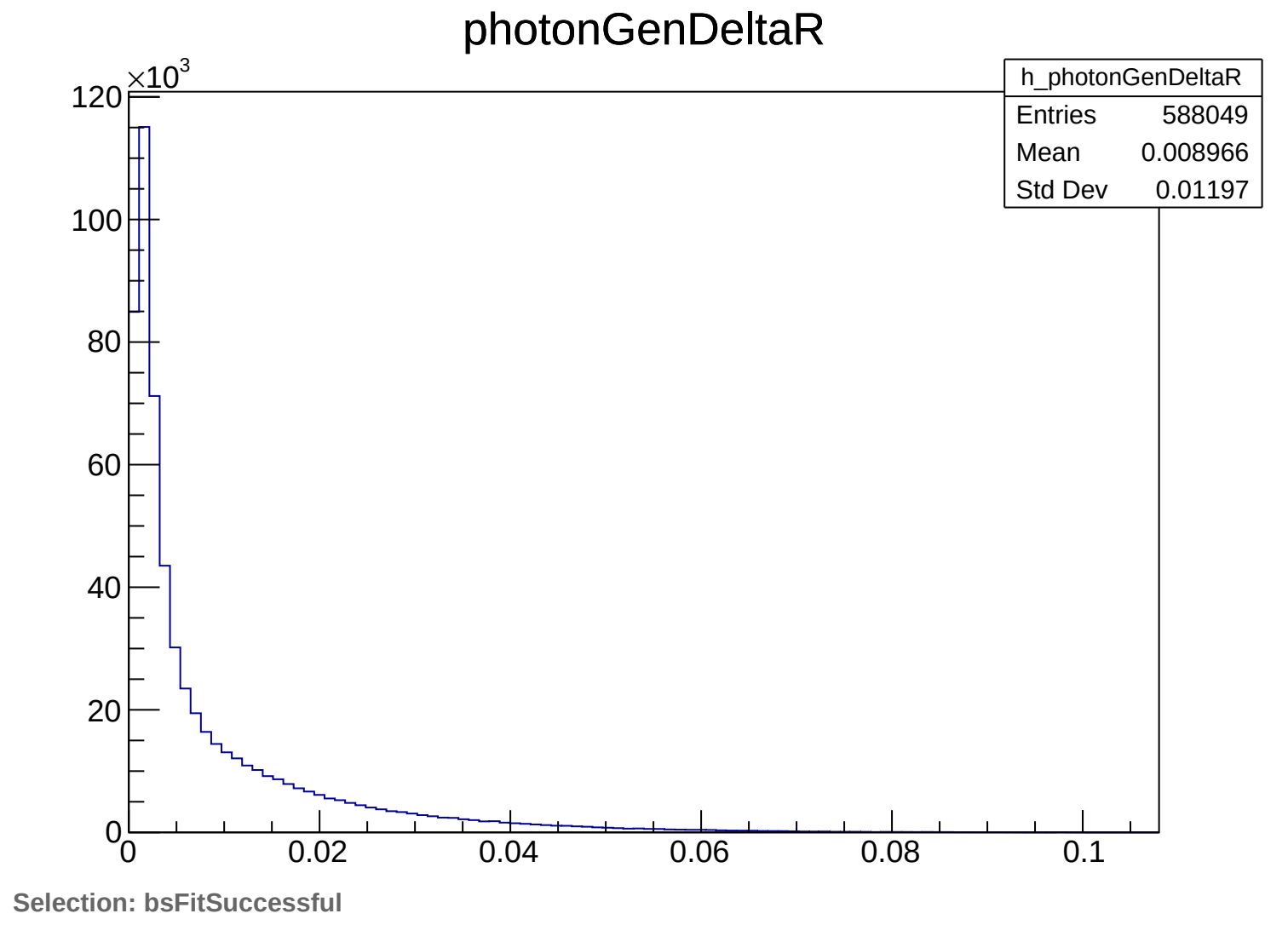

photonDimuonPhi
h_photonDimuonPhi
Entries 588061
Mean −0.01962
Std Dev 1.806
7000
6000
5000
4000
3000
2000
1000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

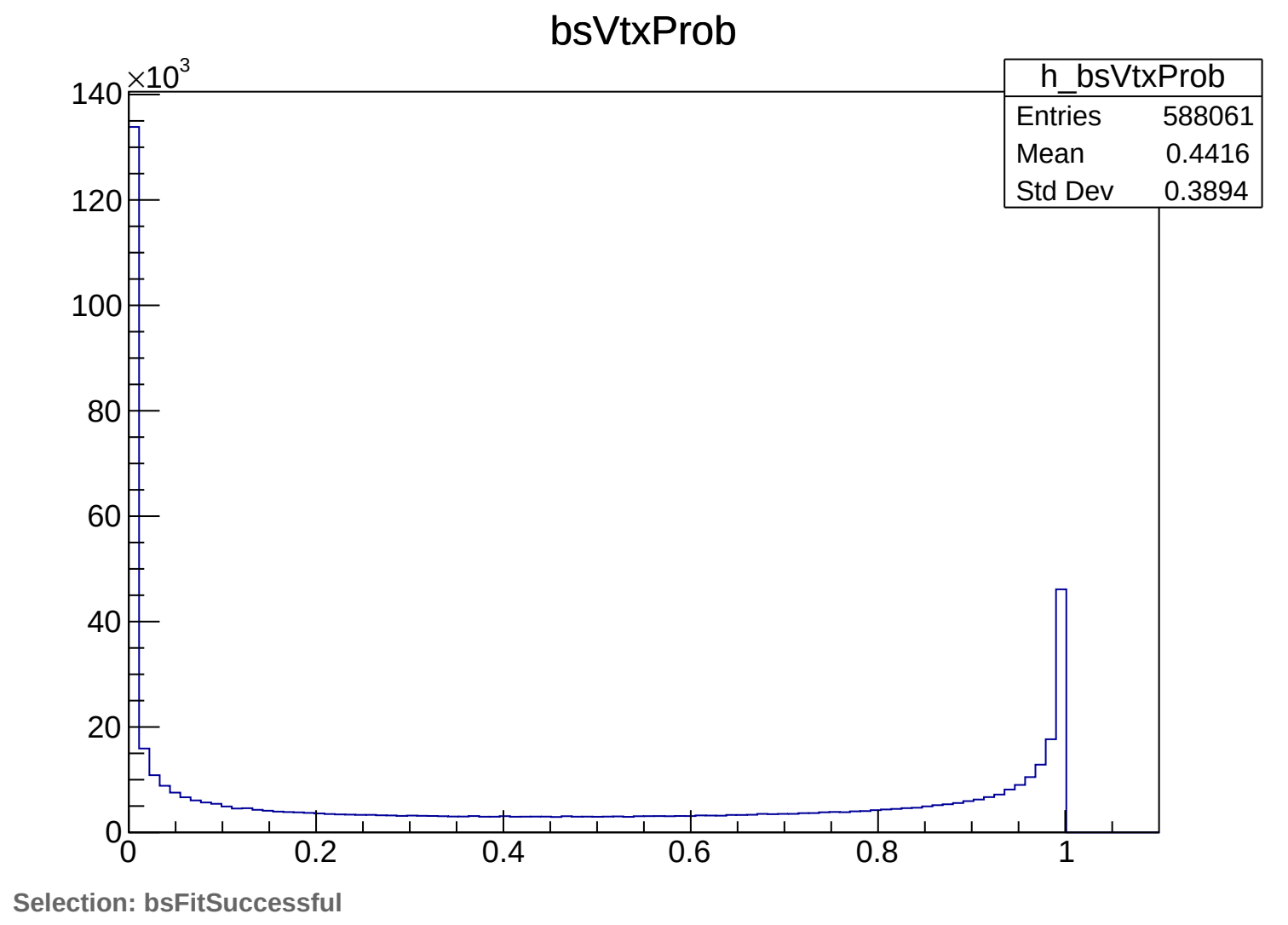

photonDimuonLxy
h_photonDimuonLxy
Entries 588061
Mean 0.2978
Std Dev 0.3553
×10^3
240
220
200
180
160
140
120
100
80
60
40
20
0
0
2
4
6
8
10
12
14
Selection: bsFitSuccessful

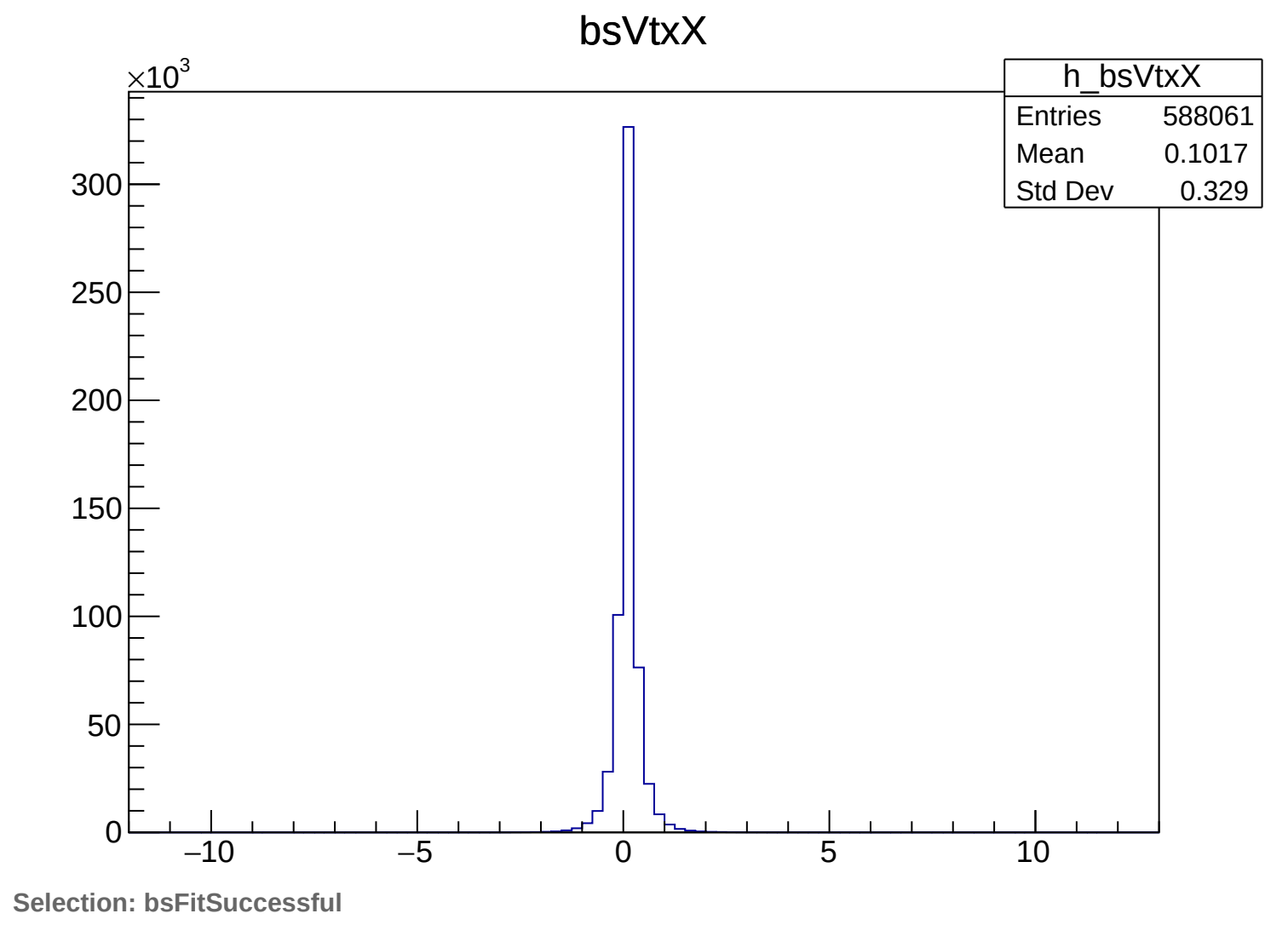

photonDimuonLxySig
h_photonDimuonLxySig
Entries 588061
Mean 45.61
Std Dev 55.5
×10^3
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
1400
Selection: bsFitSuccessful

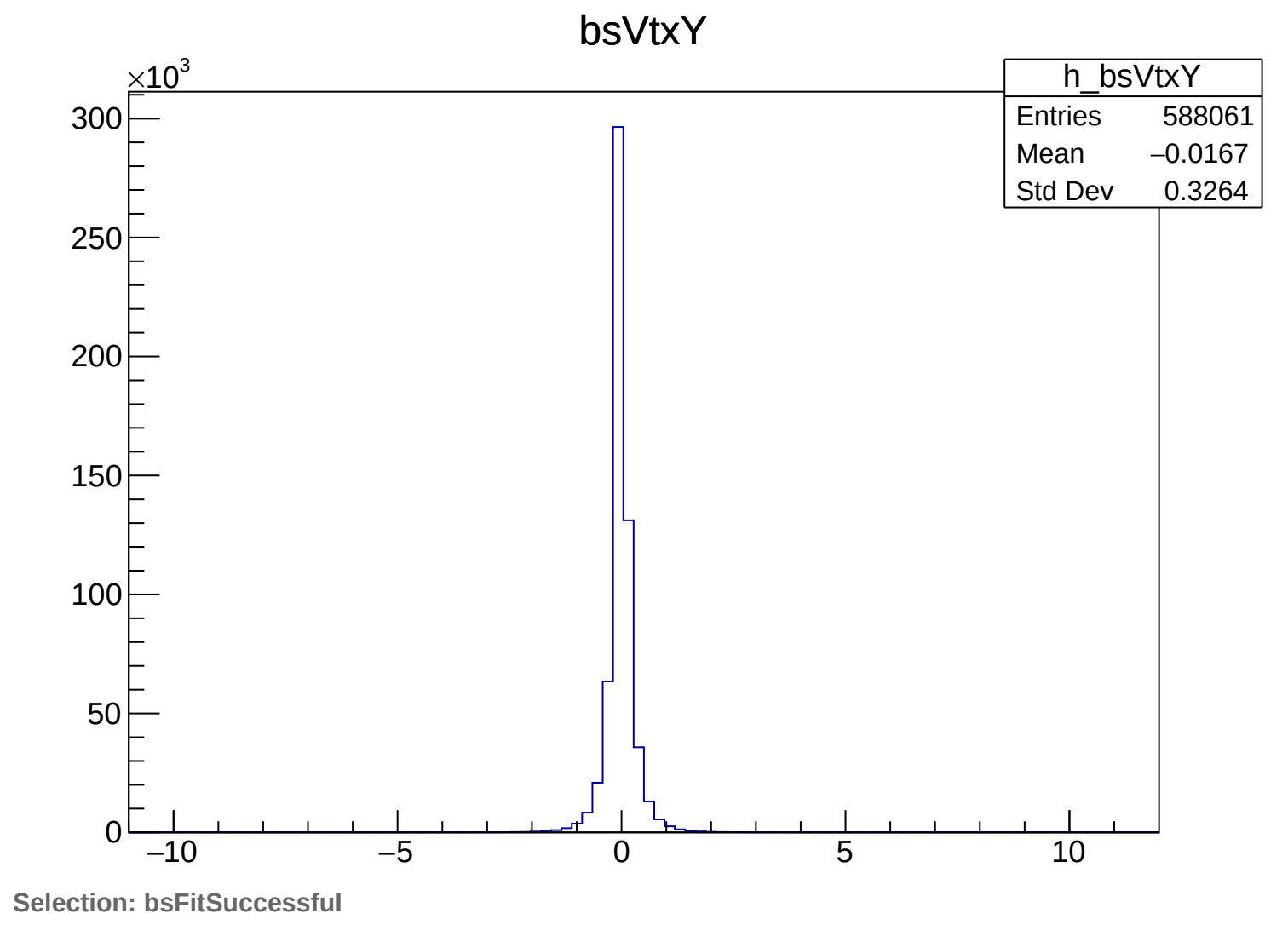

photonDimuonCosAlpha
×10^3
h_photonDimuonCosAlpha
Entries 588061
Mean 0.9942
Std Dev 0.08267
500
400
300
200
100
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful

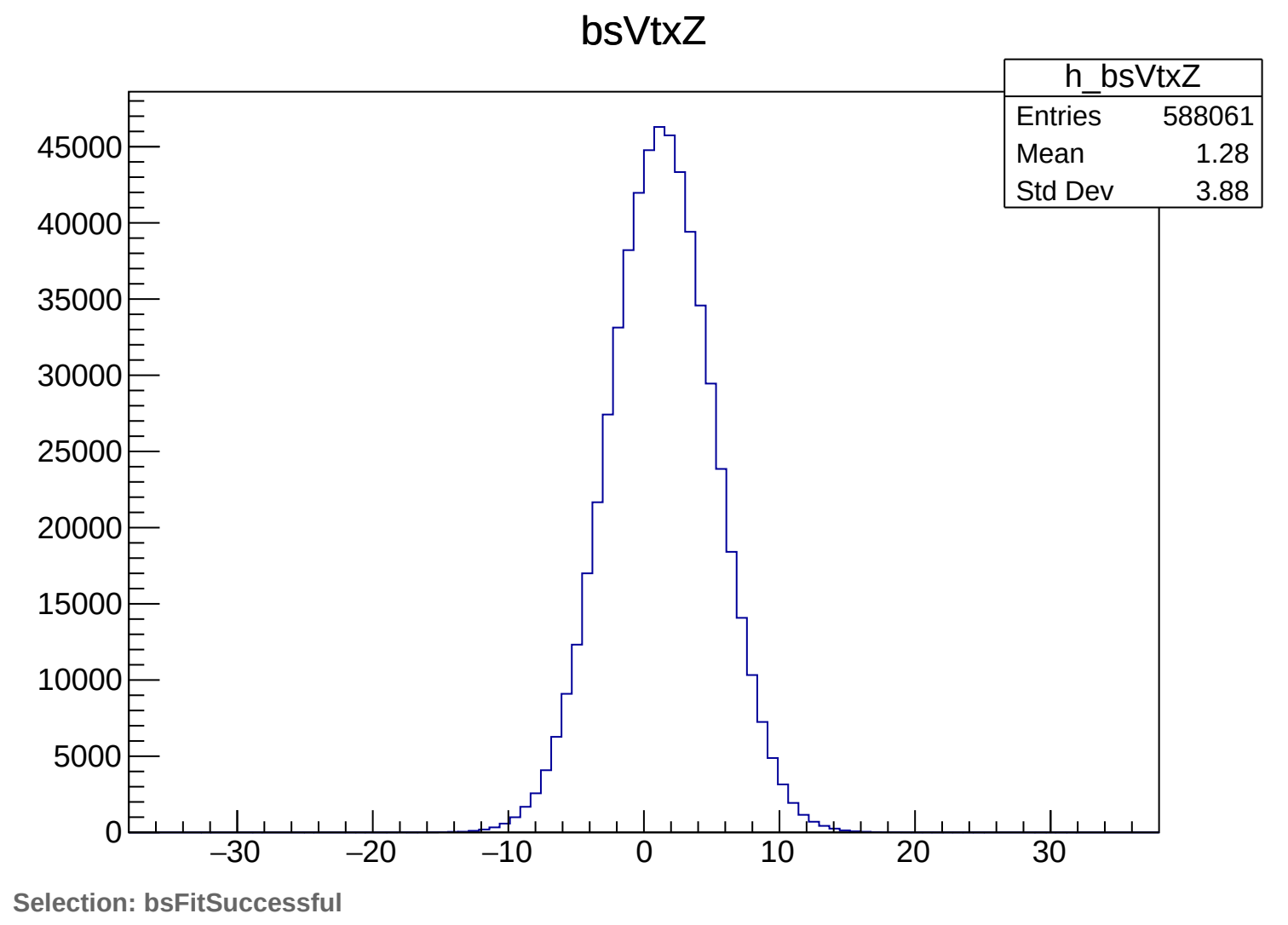

photonDimuonCosAlpha2D
×10^3
600
500
400
300
200
100
0
−1
−0.5
0
0.5
1
h_photonDimuonCosAlpha2D
Entries 588061
Mean 0.9939
Std Dev 0.08741
Selection: bsFitSuccessful

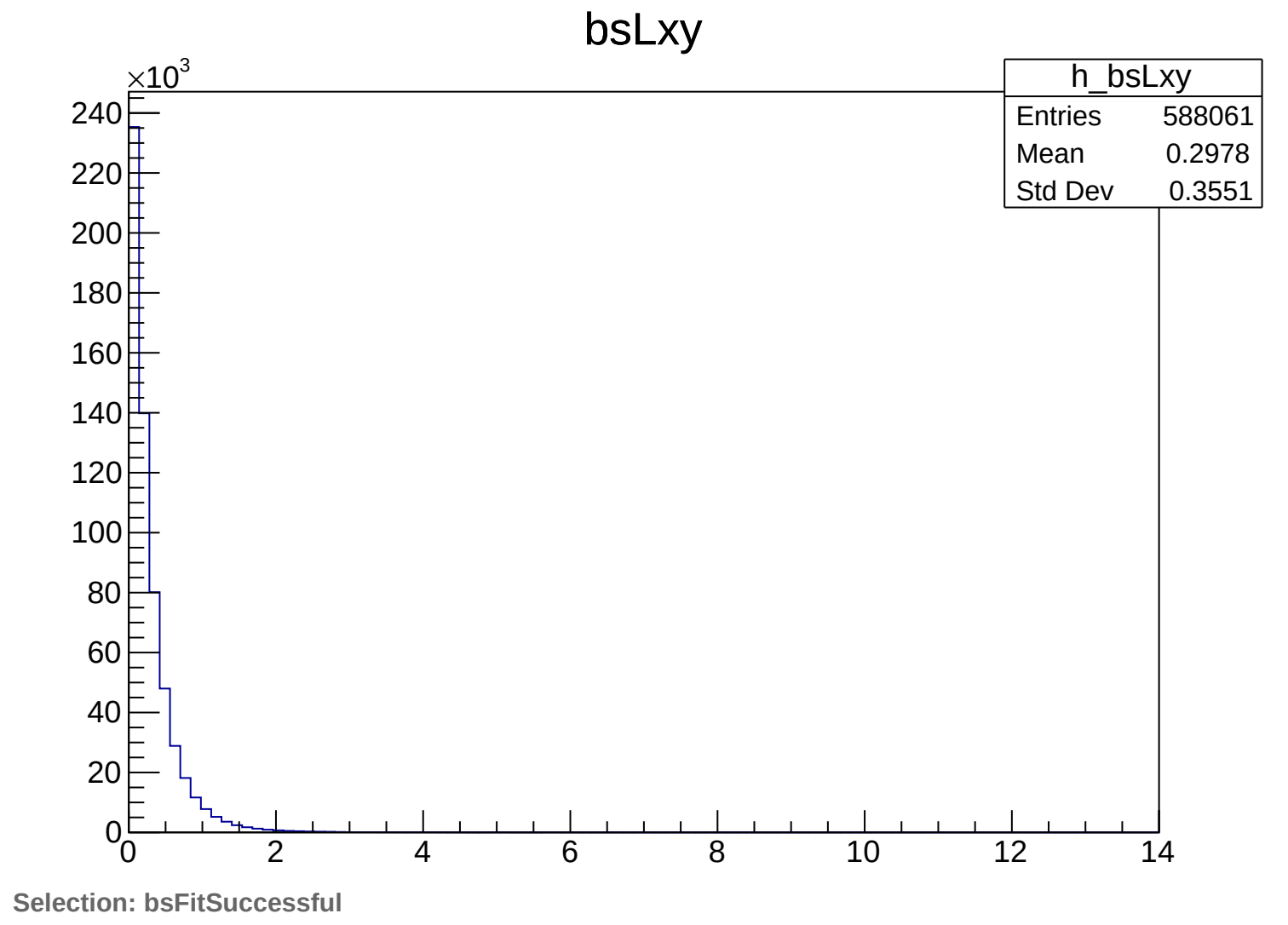

photonGenDeltaR
×10^3
120
100
80
60
40
20
0
0
0.02
0.04
0.06
0.08
0.1
h_photonGenDeltaR
Entries 588049
Mean 0.008966
Std Dev 0.01197
Selection: bsFitSuccessful

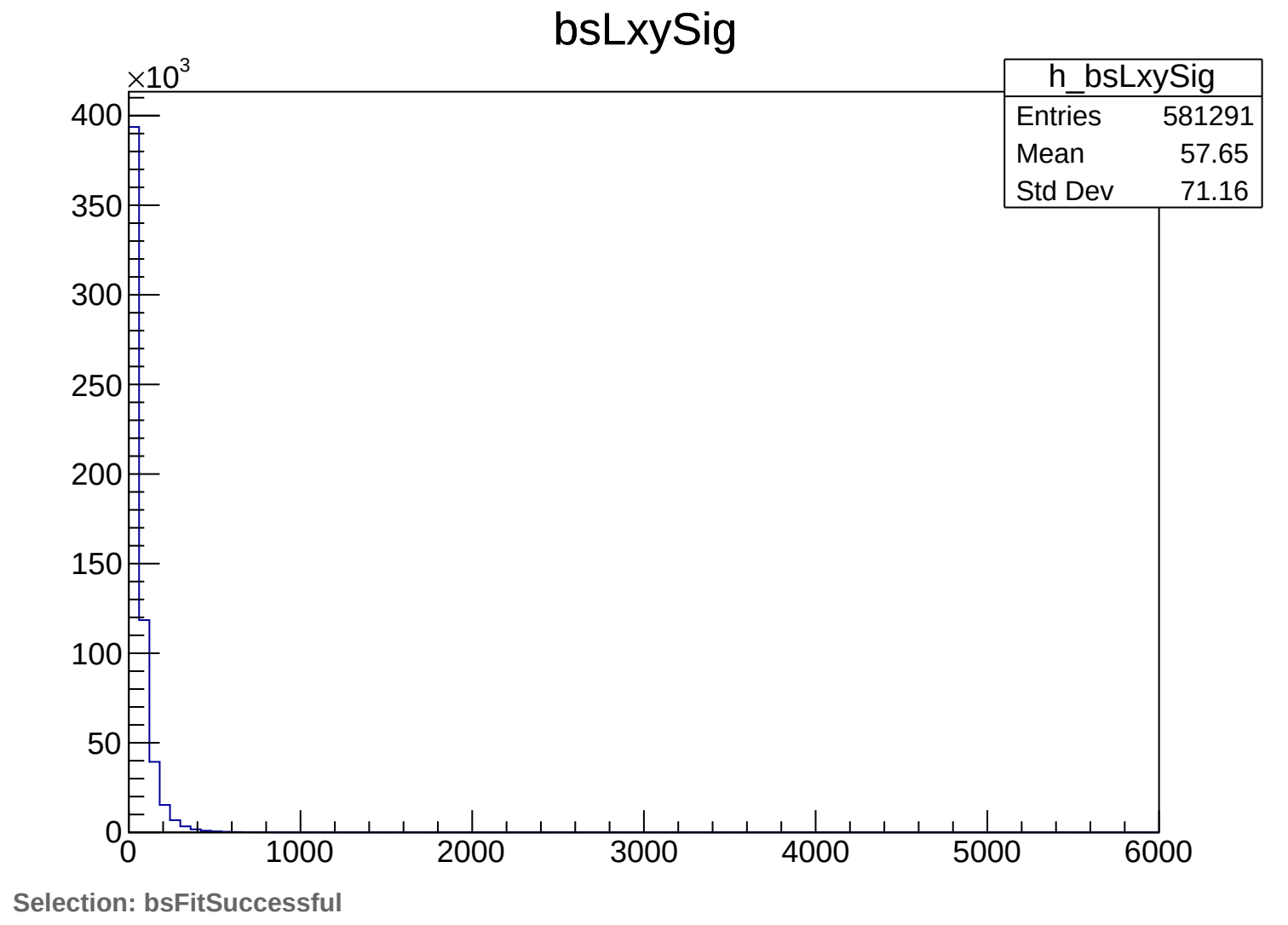

bsVtxProb
h_bsVtxProb
Entries 588061
Mean 0.4416
Std Dev 0.3894
×10^3
140
120
100
80
60
40
20
0
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful

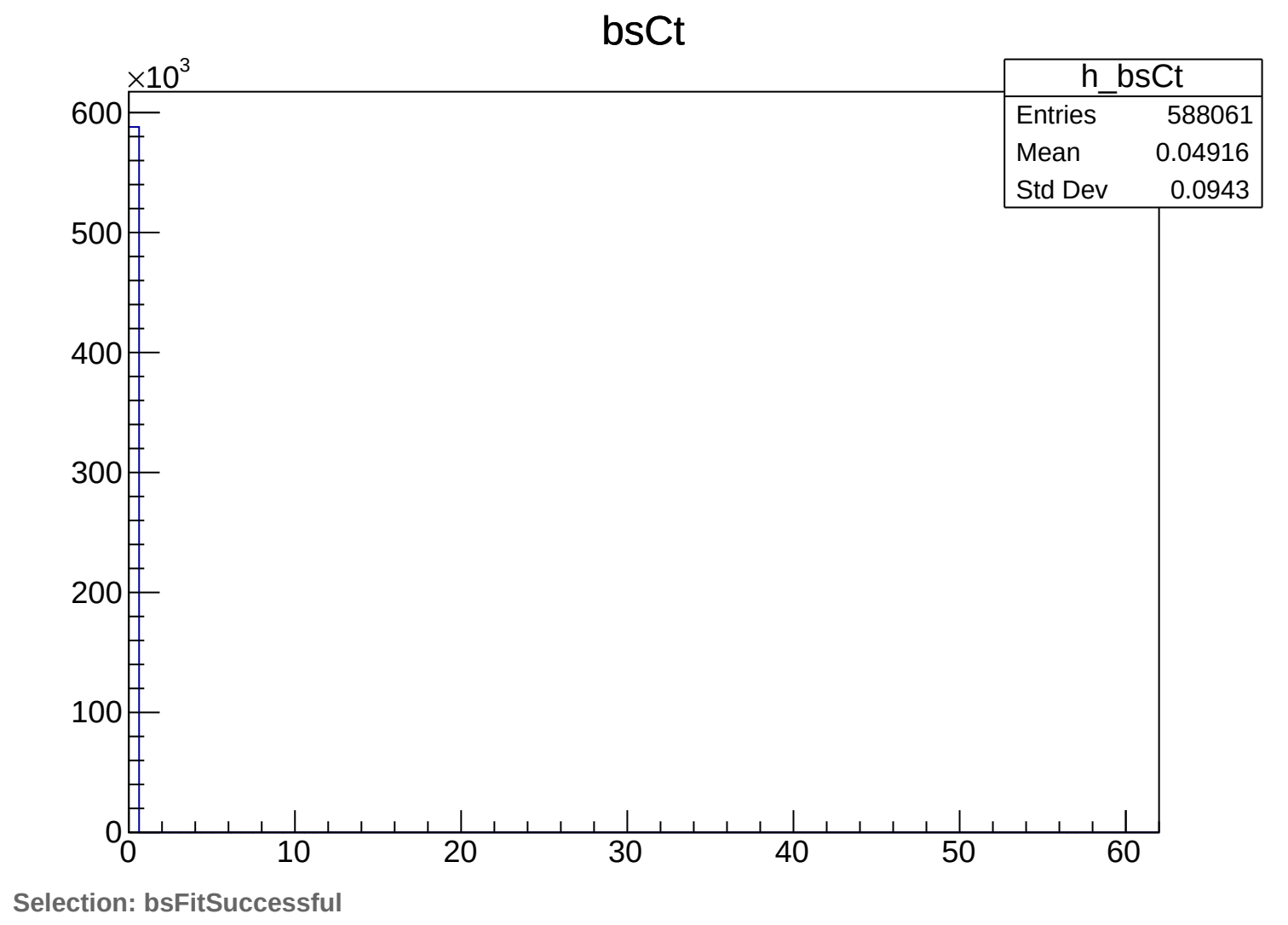

bsVtxX
×10^3
h_bsVtxX
Entries 588061
Mean 0.1017
Std Dev 0.329
300
250
200
150
100
50
0
−10
−5
0
5
10
Selection: bsFitSuccessful

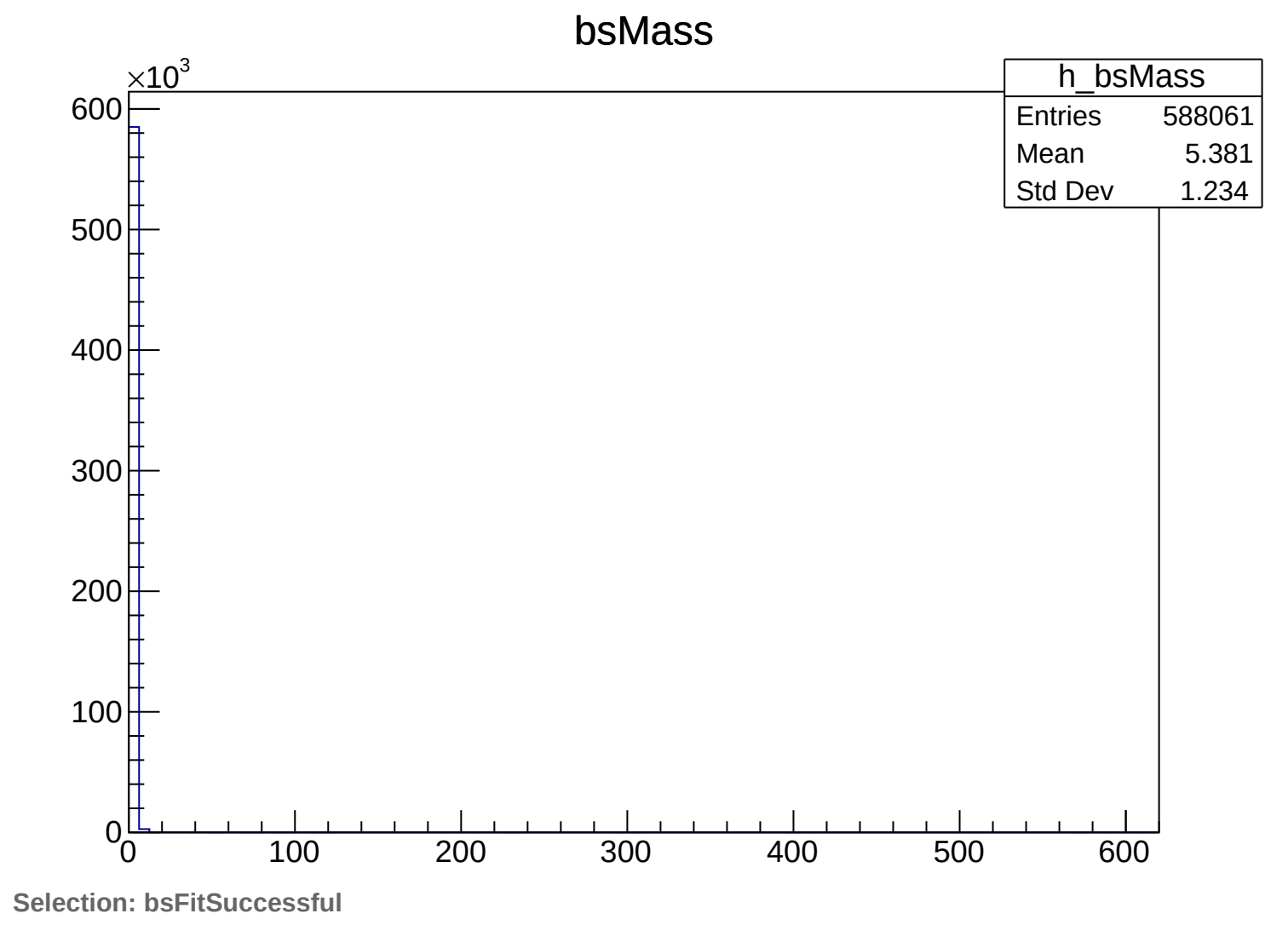

bsVtxY
×10^3
300
250
200
150
100
50
0
−10
−5
0
5
10
h_bsVtxY
Entries 588061
Mean −0.0167
Std Dev 0.3264
Selection: bsFitSuccessful

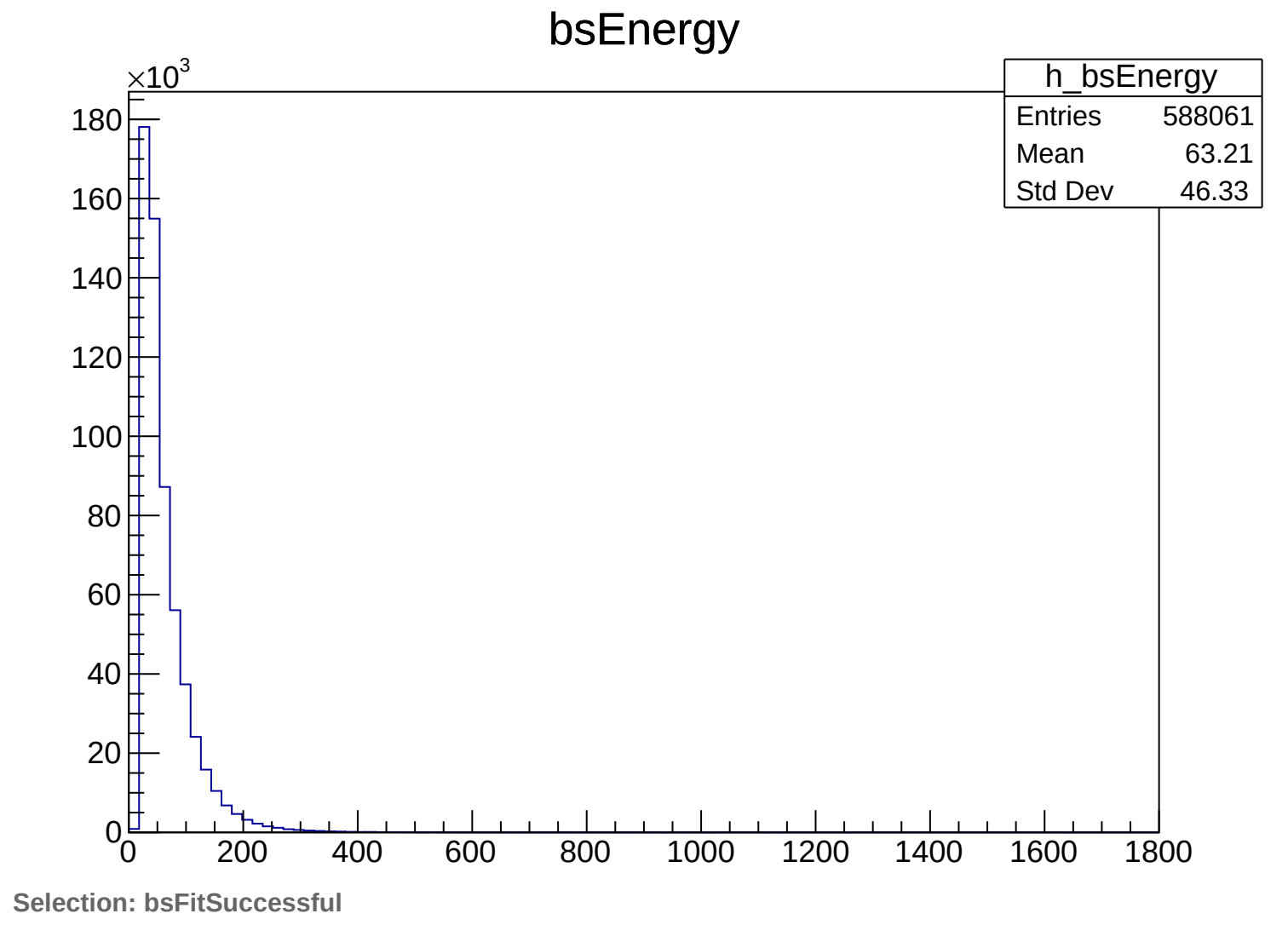

bsVtxZ
h_bsVtxZ
Entries 588061
Mean 1.28
Std Dev 3.88
45000
40000
35000
30000
25000
20000
15000
10000
5000
0
–30
–20
–10
0
10
20
30
Selection: bsFitSuccessful

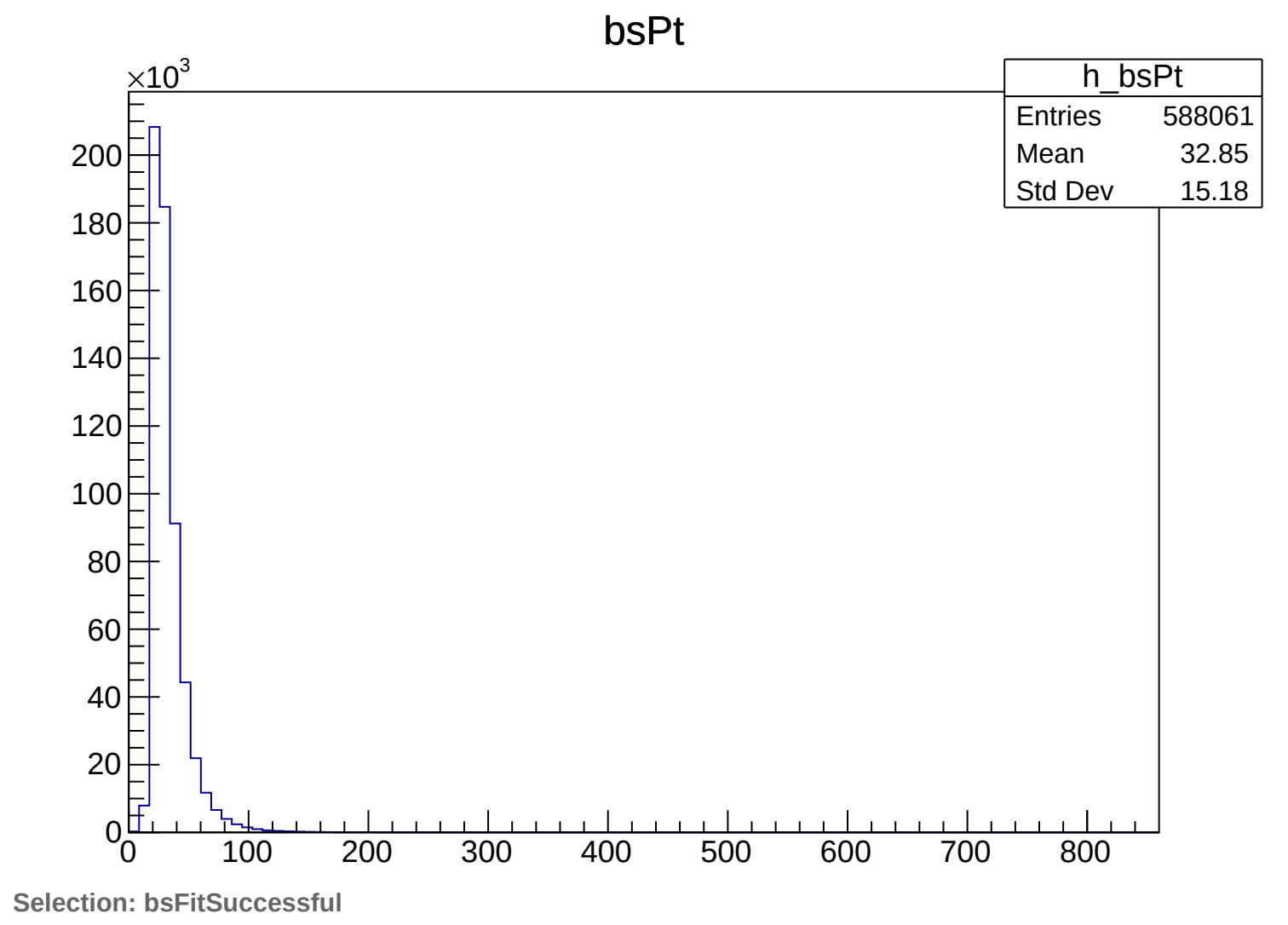

bsLxy
×10^3
h_bsLxy
Entries 588061
Mean 0.2978
Std Dev 0.3551
240
220
200
180
160
140
120
100
80
60
40
20
0
0
2
4
6
8
10
12
14
Selection: bsFitSuccessful

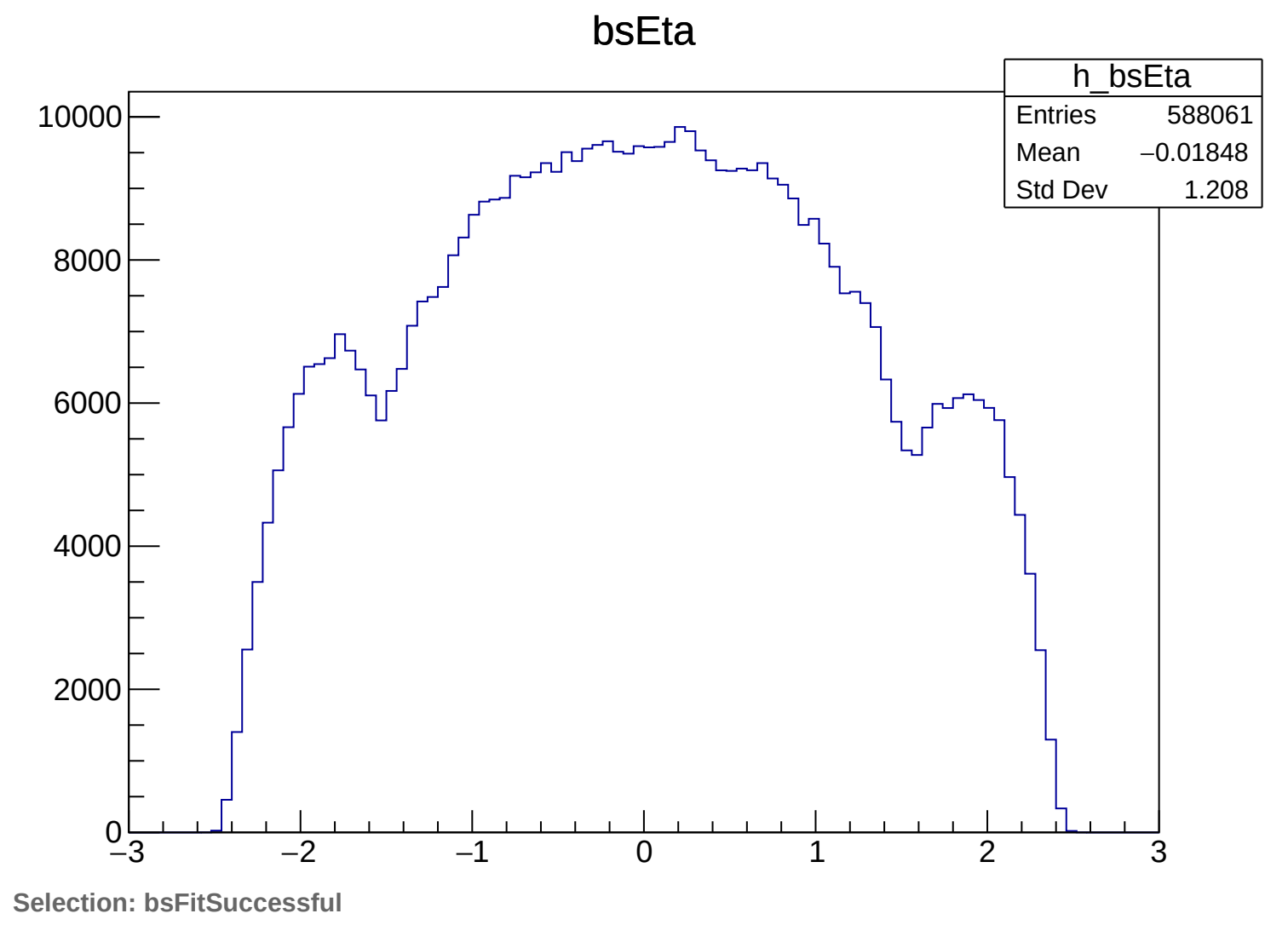

bsLxySig
×10^3
400
350
300
250
200
150
100
50
0
0
1000
2000
3000
4000
5000
6000
h_bsLxySig
Entries 581291
Mean 57.65
Std Dev 71.16
Selection: bsFitSuccessful

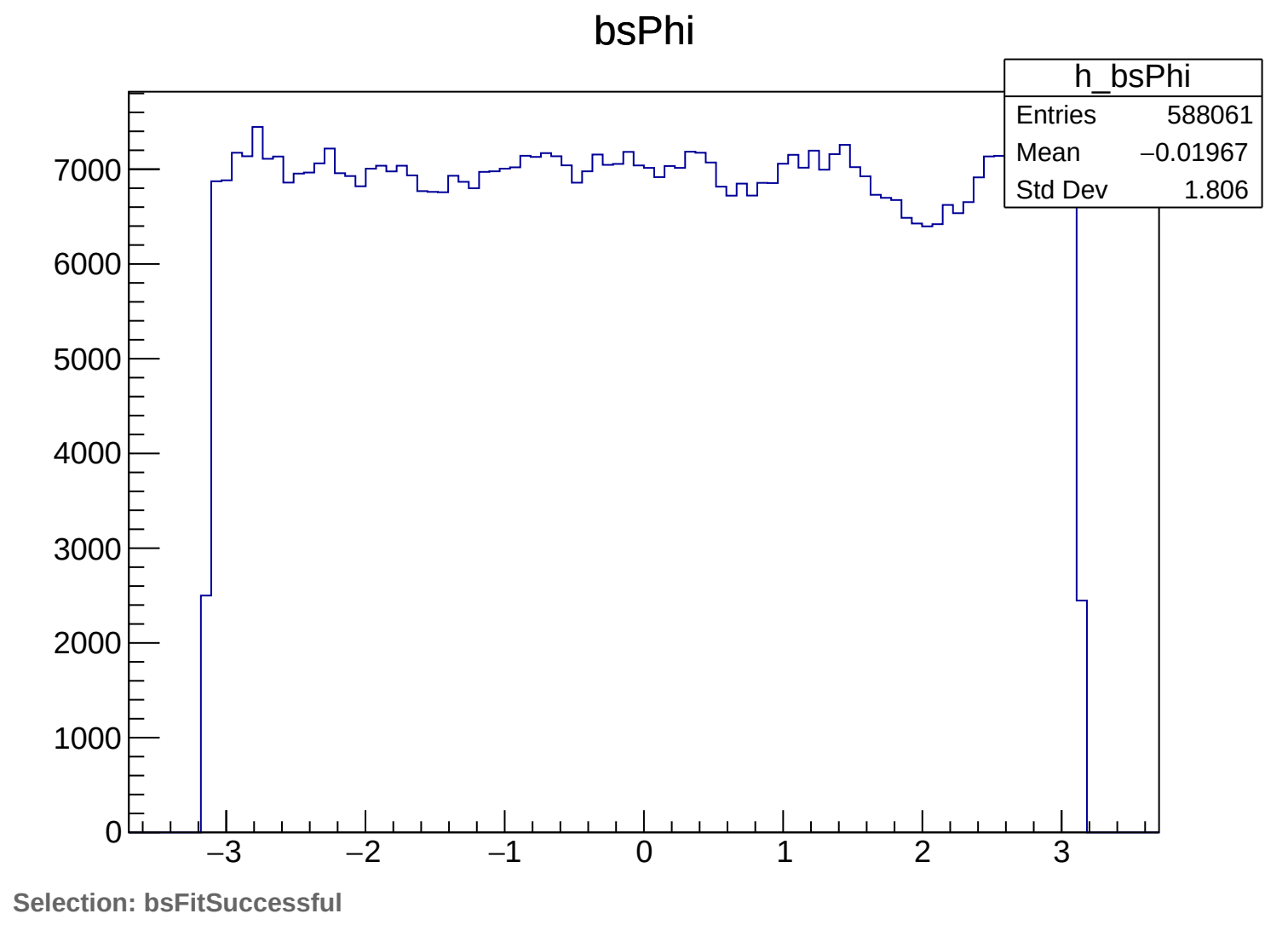

bsCt
h_bsCt
Entries 588061
Mean 0.04916
Std Dev 0.0943
×10^3
600
500
400
300
200
100
0
0
10
20
30
40
50
60
Selection: bsFitSuccessful

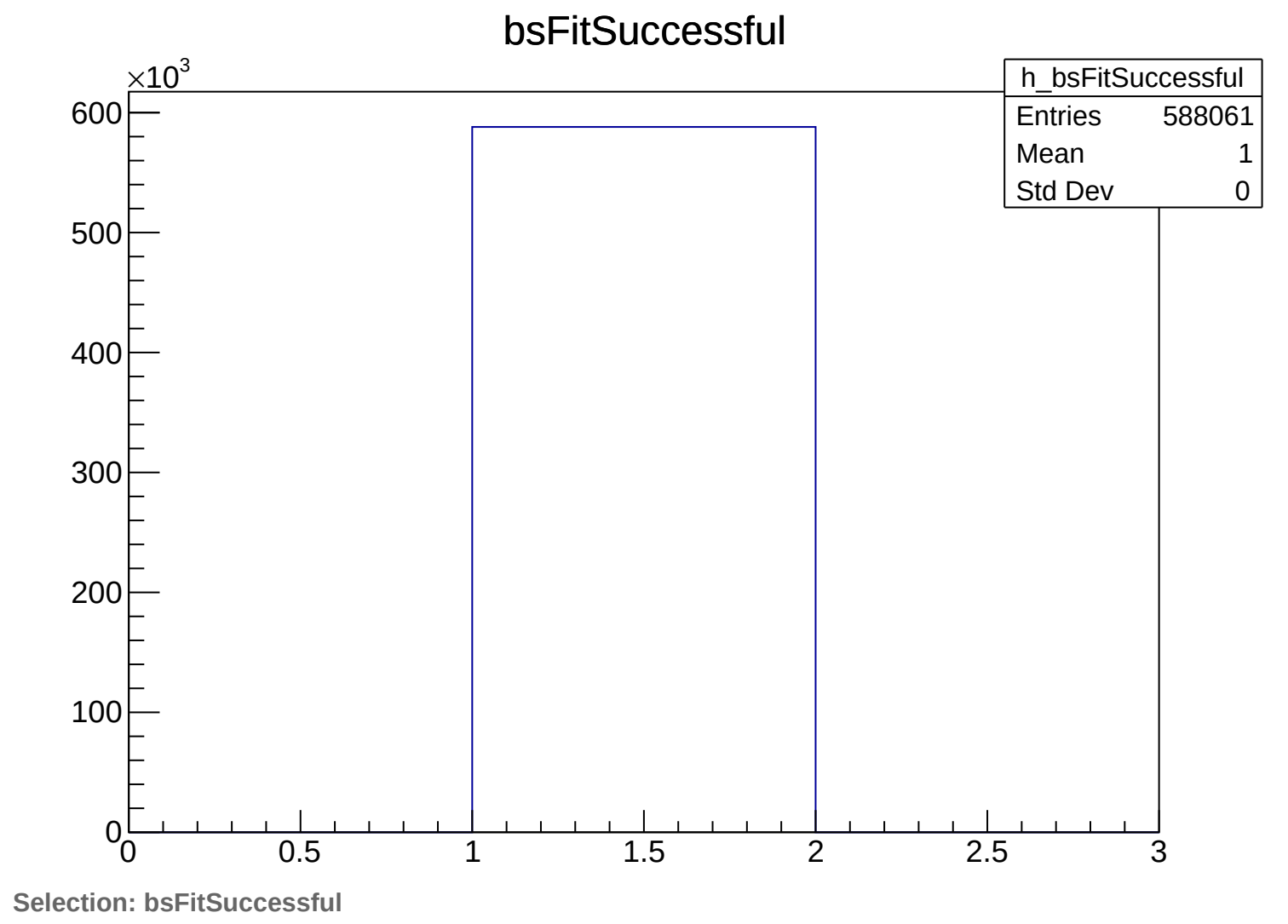

bsMass
×10^3
600
500
400
300
200
100
0
0
100
200
300
400
500
600
h_bsMass
Entries 588061
Mean 5.381
Std Dev 1.234
Selection: bsFitSuccessful


bsEnergy
×10^3
h_bsEnergy
Entries 588061
Mean 63.21
Std Dev 46.33
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
1400
1600
1800
Selection: bsFitSuccessful


bsPt
×10^3
h_bsPt
Entries 588061
Mean 32.85
Std Dev 15.18
200
180
160
140
120
100
80
60
40
20
0
0
100
200
300
400
500
600
700
800
Selection: bsFitSuccessful


bsEta
h_bsEta
Entries 588061
Mean −0.01848
Std Dev 1.208
10000
8000
6000
4000
2000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful


bsPhi
h_bsPhi
Entries 588061
Mean −0.01967
Std Dev 1.806
7000
6000
5000
4000
3000
2000
1000
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful


bsFitSuccessful
h_bsFitSuccessful
Entries 588061
Mean 1
Std Dev 0
×10^3
600
500
400
300
200
100
0
0
0.5
1
1.5
2
2.5
3
Selection: bsFitSuccessful